

Economics of Money and Banking

Lecture notes

based on the course by Perry G. Mehrling
(Barnard College, Columbia University; Coursera and INET)

Personal study notes of Alessio Martini

These notes are a personal, non-commercial study aid derived from the video lectures of *Economics of Money and Banking* by Perry G. Mehrling, filmed at Barnard College (Columbia University) and published by ColumbiaLearn on YouTube, on Coursera, and by the Institute for New Economic Thinking. Lecture content is paraphrased and re-derived from the (automatic) transcripts of the videos, not transcribed; any mistakes are mine. All rights on the original lectures belong to their authors.

How to use these notes

Each chapter covers one *lecture* of the course (22 in all), and its sections follow the short video segments into which every lecture was cut. Every chapter opens with a **Sources** box listing the segments (each one a link to its video), the reading and the newspaper article discussed in class. Part **I** (lectures 1–12) is the “Introduction” of the course: the payment system and the money market. Part **II** (lectures 13–22) extends the same “money view” to foreign exchange and capital markets.

Boxes.

- **Definitions** (blue), **propositions** (red), **exercises** and **remarks** are marked with a coloured bar on the left. A definition box titled *New concepts* marks the first time a concept appears.
- **Key formulas** boxes collect what to remember.
- **Intuition** boxes give the reading of a mechanism in a physicist’s language.
- **In the markets** boxes say where a mechanism shows up in practice.
- **Case study** boxes summarise the *Financial Times* article that opens most lectures, or a historical episode.
- **Reading** boxes summarise the assigned reading of the week.
- **Beyond the core** blocks (grey bar on the left, smaller type) hold material that can be skipped on a first reading.

Balance sheets. The basic tool of the course is the balance sheet, drawn as a “T-account”: assets on the left, liabilities (and net worth) on the right. A transaction is written as the *changes* it makes to the balance sheets of everyone involved:

Bank		Borrower	
+Loan	+Deposit	+Deposit	+Loan

The right column. The page is laid out for a screen: the text is always on the left and a column on the right holds **examples**, **course notes**, **history** boxes and **asides** (curiosities), your **notes** with their **answers**, and **feedback** on your solutions.

Video timestamps. A tag such as [▶ L2.5@3:10](#) means: lecture 2, video segment 5, at minute 3:10 of that segment. Every tag is a link to the video at that second.

Your annotations. While reading, write directly in the chapter files. The macro `\note{...}` prints a comment in the right column next to the text where it is written: a doubt, an idea, a reminder. When a doubt is answered, the note and its answer become one box in the right column. Requests about the notes themselves are collected in the chapter *Requests about these notes*, which also lists how read and unread notes are marked.

Requests about these notes

Your margin notes are of two kinds: notes on the *content* (a doubt, an idea about a mechanism), which stay next to the text they refer to, and requests about the *notes themselves* (layout, boxes, conventions), which do not depend on where they were written. The second kind is collected here, with what was done about it, so that the chapters read more smoothly. Write new ones anywhere with `\note{...}` and they will be moved here, or add them directly at the end of this list.

How the margin notes are marked.

- `\note{...}`: a *new* note, not yet read. “Read my notes” means these, and only these.
- A note that has been answered becomes one box with the question on top and a dark yellow **ANSWER** bar below it (`\begin{answer}{question}[topic] ... \end{answer}`). A note with several questions gets one ANSWER bar per question. A question asked a second time gets a short answer that links to the first one.
- `\note*{...}`: a request that has been carried out (green DONE tag).
- `\note+{...}`: a comment that has been read and needs no answer (blue SEEN tag).
- `\feedback{...}`: a comment on one of your own solutions (indigo).

The list

DONE

All chapters, exercises. esercizi brevi/a sé stanti lungo il testo, quelli collegati o complessi alla fine

WHAT WAS DONE

Standalone exercises now follow the section they practise (right before the next heading); connected or integrative exercises stay together at the end of the chapter, in the original order. In chapters where none were left, the final Exercises section was removed.

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Part I

Introduction: payments and the money market

Chapter 1

The Four Prices of Money

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L1.1 The Big Picture	19:26	Where the course comes from: Stigum, the <i>Financial Times</i> , the history of monetary and financial thought; the “money view”; dialogue between central banking and modern finance.
L1.2 Prerequisites?	7:22	IS/LM and the Fisher diagram: the two textbook ways of teaching money and banking, and why this course starts elsewhere.
L1.3 What is a Bank, a Shadow Bank, a Central Bank	12:10	Balance sheets as the analytical tool; solvency versus liquidity; shadow banks; the Fed’s balance sheet in 2012.
L1.4 Central Themes	13:09	Plan of the course; the four prices of money; Bagehot’s <i>Lombard Street</i> ; “all banking is a swap of IOUs”.
L1.5 Reading: Allyn Young	3:10	Presentation of the first reading.

Files. Mehrling’s lecture notes [materials/mehrling-notes/Lec-01-The-Four-Prices-of-Money.pdf](#) [Notes 1 p.1]; captions [materials/week1/srt/1-*.srt](#); transcripts [materials/transcripts/L01-P0*.txt](#). **Reading.** Allyn Young, four chapters from *The Book of Popular Science* (1924), file [AllynYoung.pdf](#), with the study questions [AllynYoung\(study\).pdf](#); discussed in [section 1.5](#) and again in lecture 3.

Motivation

The first lecture is an orientation. It says where the course stands among the ways money and banking are usually taught, introduces the one analytical tool that will be used everywhere (the balance sheet), and names the four prices that the whole course tries to explain:

- **par**, the price of one kind of money in terms of another kind today;
- **interest**, the price of money today in terms of money tomorrow;
- the **exchange rate**, the price of domestic money in terms of foreign money;
- the **price level**, the price of money in terms of goods.

The thread of the lecture is that the standard textbooks describe the banking system of 1950, while the system of today is *global* and *market-based*, and the problems it runs into are problems of *liquidity*, not only of solvency. For a physicist, the useful attitude is that of an experimentalist entering a new lab: the course does not start from a model and apply it, but starts from what practitioners actually do and tries to extract the regularities [► L1.2 @ 4:41].

1.1 The big picture

Where the knowledge comes from. Mehrling has never traded in the money markets: his knowledge is “book learning”, but from practitioners’ books, above all Marcia Stigum’s *The Money Market*, and from the *Financial Times*, rather than from academic journals [► L1.1 @ 0:29]. His academic work is in the history of monetary and financial economics, in two books: *The Money Interest and the Public Interest*, a history of American monetary thought, and *Fischer Black and the Revolutionary Idea of Finance*, a history of the rise of modern finance [► L1.1 @ 0:50]. These are the intellectual foundations of the course, and they explain why the readings include authors such as Allyn Young, Walter Bagehot, Ralph Hawtrey and Edward Shaw [► L1.1 @ 1:10].

A course about the present. Although its foundations are historical, this is not a course in the history of economic thought: it uses that history as a frame to understand the world as it is now [► L1.1 @ 1:31]. The big fact to understand is *financial globalization* (or “global financialization”). The lesson of the history of monetary thought is that every generation has to work out for itself the world in front of it: some ideas carry over from the past, but the system is not constant, and today it is “a strange new world” [► L1.1 @ 1:51]. This is why Mehrling does not use the standard money and banking textbooks: they evolved from the textbooks of 1950, and a bank in 1950 was a different thing from a bank today [► L1.1 @ 2:53]. Better to put them in the library and start again from the real world, accepting that the result is a little “looser” but attached to reality [► L1.1 @ 3:14].

Listening to the bankers. Being attached to reality means listening to practitioners and trying to understand their world [► L1.1 @ 3:35]. Practitioners know their own piece very well (repo, Eurodollars, . . .) but have no picture of how the system as a whole works; an academic can add value by noticing the patterns and the connections between the pieces [► L1.1 @ 3:55]. This is not easy: there will be no “take the first derivative and set it equal to zero” in this course [► L1.1 @ 4:36].

INTUITION: Reading Stigum as an anthropologist

Stigum’s *The Money Market* is the desk reference of people who trade the money markets, “the bible” of the field [► L1.1 @ 4:56]. Mehrling’s advice is to read it as an anthropologist reads the sacred text of a foreign tribe [► L1.1 @ 5:58]: not to learn to trade, but to understand how the members of the tribe think about their world. Read for every detail, it “can blow your mind”; read for the way of thinking, it is a window into a rather special world [► L1.1 @ 6:19].

Reading the Financial Times. In a sense the whole course is about learning to read the *Financial Times* (FT) and to “translate it into English” [► L1.1 @ 7:21]; the final exam contains exactly that exercise, an article to be explained to one’s roommate [► L1.1 @ 7:42]. Every

ASIDE: Mehrling’s books

Perry Mehrling (Barnard College, Columbia University, at the time of the lectures; later Boston University) wrote *The Money Interest and the Public Interest: American Monetary Thought, 1920–1970* (Harvard University Press, 1997), *Fischer Black and the Revolutionary Idea of Finance* (Wiley, 2005) and *The New Lombard Street: How the Fed Became the Dealer of Last Resort* (Princeton University Press, 2011), the book that accompanies this course.

lecture after this one starts with an article of the day. Three examples from the paper of that day show the kind of question the course is about [► L1.1 @ 8:23]:

- (a) German banks split over the role of the European Central Bank (ECB): should the ECB buy the government debt of Spain or Italy to support its price? What does it even mean for a central bank to buy the debt of a state [► L1.1 @ 8:43]?
- (b) Chinese steel mills brace for a slowdown in demand. The monetary angle: in China, inventories of iron ore had been used as *collateral* for borrowing; as the banks call in the loans, the borrowers dump the collateral, the beginning of a financial crisis [► L1.1 @ 9:04].
- (c) Foreign exchange brokers back plans of the trading platform EBS to rein in “predatory” traders: a battle between algorithmic high-frequency traders (“cyborgs”) and human dealers, who lose money to them [► L1.1 @ 9:45].

1.1.1 The money view and its ancestors

Asked which school of economics he belongs to (saltwater or freshwater, Keynesian or monetarist), Mehrling answers that he identifies with the *money view*, a name he invented himself [► L1.1 @ 10:47]. It has two lines of intellectual ancestors [► L1.1 @ 11:08], shown in [fig. 1.1](#).

- **The American line.** It starts with Allyn Young, because central banking in the United States starts in his time (the Fed was founded in 1913), when there was no tradition of central banking thought to draw on [► L1.1 @ 11:49]. Then Alvin Hansen and Edward Shaw (the three generations of Mehrling’s first book), and Hyman Minsky, who would have been the missing last chapter of that book [► L1.1 @ 12:09].
- **The British line.** Britain had a central bank for a long time and was the centre of the world economy for a long time, so it has a long tradition of thinking about central banking [► L1.1 @ 12:50]: Walter Bagehot, Ralph Hawtrey, Richard Sayers (who taught at the London School of Economics, LSE) and Charles Goodhart [► L1.1 @ 13:10]. Almost all of them talked to bankers and central bankers: Goodhart spent most of his career at the Bank of England [► L1.1 @ 13:30], and to teach at the LSE is to be a mile from the City of London, as teaching in Manhattan is to be a mile from Wall Street [► L1.1 @ 13:51].

Some of them called themselves Keynesians (Hansen, in a way Minsky), others leaned towards monetarism (Shaw, Hawtrey); Mehrling reads them as one continuous stream of thought that kept its eye on the real world of its own time [► L1.1 @ 14:11].

ASIDE: Stigum

Marcia Stigum (1934–2005) first published *The Money Market: Myth, Reality, and Practice* in 1978; the fourth edition (with Anthony Crescenzi, McGraw-Hill, 2007) is the one mentioned in class, with a new chapter on Eurodollars [► L1.1 @ 5:16]. Mehrling recalls that when he started teaching the course, Stigum was the whole text, with no lecture notes.

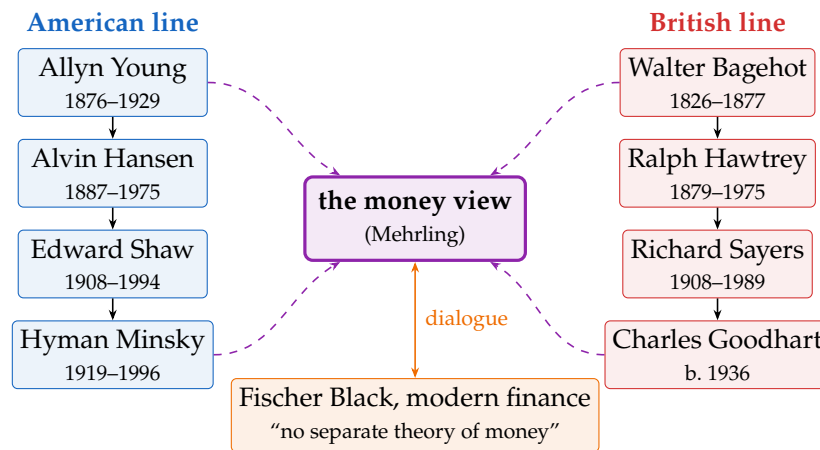


Figure 1.1: The intellectual ancestors of the money view [► L1.1 @ 11:28], and its opponent in the dialogue that the course sets up [► L1.1 @ 16:14].

1.1.2 The challenge of modern finance

Mehrling also learned modern finance, through his biography of Fischer Black. Finance holds that there is no such field as monetary economics [► L1.1 @ 14:53]: in an efficient market every asset trades at its efficient price, so there is nothing for liquidity to distort. In “straight-ahead” finance there is no *fire sale*: if a bank had to sell its loans, a price just a tiny bit below their true value would be enough for someone to snatch them up, so prices never move far from value and there are no runs on banks [► L1.1 @ 15:13]. The logical end of this view is that there is no reason for central banks to exist [► L1.1 @ 15:54].

The course is, in Mehrling’s words, a *dialogue* between the classical central banking perspective and the modern finance perspective, two ideas that seem contradictory, rubbed against each other to make sparks for new economic thinking [► L1.1 @ 16:14]. The opposition is not Keynesian versus monetarist [► L1.1 @ 16:55].

Where Mehrling stands. Pressed by the student, he states the position he will defend for the rest of the course [► L1.1 @ 17:38]: there *is* a role for central banks in setting bounds on the system, a modern version of the role that Bagehot had in mind in 1873 (“lend freely at a high rate against good security”) [► L1.1 @ 17:59]. That rule does not quite work in the modern world, and the subtitle of his book, *How the Fed Became the Dealer of Last Resort*, signals the update. Financial markets do not work perfectly; they run into problems, and banking is all about these problems, which are *liquidity* problems rather than problems of asymmetric information [► L1.1 @ 18:20]. The most compelling work on them comes not from academic journals, “fighting old battles”, but from the Bank of England and the Bank for International Settlements (BIS), the central bankers’ bank [► L1.1 @ 19:01].

1.2 Prerequisites: two textbook approaches

The formal prerequisites are intermediate macroeconomics and intermediate microeconomics [► L1.2 @ 0:08]. The course makes little use of them, but other money and banking courses start from them, and at the end of the course the money view will be connected to both [► L1.2 @ 0:29]. It is worth seeing what they are, because they are what the course is *not*

doing.

Intermediate macro: IS/LM. The macro story determines national income Y and the *nominal* interest rate i at the crossing of two curves ▶ L1.2 @ 1:15]:

- the IS curve (“investment = saving”): the pairs (Y, i) at which the market for goods clears; a higher rate depresses investment, so the curve slopes down;
- the LM curve (“liquidity preference = money supply”): the pairs at which the demand for money equals the supply; higher income raises the demand for money, so the rate must rise to keep it equal to a given supply, and the curve slopes up.

Monetary policy is then a shift of the LM curve. A whole course can be built on the foundations of the LM curve, the Taylor rule, or the argument that with inflation targeting the LM curve should be horizontal ▶ L1.2 @ 1:35]. This course instead questions the very notion of a stable money demand curve and money supply curve, at a more fundamental level ▶ L1.2 @ 1:55].

Intermediate micro: the Fisher diagram. The micro story determines the *real* interest rate in a two-period general equilibrium ▶ L1.2 @ 2:16], shown in fig. 1.2. With wealth W in units of today’s goods, consumption c_1 today and c_2 tomorrow, and real rate r , the budget line is

$$c_1 + \frac{c_2}{1+r} = W, \quad \frac{dc_2}{dc_1} = -(1+r), \quad (1.1)$$

so its slope is $-(1+r)$ in the (c_1, c_2) plane (Mehrling’s notes write $-1/(1+r)$, which is the slope with the axes exchanged). Maximising utility $u(c_1, c_2)$ on this line gives the first-order condition

$$\frac{\partial u / \partial c_1}{\partial u / \partial c_2} = 1+r,$$

the tangency between an indifference curve and the budget line ▶ L1.2 @ 2:36]. Two people with the same wealth and the same production possibilities but different tastes choose different points: one consumes more than she produces today and borrows, the other consumes less and lends ▶ L1.2 @ 2:58]. Financial markets allow this inter-temporal trade, and the real interest rate is the price that clears it ▶ L1.2 @ 3:19]. The diagram is named after Irving Fisher ▶ L1.2 @ 3:39]. A money and banking course built on it treats banks as intermediaries between savers and investors, needed because of some imperfection such as asymmetric information ▶ L1.2 @ 4:00].

HISTORY: Bagehot’s *Lombard Street* (1873)

Walter Bagehot (1826–1877), editor of *The Economist*, published *Lombard Street: A Description of the Money Market* in 1873, after the panic that followed the failure of the discount house Overend, Gurney & Co. in 1866. His prescription for the Bank of England in a panic is usually summarised as “lend freely, at a high rate, against good collateral”; in his own words, loans “should only be made at a very high rate of interest” and “on all good banking securities, and as largely as the public ask for them”. Lombard Street, in the City of London, takes its name from the Lombard (northern Italian) merchants and bankers who settled there in the Middle Ages.
References: W. Bagehot, *Lombard Street*, London: Henry S. King, 1873, ch. VII; Mehrling, *The New Lombard Street*, 2011, ch. 1.

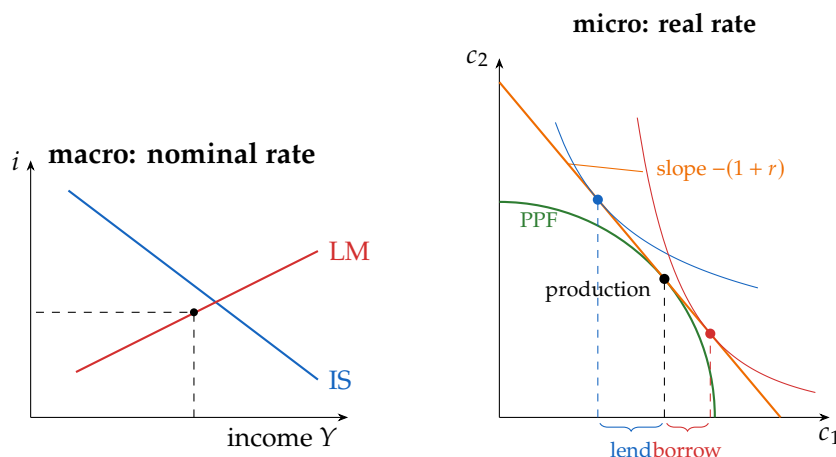


Figure 1.2: The two textbook starting points [► L1.2 @ 1:15]. Left: IS/LM fixes income and the nominal interest rate. Right: the Fisher diagram; two agents produce at the same point of the production possibility frontier (PPF) and trade along the budget line to the tangency with their own indifference curve: the one who consumes less today lends, the other borrows.

The strategy of this course. The course does not start from an economic apparatus and ask how to apply it to banking: it starts from the concrete realities of banking and tries to draw from them, inductively, how the system works and what regularities it has [► L1.2 @ 4:41]. Mehrling warns that the first two weeks feel disorienting, because there is nothing familiar to grab onto; it comes together later [► L1.2 @ 5:02]. His own way of reading the FT is to ask what Bagehot, Minsky or Fischer Black, whose biographies he wrote, would make of it [► L1.2 @ 6:23].

Exercise 1.1 (The Fisher diagram). An agent has endowment $(y_1, y_2) = (100, 0)$ and utility $u = \ln c_1 + \beta \ln c_2$.

- Find the optimal (c_1, c_2) as a function of r and β , and the amount lent.
- A second agent has endowment $(0, 110)$ and the same utility. For $\beta = 1$, find the real rate r that clears the market for loans between the two.

ASIDE: The shelf in the library

Monetary economics is a field of big books: to get anywhere one has to write a treatise. Mehrling chose it after seeing the shelf of those books in the library and thinking that a life spent reading them from left to right would be the best life ever [► L1.2 @ 7:04].

1.3 What is a bank, a shadow bank, a central bank

The analytical apparatus of the course is, largely, the balance sheet [► L1.3 @ 0:07]. Not balance sheets as accounting law (what GAAP prescribes), but as a structure that forces rigorous thinking about the economy [► L1.3 @ 0:27]. We introduce three new concepts.

Definition 1.1 (New concepts: balance sheet, solvency, liquidity).

- A *balance sheet* lists what an agent owns, its *assets* (left side), and what it owes, its *liabilities* (right side). *Net worth* (equity) is the balancing item,

$$\text{net worth} = \text{assets} - \text{liabilities},$$

so that the two sides always have the same total. Drawn as a “T-account”, the balance sheet has assets on the left and liabilities on the right [► L1.3 @ 0:49].

- An agent is *solvent* if the market value of its assets exceeds the market value of its liabilities, i.e. net worth is positive [► L1.3 @ 2:37]. (From Latin *solvere*, to loosen, hence to pay off a debt.)
- An agent is *liquid* if it can make the payments it has to make when they come due: it has cash, or can obtain it [► L1.3 @ 3:23]. (From Latin *liquidus*, flowing: liquid assets flow easily into means of payment.)

ASIDE: GAAP

GAAP, *Generally Accepted Accounting Principles*, are the accounting standards set in the United States by the Financial Accounting Standards Board. The course ignores them: its balance sheets are a tool of economic analysis, not of legal reporting.

A bank. A generic bank holds loans, securities (for example Treasury bills) and cash reserves, and owes deposit accounts and other borrowing (bonds, short-term money market borrowing); net worth balances the two sides [► L1.3 @ 1:09]. “Other borrowing” will be a very big part of the course [► L1.3 @ 1:56].

Bank	
Loans	Deposits
Securities	Other borrowing
Cash reserves	Net worth

Two issues arise [► L1.3 @ 2:17]. The first is *solvency*: are the assets worth more than the liabilities? Much of the discussion after 2008, and in the euro crisis, is about this; perhaps entire national banking systems are insolvent [► L1.3 @ 2:37]. The second, which is what makes banks different from other corporations and which the course puts first, is *liquidity* [► L1.3 @ 2:58]. Deposits are payable on demand: when a depositor pays someone at another bank, the bank must deliver cash reserves. Does it have them? If not, can it acquire them, by borrowing in the money market, by borrowing from the central bank (the Fed’s “discount window”), by selling or *repo*-ing a security [► L1.3 @ 3:44]? Or must it sell its loans at a *fire-sale* price, below their value?

INTUITION: Liquidity can become solvency

A fire sale turns a liquidity problem into a solvency problem: assets sold below their value reduce net worth, and enough of them can make it negative [► L1.3 @ 4:04]. Most economists approach banking from solvency; the course approaches it from liquidity, because that is how bankers think: they face the liquidity constraint every day, when they have to clear their accounts [► L1.3 @ 4:25]. This is why the course starts with the payment system. A rough analogy: solvency is a statement about stocks (values at one instant), liquidity a statement about flows (cash in versus cash out, every day). A system can have enough “energy” in total and still fail because it cannot deliver it at the rate required.

A shadow bank. Mehrling’s research group had just finished a paper on shadow banking, discussed at the end of the course (lecture 21) [► L1.3 @ 4:45]. A shadow bank, in the sense of the course, is an entity that holds, for example, residential mortgage-backed securities (loans turned into a kind of bond), strips most of the risk out of them with an interest rate swap and a credit default swap, and uses what is left, a short-term and relatively risk-free asset, as collateral to borrow in the *wholesale* money market [► L1.3 @ 5:06]. It borrows not from households but from money market mutual funds and large corporate investors, through repurchase agreements (*repo*), Eurodollar borrowing, asset-backed commercial paper [► L1.3 @ 6:15].

Definition 1.3 (New concepts: shadow bank, wholesale money market). A shadow bank is an entity that does what a bank does (holds long-term credit and finances it with short-term liabilities that its creditors regard as money-like) outside the regulated deposit-taking banking system. Its funding comes from the *wholesale money market*, the market where institutions, not households, lend to each other for short terms, often against collateral. The term “shadow banking” was coined by the economist Paul McCulley (PIMCO) in 2007.

Example 1.2 (A fire sale). A bank has loans 100, reserves 5; deposits 95, net worth 10. Depositors withdraw 20. Reserves cover 5; the bank must raise 15 by selling loans, and in a panic buyers pay only 60% of face value: it sells loans of face value 25. Now loans 75, reserves 0, deposits 75, net worth 0. A run of 20 out of 95 wiped out all the capital, although before it the bank was solvent by 10.

“Shadow bank”

Mortgage-backed security	Money market borrowing:
tranche	repurchase agreements
Interest rate swap	asset-backed commercial paper
Credit default swap	auction rate securities
	Eurodollar borrowing

The instruments on this balance sheet are only named here; each gets its own lecture (repo in lecture 7, Eurodollars in lecture 8, swaps in lectures 19–20). “It’s not rocket science” [► L1.3 @ 6:36]. The point now is that this model of credit became dominant: the amount of credit in the US that ran through it became larger than the amount that ran through traditional banks; the textbooks are all about banks, reality increasingly about shadow banks [► L1.3 @ 6:56]. Again there is an issue of solvency and an issue of liquidity, but liquidity is harder: it means being able to *roll over* the short-term funding. If the money market freezes, the shadow bank cannot roll over and has to liquidate its assets; much of the financial crisis of 2007–2009 was exactly this [► L1.3 @ 7:38]. The regulatory system was built for traditional banks; the Dodd–Frank Act of 2010 “doesn’t touch” shadow banking [► L1.3 @ 8:20].

The central bank is a bank. The central bank is also a bank, at a higher level [► L1.3 @ 8:40]. Its balance sheet is published every week (the Fed’s release H.4.1, “Factors Affecting Reserve Balances of Depository Institutions”). The release shown in class, of 30 August 2012, gives, in round numbers [► L1.3 @ 9:00]:

Federal Reserve, 30 Aug 2012 (\$ billion)

Treasuries (no bills)	≈1,600	Currency in circulation	≈1,000
Agency debt (Fannie Mae, Freddie Mac)		Reserve balances of banks	≈1,500
Mortgage-backed securities	852	Other	
Maiden Lane (AIG, Bear Stearns)			
Central bank liquidity swaps			
Total	≈2,800	Total	≈2,800

- The Fed had sold all its short-term Treasury bills and held only notes and bonds: it was moving into long-term debt [► L1.3 @ 9:00].
- It held mortgage-backed securities, \$852 billion of them, for the first time in its history: a legacy of the crisis [► L1.3 @ 9:21]. The “Maiden Lane” entries are what was left of the rescues of Bear Stearns and AIG.
- *Central bank liquidity swaps* are loans of dollars from the Fed to other central banks: the ECB, the Swiss National Bank, the Bank of Japan [► L1.3 @ 10:03].

- The total was about \$2.8 trillion, against less than \$1 trillion before the crisis: the balance sheet roughly tripled [► L1.3 @ 11:04].
- For each of the three balance sheets the notes ask the same two questions: solvency versus liquidity, and monetary control; and they conclude that the centrality of liquidity is what makes banks special and different from other corporations [Notes 1 p.3].
- On the liability side, currency (“green pieces of paper”) was about \$1 trillion, and the reserve balances that banks hold as deposits at the Fed were \$1.5 trillion, against about \$50 billion before the crisis (as said in class), an order of magnitude change [► L1.3 @ 11:24].

This transformation of the Fed’s balance sheet is what “QE1, QE2, QE3” (quantitative easing) refer to.

INTUITION: The central bank does not “print money”

Textbooks write money demand = money supply, $M^d = M^s$, as if the central bank decided a quantity. Behind the symbols are balance sheets: the central bank is a bank with assets and liabilities, and when it “expands the money supply” it expands *both sides* of its balance sheet at once, for example buying a security (asset) and paying for it by crediting a bank’s reserve account (liability) [► L1.3 @ 10:23]. This is a classic banking operation, and it happens at every level of the system, not only at the top.

Fed		Bank A	
+Security	+Reserves of Bank A	−Security	
		+Reserves	

Exercise 1.2 (Three ways to get a deposit). Using T-accounts for you, your bank and (where needed) a second person and a second bank, record:

- you deposit \$100 of currency;
- you deposit a check for \$100 written by someone with an account at another bank;
- the bank grants you a loan of \$100 by crediting your deposit account.

In which case does the total quantity of deposits in the economy increase? (From the study questions on Young.)

Exercise 1.3 (Liquidity and solvency). A bank has loans 200, securities 50, reserves 10; deposits 230, net worth 30. Depositors withdraw 40.

- The bank sells securities at their market value to pay. Write the new balance sheet. Is the bank solvent? Liquid?
- The securities can only be sold at 80% of their value and the loans at 50%. Which assets does the bank sell first, and what is its net worth afterwards?
- Name two ways the bank could meet the withdrawal without selling any asset.

Exercise 1.4 (Quantitative easing in T-accounts). The Fed buys \$100 billion of mortgage-backed securities from a primary dealer, which holds its account at Bank B. Write the changes on the balance sheets of the Fed, Bank B and the dealer. Which item of the Fed's liabilities grows? Relate your answer to the reserve balances of [section 1.3](#).

Exercise 1.5 (A shadow bank in a freeze). A shadow bank holds mortgage-backed securities worth 100, financed by overnight repo of 95 and capital of 5. One morning lenders agree to roll over only 80.

- (a) How much must the shadow bank raise, and from what?
- (b) If buyers pay only 90% of value for the securities, is the shadow bank still solvent after the sale?
- (c) Compare with the bank of [definition 1.1](#): which liability of a traditional bank plays the role of the repo?

1.4 Central themes

1.4.1 The plan of the course

The course is about banking *phenomena*, not about banks as legally defined institutions [[Notes 1 p.4](#)]. Mehrling's notes name four functions of banking [[Notes 1 p.4](#)]:

1. **clearing**: the payment system. Understanding payments is the best starting point to understand *liquid assets*;
2. **market making**: once liquid assets are understood, *liquid markets*. These first two functions are the subject of the first half of the course, up to the midterm;
3. **advance clearing**: how finance (forwards, futures, swaps and other derivatives) enters the picture;
4. **intermediation** (channelling savings to investment), the function textbooks start from; here it comes last.

In class the plan is told in the order of the lectures [[► L1.4 @ 0:08](#)], summarised in [fig. 1.3](#): clearing and market making; then international money and banking, the same two ideas extended to the global system [[► L1.4 @ 0:28](#)]; then banking as advance clearing [[► L1.4 @ 0:48](#)]; at the end regulation, what to do about shadow banking, and “touching the elephant”, the connection of the money view with what one learns in economics and in finance [[► L1.4 @ 1:09](#)].

1.4.2 The four prices of money

In a sense the whole course is about understanding where four prices come from [[► L1.4 @ 1:49](#)].

Definition 1.4 (New concepts: the four prices of money).

- (i) *Par*: the price of one kind of money in terms of another kind of money, today, normally fixed at one for one (a deposit of \$107 at a bank exchanges for \$107 of currency) [[► L1.4 @ 2:34](#)]. (Latin *par*, equal.)
- (ii) *Interest*: the price of money today in terms of money tomorrow; an interest rate i

means that 1 today exchanges for $1 + i$ in one period [► L1.4 @ 2:14]. Its markets: Fed funds, Eurodollars, repo. (Medieval Latin *interesse*, “to be between, to matter”: the compensation due to a lender for the loss of the use of the money.)

- (iii) *Exchange rate*: the price of domestic money in terms of foreign money, or in terms of the world reserve money [► L1.4 @ 4:58].
- (iv) *Price level*: the price of money in terms of commodities, i.e. the inverse of the purchasing power of money [► L1.4 @ 5:59].

Economics focuses on one of them, the interest rate, the price of money in the sense of a deposit transforming money today into money tomorrow [► L1.4 @ 2:14]. The other three deserve more attention than they usually get.

Par, the forgotten price. Par is a more primitive price than interest, and it is neglected [► L1.4 @ 2:34]. In the bank balance sheet of [section 1.3](#) deposits (private money, a promise of the bank) sit on one side and cash reserves (central bank money) on the other, and we take for granted that they trade at par [► L1.4 @ 2:55]. In financial crises par is sometimes broken: bank money trades at a different rate from central bank money [► L1.4 @ 3:16]. The system is *hybrid*, part private money and part state money, and par masks the hybridity: it makes the two look like the same thing [► L1.4 @ 3:36]. When the system works well they *are* the same; but why does it work well, how do we make it work well, and what do we do in a crisis [► L1.4 @ 3:56]? Par is key to the payment system. If a Californian seller charged a buyer in New York an extra 5% because there is no par clearing between the two states, we would be outraged; yet this is how it was in the United States before the Fed [► L1.4 @ 4:17].

INTUITION: Par is a fixed price, and fixed prices are hard

Par clearing had to be created by institutions, and it is not easy: a price fixed at one for one, which must not budge whatever the pressure, is like rent control, and economists know how hard such prices are to hold [► L1.4 @ 4:37]. The rest of the course explains what holds it: the central bank, the clearing system, the dealers who stand ready to trade at that price.

The exchange rate. It is particularly obscured for people living in the United States, because the dollar is the world’s best money [► L1.4 @ 5:18]. Americans never worry that “the dollar” could fall below par with the *international* dollar, the dollar actually used as world reserve [► L1.4 @ 5:39]. Yet there are two dollars, again a private one and a state one: the international dollar consists largely of dollar deposits at banks outside the US, mostly European, that have no account at the Fed. These are the *Eurodollars* (lecture 8).

The price level. Economists usually treat it with the quantity theory of money, $MV = PY$ (or $MV = PT$), imagining that the price level moves one for one with the quantity of money [► L1.4 @ 5:59]. The course comes to this only at the end [► L1.4 @ 6:20]; most of the attention goes to par, interest and the exchange rate.

IN THE MARKETS: The FT market data page

Two or three pages from the back of the FT there is a “horrific” page of tiny numbers [► L1.4 @ 6:40]: market interest rates (US dollar LIBOR, US dollar certificates of deposit, US overnight repo, Federal funds effective rate), LIBOR in pounds and Swiss francs,

exchange rates, interest rate swap rates [► L1.4 @ 7:00]. By the end of the course every number on that page should be understandable, and explainable to a roommate: it is not an arcane world reserved to partners of investment banks, but something the average citizen can and should understand [► L1.4 @ 7:20].

Exercise 1.6 (Which price of money?). For each of the following, say which of the four prices of money is involved:

- (a) a money market fund announces that each share is now worth \$0.97 instead of \$1;
- (b) the Fed funds effective rate is 0.15%;
- (c) a tourist pays 1.30 dollars for one euro;
- (d) the consumer price index rises by 2% in a year;
- (e) before 1914 a New York bank paid a check drawn on a Chicago bank at \$99.85 per \$100.

HISTORY: Exchange charges before the Fed

Before the Federal Reserve, checks drawn on distant banks were often paid at less than face value: the paying bank deducted an “exchange charge” to cover the cost of shipping currency. Universal par clearing was one of the aims of the Federal Reserve Act (1913); the Fed’s campaign for it lasted into the 1920s, and some small banks kept charging non-par fees for decades.

References: J. Jessup, “The Federal Reserve and Par Clearing”, Federal Reserve Bank of St. Louis; Young, “Mobilizing Banking Credits” (the reading of this week).

1.4.3 Lombard Street, then and now

The inspiration of the course is Bagehot’s *Lombard Street* (1873), and Mehrling’s *New Lombard Street* is an homage to it [► L1.4 @ 7:41]. Bagehot wrote a bestseller to explain the money market to his fellow citizens after a series of financial crises in which the Bank of England had done what it had promised not to do; he argued that it had done the right thing: lend freely, at a high rate, against good collateral [► L1.4 @ 8:01]. The 2007–2009 crisis was a crisis of the shadow banking system, and the Fed too did things it had sworn it would never do [► L1.4 @ 8:21]. To get out of a global crisis, citizens need to understand how the system works, so that every proposal does not get stuck in “pro-banker” versus “anti-banker” politics; the course takes no side, and approaches the system as scientists and as citizens [► L1.4 @ 8:42].

Globalization: back to the 19th century. The world has changed: it is a global financial world [► L1.4 @ 9:23]. The world of the 1950s textbooks had hardly any private capital markets, and was not global, because the global financial markets of the 19th century had broken down [► L1.4 @ 9:43]. In terms of globalization the present resembles the world before the First World War, when the pound sterling was king and the Bank of England acted as central bank of the whole world, more than the world after the Second [Notes 1 p.4]. Existing theory was developed for a postwar world (government debt) and a post-Depression one (government guarantees, regulation), hence its emphasis on intermediation and monetary policy; practical central bankers know the new world better than academic economists, and money market traders know it best of all [Notes 1 p.4]. Hence

Bagehot is a better guide for today than Irving Fisher [► L1.4 @ 10:03]:

Fisher lived in a simpler, closed world; Bagehot lived in a global financial system organised around the pound. He did not know derivatives or high-frequency trading, but he would know how to start thinking about them, because he understood how money markets work [► L1.4 @ 10:24].

Finance: all banking is a swap of IOUs. The second novelty is the integration of finance. Connecting money and finance, the subjects of Mehrling’s two books, which people regard

as separate fields, occupied him for ten years [► L1.4 @ 10:44]. The motto of the course is the connection [► L1.4 @ 11:04]:

KEY FORMULAS

All banking is a swap of IOUs. [► L1.4 @ 11:25]

Bank		Borrower	
+Loan (borrower's IOU)	+Deposit (bank's IOU)	+Deposit (bank's IOU)	+Loan (borrower's IOU)

A loan is an exchange of promises: the borrower's promise to pay later against the bank's promise to pay on demand.

By “swap” Mehrling means interest rate swaps and credit default swaps, but also ordinary loans [► L1.4 @ 11:25]. Finance believes it knows everything about swaps; the money view looks at them differently: every swap has a *funding* piece and a *liquidity* piece, and the liquidity piece drives the action [► L1.4 @ 11:45]. The course is about money markets and liquidity, connected to solvency, but starting from where Bagehot would start: the money market as a global system of clearing [► L1.4 @ 12:06]. The aim is monetary theory for the real world of financial globalization, using as guideposts the great texts of the past and the “great texts of the present”, Stigum and the FT [► L1.4 @ 12:26].

ASIDE: IOU

IOU is a phonetic abbreviation of “I owe you”, used since the 17th century for an informal written acknowledgement of a debt. In the course it means any promise to pay, from a handwritten note to a Treasury bond or a deposit.

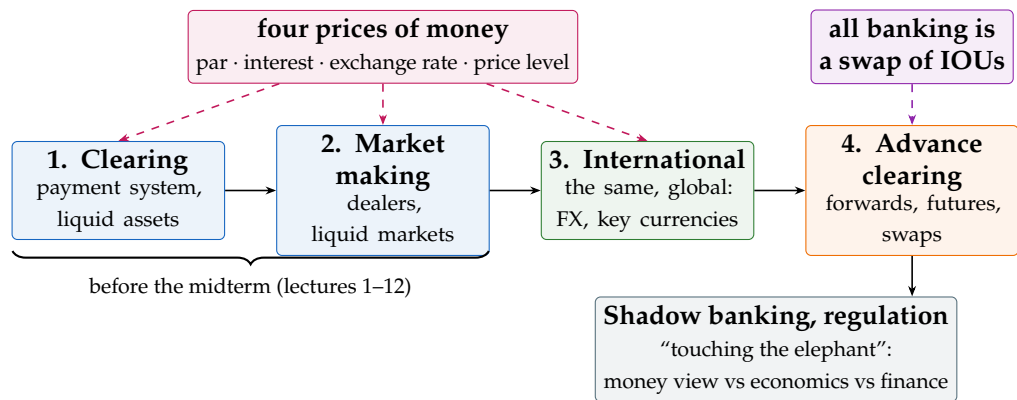


Figure 1.3: Map of the course [► L1.4 @ 0:08]: the four functions of banking in the order in which they are treated, the four prices that the course explains, and the motto that connects money and finance.

1.5 Reading: Allyn Young

The first reading is by Allyn Young, one of the authors who started Mehrling on his path: the first three chapters of *The Money Interest and the Public Interest* are a biography of Young [► L1.5 @ 0:00].

READING: Allyn Young, four encyclopedia chapters (1924)

Young wrote the chapters, unsigned, for an encyclopedia, *The Book of Popular Science*, in 1923–1924, just after the First World War [▶ L1.5 @ 0:00]: “The Mystery of Money”, “The Monetary System of the United States”, “Mobilizing Banking Credits”, “Dear and Cheap Money”. The Fed had been founded in 1913, and the United States had no tradition of thinking about central banking; Americans were traditionally suspicious of it [▶ L1.5 @ 0:40]. Young writes about the Fed when it is a new institution whose role is unclear: he is a friend of the president of the Federal Reserve Bank of New York and advises him, while open market operations and stabilisation policy are being invented [▶ L1.5 @ 1:15]. He compares the Fed with the Bank of England and predicts that London will remain the centre of world finance; he was wrong: it was the end of sterling’s hegemony, although the dollar became the world reserve currency only after the Second World War, after the Great Depression [▶ L1.5 @ 1:15]. The chapters are easy to read but deep: on a second reading one notices how carefully Young steers away from minefields [▶ L1.5 @ 2:19]. Their spirit is the spirit of the course: having lived through a crisis we do not fully understand, in a system that is changing, we have to figure it out, as he did in 1924 [▶ L1.5 @ 1:48].

Study questions (AllynYoung(study).pdf): where does Young draw the line between money and credit? Represent with T-accounts the three ways to obtain a bank deposit (Exercise 1.2). How is Young’s hostility to government “fiat” money and his sympathy for bank “credit money” consistent with the hierarchy of money (lecture 2)? The reading is analysed in detail in lecture 3.

Key formulas and ideas

KEY FORMULAS

Balance sheet	$\text{assets} = \text{liabilities} + \text{net worth}$ (always balances)
Solvency	$\text{net worth} = \text{market value of assets} - \text{market value of liabilities} > 0$ (a stock)
Liquidity	$\text{cash available or obtainable} \geq \text{payments due, every day}$ (a flow)
Four prices of money	par (money vs money, today); interest (money today vs money tomorrow, $1 \mapsto 1 + i$); exchange rate (domestic vs foreign money); price level (money vs goods)
Fisher diagram	$c_1 + c_2/(1 + r) = W$, slope $-(1 + r)$; optimum: $u_1/u_2 = 1 + r$ (1.1)
Central bank	expanding the money supply = expanding both sides of its balance sheet
Motto	all banking is a swap of IOUs

HISTORY: Allyn Young (1876–1929)

Allyn Abbott Young taught at Cornell, Stanford and Harvard; in 1927 he moved to the London School of Economics, where he died of pneumonia in March 1929, during an influenza epidemic, at 52. He is best known today for “Increasing Returns and Economic Progress” (*Economic Journal*, 1928). The friend at the New York Fed was Benjamin Strong, its first governor (1914–1928), the dominant figure of the early Fed. Young’s encyclopedia chapters are reprinted in P. Mehrling and R. Sandilands (eds.), *Money and Growth: Selected Papers of Allyn Abbott Young*, Routledge, 1999, from which the course reading is taken. References: Mehrling, *The Money Interest and the Public Interest*, 1997, ch. 1–3; Mehrling and Sandilands, 1999.

Chapter 2

The Natural Hierarchy of Money

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L2.1 FT: The Eurocrisis, Liquidity vs. Solvency	10:05	Draghi's bond purchases (a liquidity operation) versus Soros's European Fiscal Authority (a solvency operation), in T-accounts.
L2.2 Hierarchy of Financial Instruments	9:39	Money versus credit; gold, currency, deposits, securities; where the line between money and credit is drawn depends on the point of view.
L2.3 Hierarchy of Financial Institutions	6:37	The same hierarchy in balance sheets; inside and outside money.
L2.4 Dynamics of the Hierarchy	6:08	The pyramid of credit; expansion and contraction at every time scale, in quantity and in quality.
L2.5 Discipline and Elasticity	8:49	Scarcity of ultimate money and elasticity of derivative credit; currency principle versus banking principle.
L2.6 Hierarchy of Market Makers	9:17	Central bank, banks and security dealers as market makers; the prices of money as the links between layers.
L2.7 Managing the Hierarchy	18:03	Lender of last resort, origin of monetary policy, the term structure and the "artificial hierarchy"; Operation Twist.

Files. Mehrling's lecture notes [materials/mehrling-notes/Lec-02-The-Natural-Hierarchy-of-Money.pdf](#) [Notes 2 p.1] (dated 14 Sept 2009); captions [materials/week1/srt/2-*.srt](#); transcripts [materials/transcripts/L02-P0*.txt](#). **Reading.** Allyn Young ([section 1.5](#)); the study questions ask to place Young's "The Mystery of Money" in the framework of this lecture.

Motivation

Mehrling calls this lecture "the whole course": everything that follows hangs on its picture, like ornaments on a tree [► [L2.2 @ 0:08](#)][► [L2.4 @ 5:47](#)]. Its first sentence is the thesis:

Always and everywhere, monetary systems are hierarchical.

The lecture builds the hierarchy in four steps, each adding a dimension to the previous

one:

1. a hierarchy of *instruments* (gold, currency, deposits, securities), [section 2.2](#);
2. a hierarchy of *institutions* that issue and hold them, [section 2.3](#);
3. the *dynamics* of the hierarchy, which expands and contracts at every time scale, and the two principles behind it, discipline and elasticity, [sections 2.4](#) and [2.5](#);
4. a hierarchy of *market makers* and of the *prices of money* that connect the layers, [section 2.6](#), and the central bank's attempt to *manage* it, [section 2.7](#).

Four mental obstacles. The lecture comes so early because of four habits of thought, partly from economics courses, that stand in the way [[► L2.2 @ 0:48](#)]:

- (a) banks expand both sides of the balance sheet at once, which looks like a free lunch, “the alchemy of banking, creating something from nothing”, and good economics students resist it [[► L2.2 @ 1:09](#)];
- (b) money is usually thought of as a positive quantity, a “stuff”; seeing it as somebody's *liability* is hard (this is the inside/outside money distinction) [[► L2.2 @ 1:50](#)];
- (c) money is thought of as a creature of the nation-state; in this lecture the state does not appear, the central bank is just a bankers' bank (the state enters in lecture 3) [[► L2.2 @ 2:11](#)];
- (d) general equilibrium puts all goods at the same level, linked by relative prices; monetary instruments are *not* at the same level, they only look so, until a crisis shows that things that seemed equivalent are not [[► L2.2 @ 2:31](#)].

Students without economics background often find the start easier, because they have nothing to unlearn. Nothing here contradicts the economics one knows; it is integrated with it at the end of the course [[► L2.2 @ 3:33](#)].

2.1 FT: the eurocrisis, liquidity versus solvency

CASE STUDY: FT, September 2012: Draghi and Soros

The ECB under Mario Draghi has just announced that it is prepared to support the value of the short-term debt of the “peripheral” euro countries (Spain, Italy, . . .), to the dismay of some German central bankers [[► L2.1 @ 0:27](#)]. In the same issue of the FT, George Soros, under the title “Lead or leave currency bloc, Soros tells Germany”, argues that the ECB is not doing enough and makes his own proposal [[► L2.1 @ 0:48](#)]. Mehrling uses the two plans to consolidate the distinction of lecture 1: Draghi's is a *liquidity* operation, Soros's a *solvency* operation [[► L2.1 @ 1:08](#)].

Draghi: a liquidity operation. Investors, banks among them, hold short-term debt (*bills*: short term; *bonds*: usually long term) of the peripheral countries, which pays much higher rates than German or French debt [[► L2.1 @ 1:54](#)]. The ECB says it will cap those rates, without announcing an exact cap, by entering the market as a buyer of the bills, and it pays for them, at least at first, by expanding its own balance sheet: “creating money” [[► L2.1 @ 2:34](#)].

ECB		Bondholders	
+Peripheral bills	+Money (reserves)	–Peripheral bills	
		+Money	

This expansion of both sides of the balance sheet is what people mean by “monetizing government debt” [► L2.1 @ 2:56]. It supports the price of the bills and drags down the rates the periphery pays: a liquidity operation [► L2.1 @ 3:18].

Soros: a solvency operation. Soros says that this only buys time: the fundamental problem is that some peripheral countries have too much debt, a debt-to-GDP ratio above 60% [► L2.1 @ 3:58]. His plan [► L2.1 @ 4:40]:

1. a *European Fiscal Authority* (EFA), a fiscal unit of Europe as a whole, “essentially like the US Treasury”, buys the “excess” long-term bonds of the periphery (the part above 60% of GDP);
2. it finances the purchase by issuing its own short-term debt, a liability of Europe as a whole: *Euro bills*;
3. the ECB accepts Euro bills as the best collateral in the world, and may at first buy some of them by expanding its balance sheet, to get the operation going [► L2.1 @ 5:44];
4. once bought, the excess bonds can be forgiven: the EFA’s net worth takes the hit, but Europe as a whole is solvent where Portugal is not; the debt has moved to the larger, solvent entity [► L2.1 @ 6:50].

HISTORY: “Whatever it takes” and OMT (2012)

On 26 July 2012, in London, Draghi said that “within our mandate, the ECB is ready to do whatever it takes to preserve the euro. And believe me, it will be enough.” On 6 September 2012 the ECB Governing Council announced *Outright Monetary Transactions* (OMT): unlimited purchases of government bonds with 1 to 3 years to maturity, on the secondary market, for countries under an EU assistance programme. OMT was never actually used, but spreads fell sharply after the announcement. In 2015 the European Court of Justice ruled it compatible with the EU treaties. *References:* ECB press release “Technical features of Outright Monetary Transactions”, 6 Sept 2012; CJEU, case C-62/14 *Gauweiler*, 16 June 2015.

EFA		ECB		Bondholders	
+Excess bonds	+Euro bills	+Euro bills	+Money	–Excess bonds	
				+Money	

“You are going to get used to this pattern of thinking”: this minus and that plus are the same thing, every transaction has two sides and both are booked [► L2.1 @ 6:29].

INTUITION: Money that appears and disappears

Banks all over Europe do not trust each other and hold huge reserves at the ECB, earning nothing, as the US banks held \$1.5 trillion at the Fed [► L2.1 @ 7:50]. Soros expects them to swap those reserves for Euro bills. Then the ECB’s balance sheet shrinks back: it was used only temporarily to distribute the bills, the money disappears, and in the end no debt has been monetized; excess reserves have simply been replaced by Euro bills [► L2.1 @ 8:31]. Money is created when a balance sheet expands and destroyed when it contracts.

ECB		Banks	
–Euro bills	–Reserves of banks	–Reserves	
		+Euro bills	

There is no European treasury, so the job is left to the ECB. “When central banks do very bold things, it’s because somebody else isn’t doing their job” [► L2.1 @ 9:12]. Understanding the mechanics allows a saner political discussion than throwing around words like “hyperinflation” [► L2.1 @ 9:53].

Exercise 2.1 (Draghi versus Soros in T-accounts). Peripheral governments have issued 100 of excess long-term bonds, held by banks.

- Record Soros’s plan step by step: the EFA buys the bonds with Euro bills; the ECB buys 100 of Euro bills from the EFA’s buyers with new reserves; banks then swap 100 of excess reserves for Euro bills from the ECB.
- What is the final change in the ECB’s balance sheet? In the money supply?
- The EFA forgives the bonds. Whose net worth falls, and by how much?

2.2 The hierarchy of financial instruments

Start from the familiar distinction between money and credit [► L2.2 @ 4:18].

Definition 2.1 (New concepts: money and credit). *Money* is the means of final settlement: what debts are cancelled in. *Credit* is a promise to pay money: a means of *delaying* final settlement. Money is not just “more expensive” than credit, it is *qualitatively* better [► L2.2 @ 4:18]. (“Credit” from Latin *credere*, to believe, to trust: a credit is a promise one trusts.)

The distinction is fine as far as it goes, but it goes both too far and not far enough [Notes 2 p.1]. Not far enough, because it has only two layers; too far, because what counts as final settlement depends on the layer: *what looks like money at one level looks like credit from the level above*. To see this, take a world where the ultimate money is unproblematic: a gold standard [► L2.2 @ 4:59], with four layers [► L2.2 @ 5:42]:

- Gold**, the ultimate money, the international means of payment.
- National currency**, a liability of the central bank and a promise to pay gold at the mint parity.
- Bank deposits**, liabilities of private banks, promises to pay currency on demand: twice-removed promises to pay gold.
- Securities** (and loans of every kind), promises to pay currency at some date in the future: even more attenuated promises.

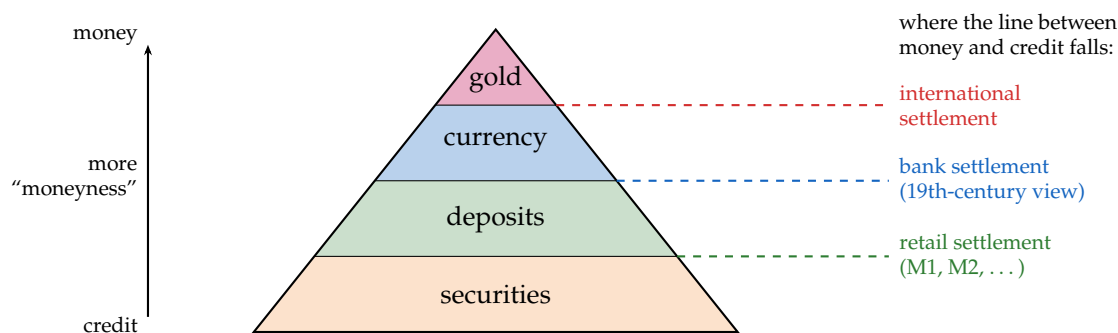


Figure 2.1: The simple hierarchy of instruments under a gold standard [► L2.2 @ 5:42]. Each layer is a promise to pay the layer above. The line between money and credit depends on who is settling [► L2.2 @ 7:05].

Where is the line between money and credit? It depends on the point of view [► L2.2 @ 7:05]:

- **Banks.** Banks clear payments among themselves in national currency, so for a bank settling its accounts at the end of the day currency (“high-powered money”) is money and deposits are credit. This was the typical 19th-century view [► L2.2 @ 7:26].
- **Households and firms.** They pay with deposits, so deposits and everything above are money and securities are credit. This is what textbooks mean by the money supply, with the ambiguity reflected in the many definitions M1, M2, M3 [► L2.2 @ 7:46].
- **Countries.** Your own currency is no use in paying other countries: they want their own currency, or the international means of settlement, gold under a gold standard, perhaps SDRs (the IMF’s Special Drawing Rights) today. Then only gold is money [► L2.2 @ 8:07]. (The US is the exception, because of the international role of the dollar.)

So what counts as money depends on who you are and on the layer you are talking about. The course starts from concepts, not from definitions: in each situation, ask which layer we are at, and what is money and what is credit there [► L2.2 @ 8:27]. “Once you get used to thinking about a hierarchy, you’ll see it everywhere” [► L2.2 @ 9:08].

◊ **BEYOND THE CORE: More layers than four**

Even with four layers the picture is much simpler than the real world [Notes 2 p.2]. Currency and high-powered money are not the same: high-powered money includes the deposits of banks at the central bank. Deposits are not homogeneous, and there are deposit-like instruments, such as money market mutual fund (MMMF) accounts, that are not the liability of any bank. Securities are even more heterogeneous, by maturity and credit quality. All this reinforces the point: avoid sterile debates about what is money and what is credit, and stand on the hierarchical character of the system.

Exercise 2.2 (Where is the money?). For each agent, say which instruments are money and which are credit from its point of view, and why:

- a household paying its rent;
- a bank settling its payments with other banks at the end of the day;
- the central bank of a small country paying for imports under a gold standard;
- the same central bank today.

2.3 The hierarchy of financial institutions

The same hierarchy can be seen through the institutions that issue and hold the instruments, i.e. through balance sheets [► L2.3 @ 0:08]:

- the **central bank**, a bankers’ bank, holds reserves of the international money (gold) and issues the national money (currency) as its liability [► L2.3 @ 0:35];
- the **banking system** holds currency (or deposits at the central bank) as reserves and issues deposits, the private money supply [► L2.3 @ 1:15];

COURSE NOTE: Hawtrey

The money/credit distinction is the framework of Ralph Hawtrey’s *Currency and Credit* (1919; Mehrling’s notes [Notes 2 p.1] cite the 1923 edition).

ASIDE: Not a cloakroom ticket

A currency backed by gold reserves does not *represent* gold: it is not a cloakroom ticket for gold held somewhere on the holder’s behalf. Reserves only make the promise to pay more credible. This holds even with 100% reserves (a currency board): there is still a promise, and a promise can be broken [Notes 2 p.1].

- the **private sector** holds deposits, and also securities (through pension funds, insurance companies), and issues securities [► L2.3 @ 1:57].

Central bank		Banking system		Private sector	
Gold	Currency	Currency	Deposits	Deposits Securities	Securities

Much is left out: government debt, the main asset of a modern central bank; loans, the main asset of banks; and the shadow banking system altogether. This is 19th-century, gold-standard banking [► L2.3 @ 2:40].

Every item appears twice, except gold [► L2.3 @ 3:01]. Each instrument is an asset of one balance sheet and a liability of another, except gold, which is no one’s liability.

Definition 2.2 (New concepts: outside and inside money). *Outside money* is a financial asset that is no one’s liability (gold). *Inside money* is money that is also someone’s liability, a form of credit: currency, deposits, the liabilities of shadow banks [► L2.3 @ 3:21]. If all balance sheets are consolidated into one, inside money cancels (asset of one, liability of another) and only outside money remains.

Almost all money is inside money: “it’s not stuff, it’s not a heap of anything, it’s somebody’s promise” [► L2.3 @ 4:02]. The terms come from John Gurley and Edward Shaw, *Money in a Theory of Finance* (1960), a book about banking as intermediation, of which the course reads some chapters [► L2.3 @ 4:24]. Surprisingly, Gurley and Shaw treat *currency* as outside money: they think of a single nation-state with fiat currency, and consolidate only the private sector, so currency has no counterpart in their accounts [► L2.3 @ 5:06]. Off the gold standard, the “ontological status” of currency is indeed less clear [► L2.3 @ 5:26]. The course takes a global view: every national money must be convertible, at some exchange rate, into an international money, so currency is always treated as *inside* money [► L2.3 @ 5:46].

INTUITION: Consolidation hides what matters

Consolidating all balance sheets makes the whole edifice of credit vanish, leaving a tiny quantity of gold. But the consolidation “misses the entire point” [Notes 2 p.3]: interest rates and GDP depend on the gross quantity of credit, on who issues it, who holds it, and where it lies in the hierarchy. Standard macro models keep only a quantity of money, usually treated as an outside asset. A physicist would say: the total charge is zero, but the field depends on where the charges are.

Exercise 2.3 (Consolidation). Write the balance sheets of [section 2.3](#) with numbers: gold 10; currency 100, held by banks; deposits 1000, held by the private sector; securities 5000, issued by the private sector and held half by the private sector itself.

- (a) Consolidate the three sectors. What remains?
- (b) Consolidate only the private sector and the banking system, as Gurley and Shaw do. What is “outside” money now?
- (c) Compute the “branching ratios” deposits/currency and currency/gold. What happens to them in an expansion?

2.4 The dynamics of the hierarchy

The pyramid. Take every balance sheet in the economy and record each instrument as a liability on the left and as an asset on the right, at its level of the hierarchy [► L2.4 @ 0:07]. The result is a symmetric pyramid centred on zero (fig. 2.2): the vertical axis is quality, from credit at the bottom to money at the top; the horizontal axis is quantity [► L2.4 @ 1:56]. The tip is drawn at zero because the quantity of gold is vanishingly small compared with the credit below it, “gold is really even just in the mind” [► L2.4 @ 1:15]: a pyramid of promises to pay, all the way up [► L2.4 @ 1:35].

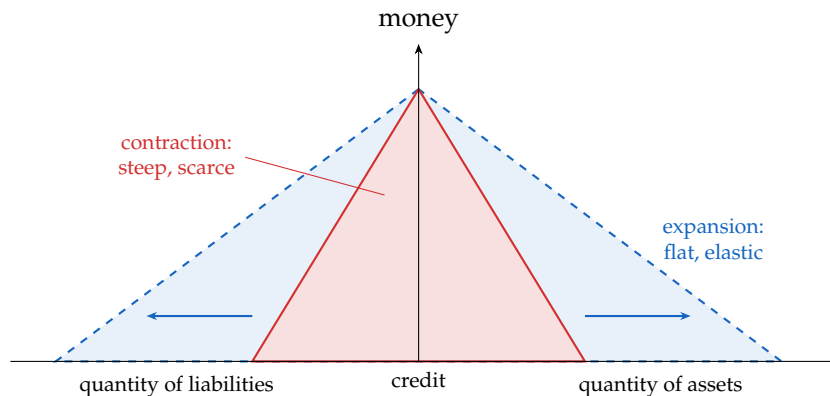


Figure 2.2: The hierarchy as a pyramid centred on zero [► L2.4 @ 0:07]: liabilities to the left, assets to the right, quality on the vertical axis. In an expansion the pyramid flattens (more credit per unit of money); in a contraction it becomes steep again [► L2.4 @ 3:21].

The system breathes. The pyramid adds a third dimension: the system is dynamic, at every time scale [► L2.4 @ 2:17]:

- within the day, payments are cleared with expansion and contraction of credit (daylight overdrafts);
- over the business cycle, credit booms and busts;
- over the rise and fall of nations, wars and depressions.

“At all time scales, this system is breathing”: its expansions and contractions are “the beating heart of the financial system” [► L2.4 @ 2:59]. In an expansion the pyramid becomes flatter and wider: the amount of credit for a given amount of money (a “branching ratio”) increases; this is *irrational exuberance*. In a contraction it shrinks and becomes steeper: *financial crisis* [► L2.4 @ 3:21]. The system expands until it hits some limit, collapses until it hits another, and goes back and forth [► L2.4 @ 4:04].

Quantity and quality. The fluctuation has the two dimensions of the picture [► L2.4 @ 4:24]:

1. **quantity:** credit grows in a boom and collapses in a crisis (“deleveraging”);
2. **quality:** in a boom credit starts to look like money, moves up the hierarchy, becomes more liquid and usable for payments; in a contraction you discover that what you hold is credit, not money: that gold and currency are not the same thing, that deposits and currency are not the same thing [► L2.4 @ 4:44].

The quality dimension shows up in prices: *credit spreads* (the excess of a borrower's interest rate over the rate on the best credit) become very flat in a boom and blow out in a crisis [► L2.4 @ 5:26].

Definition 2.3 (New concepts: *moneyiness*, *credit spread*, *deleveraging*). The *moneyiness* of an instrument is how close it is to money, i.e. how readily it is accepted in settlement at par. A *credit spread* is the difference between the interest rate paid by a borrower and the rate on a reference of the highest quality. *Deleveraging* is the reduction of debt relative to assets or capital (from *leverage*, the lever that multiplies the effect of a given capital).

IN THE MARKETS: Daily, cyclical, secular

A single trading day, a business cycle and a century of history show the same pattern of expansion and contraction: the hierarchy flattens as everyone accepts everyone's promises, then steepens as the promises are tested. The 2007–2009 crisis is the reassertion phase at business-cycle frequency; the end of sterling as world money, which Young did not foresee (section 1.5), is the same phenomenon at the secular scale.

2.5 Discipline and elasticity

Two principles drive the fluctuation [► L2.5 @ 0:08]:

KEY FORMULAS

scarcity of ultimate money (discipline)	↔	currency principle
elasticity of derivative credit (elasticity)	↔	banking principle

1. **Scarcity (discipline).** There is only so much gold at the top of the pyramid [► L2.5 @ 0:36]. More generally, no one can, by their own action, increase the quantity of money at a level above their own: governments cannot create gold, banks cannot create government currency, you and I cannot create currency, because our liabilities are not currency [► L2.5 @ 2:41].
2. **Elasticity.** Credit is called *derivative* because it is a promise to pay money, denominated in money but not money itself; the word should get us used to derivatives before swaps appear [► L2.5 @ 0:58]. Agents at any level can expand credit at their own level: if I buy something from you and you accept my IOU, the trade happens, and the scarcity of gold has nothing to do with it [► L2.5 @ 2:00]. If banks accommodate us, our credit expansion becomes an expansion of deposits; if the central bank accommodates the banks, of currency too [► L2.5 @ 3:01].

In a crisis discipline “smacks you”: this is money and that is not; in a boom elasticity says everything is money [► L2.5 @ 1:39]. The system is always some balance between discipline at the top and elasticity at the bottom, and the shifting balance shows up in prices and in output: it is a way to understand the fundamental instability of credit [► L2.5 @ 3:22]. Assuming it away because it makes the mathematics easier takes us farther from reality, not closer [► L2.5 @ 4:03].

The same swing in monetary theory. One can build a monetary theory on either principle [► L2.5 @ 4:44]:

- from the top, on the scarcity of ultimate money: the *currency principle*; monetarism can be read this way [► L2.5 @ 5:04];
- from the bottom, on the elasticity of credit: the *banking principle* [► L2.5 @ 5:28].

Internationally the same opposition is *metallism* versus *chartalism*, which come later in the course [► L2.5 @ 6:33]. When scarcity of ultimate money is the issue, monetarism looks right; when elasticity is the issue, the Keynesians look right; monetarists are in the ascendant when we need more discipline, Keynesians when we need more elasticity [► L2.5 @ 6:09]. This makes economists look fickle, but Mehrling's claim is that *both* theories are right: each captures something true of the system, neither the whole truth [► L2.5 @ 7:36]. This is his answer to "which school are you" (section 1.1): embrace the fluctuation, of practice and of theory, and ask where in the system there is too much scarcity or too much elasticity right now; that makes for a healthier policy dialogue [► L2.5 @ 8:17].

2.6 The hierarchy of market makers

Put the institutions back on the hierarchy of instruments: each of them *straddles* two layers, and each is, in its own way, a market maker [► L2.6 @ 0:07]. In the private sector, focus on security dealers.

Definition 2.4 (New concepts: market maker, dealer, bid–ask spread). A *market maker* (or *dealer*) stands ready to buy or to sell an asset at prices it quotes, and absorbs the difference between purchases and sales in its own inventory. It quotes two prices: the *bid*, at which it buys, and the *ask* (or offer), at which it sells; the *bid–ask spread* is the difference, its compensation [► L2.6 @ 1:08]. A dealer trades for its own account, unlike a *broker*, who only matches buyers and sellers.

- **Security dealers.** If you want to buy Apple stock a dealer sells it to you, if you want to sell it a dealer buys it; there are liquid markets in Apple stock because dealers quote prices [► L2.6 @ 1:28]. They knit together deposits and securities, and determine the price of securities, for which the interest rate stands in [► L2.6 @ 1:49]. In Mehrling's notes [Notes 2 p.6], they hold inventories of securities and money (in practice, of repos and reverse repos, which give access to securities and deposits held elsewhere).
- **Banks** are dealers too, quoting a two-way price in money: give Citibank cash and it gives you a deposit, and the other way round [► L2.6 @ 2:51]. The price they maintain is *par*, one for one, a single price rather than two; and keeping a price fixed is a challenge [► L2.6 @ 3:12].
- **The central bank** is a dealer at a higher level, between international and domestic money: if you need international money to buy abroad, it sells it to you; if you want to change it into domestic money, it buys it. The price made there is the *exchange rate* [► L2.6 @ 3:33].

HISTORY: Currency School versus Banking School

The names come from the British debate of the 1830s–1840s on the Bank of England's note issue. The Currency School (Samuel Jones Loyd, later Lord Overstone; Robert Torrens; George Warde Norman) held that bank notes should vary exactly as a fully metallic currency would, one for one with gold. The Banking School (Thomas Tooke, John Fullarton) held that the note issue adapts to the "needs of trade" and cannot be overissued if notes are convertible. The Currency School won: Peel's Bank Charter Act of 1844 split the Bank into an Issue Department, whose notes beyond a fixed "fiduciary issue" of £14 million had to be backed by gold, and a Banking Department. The Act had to be suspended in the crises of 1847, 1857 and 1866.

References: F. W. Fetter, *Development of British Monetary Orthodoxy 1797–1875*, Harvard UP, 1965; Young, "Dear and Cheap Money" (course reading).

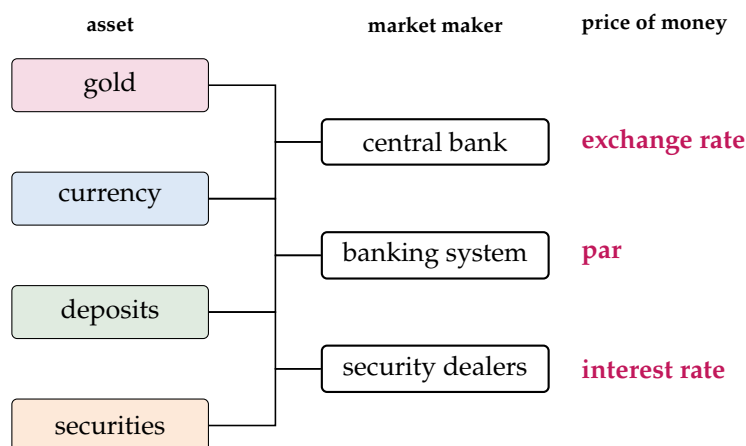


Figure 2.3: The simple hierarchy of market makers [► L2.6 @ 0:07]: each market maker has one leg in each of two adjacent layers (the two sides of its balance sheet), and the price at which it trades them is one of the prices of money of [definition 1.4](#).

From quality to quantity. The prices of money are themselves hierarchical, one between each pair of layers [► L2.6 @ 4:15]. As long as these prices exist, the hierarchy is hidden: there is a price at which securities exchange for deposits, one at which deposits exchange for currency, and the qualitative difference looks like a quantitative similarity [► L2.6 @ 4:56]. Economics abstracts from money to see good A and good B exchanging at a relative price; applied to money, the same impulse loses the hierarchy [► L2.6 @ 4:35]. Economics prices a kidney and an army tank in the same way; the hierarchy of money is a reminder that some things are different in kind [► L2.6 @ 5:57]. In normal times the approximation is not so bad; in a crisis it fails.

Market makers under stress. As the system swings between elasticity and scarcity, the market makers stand in the middle, one leg on each level [► L2.6 @ 6:38]. When the hierarchy reasserts itself they are pulled in two directions: long the lower layer and short the upper one, if the upper one becomes scarce it rises in value, and they can become insolvent, but “liquidity kills you quick”: if you cannot settle your books at the end of the day, you are done [► L2.6 @ 7:19]. All three face this constraint, which is where the course starts next [► L2.6 @ 8:00].

- Security dealers have it easier, because in a crisis they can change their price: asset prices plunge [► L2.6 @ 8:00].
- Banks cannot: a bank offering “80 cents on the dollar” for its deposits would not be a bank any more [► L2.6 @ 8:20].
- The same holds for a fixed exchange rate or the gold parity [► L2.6 @ 8:41].

In the worst case the market makers lose all their money and stop making markets, and one cannot move from one level of the hierarchy to another at any price, because there is no dealer: that is a financial crisis [► L2.6 @ 9:01].

INTUITION: The hierarchy is natural, not imposed

Mehrling calls the hierarchy *natural* (notes, section IV): the institutions are hierarchical *because* the underlying system is, not the other way round; the hierarchy is not imposed

from the top by the state or the central bank, it grows from the ground up. (He adds that he does not mean self-organisation in the sense of Austrian economists such as Carl Menger.) The market makers are what turn qualitatively different layers into a continuum of prices.

Exercise 2.4 (Market makers in a crisis). For each of the three market makers of [fig. 2.3](#), describe what happens to its balance sheet when the layer above becomes scarce, and what options it has: change its price, use reserves, borrow from the layer above, stop making the market.

2.7 Managing the hierarchy

So far the market makers have been profit maximizers: dealers and banks do not create liquidity to be nice, they do it to make money, and the course will analyse the source of that profit [[► L2.7 @ 0:08](#)]. One entity is different. Once the central bank becomes an agent of the state rather than a private bankers' bank, it need not maximize (short-term) profit [[► L2.7 @ 0:49](#)]. Sitting at the top, it notices that when things get really bad what everybody wants is its own liabilities, the best in the system, and that it can help [[► L2.7 @ 1:10](#)]. This happened gradually: the Bank of England was a private bank for a long time; in the US, J. P. Morgan acted as a private central bank in 1907 [[► L2.7 @ 1:30](#)].

Definition 2.5 (New concepts: lender of last resort). A *lender of last resort* is an institution that, when the scarcity of money threatens to make the market makers stop making markets, introduces elasticity from the top: it lends its own liabilities (expands the money supply) against securities taken as collateral, and so rejoins the layers of the hierarchy that have come apart [[► L2.7 @ 2:35](#)]. ("Last resort": the lender to whom one turns when all other sources of funding are gone.)

The widening remit of central banks [[► L2.7 @ 3:17](#)]:

1. **defend the exchange rate** (the mint parity), the most primitive job; central banks that did only this and "let the rest of the economy go to hell" learned that they would be sucked down with it [[► L2.7 @ 3:59](#)];
2. **help banks maintain par**: in a run, people want currency instead of deposits, and the central bank can create currency: financial stability, the lender of last resort [[► L2.7 @ 4:21](#)];
3. **monetary policy**: once the central bank has taken responsibility for the crisis, it would rather prevent it, leaning against the wind: raising rates to impose scarcity when the economy is all elasticity, lowering them to inject elasticity when it is all discipline. Counter-cyclical policy comes from taking responsibility for the crisis [[► L2.7 @ 6:07](#)];
4. **regulation**: making sure banks do not get overextended or "buy crap assets" comes from the same source [[► L2.7 @ 7:29](#)];
5. at the extreme, the ambition to eliminate the business cycle altogether by creating an artificial hierarchy [[► L2.7 @ 14:16](#)].

Bagehot's point was that the Bank of England, a private central bank, had in past crises been acting as a lender of last resort without intending to, and should bring this to

consciousness and embrace it: in a crisis, lend freely, at a high rate, against good collateral. It was very controversial when published, then became orthodoxy [► L2.7 @ 5:27].

Policy rate and market rates. Most central banks intervene by setting a very short-term interest rate (in the US the Fed funds rate, an overnight rate) [► L2.7 @ 8:18]. Rates at different maturities form the *term structure* (fig. 2.4): at the short end, overnight money, set by policy; at the long end, 30-year Treasury bonds, set by private security markets. “So who wins?” [► L2.7 @ 8:39]

Definition 2.6 (New concepts: term structure, expectations hypothesis). The *term structure of interest rates* (the yield curve) is the interest rate as a function of maturity. The *expectations hypothesis* says that a long rate is the expected average of the future short rates over its life: with R_n the yield on an n -period bond and r_t the one-period rate at t ,

$$(1 + R_n)^n = \mathbb{E} \left[\prod_{t=0}^{n-1} (1 + r_t) \right], \quad \text{approximately} \quad R_n \approx \frac{1}{n} \sum_{t=0}^{n-1} \mathbb{E}[r_t].$$

It looks like an arbitrage condition (rolling over overnight loans for 30 years versus lending for 30 years), yet it is rejected empirically “basically all the time” [► L2.7 @ 10:01]. Mehrling will argue that the reason is liquidity.

ASIDE: A copy of *Lombard Street*

Mehrling shows his used copy of Bagehot’s *Lombard Street*, inscribed “Merry Christmas to father from Ed, 1907”: the year of a financial crisis, and Ed was giving his father the key to understanding it [► L2.7 @ 4:44].

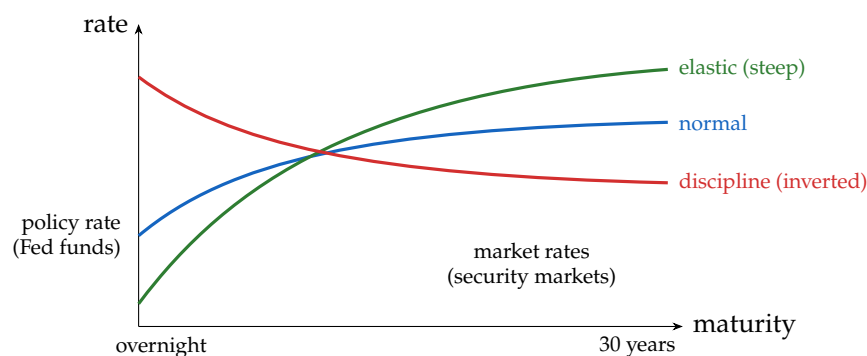


Figure 2.4: The term structure of interest rates [► L2.7 @ 8:39]: the central bank sets the overnight end, security markets the long end. To impose discipline it pushes short rates above long rates (inverted curve); to make the pyramid more elastic it pushes them far below [► L2.7 @ 12:09].

The artificial hierarchy. Policy tries to move the market by moving the short rate, but the market constrains what policy can do; and if policy simply follows the market it accepts the hierarchy as it is. “Monetary policy is all about creating an artificial hierarchy of money and credit” [► L2.7 @ 11:03]. In the term structure overnight money at the Fed plays the role of currency, long bonds the role of securities [► L2.7 @ 11:28]. To impose discipline the central bank makes short-term money “ridiculously expensive”, more than long-term money: an inverted (negatively sloped) curve. To make the pyramid more elastic it lowers short rates far below long rates [► L2.7 @ 12:09]. Choosing the point between elasticity and discipline is monetary policy, with its politics [► L2.7 @ 12:52].

INTUITION: A thin reed

The whole of monetary policy hangs on an overnight interest rate, “a very thin reed”, while the credit pyramid is enormous. How can the “puny little Fed” have any influence on it? Where exactly is the transmission mechanism? This is a serious question that the course will be able to ask precisely once inside the system [► L2.7 @ 13:12]. Mehrling’s notes [Notes 2 p.8] add: the main policy instrument is the overnight rate, and there can be considerable “slippage” between it and the long rates on which investment and spending depend.

A brave new world. The central question of the course is the proper role of the central bank today [► L2.7 @ 14:36]. Draghi, putting a floor under the price of peripheral bills by buying them himself, was acting as a *security dealer*, a role nobody had anticipated [► L2.7 @ 14:57]. The Fed, which had never held a mortgage-backed security since 1913, bought about a trillion dollars of them in a few months [► L2.7 @ 15:19]. Central banks are reacting, making it up: we live in a “Bagehot moment”. Bagehot looked back at what the Bank had done and told it to embrace it; we have no simple rule like his for today’s global, shadow-banking system, and the course is part of the search for one [► L2.7 @ 16:00].

IN THE MARKETS: Operation Twist

Asked about *Operation Twist* (2011–2012: the Fed sold short-term Treasuries and bought long-term ones, to lower long rates without expanding its balance sheet), Mehrling answers that by buying long-term bonds instead of only fixing the overnight rate the Fed is trying to influence the shape of the pyramid directly; whether it works, and why, is much debated [► L2.7 @ 17:28]. The name recalls the 1961 operation of the same kind, which “twisted” the yield curve.

Exercise 2.5 (The expectations hypothesis). The overnight rate is 1% today and is expected to rise by 1 percentage point a year for the next 4 years, then stay constant.

- Using the approximate expectations hypothesis, compute the 2-, 5- and 10-year yields and sketch the term structure.
- The observed 10-year yield is 1 percentage point higher than your answer. Give two possible explanations.

HISTORY: Punch bowl and string

William McChesney Martin, Fed chairman 1951–1970, said in a speech of October 1955 that the Fed is “in the position of the chaperone who has ordered the punch bowl removed just when the party was really warming up”: raising rates in a boom. “Pushing on a string”, for the impotence of lower rates in a slump, is usually attributed to Keynes, but the first recorded use is in the 1935 House hearings on the Banking Act, in an exchange between Congressman T. Alan Goldsborough and Fed chairman Marriner Eccles. Mehrling’s notes [Notes 2 p.7] quote both, attributing the second to Keynes as is customary.

References: W. McC. Martin, address to the New York Group of the Investment Bankers Association, 19 Oct 1955; US House, *Banking Act of 1935*, Hearings, March 1935.

Key formulas and ideas

KEY FORMULAS

Hierarchy	gold > currency > deposits > securities: each layer a promise to pay the layer above
Money vs credit	means of final settlement vs promise to pay; the line depends on the layer
Inside / outside	inside money = someone's liability (cancels on consolidation); outside = no one's (gold)
Dynamics	expansion: flatter pyramid, more credit per unit of money, credit looks like money; contraction: steeper, spreads widen
Two principles	scarcity of ultimate money (currency principle) vs elasticity of derivative credit (banking principle)
Market makers	central bank: exchange rate; banks: par; security dealers: interest rate
Expectations hypothesis	$(1 + R_n)^n = \mathbb{E} \prod_t (1 + r_t)$, i.e. $R_n \approx \frac{1}{n} \sum_t \mathbb{E} r_t$ (empirically rejected)
Monetary policy	creating an artificial hierarchy by setting the overnight rate

Chapter 3

Money and the State: Domestic

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L3.1 FT: Quantitative Easing and the Fed	7:47	The Fed's balance sheet 2007–2010; QE; the expected QE3.
L3.2 Allyn Young: Money and Economic Orthodoxy	9:09	Where Young stands against the orthodoxy of economists and against outside opinions.
L3.3 National Banking System before the Fed	3:22	The pyramid of reserves: country banks, reserve cities, New York.
L3.4 Civil War Finance, bonds and loans	8:49	Bond sales, the \$150 million bank loan of 1861, suspension of gold payments.
L3.5 Civil War Finance, legal tenders	7:07	The greenbacks: state money inserted into the hierarchy.
L3.6 National Banking System, origins	6:30	Bonds sold to banks, bank notes backed by bonds; a fixed quantity of base money.
L3.7 National Banking System, instability	5:57	Seasonal demand for currency, call loans, crises; 1907 and clearinghouse certificates.
L3.8 Federal Reserve System, plan	6:32	Discounts and rediscounts: elastic reserves and elastic currency backed by loans to Main Street.
L3.9 Federal Reserve System, actual	6:48	War finance fills the Fed with Treasury bills; open market operations; state money versus private money.

Files. Transcripts [materials/transcripts/L03-P0*.txt](#); Mehrling's notes [\[Notes 3 p.1\]](#). **Reading.** Allyn Young's four chapters ([AllynYoung.pdf](#), see [section 1.5](#)); this lecture translates their history of American money into balance sheets.

Motivation

Lecture 2 left the state out: the central bank was just a bankers' bank [[▶ L2.2 @ 2:11](#)]. Here the state comes in, through the history that Allyn Young tells: how the United States financed the Civil War, how that war finance produced the National Banking System, why that system was unstable, and how the Federal Reserve was designed to fix it and what it

became instead. The history serves two purposes [► L3.9 @ 3:37]:

1. practice with balance sheets: every episode is a set of T-accounts;
2. a conceptual point: the monetary system is a *hybrid* of state money and private money, and a theory built on only one of the two misses half of it.

A recurring pattern: *the history of American central banking is a history of war finance* [► L3.8 @ 0:07]. Behind the scenes are the two bogeymen of American monetary experience, Big Finance and Big Government, and the populist demand for easy money that comes from neither Wall Street nor Washington [Notes 3 p.1]. Mehrling's notes tell the story in three acts [Notes 3 p.1]:

Act 1	Civil War finance	greenback era
Act 2	National Banking System	currency principle, scarce money
Act 3	Federal Reserve System	banking principle, elastic credit

3.1 FT: quantitative easing and the Fed

CASE STUDY: FT, September 2012: "Fed's expected QE move weighs on the dollar"

Global stocks rallied and the dollar was sold amid expectations that the Fed would announce a third round of quantitative easing, and that Germany's constitutional court would allow Europe's new bailout fund (the court did, with reservations) [► L3.1 @ 0:28]. "QE" is a word not yet in the textbooks: it was invented during the crisis [► L3.1 @ 1:09]. From the article (D. Sherlock, "Global overview", reading for the lecture): the dollar index fell 0.5% to below 80 for the first time since May; sterling reached a four-month high above \$1.60; BNY Mellon reported dollar outflows at more than three times the past year's average; Moody's was considering lowering the US rating from Aaa to Aa1 unless budget negotiations reduced the debt/GDP ratio. Not everyone was convinced: for one economist "the promise of QE has proven to be as effective as the actual execution", and extending the guidance on how long short rates would stay low would be a more effective tool than more QE. Spanish two-year yields fell below 3%, from above 7% in July, on the ECB's plans to buy short-dated debt of troubled sovereigns.

The Fed's balance sheet in the crisis. The chart shown in class plots the Fed's balance sheet from January 2007 to March 2010, assets as positive numbers and liabilities as negative ones, with two vertical lines: the collapse of Bear Stearns (March 2008) and of Lehman Brothers and AIG (September 2008) [► L3.1 @ 1:52]. It shows three phases.

1. **Interest rates.** The crisis began in August 2007, and the Fed first lowered its rate from 5% to 2% in steps, trying to make the system more elastic [► L3.1 @ 2:33]. It was not enough: Bear Stearns failed [► L3.1 @ 2:53].
2. **Changing the composition of assets.** Before the crisis the Fed held essentially Treasury bills, against currency [► L3.1 @ 3:13]. After Bear Stearns it sold about half of its bills and lent the proceeds directly to banks and broker-dealers to support financial markets; liabilities did not change [► L3.1 @ 3:34].
3. **Expanding both sides.** After Lehman and AIG this was not enough either, and the Fed expanded its own liabilities: the balance sheet roughly doubled in a few weeks

[► L3.1 @ 3:54]. First came short-term loans, to domestic and to international banks: the *central bank liquidity swaps* with the ECB, the Bank of Japan and others reached about \$600 billion, because the problem had become global [► L3.1 @ 4:34]. As these facilities ran off around the turn of the year, the Fed started buying mortgage-backed securities, about a trillion dollars of them [► L3.1 @ 5:15].

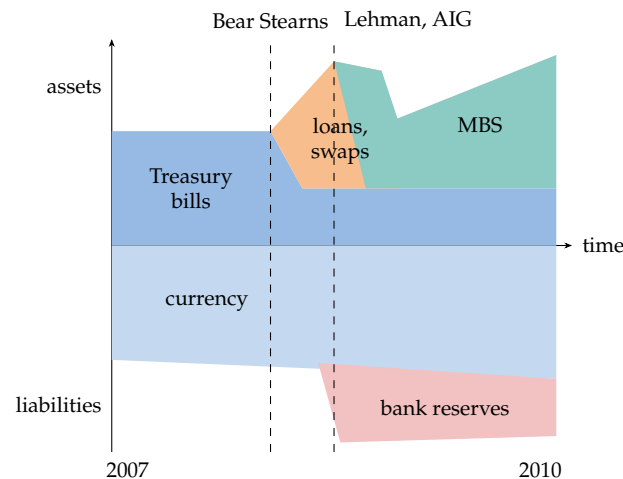


Figure 3.1: Schematic of the Fed's balance sheet in the crisis, after the chart shown in class [► L3.1 @ 2:12] (shapes only indicative, not data): first a change of composition (bills sold, loans made), then an expansion of both sides (loans, swaps, then mortgage-backed securities, against reserves).

QE3: two ways to lower long rates. At Jackson Hole the Fed had signalled that the economy was softening. Short rates were already near zero; the question was how to lower the long end of the term structure further [► L3.1 @ 5:36]:

1. **Announce** that short rates will stay low for a very long time: if the market believes it, long rates fall, by the expectations hypothesis (definition 2.6) [► L3.1 @ 6:38];
2. **buy long bonds**, expanding the balance sheet on both sides, which pushes up their prices: a capital gain for whoever owns them, hence the excitement of the market [► L3.1 @ 6:59].

In practice the Fed would probably do both [► L3.1 @ 6:59].

Definition 3.1 (New concepts: quantitative easing). *Quantitative easing* (QE) is the purchase by a central bank of large quantities of assets, especially long-term ones, paid for by creating reserves, i.e. by expanding both sides of its balance sheet, when the short-term policy rate is already near zero. "Quantitative" because the policy is stated as a quantity of purchases rather than as a rate.

Exercise 3.1 (QE in T-accounts). The Fed buys 100 of long-term Treasury bonds from a pension fund that banks with Bank A. Write the changes in the three balance sheets and explain why the market price of bonds is expected to rise.

COURSE NOTE: QE1, QE2, QE3

In class Mehrling calls the expansion after Lehman "QE1" and the purchases of mortgage-backed securities "QE2" [► L3.1 @ 4:14]. In the usual numbering the first program of large-scale asset purchases (QE1, from November 2008, extended in March 2009) included the MBS purchases, about \$1.25 trillion by March 2010; QE2 (November 2010) was \$600 billion of long-term Treasuries; QE3, announced on 13 September 2012, the day after this lecture's FOMC meeting, was open-ended MBS purchases of \$40 billion a month. Mehrling also says "March of 07" for Bear Stearns, which failed in March 2008.

3.2 Allyn Young against orthodoxy

Young wrote in 1924, after the First World War; he had been the economist of the American delegation at Versailles, a contemporary of Keynes but a much quieter personality [► L3.2 @ 0:50]. His encyclopedia chapters read smoothly, and do not cite the literature they answer, so it is easy to miss that he takes position on many controversial issues [► L3.2 @ 1:31]. Mehrling lists six.

Against economic orthodoxy [► L3.2 @ 1:54]:

1. **Barter.** Economics tells the story that money was invented to escape the inconvenience of barter (a trope Mehrling attributes to Jevons). Young: there was never a period of barter; money has been there ever since trade, it is “baked into the cake” [► L3.2 @ 1:54].
2. **Growth.** A developing economy needs a proper financial system. Orthodoxy then and now disagrees: in the Solow growth model growth comes from capital accumulation, population and productivity, technical change is a residual, and any growth due to finance is dropped into that residual; money is a veil to look through [► L3.2 @ 2:35].
3. **The currency principle.** Most economists think money gets its value from scarcity relative to demand, as in the neoclassical theory of value, and fear over-issue [► L3.2 @ 3:59]. Young criticises the currency principle: the National Banking System was built on it, limiting both note and deposit issue, and the lack of elasticity caused periodic crises; the Fed arose to give elasticity [► L3.2 @ 4:40].

Young was professor of monetary economics at Harvard and adviser to Benjamin Strong of the New York Fed, but his view was a minority one among economists, then and now, though not inside central banks [► L3.2 @ 5:22].

Against outside opinions [► L3.2 @ 5:43]:

4. **Chartalism.** Against the idea that the dollar has value because the state declares it legal tender [► L3.2 @ 6:04].
5. **Efficient markets.** Young points out how the inefficiency of the monetary system made asset prices swing apart from fundamental values; he is against speculation and for public policy to control bubbles [► L3.2 @ 6:45].
6. **Populism.** Above all he defends the Federal Reserve against those who saw it as a behemoth, and, against William Jennings Bryan, he wants the gold standard: the dollar is valuable because it is convertible into gold [► L3.2 @ 7:47]. This fits the hierarchy of lecture 2, with gold at the top. But the gold standard does not work perfectly by itself: it needs better financial institutions [► L3.2 @ 8:48].

3.3 The National Banking System before the Fed

The issue of gold versus silver versus fiat money is left to the lectures on international money [► L3.3 @ 0:07]. Young’s chapters contain a table of the reserves of national banks before the Fed, in three groups [► L3.3 @ 0:30]:

1. **Central reserve cities:** New York, Chicago, St. Louis, required to hold 25% of their deposits in “lawful money” (currency, or gold) [► L3.3 @ 0:50]. Deposits in the table:

ASIDE: Why “QE1”

For someone “of a certain age”, QE1 and QE2 are ocean liners, the *Queen Elizabeth* and the *Queen Elizabeth 2*; Mehrling guesses the joke was coined by the *Financial Times*, a British paper [► L3.1 @ 4:14].

HISTORY: Bryan’s “Cross of Gold” (1896)

At the Democratic national convention in Chicago, on 9 July 1896, William Jennings Bryan closed his speech for the free coinage of silver with: “you shall not crucify mankind upon a cross of gold”. He won the nomination but lost the election to William McKinley; the Gold Standard Act of 1900 put the US formally on gold.
References: *Official Proceedings of the Democratic National Convention*, 1896; Gold Standard Act, 14 March 1900.

\$825 million in New York, \$262 million in Chicago, \$116 million in St. Louis; New York held \$218 million of lawful money, 26.8% [► L3.3 @ 1:52]. To some extent the central bank of the US was the collection of New York banks, “J. P. Morgan and his buddies”, which also faced Europe in international transactions [► L3.3 @ 1:11].

- 2. **Other reserve cities** (Boston, San Francisco, Kansas City, . . .): also 25%, but partly held as deposits in New York, “due from reserve agents” [► L3.3 @ 2:12].
- 3. **Country banks**, about six thousand small banks: 15% reserves, partly cash (\$199 million) and partly deposits in reserve cities or New York (\$226 million) [► L3.3 @ 2:32].

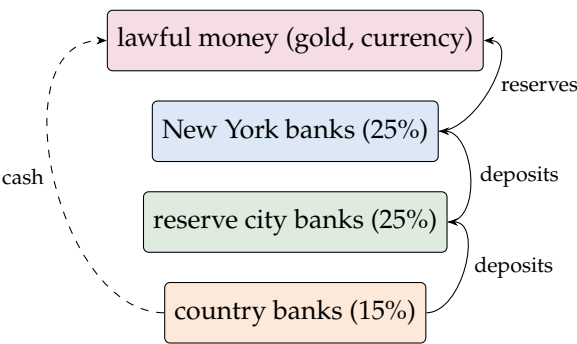


Figure 3.2: The layering of reserves in the National Banking System [► L3.3 @ 2:53]: country banks keep reserves as deposits in a reserve city, reserve city banks keep them in New York, New York banks keep lawful money. “Hierarchy, you can see that.”

3.4 Civil War finance

The problem of war finance: the government must mobilise as many resources as it can to win the war [► L3.4 @ 0:07]. It taxes as much as it can, it conscripts labour, and then it asks for volunteers: it sells bonds [► L3.4 @ 0:49]. Salmon P. Chase, Lincoln’s Secretary of the Treasury, tried three methods in turn.

3.4.1 Stage 1: bonds and loans

Version 1: bonds sold to the public. The government issues bonds and receives deposits, which it then spends on war goods; within the banking system only the ownership of deposits changes, and the total quantity of deposits does not [► L3.4 @ 1:33].

Government		Private sector		Banks	
+Deposits (G)	+Bonds	+Bonds		–Deposits (private)	
		–Deposits		+Deposits (G)	

Spending the deposits on war goods transfers them back to the private sector [► L3.4 @ 2:56].

HISTORY: Young at Versailles

In 1918–1919 Allyn Young headed the economic division of the Inquiry and then of the American Commission to Negotiate Peace in Paris; Keynes was there for the British Treasury.
References: C. P. Blich, *Allyn Young: The Peripatetic Economist*, Macmillan, 1995.

Version 2: a loan from the banks. If the public will not buy the bonds, or only at a high rate, the government can borrow directly from the banking system [► L3.4 @ 3:16]. A loan is like a bond, a promise to pay at a future date, but not a security traded in the open capital market [► L3.4 @ 3:57]. It is a swap of IOUs: the banks say to the government “I owe you \$150 million” (a deposit), the government says “I owe you \$150 million” (the loan) [► L3.4 @ 4:18]. This is what Chase did in August 1861 with the big New York bankers [► L3.4 @ 4:59].

Government		Banking system	
+Deposits \$150m	+Loan \$150m	+Loan \$150m	+Deposits (G) \$150m

The deposits are bank money “created from thin air”, an expansion of both sides of the balance sheet; “you might call this quantitative easing”. There is no central bank and no private sector in this operation [► L3.4 @ 5:20].

Chase takes the gold. Instead of spending the deposits at home, Chase withdrew them: deposits are promises to pay on demand, and he demanded gold [► L3.4 @ 6:21]. He wanted the gold to buy war material abroad, where dollars were not wanted; this turned out to be key, because the South had to barter cotton for European war material, while the North could pay in gold [► L3.4 @ 7:23].

Government		Banking system	
-Deposits		-Gold	-Deposits (G)
+Gold			

With their gold reserves gone, the banks could not pay any other deposit in gold and suspended specie payments: the United States left the gold standard [► L3.4 @ 8:03]. Deposits were no longer promises to pay gold: so what were they? Chase’s next move answered the question [Notes 3 p.3]. The theme of Young’s account is the maximal use of existing money and credit institutions to finance the war, testing their limits; the stress broke the whole banking system free of gold, which it regained only seventeen years later [Notes 3 p.2].

Definition 3.2 (New concepts: specie, suspension, legal tender). *Specie* is coined metal (Latin *in specie*, “in kind”); *specie payments* are the conversion of notes and deposits into coin on demand, and their *suspension* is the refusal to convert. *Legal tender* is money that, by law, a creditor cannot refuse in payment of a debt (Latin *tendere*, to offer).

3.4.2 Stage 2: the greenbacks

Banks would not fall for it twice [► L3.5 @ 0:08]. Chase’s second method was legal tender notes, the *greenbacks*. Mehrling shows a dollar authorized by the act of 3 March 1863 (the third Greenback act), with the words “this note is a legal tender for one dollar”: you cannot get gold for it, only another dollar bill, and within the US you cannot refuse it in payment of a debt [► L3.5 @ 0:29].

The government buys war goods by paying with notes it prints [► L3.5 @ 2:56]; the private sector deposits some of them (say 100) in banks, where they become the banks’ reserves [► L3.5 @ 4:21]:

COURSE NOTE

The notes say the return to gold came “17 years later in 1878” [Notes 3 p.2]. The Specie Resumption Act was passed in January 1875 and convertibility was resumed on 1 January 1879, seventeen years after the suspension of December 1861.
Reference: W. C. Mitchell, *Gold, Prices, and Wages under the Greenback Standard*, University of California, 1908; Federal Reserve History, “Specie Resumption Act”.

Government		Private sector		Banks	
+War goods	+Legal tenders 400	–War goods		+Legal tenders 100	+Deposits 100
		+Legal tenders 300			
		+Deposits 100			

Depositors now withdraw legal tenders, not gold. In the language of lecture 2, Chase inserted a layer of state money between gold and deposits, at the top of the US hierarchy [► L3.5 @ 5:09].

Consequences. Legal tenders could not buy foreign goods: foreigners ask their gold value [► L3.5 @ 5:51]. Throughout the war their value kept falling against gold, to about 50 cents on the dollar (a fall of the exchange rate), and domestic prices rose, because the government was adding purchasing power instead of transferring it from the private sector as a bond sale does [► L3.5 @ 6:11]. Young is “aghast” at this, while understanding that in war you do what it takes; the better way is the third one [► L3.5 @ 4:01].

Mehrling’s notes add three points:

- he says “legal tender” rather than “fiat money” on purpose: the notes were liabilities expected to be redeemed in gold at some future date [Notes 3 p.3]; Mitchell explained their depreciation by market expectations of a return to convertibility, which shifted whenever the North won or lost a battle [Notes 3 p.4];
- after the war, once the North was creditworthy, bonds were sold to buy gold and the gold used to retire legal tenders, bringing the dollar back to its prewar parity and deflating prices to prewar levels [Notes 3 p.4];
- Young stresses the cost: legal tenders bought only 50 cents of gold during the war and were redeemed at par afterwards, besides the damaging shifts of wealth between creditors and debtors from inflation and then deflation. In each later war (World War I and II) the money and credit system was again used to the maximum, but it was more developed, so it provided more finance with less inflation [Notes 3 p.4].

3.4.3 Stage 3: the National Banking System

In 1863 Chase went back to bonds [► L3.6 @ 0:09]. He asked the banks to buy, at full price, bonds with a low interest rate, paying with deposits (no gold withdrawals this time: the banks had none) [► L3.6 @ 0:58]. To sweeten the deal, banks could *issue bank notes using the bonds as collateral* [► L3.6 @ 1:39]. The government spends the deposits on war goods; then, when the public wants to withdraw its deposits in currency, the banks print their own notes instead of paying out their reserves [► L3.6 @ 2:20]:

Step 1: bonds for deposits, deposits spent on war goods.

Government		Banks		Private sector	
+War goods	+Bonds	+Bonds	+Deposits	–War goods	
				+Deposits	

HISTORY: The loans of 1861 and the suspension

In August 1861 the associated banks of New York, Boston and Philadelphia agreed to lend the Treasury \$150 million in three instalments, paying in coin as the government drew on it. Chase insisted on withdrawing the coin rather than leaving it on deposit. After the Trent affair raised fears of war with Britain, the New York banks suspended specie payments on 30 December 1861, followed by the Treasury.

References: W. C. Mitchell, *A History of the Greenbacks*, Chicago UP, 1903, part I ch. 2; B. Hammond, *Sovereignty and an Empty Purse: Banks and Politics in the Civil War*, Princeton UP, 1970.

COURSE NOTE: Numbers

In class: \$400 million authorized, “almost 400 million” issued, falling to “50 cents on the dollar” [► L3.5 @ 0:50]. Historically the three Legal Tender Acts (25 February 1862, 11 July 1862, 3 March 1863) authorized \$450 million, of which about \$431 million were outstanding at the peak (1864), and the greenback reached its low in July 1864, at about 35 cents in gold (Mitchell, 1903).

ASIDE: Wesley Clair Mitchell

In these passages Young draws on his close friend Wesley Clair Mitchell, author of *A History of the Greenbacks* (1903), a long-time professor at Columbia and one of the founders of the National Bureau of Economic Research (1920), the great American institutionalist and quantitative economist [► L3.5 @ 1:55].

Step 2: the public withdraws its deposits in bank notes.

Banks		Private sector	
	–Deposits	–Deposits	
	+Bank notes	+Bank notes	

The bonds were deposited with a central authority as collateral: if the bank failed, a note holder did not need to go to the bankrupt bank, he could get a government bond for his note [► L3.6 @ 4:48]. In effect the government borrowed from the banks by allowing them to create money: notes one step removed from legal tenders, promises to pay legal tender, private money [► L3.6 @ 5:30].

Why would banks accept bonds paying 2%, below the market rate? Because they could issue notes paying 0% against them, and the difference was profit; and a tax on every note issue backed by anything else kept them from backing notes with higher-yielding assets [Notes 3 p.4][Notes 3 p.5]. The overriding aim was to strengthen the market for government bonds; the lasting effect was to fix the money supply [Notes 3 p.4].

A fixed base. The number of such bonds was fixed, so at the end of the war they became the money supply: together with the legal tenders and gold (after resumption), a *fixed supply of base money* after 1863, which did not expand or contract with the business cycle [► L3.6 @ 5:50].

Exercise 3.2 (Three ways to finance a war). The government needs 100 of war goods. Write the T-accounts of government, banks and private sector when it

- (a) sells 100 of bonds to the public;
- (b) borrows 100 from the banks and spends the deposits;
- (c) prints 100 of legal tenders, of which the public deposits 40 in banks.

In each case, what happens to the total of deposits and of currency? Which method adds purchasing power?

3.5 The instability of the National Banking System

Most deposits were in the country banks, \$2,627 million in Young’s table, whose reserves were mostly deposits in the money centres [► L3.7 @ 0:07]. The United States was an agricultural country, and the demand for currency had a strong seasonal cycle: at harvest time farmers withdrew notes to move the crops; Young mentions \$50 million [► L3.7 @ 0:29].

The seasonal squeeze [► L3.7 @ 1:09]:

- 1. farmers withdraw \$50 million in cash from country banks;
- 2. country banks replenish their cash by drawing \$50 million on their deposits in New York;
- 3. New York banks held total reserves of \$221 million: they lose 25% of them and fall far below the required 25% [► L3.7 @ 1:30];
- 4. to rebuild reserves they call in their *call loans* to stock-market speculators: in slack

COURSE NOTE

The tax was the 10% federal tax on state bank notes enacted in 1865 (effective 1866), which drove state bank notes out of circulation. Mehrling’s notes also remark that historians found that note issue never reached the level of outstanding eligible bonds, so banks could in principle have issued more: the point is that they did not, and inelasticity was the problem [Notes 3 p.4].
Reference: Federal Reserve History, “The National Banking Era”; *Veazie Bank v. Fenno*, 75 U.S. 533 (1869).

COURSE NOTE: Which bonds?

Mehrling describes “special two percent bonds” [► L3.6 @ 0:58]. The National Currency Act (25 February 1863) and the National Bank Act (3 June 1864) required national banks to deposit US bonds with the Comptroller of the Currency to secure their notes (notes up to 90% of the bonds’ value); in the 1860s these were mostly the war bonds paying 5–6%. The 2% “consols of 1930”, used for note issue until 1935, date from the Gold Standard Act of 1900. Also, the 25%/15% reserve requirements were set in 1863–64; Chicago and St. Louis became central reserve cities only in 1887.

seasons they had lent idle reserves overnight to brokers, and at harvest they want the money back [► L3.7 @ 3:09];

- 5. the speculators sell what they bought: stock prices fall, interest rates rise [► L3.7 @ 3:50];
- 6. the banks also try to draw reserves from the world money market, London, disrupting the world gold market [► L3.7 @ 4:13].

“The United States is this big child in the world monetary system, and every fall it throws a little fit” [► L3.7 @ 4:41].

The notes add that reserves were not just inelastic but *perversely* elastic, because of the reserve requirements (25% all in cash for central reserve cities, 25% partly in central-city deposits for reserve cities, 15% partly in city deposits for country banks): a drain on reserves forced banks to contract. The result was a seasonal pattern of interest rates, as the harvest expansion of credit took place on a fixed reserve basis, and periodic financial crises from 1873 on, whenever banks had to pay cash they did not have. Young’s diagnosis: the problem was the inelasticity of reserves, and the answer was to make reserves a form of credit [Notes 3 p.5][Notes 3 p.6].

READING: O. M. W. Sprague, *History of Crises under the National Banking System* (1910)

The classic study of the crises of 1873, 1884, 1890, 1893 and 1907, written for the National Monetary Commission, cited in Mehrling’s notes as the reference on the pattern described here [Notes 3 p.5].

Country banks		New York banks	
–Deposits in NY 50	–Deposits 50	–Reserves 50 later: –Call loans	–Deposits of country banks 50

Definition 3.3 (New concepts: call loan, clearinghouse). A *call loan* is a loan repayable on demand (“at call”), typically to a stockbroker against securities. A *clearinghouse* is an association of banks that settles the claims of its members on each other in one place, netting them (from *clearing*, settling accounts so that they are “clear”).

1907. Europe eventually “got sick of this” and sent help, Paul Warburg, to help create a central bank; there was also domestic agitation for one [► L3.7 @ 5:02]. The key moment was the crisis of 1907, when J. P. Morgan showed how to get out of a crisis: the clearinghouse of the New York banks issued *clearinghouse loan certificates*, a substitute for money, creating elastic reserves, but privately and only for its members. The Fed’s idea was to do the same for everyone, with a central bank [► L3.7 @ 5:23]. (Clearinghouses are the subject of lecture 5.)

Exercise 3.3 (The seasonal squeeze). Using the numbers of section 3.5, suppose New York banks have deposits of 825 and reserves of 221, and country banks draw 50.

(a) Compute the New York reserve ratio before and after.

(b) How many call loans must be called (or gold imported) to restore 25%?

(c) Redo (b) when a Reserve Bank discounts loans for the New York banks.

Exercise 3.4 (Bank notes backed by bonds). A national bank buys 100 of government bonds with deposits and may issue notes up to 90% of their value. Write its balance sheet after the public withdraws 90 of deposits in notes. Why could the total of notes not respond to the seasonal demand for currency?

3.6 The Federal Reserve System: plan and reality

3.6.1 The plan

Attempts by the bankers themselves to create a central bank were not trusted; President Wilson got it through, just in time to use it for First World War finance [► L3.8 @ 0:07]. Young describes three levels [► L3.8 @ 0:50]:

- 1. **member banks**, ordinary banks making loans to farmers and merchants, “Main Street”, by expanding their own deposits [► L3.8 @ 1:32];
- 2. the **Federal Reserve Banks**, located in the same cities as the old reserve-city banks (Boston, San Francisco, . . .);
- 3. the **Federal Reserve** itself at the top.

What holds member banks back from lending is the fear that deposits are withdrawn and reserves run out [► L3.8 @ 2:13]. The plan removes the fear: a bank short of reserves can get more from its Reserve Bank by using its loans as collateral (a *discount loan*), or by selling them to the Reserve Bank if they are in the right form [► L3.8 @ 2:33].

Member bank		Reserve bank	
+Loans to Main Street	+Deposits	+Discount loan	+Reserves of member bank
+Reserves	+Discount loan		

“The alchemy of banking once again”: the member bank lends without fear, because its loans can be turned into reserves, and the Reserve Bank creates the reserves from thin air by expanding its own balance sheet: *elastic reserves*, unlike the National Banking System [► L3.8 @ 3:41]. If depositors want cash instead, Reserve Banks can *rediscount* at the Federal Reserve, which prints notes: *elastic currency* [► L3.8 @ 4:21]. (This is Young’s description, not how it works today.)

Definition 3.4 (New concepts: discount, rediscount). To *discount* a bill is to buy it before maturity for less than its face value; the difference is the interest. A central bank *discount loan* is a loan to a bank against eligible paper, and to *rediscount* is to discount again paper that has already been discounted once (by the member bank, then by the Reserve Bank). The rate charged is the *discount rate*.

The whole idea was a seasonal one: in an agricultural country the demand for currency and deposits fluctuates with the seasons, so the supply of currency and of reserves must fluctuate too, backed by loans to Main Street [► L3.8 @ 5:26]. The Federal Reserve Act says the assets of the Reserve Banks are to be these discounts; that is how Wilson sold it [► L3.8 @ 6:08].

HISTORY: From 1907 to 1913

The panic of October 1907 started with the failed attempt to corner United Copper and the run on the Knickerbocker Trust; J. P. Morgan organised the rescue, and clearinghouses across the country issued loan certificates. The Aldrich–Vreeland Act (1908) created a National Monetary Commission and emergency currency; Paul Warburg, a German banker who had settled in New York in 1902, became the main advocate of a central bank. The Federal Reserve Act was signed by President Woodrow Wilson on 23 December 1913; the twelve Federal Reserve Banks opened in November 1914.

References: R. F. Bruner and S. D. Carr, *The Panic of 1907*, Wiley, 2007; Federal Reserve Act, 1913.

3.6.2 The reality

The Fed's balance sheet before the 2007 crisis held hardly any loans to Main Street: its main asset was Treasury bills [► L3.9 @ 0:08]. This was true almost from the start, because the US went to war soon after the Fed was created, and at the end of the war the Reserve Banks, and the member banks too, were "stuffed full of Treasury bills" [► L3.9 @ 0:28]. In the war the Fed served as prime dealer for distributing government bonds, taking onto its balance sheet whatever banks and private bondholders would not absorb; after it, instead of the planned elasticity driven by the discount of private credit, it used open market operations, outright purchases and sales of government securities, to adjust money supply to money demand [Notes 3 p.6][Notes 3 p.7]. Young, writing in the 1920s, treats this as temporary, and imagines *open market operations* as a temporary way to provide elasticity [► L3.9 @ 0:48].

Definition 3.5 (New concepts: *open market operations*). *Open market operations* are purchases and sales of securities (classically Treasury bills) by the central bank in the open market, paid for by creating or destroying reserves. They were developed by the New York Fed under Benjamin Strong in the 1920s [► L3.9 @ 1:34].

The importance of open market operations is a historical accident: the Fed was never supposed to lend to the government [► L3.9 @ 2:15]. The framers had the greenbacks in mind: give the government the power to print money and it will print money; so only loans to Main Street were to back the expansion of reserves and notes. Then came the First World War, "the cat was out of the bag", and everybody got used to a Fed full of Treasury bills, so much so that in 2008 people found it weird when the Fed sold them and lent to banks, to the private sector, in its commercial paper operations [► L3.9 @ 2:35]. (Mehrling adds that the textbook account of open market operations changing the money supply is not how they actually work: repo is involved, lecture 7 [► L3.9 @ 1:54].)

3.6.3 State money and private money

The conceptual point [► L3.9 @ 3:58]: the system is a *hybrid*. The central bank is partly a bankers' bank; in war it is the government's bank, helping it sell its bonds, as in both world wars, and that is its job, not a perversion of it; in normal times it facilitates private capital markets; and it moves back and forth between these roles [► L3.9 @ 4:18].

INTUITION: Two theories, both partly true

One can build a monetary theory in which the state prints money and everything else is a promise to pay it (chartalism), or one in which everything is private credit and the state is just a borrower (metallism, starting from gold and private pledges). Neither is any good alone, because the system is a hybrid that moves back and forth between the two [► L3.9 @ 4:40]. As with discipline and elasticity (section 2.5), Mehrling is neither chartalist nor metallist, and uses both [► L3.9 @ 5:40]. The balance sheet approach is the bridge: debits and credits work the same way for state money and for private money [► L3.9 @ 6:22].

The notes conclude: American monetary history is a story of getting the public/private balance right, which took a while and is still ongoing; the history of monetary thought is likewise a story of getting the balance between the two dimensions right for its own time. Sometimes the state dimension dominates (in war), sometimes the private one, but both are always present [Notes 3 p.7].

Key formulas and ideas

KEY FORMULAS

Bond sale to public	government +deposits/+bonds; only ownership of deposits changes, total deposits unchanged
Loan from banks	swap of IOUs: bank +loan/+deposit; both sides expand (new bank money)
Legal tenders	state money inserted between gold and deposits; adds purchasing power: depreciation and inflation
National bank notes	bank notes backed by government bonds; fixed base money, inelastic
Seasonal squeeze	farmers' cash demand → drain on New York reserves → call loans called → stocks fall, rates rise
Fed plan	discount and rediscount of loans to Main Street: elastic reserves, elastic currency
Fed reality	war finance: Treasury bills as main asset; open market operations
Hybrid	state money and private money; chartalism and metallism each partly true

Chapter 4

The Money View, Micro and Macro

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L4.1 FT: Dealer of Last Resort	5:23	Bernanke's QE3 and Draghi's conditional promise as dealer-of-last-resort actions.
L4.2 Reading: Hyman Minsky	3:07	Minsky and the financial instability hypothesis; Copeland's flow of funds.
L4.3 Payments: Money and Credit	5:36	Paying with money, with pure credit, with a bank's IOU.
L4.4 Payments: Discipline and Elasticity	4:13	The payment system is a credit system.
L4.5 The Survival Constraint	3:35	Every agent as a bank: cash inflow must cover cash commitments.
L4.6 Sources and Uses Accounts	6:55	The sources-and-uses matrix and its two rules.
L4.7 Payment Example: Money and Credit	10:05	A coffee paid in cash, a dinner paid by credit card.
L4.8 Flow of Funds Accounts	10:10	Copeland's macro accounts; statistical discrepancies; what they miss.
L4.9 The Survival Constraint, redux	2:40	Three ways to meet a cash shortfall.
L4.10 Liquidity, Long and Short	9:42	Time pattern of cash flows and commitments; price-insensitive borrowers.
L4.11 Financial Fragility, Flows and Stocks	6:23	The scales of cash flows and commitments; the money market as stress detector.

Files. Transcripts `materials/transcripts/L04-P*.txt`; Mehrling's notes [Notes 4 p.1], which point to the Fed's Z.1 release and its notable features in 2012: household deleveraging, a switch from credit to money, the largest instrument discrepancy in repo, sectoral discrepancies. **Reading.** Mehrling's article on Hyman Minsky (not among the course files); Minsky, *Stabilizing an Unstable Economy* (1986), quoted in class.

Motivation

Lectures 1–3 drew the hierarchy with balance sheets, which are *stocks*. This lecture introduces the other half of the method: *flows* of cash. Its central idea, taken from Hyman Minsky, is to look at every economic unit “as if it were a bank”, facing every day a *survival constraint*: cash inflows must cover cash commitments. The lecture develops it at two levels:

1. **micro**: payments with money and with credit, the sources-and-uses accounts of a

single agent, the survival constraint and the ways to meet it;

2. **macro:** Morris Copeland's flow of funds accounts, and the money market interest rate as the place where the balance of the whole economy's cash flows and commitments shows up.

4.1 FT: dealer of last resort

CASE STUDY: FT, September 2012: QE3 and Münchau on the ECB

The FOMC has announced that it will keep buying mortgage-backed securities until employment improves: "quite astonishing" open-ended support, prepared by Bernanke's remarks at Jackson Hole [▶ L4.1 @ 0:07]. In the same week Draghi said he would do whatever is necessary: central bankers are determined to use their balance sheets to prevent anything bad [▶ L4.1 @ 0:48]. In the FT, Wolfgang Münchau argues that QE would be the right policy for Europe too [▶ L4.1 @ 1:09]. He points out that Draghi will buy Spanish or Italian bonds only if those countries *ask*, which a political leader may be reluctant to do, while Bernanke simply announced that he would buy [▶ L4.1 @ 1:30].

Promises and purchases. If Spain and Italy do not ask, why should the price of their bonds move? Only through expectations, and Münchau, with Mehrling, considers that a weak reed [▶ L4.1 @ 2:32]. The course looks at what central banks actually do [▶ L4.1 @ 2:53]. Mehrling will call both actions *dealer of last resort* actions: the central bank promises to buy bonds at a particular price, so that no one will ever sell them for less; buying at a high price lowers their yield [▶ L4.1 @ 3:16]. For this to work the promise must be credible, i.e. the central bank must actually buy. Bernanke is buying, and the market price will move to his price at once; Mehrling predicts that the price of Spanish and Italian bonds will not move all the way to Draghi's level until Draghi actually buys [▶ L4.1 @ 3:57].

INTUITION: Why central banks are so bold

Central banks are doing things they have never done, and boldly, because the fiscal authorities are not: in Europe they are caught up in austerity and in blaming Greece, in the US Republicans and Democrats cannot agree. The central banks are trying to hold the system together through the election cycle until fiscal authorities can sort things out, "not a job that any central banker ever signed up for" [▶ L4.1 @ 4:38].

4.2 Reading: Hyman Minsky

Mehrling knew Hyman Minsky, who belongs to the American tradition he calls his own (Wesley Clair Mitchell, Edward Shaw, Morris Copeland) [▶ L4.2 @ 0:07]. When the crisis started, the *New Yorker* ran a "Talk of the Town" piece calling it a "Minsky moment"; for a man who grew up in New York, after decades of neglect by economists, that would have been immortality [▶ L4.2 @ 0:28].

COURSE NOTE: OMT conditionality

Outright Monetary Transactions ([section 2.1](#)) require that the country first request assistance from the European Stability Mechanism and accept its conditions; this is the "asking" that Münchau worried about.

READING: Mehrling on Minsky

Minsky’s *financial instability hypothesis*: there is something inherently unstable about financial systems, and the role of the central bank is to put bounds on that instability, to keep it from flying off on the upside or on the downside; this is not easy [► L4.2 @ 1:10]. It is a very different vision from the technocratic one of moving the Fed funds rate by another 25 basis points to get it just right; Minsky’s was a mentality prepared for financial crisis [► L4.2 @ 1:30]. A common misreading, Mehrling warns, takes Minsky’s hedge, speculative and Ponzi units to be about leverage; read through Copeland, they are about the *alignment in time* of cash flows and commitments (section 4.8) [► L4.10 @ 0:07].

Three images of the macroeconomy. Macroeconomics courses give two images of how money enters the economy [► L4.2 @ 2:31]:

- 1. the national income and product accounts (NIPA), $C + I + G + NX = Y$;
- 2. the quantity theory, $MV = PT$.

A third, proposed by Morris Copeland in 1952, is the *flow of funds*: used on Wall Street and inside the Fed, but never turned into a proper macroeconomic theory, which many think is what should be attempted [► L4.2 @ 2:52].

4.3 Payments: money and credit

Suppose there are only two people, you and I; each produces what the other wants, but not at the same time, so barter does not work [► L4.3 @ 0:07]. There are three ways to arrange the trade (I buy goods from you):

- 1. **Money**: you give me goods, I give you a token; only existing money changes pocket [► L4.3 @ 0:56].
- 2. **Pure credit**: I write you a note, “I owe you”; this requires trust, as between friends who take turns paying lunch [► L4.3 @ 1:39].
- 3. **Bank credit**: if we do not trust each other but both trust a third party, the bank accepts my IOU and issues its own IOU, which you accept [► L4.3 @ 3:42].

HISTORY: The Minsky moment

Hyman P. Minsky (1919–1996), born in Chicago, taught at Washington University in St. Louis and ended his career at the Levy Economics Institute of Bard College. The phrase “Minsky moment” was coined by Paul McCulley in 1998 for the Russian crisis; John Cassidy’s *New Yorker* piece “The Minsky Moment” appeared on 4 February 2008. Mehrling’s “grew up in New York” refers to Minsky’s youth: he attended high school in New York City. References: H. P. Minsky, *Stabilizing an Unstable Economy*, Yale UP (Twentieth Century Fund), 1986; J. Cassidy, *The New Yorker*, 4 Feb 2008.

(1) Money:

Me		You	
+Goods		−Goods	
−ΔM		+ΔM	

(2) Pure credit:

Me		You	
+Goods	+ΔIOU	−Goods	
		+ΔIOU	

(3) Bank credit:

Me		Bank		You	
+Goods	+IOU	+IOU (mine)	+M	−Goods	
				+M	

The difference between (1) and the other two: with money the quantity of money does not change, existing pieces of paper move; with credit the quantity of outstanding IOUs *increases* with the transaction [► L4.3 @ 3:01]. In (3) the bank's IOU is called *M* because it is used as money, but it is a liability of the bank created by expanding its balance sheet: money is a form of credit, and it expands and contracts with the pattern of trade [► L4.3 @ 4:45]. This stripped-down picture is how the modern economy works [► L4.3 @ 5:27].

4.4 Discipline and elasticity in payments

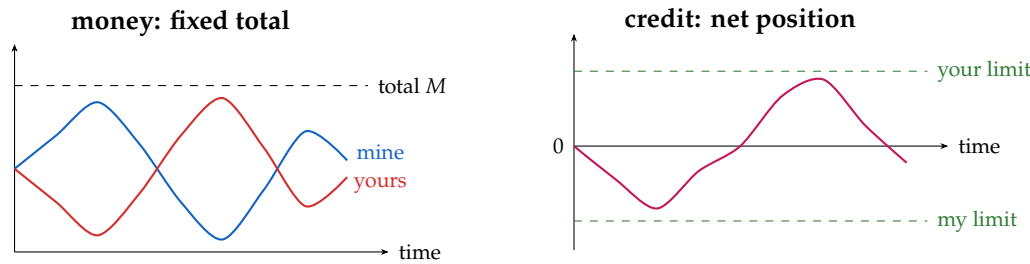


Figure 4.1: Payments with a fixed quantity of money (left: balances only move between pockets, and a trade is impossible when the buyer has run out) and with credit (right: the net position starts at zero and expands and contracts within credit limits) [► L4.4 @ 0:28].

This is where discipline and elasticity ([section 2.5](#)) become concrete [► L4.4 @ 0:07].

- With a **fixed quantity of money**, my balance goes up when I sell to you and down when I buy, while the total does not change. If there is little money, people run out, and some mutually advantageous trades are impossible: that is *discipline* [► L4.4 @ 0:28].
- With **pure credit**, positions start at zero and expand and contract with the pattern of payments, within each party's credit limit; there is no money, only credit that comes and goes [► L4.4 @ 1:35].
- With a **bank**, I can borrow up to my limit at the bank, and I am willing to hold large positive balances with the bank, much more than I would lend to you; the bank does this with many people at once: a multilateral credit system [► L4.4 @ 2:58].

Proposition 4.1 (The payment system is a credit system). *The payment system is a credit system, not a money system, and it has to be one in order to have enough elasticity: people who want to make mutually advantageous trades must be able to make them even if none of them has money (gold, the thing fixed in quantity) at the moment [► L4.4 @ 3:39]. When looking at the actual system, expect to find credit, even where it does not look like credit [► L4.4 @ 4:00].*

4.5 The survival constraint

Mehrling reads from Minsky's *Stabilizing an Unstable Economy* (1986), p. 198 [► L4.5 @ 0:07]: to analyse how financial commitments affect the economy one must look at economic units in terms of their cash flows; the cash flow approach looks at all units, households, corporations, state and municipal governments, even national governments, *as if they were banks* [► L4.5 @ 0:28]. That is: every unit is a money-flow operation, cash in and cash out, and the goods side is secondary [► L4.5 @ 0:49]. The macro side of this view comes from Copeland ([section 4.7](#)); the micro side is the survival constraint [► L4.5 @ 1:09].

Definition 4.2 (New concepts: survival (reserve) constraint). The *survival constraint* (Minsky), or *reserve constraint*, of an agent is the requirement that in every period

$$\text{cash inflow} \geq \text{cash outflow (cash commitments)},$$

where cash commitments include promises already made, such as debt repayments. If it fails, the agent defaults: “it doesn’t matter if you have lots of wealth, if you have no cash” [► L4.5 @ 2:10].

The binding constraint is not the budget constraint of economics, and certainly not an *intertemporal* budget constraint: it is a liquidity constraint [► L4.5 @ 1:50]. “Liquidity kills you quick”: a bank can go on doing business for a long time while insolvent, many do, but cannot survive one day illiquid. If everyone is a bank, this is what focuses the mind [► L4.5 @ 2:51].

INTUITION: Budget constraint versus survival constraint

The intertemporal budget constraint, $\sum_t c_t / (1 + r)^t \leq W$ (eq. (1.1)), restricts present values: any timing of payments is allowed as long as the totals balance, because one can always borrow against future income at the rate r . The survival constraint is a constraint at *each* date: a pointwise condition rather than an integral one. Two agents with the same present values can differ completely in whether they survive.

Three ways to meet a shortfall. If cash inflow does not cover a commitment, there are three choices [► L4.9 @ 0:07]:

1. **dishoard**: run down cash reserves;
2. **borrow**: find someone to lend more cash;
3. **sell assets**, at whatever price one can get: a fire sale.

(One could also sell goods; the course focuses on financial assets [► L4.9 @ 0:49].) Only the first is dependable in a crisis: the others need someone to take the other side of the trade, at a good price, and in a crisis they may not want to [► L4.9 @ 1:09]. This is why reserves matter: at the end of the day it is cash that meets commitments. But nobody wants to hold only money; credit lets us escape the constraint by pushing it into the future: we live to fight another day, but that day the borrowing comes due and we face the constraint again [► L4.9 @ 1:52]. *Credit gives us elasticity today but discipline tomorrow* [► L4.9 @ 2:32].

Exercise 4.1 (Time patterns). A firm expects cash inflows of 10, 10, 10, 10 and has debt of 24.

- (a) Give a repayment schedule that never violates the survival constraint and one that violates it at t .
- (b) In the second case, how much must it roll over at t ? At what rate does it become insolvent if it must pay interest i on the rolled-over amount for three periods?
- (c) Explain why its demand for funds at t is price-inelastic.

4.6 Sources and uses of funds

The *flow of funds* accounts are real accounts, collected by the Fed and watched closely by financial advisers; start from their micro foundation [► L4.6 @ 0:06]. For one agent, distinguish *uses* from *sources* of funds, and the goods-and-services account (“above the line”) from the financial account (“below the line”), which has three categories: financial assets, financial liabilities and money, a hierarchy of credit and money in which money is the ultimate means of payment [► L4.6 @ 0:48].

	Uses (cash out)	Sources (cash in)
Goods and services (incl. labour)	expenditure	receipts
Financial assets	accumulation	decumulation
Financial liabilities	repayment	borrowing
Money	hoarding	dishoarding

Figure 4.2: The sources-and-uses framework [► L4.6 @ 1:53]. Every entry is a *flow* over a period, not a stock: this is not a T-account. Above the line, goods; below the line, the financial account.

- Selling anything, including labour, is a receipt, a source of funds; buying is an expenditure, a use. Unlike the national accounts, which count only final sales (value added), here every sale counts: it is a source of cash for the seller even if it adds no value for the economy [► L4.6 @ 2:14].
- Selling a financial asset is a source (decumulation), buying one a use (accumulation) [► L4.6 @ 2:58].
- Borrowing is a source, repaying a use [► L4.6 @ 3:19].
- Taking cash out of one’s pile is dishoarding (a source), adding to it hoarding (a use) [► L4.6 @ 3:41].

The framework is complete: every transaction fits in one of the categories, which lets one sort out complicated transactions [► L4.6 @ 4:22].

Proposition 4.3 (The two rules of the flow of funds).

- For each agent, every use has a corresponding source and vice versa: nothing can be bought, no asset accumulated, no debt repaid, no cash hoarded, without the money coming from somewhere [► L4.6 @ 4:42].*
- Every agent’s use is some other agent’s source and vice versa: my receipt is somebody’s expenditure, my purchase of an asset is somebody’s sale [► L4.6 @ 5:45].*

Consequently each transaction requires at least four entries, two for each of two agents [► L4.7 @ 1:56]. Rule (ii) is what turns the framework into macroeconomics: the economy is a web in which everything is connected [► L4.6 @ 6:28].

INTUITION: Double entry, twice

Rule (i) is the conservation law of each agent’s cash (inflow = outflow, with hoarding as the “storage” term); rule (ii) is the conservation law between agents (every outflow

is someone's inflow). Together they make the matrix of all agents' flows have zero row sums (each type of flow balances across agents) and equal column sums for each agent (uses = sources), as Kirchhoff's laws do for currents at nodes and around a circuit.

4.6.1 Two examples

A coffee paid in cash. Before class Mehrling bought a \$2 coffee at Oren's Daily Roast, in cash [► L4.7 @ 0:08]:

	Mehrling		Oren's	
	uses	sources	uses	sources
goods	coffee 2		coffee 2	
money	dishoard 2		hoard 2	

Four entries, both rules satisfied [► L4.7 @ 0:48]. The point is to build intuition for the inside character of money, for the interlocking of the monetary system, for its money flows; these intuitions are not natural for economists, trained to see money as a veil [► L4.7 @ 1:56].

A dinner paid by credit card. After class, dinner at Virelli, \$20, charged to MasterCard [► L4.7 @ 2:38]. Three parties, MasterCard in the middle as the bank of [section 4.3](#) [► L4.7 @ 3:25]. In stages [► L4.7 @ 3:46]–[► L4.7 @ 9:04]:

1. **Dinner.** Mehrling's expenditure, Virelli's receipt. Signing the slip is not paying but *borrowing*: an IOU to MasterCard (Mehrling borrows, MasterCard accumulates). Virelli lets him go because it gets MasterCard's IOU (MasterCard borrows, Virelli accumulates) [► L4.7 @ 4:33]. Two new credits, out of thin air.
2. **Merchant settlement.** The next day or week Virelli presents its claim: Virelli decumulates and hoards cash, MasterCard dishoards and repays [► L4.7 @ 7:00]. For Virelli it is as if it had been paid in cash.
3. **Cardholder settlement.** At the end of the month Mehrling repays (disharding cash, repaying debt), and MasterCard decumulates the IOU and hoards cash [► L4.7 @ 8:24].

	Mehrling		MasterCard		Virelli	
	uses	sources	uses	sources	uses	sources
goods	1: dinner				1: dinner	
assets			1: IOU _M	3: IOU _M	1: IOU _{MC}	2: IOU _{MC}
liabilities	3: repay	1: IOU _M	2: repay	1: IOU _{MC}		
money	3: dishoard		3: hoard	2: dishoard	2: hoard	

Credit expanded, two credits at once, and then collapsed in two phases; at the end there is no credit left, and cash settled the debts: the elasticity came from credit, the discipline from the debts having to be paid [► L4.7 @ 9:26].

Exercise 4.2 (Sources and uses). Record in the sources-and-uses framework, for all agents involved:

- (a) you receive a \$3000 salary paid into your bank account;

- (b) you buy \$1000 of shares from another investor, paying from your account;
- (c) you repay \$500 of a bank loan;
- (d) a firm sells a used machine to another firm on 30-day trade credit.

Check both rules in each case.

Exercise 4.3 (The credit card, completed). Write the three stages of the dinner example as changes in T-accounts (stocks) for Mehrling, MasterCard and Virelli, assuming each pays through a bank. What is the total quantity of credit outstanding after each stage?

4.7 The flow of funds accounts

Morris Copeland, *A Study of Moneyflows in the United States* (NBER, 1952), proposed these accounts as a framework for macroeconomic analysis [► L4.8 @ 0:07]. They apply the sources-and-uses framework to whole sectors, with aggregation and netting, which is what makes them hard to read for people who do not have the framework in mind [► L4.8 @ 0:27].

- **Columns:** sectors (households and nonprofit organizations, nonfinancial business, state and local governments, federal government, domestic financial sectors, rest of the world), each with a “uses” and a “sources” column [► L4.8 @ 0:48].
- **Rows:** first the goods and services part (about 12 lines), then the financial instruments: various kinds of money first (official reserves, SDRs, Treasury currency, deposits; lines 14–22), then forms of credit (from line 23), in the reverse order of the hierarchy [► L4.8 @ 1:09].
- **Checks:** by rule (i) each sector’s uses and sources columns must have the same sum; by rule (ii) each row must sum to zero. They do not quite: the *statistical discrepancies* show where the net has missed something [► L4.8 @ 1:50].
- **Negative entries:** the published accounts record repayment as a negative source (a reduction of borrowing) and decumulation as a negative use; this moves entries to the other side but keeps the rules [► L4.8 @ 3:12].

Copeland against Keynes and the monetarists [► L4.8 @ 4:54]:

- $C + I + G = Y$ ignores purchases and sales of financial assets and of existing goods (old houses, a very big market), and much of what happens in the monetary and financial system [► L4.8 @ 5:15];
- $MV = PT$ treats all transactions as made with money, “as if the whole economy were Oren’s Daily Roast”, while most transactions are credit transactions, absorbed into velocity [► L4.8 @ 5:56].

In the flow of funds the sale of goods and the sale of assets are equivalent ways of obtaining a source of funds; the national accounts shunt the latter aside. And the Keynesian framework, the notes observe, grows out of the quantity theory: $C + I + G + X - M$ is a disaggregation of MV , and Y a subset of PT ; Copeland wanted to go further [Notes 4 p.6]. Mehrling thinks Copeland was right, but he did not win: the accounts were compiled by a small division of the Board of Governors, but not used for policy, because no macroeconomic theory was attached to them; Keynesians and monetarists each had theirs. “So there’s a Nobel prize for you” [► L4.8 @ 6:17].

What the accounts miss. In the table shown, the largest discrepancy in the last column is \$115 billion on line 22, “Fed funds and security RPs”: the wholesale money market escapes the statistical net [► L4.8 @ 7:18]. Sector discrepancies are large too (\$561 billion for households and nonprofits, \$440 billion for domestic nonfinancial sectors), partly because “nonfinancial” firms are also financial: GM makes cars, but it is also a kind of bank with international operations [► L4.8 @ 7:59]. And there are no derivatives at all, no swaps, no options: they did not exist in 1952, and they are off balance sheet [► L4.8 @ 9:22]. After the crisis the Office of Financial Research was created to build new accounting structures; Mehrling proposed to “soup up” the accounts to include swaps and derivatives, as the course will do [► L4.8 @ 8:59].

Exercise 4.4 (Statistical discrepancy). In a two-sector economy households lend 100 to firms through repo, and firms report borrowing 85. Explain how the discrepancy shows up in the row and column checks of the flow of funds matrix, and where you would look for the missing flow.

COURSE NOTE: Flow of funds today
The flow of funds accounts have been published by the Fed since 1955, now as the *Financial Accounts of the United States* (release Z.1). The Office of Financial Research, created by the Dodd–Frank Act in 2010, is an office of the US Treasury Department, not of the Fed (in class it is called “a division of the Fed” [► L4.8 @ 8:59]).

4.8 Liquidity, long and short

The survival constraint applies not only today but tomorrow and after: pushing a problem into the future lets you fight another day, but you should push it where you expect cash inflows, not where you have other commitments [► L4.10 @ 0:28]. Example, over four periods $t, \dots, t + 3$, with cash inflow 10 every period (40 in all) and total commitments 20: no solvency problem, a “profit” of 20 [► L4.10 @ 1:31].

	t	$t + 1$	$t + 2$	$t + 3$	
cash inflow (all cases)	10	10	10	10	
commitments, hedge	5	5	5	5	never binds
commitments, short-term	20	0	0	0	binds today
commitments, future mismatch	0	20	0	0	binds tomorrow

Figure 4.3: Same stocks, different flows [► L4.10 @ 2:35][Notes 4 p.7]: in all cases inflows total 40 and commitments 20 (ignoring discounting), so the balance sheet is the same; only the time pattern differs. The notes call the three a liquid capital structure, an illiquid one with a current mismatch, and an illiquid one with a future mismatch.

- 1. **Hedge (liquid) financing:** commitments of 5 every period; the constraint never binds, the question “how am I going to meet it?” never arises [► L4.10 @ 2:35].
- 2. **Short-term financing:** the business is fine (10 every period) but financed with short-term debt, all due today: 20 owed, 10 coming in. One of the three choices is forced now: use reserves, roll over the debt, or sell the business [► L4.10 @ 3:48]. Short-term borrowing is the sign of vulnerability, and real banks, with demand deposits and overnight repo funding, look like this all the time; hence the importance of the overnight market, enormous and invisible to households [► L4.10 @ 4:30].
- 3. **A problem coming later** (a third case, drawn on the board with “a little leeway”): no problem today, but a future misalignment of flows and commitments that the agent, and its creditors, can see. While still liquid, the agent refinances, spreading the debt over time or prepaying part of it: *anticipated* liquidity constraints have consequences

for liquidity today, because tomorrow the markets may be frozen and a perfectly solvent agent could be “dead” [► L4.10 @ 6:00].

Price-insensitive borrowers. The economy is a crowd of agents with different time patterns of flows and commitments, all trading in the same market [► L4.10 @ 7:22]. An agent under stress must borrow now, whether the short rate is 10% or 20%: its demand is price-inelastic, and the more such agents there are, the further rates can move, for reasons that have nothing to do with time preference [► L4.10 @ 8:03]. Think of Bear Stearns or Lehman before their collapse: this is the stuff of financial crises, and all banks, with short-term liabilities that can vanish tomorrow, look like this [► L4.10 @ 8:44]. Agents with surplus cash lend it into the market. Over the business cycle the balance swings between easy and tight credit, and it shows up in the money market [► L4.10 @ 9:04].

In the notes’ words, all three types meet in the money market at every moment: those with a current mismatch are *necessitous borrowers*, who must borrow whatever it costs; those with a future mismatch may borrow today to avoid tomorrow’s problem, or wait; those with current liquidity may help the others, or do something else. “The result of all this pushing and pulling is the money rate of interest” [Notes 4 p.8]. If there is a mismatch, someone must be persuaded to give up current cash for a mere promise of future cash, and the relative price of the two is the rate of interest: hence mismatches show up as upward pressure on the short-term money rate. And since people move anticipated future problems into the present, where they are easier to handle, the state of the money market measures how well the whole structure of commitments matches the pattern of cash flows, looking forward as well as today [Notes 4 p.8][Notes 4 p.9].

Definition 4.4 (New concepts: *hedge financing, rollover*). *Hedge financing* (Minsky): commitments in every period are covered by expected cash inflows of the same period. To *roll over* a debt is to repay it with the proceeds of a new debt, pushing the commitment into the future. (Minsky also defines *speculative* units, whose near-term commitments exceed near-term inflows and must be rolled over, and *Ponzi* units, which must borrow even to pay interest.)

4.9 Financial fragility: flows and stocks

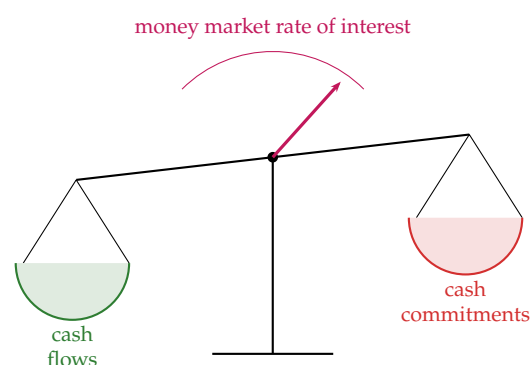


Figure 4.4: The scales [► L4.11 @ 0:07]: the balance between the pattern of cash flows and the pattern of cash commitments, across the whole economy, sets the money market interest rate.

The image, “because we’re in New York”, is the scales of Lady Liberty: one pan holds

cash flows, the other cash commitments, and the pointer is the money market interest rate [► L4.11 @ 0:07]. The Fed does something too, but behind the scenes there are balance sheet facts: promises to pay at future dates that either are or are not lined up with the prospects of whoever made them; if they are not, that person is forced to act. “It’s not about preference, it’s about being forced to do something by the promises you’ve made in the past” [► L4.11 @ 1:10]. You buy a house, then lose your job: the cash flows go away, the commitments stay [► L4.11 @ 1:52].

Proposition 4.5 (Minsky’s fragility, in the money view). *Financial fragility is the balance of cash commitments (promises to pay) relative to the cash flows actually arising from activity (wages, profits, sales). When most agents’ cash flows cover their commitments there is no crisis; when many agents’ do not, the system becomes fragile, and the crisis shows up in the money market rate of interest [► L4.11 @ 2:12].*

The money market is thus the “crisis detector”, the “stress detector”: the bouncing of its rate reflects the shifting balance between cash flows and commitments, a “sufficient statistic” for the state of the macroeconomic balance [► L4.11 @ 3:15].

Stocks are residues of flows. In a T-account, financial assets and money on the left, financial liabilities and net worth on the right; net worth is solvency [► L4.11 @ 3:57]. Each stock is the accumulation of past flows: assets bought, cash hoarded, debts incurred [► L4.11 @ 4:38]. And each looks forward too: liabilities and assets have dates, the dates of future payments; the stock value adds them up, as a discounted present value (40 and 20 in the examples) [► L4.11 @ 5:18]. The examples of fig. 4.3 have identical stocks and very different prospects: the balance sheet shows solvency; to see fragility one must look behind it, at the time patterns [► L4.11 @ 5:38].

Solvency as the outer bound. For the money view solvency matters mainly as an outer bound on each agent’s credit limit, which is somehow related to net worth: solvent agents have unused borrowing power they can mobilize to make payments. So asset price fluctuations move borrowing power, with consequences for immediate liquidity: solvency problems can easily become liquidity problems [Notes 4 p.10].

Key formulas and ideas

Example 4.6 (A factory is a bond). Finance values assets, real ones too, as the present value of future cash flows: from a financial point of view a factory is just a kind of bond [Notes 4 p.9]. A factory expected to yield a net cash flow of \$10 a year in perpetuity, with the interest rate at 5% and expected to stay there, is worth

$$\sum_{t \geq 1} \frac{10}{(1+R)^t} = \frac{10}{R} = \$200.$$

Money is easier: its face value is its present value [Notes 4 p.9].

KEY FORMULAS

Money vs credit payments	money: existing balances move, total fixed; credit: outstanding IOUs increase with the transaction
Payment system	a credit system (elasticity), settled in money (discipline)
Survival constraint	every period: cash inflow $\geq$ cash commitments (not an intertemporal budget constraint)
Shortfall	dishoard, borrow, or sell assets; only reserves are dependable in a crisis
Sources and uses	receipts/expenditure; decumulation/accumulation; borrowing/repayment; dishoarding/hoarding
Two rules	(i) each agent: uses = sources; (ii) each use is another agent's source $\Rightarrow \geq 4$ entries per transaction
Fragility	commitments vs cash flows in time; shows up in the money market rate
Stocks	accumulated past flows and discounted future flows; same stocks can hide different time patterns

Chapter 5

The Central Bank as a Clearinghouse

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L5.1 Martin Wolf on QE3	3:16	Wolf: QE3 more likely to do too little than too much; the nominal GDP gap.
L5.2 One Big Bank	8:20	A payment system as one big bank: money version and credit version; the payment matrix.
L5.3 Multiple Banks, a challenge	3:56	Many banks: paying in reserves is more constraining.
L5.4 Reading: Charles F. Dunbar	2:05	Dunbar's <i>Chapters on the Theory and History of Banking</i> (1891).
L5.5 Correspondent banking	10:24	Netting, correspondent balances as a swap of IOUs; who expands.
L5.6 Correspondent banking, system network	3:55	The hierarchical network of correspondents up to New York.
L5.7 Clearinghouse, normal operations	8:28	Clearinghouse certificates, multilateral netting, settlement.
L5.8 Clearinghouse, private lender of last resort	10:44	Clearinghouse loan certificates in crises; why the Fed replaced them.
L5.9 Central Bank Clearing	4:54	The Fed as a public clearinghouse: Fedwire, Fed funds, discount loans.
L5.10 Central Bank Cooperation	5:41	Internal and external drain; the major central banks acting as a clearinghouse.

Files. Transcripts `materials/transcripts/L05-P*.txt`; Mehrling's notes [Notes 5 p.1]. **Reading.** Charles F. Dunbar, *Chapters on the Theory and History of Banking* (1891; 2nd ed. 1901), the chapter on the checking system (not among the course files).

Motivation

This lecture opens the first block of the course, *banking as a system of payments*, and *central banking as a clearinghouse* [► L5.2 @ 0:08]. Starting from payments is unusual, but it builds the intuition for what follows: why the Fed funds market is so central, what goes on in repo and in the Eurodollar market [► L5.2 @ 0:28]. The guiding idea [► L5.2 @ 0:48]:

all the institutions and practices of the monetary system are attempts, worked out over a long time, to make a system of many banks work as if it were one big bank.

The lecture goes up the hierarchy: one big bank (section 5.2); many banks (section 5.3) linked by correspondent balances (section 5.5); the clearinghouse at the top, in normal times and in crises (sections 5.6 and 5.7); the Fed as a public clearinghouse (section 5.8); and the central banks of the world as a clearinghouse one level higher (section 5.9).

5.1 FT: Martin Wolf on QE3

CASE STUDY: FT, September 2012: “Bernanke makes a historic choice”

Martin Wolf, the FT’s chief economics commentator, writes every Wednesday [► L5.1 @ 0:09]. He argues that critics of the new Fed policy are right that it will probably fail to work as hoped, but wrong that it will do vast damage: the fear of hyperinflation is misguided, and the costs of deficient demand are far greater; the Fed “is in truth more likely to achieve too little of what it seeks than too much” [► L5.1 @ 0:50]. Mehrling agrees: it is not clear how QE translates into aggregate demand [► L5.1 @ 1:10]. Wolf’s chart of nominal GDP against its pre-crisis trend, the goal that Bernanke and Michael Woodford discussed at Jackson Hole, shows a gap that is widening, not closing; perhaps because fiscal policy is stuck, too much is being asked of the central bank, in the US and in Europe [► L5.1 @ 1:50].

5.2 One big bank

“One big bank” can mean two things, following the money/credit distinction of lecture 4 [► L5.2 @ 1:29].

The money version. The bank holds reserves (say gold) and owes deposits to people, α and β . A payment from α to β is a book entry from one account to another; no reserves move at all [► L5.2 @ 2:11]. The limit: when a deposit is run down to zero, no more purchases are possible, unless one can borrow, bilaterally, from someone who has a deposit [► L5.2 @ 2:54]. The IMF, with its Special Drawing Rights, is built on this discipline model: a fixed balance sheet, reallocated, expanded only now and then “in one big tranche” [► L5.2 @ 3:14].

The credit version. The bank has deposits and also *overdrafts* of people γ and λ , on the asset side: it uses the fact that no reserves are really needed [► L5.2 @ 3:35]. Now the balance sheet may expand or contract, depending on the pattern of payments [► L5.2 @ 4:16].

ASIDE: The Money View blog

Mehrling and his co-blogger Daniel Neilson posted on QE3 on their blog, together with a short video on Draghi and the Soros plan (section 2.1) [► L5.1 @ 1:30].

One big bank (money)		One big bank (credit)	
Reserves	Deposit α Deposit β	Overdraft γ Overdraft λ	Deposit α Deposit β
from \ to	α β	γ λ	
α	0 0	– –	
β	0 0	– –	
γ	+	+	0 0
λ	+	+	0 0

Figure 5.1: The payment matrix of the credit one-big-bank [► L5.2 @ 4:37]: change in the size of the balance sheet when the row agent pays the column agent. α, β have deposits, γ, λ overdrafts.

Reading the matrix [► L5.2 @ 4:58]:

- paying oneself, or a depositor paying a depositor: no change, exactly as in the money version;
- an overdrawn agent paying another overdrawn agent: one overdraft grows, the other shrinks, total unchanged;
- a depositor (α) paying an overdrawn agent (γ): α 's deposit falls and γ 's overdraft is partly repaid: the balance sheet *contracts* [► L5.2 @ 6:21];
- an overdrawn agent paying a depositor: the overdraft grows and the deposit grows: the balance sheet *expands* [► L5.2 @ 6:41].

The aim of a payment system is that whenever there is a willing buyer and a willing seller, the payment system does not stand in the way. The credit system can do everything the money system does and more; the actual US domestic payment system has this credit character, and the Federal Reserve System is an example of it [► L5.2 @ 7:22].

♦ **BEYOND THE CORE: What is the quantity of money?**

In the money version it is natural to call the deposits the quantity of money. With overdrafts the measure becomes ambiguous; the notes list three candidates [Notes 5 p.1]:

1. deposits, which counts only the positive balances;
2. deposits minus overdrafts;
3. deposits minus credit limits (the unused borrowing power counts as means of payment too).

This ambiguity led Fischer Black to propose not trying to measure money at all, in “Banking in a World without Money” (lecture 22) [Notes 5 p.2].

Definition 5.1 (New concepts: *overdraft*). An *overdraft* is a negative balance on a deposit account, i.e. a loan from the bank created automatically when the holder pays more than he has (one “draws over” the balance). It is an asset of the bank.

Exercise 5.1 (The payment matrix). In the credit one-big-bank, α and β have deposits of 50 each, γ and λ overdrafts of 30 each. Compute the size of the balance sheet after each of the following payments, applied in sequence: γ pays α 20; α pays λ 40; λ pays γ 10.

5.3 Many banks

There is no one big bank. In the United States there are thousands of banks, fewer than the six or seven thousand of Young’s time, some very big (Citibank, Bank of America), many small or regional; in some countries, such as the UK or Canada, there are a handful of big banks [► L5.3 @ 0:07]. The problem is how to knit them into one country-wide system in which anybody can pay anybody [► L5.3 @ 0:48].

Take bank A with depositors α, β and bank B with depositors γ, λ , all with positive deposits; the credit element now is *interbank* credit. This is the wholesale money market, behind the scenes, and its elasticity is what provides elasticity in the retail market

[► L5.3 @ 1:09]. One could pay in reserves: when α pays γ 10, bank A sends 10 of reserves to bank B [► L5.3 @ 2:31]:

Bank A		Bank B	
-Reserves 10	-Deposit α 10	+Reserves 10	+Deposit γ 10

This is *more* constraining than one big bank: not only must α have a deposit, bank A must have reserves. So banks do not generally do this within a country [► L5.3 @ 3:12].

5.4 Reading: Charles F. Dunbar

READING: Dunbar, *Chapters on the Theory and History of Banking* (1891)

This was the textbook used at Harvard College; the course reads its chapter on how the checking system works in the United States [► L5.4 @ 0:08]. It explains a payment system without a central bank (there was no Fed in 1891), which helps to understand why one was eventually created; and it is “crystal clear”: no four-colour diagrams, no workbook, no slides [► L5.4 @ 0:29]. Dunbar was the first professor of money and banking at Harvard, a former journalist who had written on banking for a Boston paper; the second edition was enlarged by O. M. W. Sprague, of the Harvard Business School, a good friend of Allyn Young [► L5.4 @ 1:09].

5.5 Correspondent banking

Collecting and netting. During the day bank A receives checks, orders to pay. It does not pay them one by one: it collects them and sorts them by *bank* (the routing number at the bottom of a check), not by customer; which account at bank B gets credited is bank B’s problem [► L5.5 @ 0:27]. Checks between customers of bank A itself need no reserve flows; for each other bank it sums what it owes (“due to B”) and what it is owed (“due from B”), and at the end of the day computes the net, which may be positive or negative [► L5.5 @ 1:29]. Netting already saves moving reserves back and forth across the street every day [► L5.5 @ 3:43].

Correspondent balances: a swap of IOUs. Banks that do business regularly can do better: bank A holds a deposit at B in its own name (a *banker’s balance*), and B holds one at A. Instead of paying for these deposits with reserves, they swap them: each says to the other “I owe you \$1000”. Both balance sheets expand, deposits “created from thin air” [► L5.5 @ 4:04].

Bank A		Bank B	
+Deposit at B	+Deposit of B	+Deposit at A	+Deposit of A

It looks like a free lunch that cannot do anything, but it does: now net payments can be made without moving reserves [► L5.5 @ 5:30]. For α (at A) to pay λ (at B) 10, there are two ways [► L5.5 @ 5:51]–[► L5.5 @ 8:16]:

HISTORY: Dunbar and Sprague

Charles Franklin Dunbar (1830–1900) edited the *Boston Daily Advertiser* before becoming, in 1871, the first professor of political economy at Harvard; he founded the *Quarterly Journal of Economics* in 1886. The second edition of his *Chapters* (1901) was edited by Oliver M. W. Sprague (1873–1953), later the author of *History of Crises under the National Banking System* (1910) for the National Monetary Commission. In class the paper is called “the Boston Globe” [► L5.4 @ 1:09].
References: F. W. Taussig, “Charles Franklin Dunbar”, *QJE* 14 (1900); Dunbar, *Chapters on the Theory and History of Banking*, 2nd ed., Putnam, 1901.

1. A draws down its deposit at B: A's balance sheet *shrinks*, B's only changes owner (B swaps an IOU to A for an IOU to λ);
2. B increases its deposit at A (A's liabilities to B grow): A's balance sheet does not change in size, B's *expands*.

(1)		Bank A		Bank B	
		–Deposit at B 10	–Deposit α 10		–Deposit of A 10 +Deposit λ 10
(2)		Bank A		Bank B	
			–Deposit α 10 +Deposit of B 10	+Deposit at A 10	+Deposit λ 10

Either way it is credit: either the debtor bank's balance sheet shrinks or the creditor bank's expands [► L5.5 @ 8:16]. In the first case total deposits, counting interbank balances, fall; in the second they rise [Notes 5 p.3]. The invention of correspondent banking moves payments *between banks* from a money system to a credit system: they are no longer constrained by the quantity of gold, only by bilateral credit limits [Notes 5 p.3].

Definition 5.2 (New concepts: **correspondent balances**). A *correspondent balance* (banker's balance) is a deposit that one bank holds at another to settle payments between them. The bank that holds the account is the *respondent*, the bank that keeps it the *correspondent* (the two "correspond", i.e. exchange letters and orders, in the old sense of the word).

Which one gets picked? Hierarchy. In practice the big city bank says to the small country bank: I have no reason to hold deposits with you; you hold a deposit with me, paid in reserves up front, or I lend it to you for a fee, and my deposits are the clearing balances [► L5.5 @ 8:37]. This is discipline for the debtor bank: to join the payment system a country bank must keep balances at an urban bank; if it runs out there is no guarantee the city bank will expand to help it, and it may charge extra. "This is the hierarchy of money showing up in the payment system" [► L5.5 @ 9:17]. Correspondent balances make the system work much more like the ideal one big bank on a credit basis; the retail customer sees none of this [► L5.5 @ 9:58]. The notes: the more central bank always takes deposits from the less central; in the US country banks held deposits at city banks, and those reserve cities survive as the cities of the twelve Federal Reserve Banks [Notes 5 p.3].

The network. Before the central bank, money-centre banks (Cleveland, Chicago, ...) had correspondent relations with banks in their hinterland; payments within a region just moved deposits at the centre bank [► L5.6 @ 0:07]. Payments between regions need a payment one layer up, between centre banks; in practice this was not done bilaterally but hierarchically, through a few New York banks at which all the money centres held accounts, with further regional layers [► L5.6 @ 1:10]. The whole structure nets as much as it

can, so that reserves do not have to flow, and builds credit relations locally and higher up, balancing elasticity and discipline from the smallest bank “in the middle of nowhere” up to J. P. Morgan in New York [► L5.6 @ 2:53]. At the top, the biggest New York banks must clear with each other: that is where the clearinghouse comes in [► L5.6 @ 3:34].

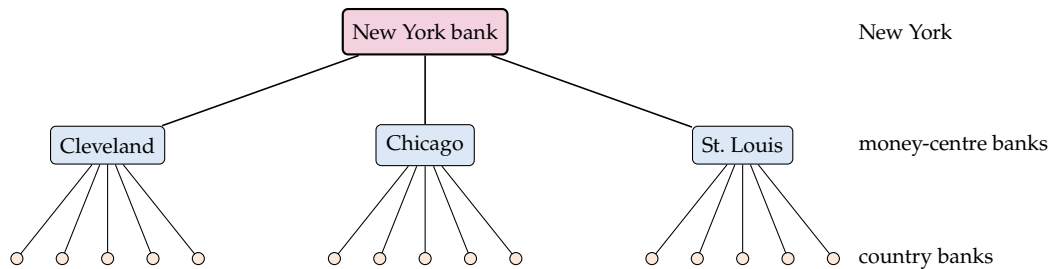


Figure 5.2: The correspondent network before the Fed [► L5.6 @ 0:07]: country banks hold balances at their money-centre bank, money-centre banks at a New York bank; a payment between two country banks in different regions moves balances at every level up to the common correspondent.

Exercise 5.2 (Correspondent balances). Country bank C holds a balance of 100 at city bank N. Customers of C pay customers of N 30 more than they receive.

(a) Write the T-accounts when the net is settled by drawing down C’s balance.

(b) The balance is now too low; N offers to lend C 50 by crediting its account. Write the T-accounts and explain why this is “discipline for the debtor bank”.

5.6 The clearinghouse in normal times

The New York Clearing House Association (NYCHA) is a mutual organisation of the big New York banks, “first among equals” J. P. Morgan [► L5.7 @ 0:08]. It was set up on the discipline model: every member deposits gold or currency, and the clearinghouse issues *clearinghouse certificates* to its credit, physical notes standing for the reserves [► L5.7 @ 0:55].

Clearinghouse	
Gold (reserves)	Clearinghouse certificates (to A, B, ...)

During the day: credit. The members agree to treat every amount due to or from another member as due to or from the clearinghouse itself [► L5.7 @ 1:58]. Each bank then has a single net position against the clearinghouse, smaller than the sum of its bilateral nets, because positives and negatives offset: *multilateral netting* [► L5.7 @ 2:40]. In Dunbar’s description these are actual checks: at the end of the day the banks add them up and meet at the clearinghouse, and those in net deficit hand clearinghouse certificates to those in net surplus; then they go home [► L5.7 @ 3:42].

INTUITION: Bilateral versus multilateral netting

With n banks and gross obligations x_{ij} (from i to j), bilateral netting settles $|x_{ij} - x_{ji}|$ for each pair, while multilateral netting settles only each bank's net position $\sum_j (x_{ij} - x_{ji})$. The sum of the multilateral nets is never larger than the sum of the bilateral ones (triangle inequality), and is usually much smaller: a ring of payments $A \rightarrow B \rightarrow C \rightarrow A$ of equal size nets to zero.

“There is a credit expansion during the day that creates elasticity, and then there's a settlement at the end of the day”: elasticity and discipline at the very highest level of the system [► L5.7 @ 4:23]. The members know they have this elasticity and can sell it to the banks below them, and so on down: the whole system operates as a credit system because the clearinghouse does, during the day. Sometimes settlement is every three days, which is why the length of the *float* matters [► L5.7 @ 5:05].

Failing to settle. A bank that has run down its certificates over several days of deficits has three options [► L5.7 @ 5:48]:

1. pay in clearinghouse certificates, if it has them;
2. borrow certificates bilaterally from another member, e.g. from J. P. Morgan, or from a member in net surplus;
3. if nobody will lend, because it is thought insolvent, default: the clearinghouse's gold goes to pay the creditors pro rata, the losses are shared among all the members, and the defaulter is expelled, “banished to the outer ring of hell”.

Proposition 5.3 (Joint credit). *The credit of a group of banks that jointly and severally guarantee an obligation is better than the credit of any individual bank: clearinghouse credit is better than the bilateral credit of its members* [► L5.7 @ 7:32].

Definition 5.4 (New concepts: clearing, settlement, netting, float). *Clearing* is the process of collecting and offsetting payment orders between banks; *settlement* is the final transfer of the means of payment (reserves) that discharges the net obligations. *Netting* replaces gross obligations by net ones. The *float* is the credit that exists between the moment a payment is credited to the payee and the moment it is settled.

Exercise 5.3 (Netting). Four banks have the following gross obligations at the end of the day (row pays column): $A \rightarrow B$ 30, $B \rightarrow C$ 40, $C \rightarrow A$ 35, $A \rightarrow C$ 10, $D \rightarrow A$ 5, $B \rightarrow D$ 15.

- (a) Compute the total gross payments, the total of bilateral nets, and the total of multilateral nets.
- (b) Which banks must deliver certificates to the clearinghouse, and how many?

5.7 The clearinghouse as private lender of last resort

In times of stress all the members need to make payments, to their correspondents or abroad, in gold; they all want to withdraw their accounts at the clearinghouse, and even the biggest banks can run out of certificates: the country banks hold their reserves in New York, and the New York banks do not have enough gold to cover them [► L5.8 @ 0:08]. Then the clearinghouse creates money, not by bilateral borrowing but by joint borrowing and

lending [► L5.8 @ 0:50]: it lends to bank A (at 6% in the example) by printing a *clearinghouse loan certificate*, a liability of the clearinghouse, guaranteed by all its members, against whatever collateral A can pledge, for example government securities [► L5.8 @ 1:32].

Clearinghouse		Bank A	
+Loan to A (6%)	+Loan certificates	+Loan certificates	+Loan from clearinghouse

Money has been created at the apex of the system: more of the best money, which circulates among banks as if it were gold. The certificates say “loan certificate”, but they are treated exactly like the ordinary certificates, since both are liabilities of the clearinghouse [► L5.8 @ 2:38]. This is a lender of last resort operation, managed by the clearinghouse; the New York clearinghouse still exists, still clears for its members, and clears much of the Eurodollar market [► L5.8 @ 3:40].

Bagehot’s rule in private hands. A committee decides [► L5.8 @ 4:21]. Six percent in 1907 was a high rate, “lend freely at a high rate against good security”: high, so that the borrower wants to repay. So high, in fact, that banks liked to keep these certificates, the safest asset in the system at 6%, and after the crisis the clearinghouse had trouble getting them back [► L5.8 @ 4:42]. Good security: the bankers know each other and would not take “garbage”; a bank without acceptable collateral had to borrow some from another bank first, unlike the Fed, which can decide to let it pass [► L5.8 @ 5:43].

Playing favourites. Young describes how, in crises, New York banks refused to pay the deposits of country banks in reserves, offering loan certificates, take it or leave it, and kept their gold to pay abroad, where the Bank of England would accept nothing but proper international money [► L5.8 @ 6:46]. It was never clear that the certificates were legal, since they were used as money (unlike the bank notes backed by government bonds of [section 3.4.3](#)) [► L5.8 @ 7:47]; as long as they stayed among bankers nobody noticed, but sometimes they circulated. After 1907 a law made the practice legal, with rules on acceptable collateral [► L5.8 @ 8:29]. Some argue that this is all that should have been done, and there is some truth in it: it was a decentralised lender of last resort, discipline in good times and elasticity in crises [► L5.8 @ 9:11]. But the New York bankers were playing favourites; the banks of the hinterland, refused payment when everybody needed it, sometimes went bankrupt to save the New York banks, and pressed for democratic control. That is how the Fed came about: “one way to understand the Fed is that it is basically a clearinghouse” [► L5.8 @ 10:13].

Exercise 5.4 (Loan certificates). Bank A owes the clearinghouse 20 at settlement and has no certificates. The clearinghouse issues 20 of loan certificates at 6% against A’s government bonds. Write the T-accounts of the clearinghouse, A, and the creditor bank B. What happens a week later when A repays? Why might B prefer not to be repaid?

5.8 The Fed as a public clearinghouse

The Federal Reserve, when on the gold standard, held gold and issued Federal Reserve notes and bank reserves (deposits of banks at the Fed) [► L5.9 @ 0:07]. It can run intraday

COURSE NOTE: History of the NYCHA

The New York Clearing House Association was founded in 1853, long before Morgan’s prominence; Morgan’s role as “first among equals” refers to the crises of the 1890s and 1907. Clearinghouse loan certificates were first issued

credit on its own books: *Fedwire*, now electronic [► L5.9 @ 0:49]. The private clearing system of the New York banks is today called *CHIPS*. CHIPS clears first, then Fedwire, and Fedwire is “real money”, deposits at the Fed, the ultimate settlement asset [► L5.9 @ 1:12].

Federal Reserve	
Gold	Federal Reserve notes
Discount loans	Bank reserves

Overdrafts and balances build up during the day. At the end of the day a bank without a positive balance at the Fed must [► L5.9 @ 1:33]:

- 1. borrow from another bank that has a surplus: the *Fed funds market* (next lecture);
- 2. or borrow from the Fed, which makes *discount loans* by expanding its own balance sheet, if the banks cannot find each other or B is unwilling: nobody forces them [► L5.9 @ 2:16];
- 3. or default.

Discount loans to troubled banks appear on the Fed’s balance sheet and are small: with thousands of banks, one fails every week or so, and whole divisions deal with them; they matter for the aggregate only in system-wide stress [► L5.9 @ 2:58]. Then the Fed does what the clearinghouse did, but legally, as monetary policy: it expands both sides of its balance sheet to create more of the best money. Bernanke goes further than any clearinghouse would: he does not only lend to banks, he buys mortgage-backed securities from whoever holds them, and lends to broker-dealers; the clearinghouse was a club, the Fed has more members and in a crisis deals with the whole economy [► L5.9 @ 3:18]. Private and public clearinghouse differ in membership, rules and priorities, but the mechanics are very much the same [► L5.9 @ 4:41].

INTUITION: One step beyond the clearinghouse

In the notes, central banking is “nothing more than one step beyond the clearinghouse”: it regularizes and strengthens the clearinghouse and erases the difference between clearinghouse certificates (backed by gold) and clearinghouse *loan* certificates (backed by members’ loans), both becoming deposits or currency of the central bank [Notes 5 p.5]. A clearinghouse can be broken by any single member who insists on redeeming its certificates in gold; a central bank cannot be broken by internal forces, because its liabilities *are* the member banks’ reserves. All payments inside the system net out on its books: it makes the system of banks work as if it were a single bank. What remains is managing relations with the rest of the world, internal and external drains, and that is the origin of the “art of central banking” [Notes 5 p.5].

Definition 5.5 (New concepts: *Fedwire*, *CHIPS*). *Fedwire* is the Federal Reserve’s real-time gross settlement system, which transfers reserve balances between banks’ accounts at the Fed. *CHIPS* (Clearing House Interbank Payments System), operated by The Clearing House since 1970, is the private system on which large-value, mostly international dollar payments are netted among member banks and then settled through their accounts at the Fed.

in New York in 1857 and 1860 and in most crises thereafter (1873, 1884, 1890, 1893, 1907). The law Mehrling refers to is presumably the Aldrich–Vreeland Act (1908), which authorised groups of national banks (“national currency associations”) to issue emergency currency against commercial paper and other securities; it was used once, in 1914. Today the association’s successor, The Clearing House, operates CHIPS.

HISTORY: The panic of 1907

In October 1907 the run on the Knickerbocker Trust Company spread to the trust companies, which were not members of the New York Clearing House. J. P. Morgan assembled bankers in his library to organise loans to the weaker institutions, and the clearinghouses of New York and of many other cities issued loan certificates; currency went to a premium over certified checks, i.e. par between currency and deposits was broken. References: Bruner and Carr, *The Panic of 1907*, 2007; Sprague, *History of Crises under the National Banking System*, 1910.

5.9 Central bank cooperation

What can kill a clearinghouse is members withdrawing their gold because they need it [► L5.10 @ 0:08]. Why would they? Distinguish [► L5.10 @ 0:28]:

Definition 5.6 (New concepts: *internal* and *external drain*). An *internal drain* is a loss of reserves to the domestic public, which wants cash; an *external drain* is a loss of reserves to foreigners, who want the international money (gold). An internal drain can be met by expanding the balance sheet at the top of the national hierarchy; an external drain comes from a level higher than the national system, and can kill a clearinghouse or a central bank.

The same holds for central banks: if foreign central banks bring Federal Reserve notes and ask for gold, the Fed can run out, because its notes are not fully covered [► L5.10 @ 1:09]. Then it *suspends* payments: “I know these were promises to pay gold, I changed my mind; I will pay some time later”: it abandons the fixed mint parity, and all central banks have done it, the Bank of England periodically [► L5.10 @ 1:51]. An internal crisis can be handled by expanding the balance sheet; when the whole country loses reserves, the only way to get elasticity is to eliminate the discipline by going off gold [► L5.10 @ 2:32], unless all the central banks agree to act as a clearinghouse and expand together, one layer higher [► L5.10 @ 2:53].

The world clearinghouse. That is what is happening now: Draghi and Bernanke announce QE; the Bank of England and the Swiss National Bank already do it; the Bank of Japan was in the FT the day before. “There are only five central banks that matter”, and whether or not they met in a room like the New York bankers, they act in concert: all expanding at the same time, so there is no external drain, “the Martians are not yet doing business with us” [► L5.10 @ 3:15]. They do what the IMF cannot, since the IMF needs all its members to agree to expand SDRs; the “C5” can do it privately, leaving the rest of the world out, like the New York bankers. Expect complaints from the rest of the world that the big central banks take care of each other and not of them [► L5.10 @ 4:37].

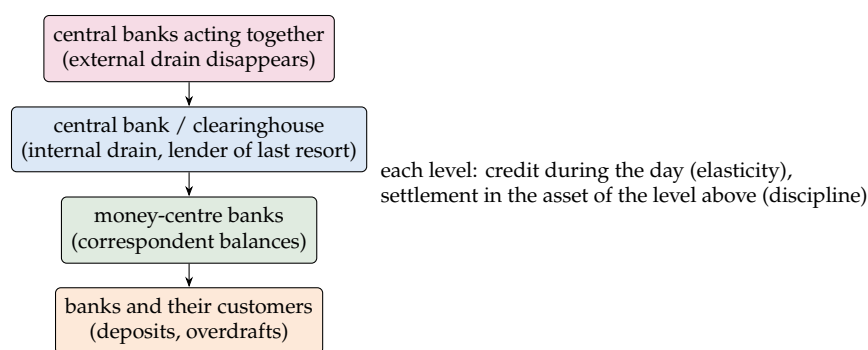


Figure 5.3: Clearing all the way up: the same mechanism of intraday credit and final settlement repeats at each level of the hierarchy, from bank customers to the central banks of the world.

Key formulas and ideas

KEY FORMULAS

One big bank	money version: payments are book entries, balance sheet fixed; credit version: depositor→overdrawn contracts, overdrawn→depositor expands
Many banks	paying in reserves is more constraining than one big bank
Correspondent balances	a swap of IOUs; a payment either shrinks the debtor bank or expands the creditor bank; hierarchy decides which
Clearinghouse	certificates against gold; intraday multilateral netting (elasticity); end-of-day settlement in certificates (discipline)
Failing to settle	pay, borrow from a member, or default (losses shared pro rata)
Loan certificates	clearinghouse lends its own liabilities against collateral at a high rate: private lender of last resort
Fed	public clearinghouse: Fedwire, Fed funds, discount loans
Drains	internal: met by expansion; external: suspension, unless central banks expand together

Chapter 6

Federal Funds, Final Settlement

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L6.1 FT: European Bank Deleveraging	5:43	European banks sell property loans to meet capital ratios; three overnight rates in the FT.
L6.2 What are Fed Funds?	5:13	An interbank market in reserves; the Fed funds rate over the cycle; excess reserves.
L6.3 Payment settlement versus Required Reserves	1:11	The two jobs of a Fed funds desk (Stigum).
L6.4 Payment elasticity/discipline, public and private	9:03	Daylight overdrafts at the Fed; CHIPS; real-time gross settlement.
L6.5 The Function of the Fed Funds Market	9:15	Bilateral overnight borrowing replaces central overnight lending; paying today by promising tomorrow.
L6.6 Payment versus Funding: an example	11:24	Buying an apartment with a mortgage: the Fed funds market and a matched-book dealer.
L6.7 Brokers versus Dealers	2:03	The difference.
L6.8 Payments Imbalances and the Fed Funds Rate	7:06	Imbalances, dealer prices, and how the Fed influences the rate through repo.
L6.9 Secured versus Unsecured Interbank Credit	5:05	Fed funds are unsecured; credit lines; stress in unsecured markets.
L6.10 Required Reserves, redux	7:20	Reserve requirements, lagged and contemporaneous reserve accounting, and why they are ineffective.

Files. Transcripts `materials/transcripts/L06-P*.txt`; Mehrling's notes [Notes 6 p.1]. **Reading.** Stigum, *The Money Market*, the chapter on the Fed funds market (quoted in class).

Motivation

The next three lectures cover the three main dollar money markets ► [L6.1 @ 3:57](#):

1. **Fed funds** (this lecture): unsecured interbank loans of reserves at the Fed, the market

for “the best money in the world”;

- 2. **repo** (lecture 7): secured loans;
- 3. **Eurodollars** (lecture 8): unsecured interbank loans of dollars offshore.

In the FT of the day their overnight rates are 0.15% (US dollar LIBOR, the Eurodollar rate), 0.29% (US overnight repo) and 0.16% (Fed funds effective): all overnight, all different. Why are they what they are? [► L6.1 @ 4:38]The thread of this lecture: the Fed funds market exists to meet the *survival constraint* of banks at the end of each day, i.e. *final settlement*.

6.1 FT: European bank deleveraging

CASE STUDY: FT, September 2012: “Europe’s banks race to offload property loans”

European banks hold property loans left over from the boom in Spain, Ireland and elsewhere, many no longer worth face value [► L6.1 @ 0:28]. A new round of regulation will require more capital against such risky loans, so that losses fall on private capital rather than the taxpayer [► L6.1 @ 1:10]. Capital requirements are ratios of capital to risky assets; rather than raise capital, the banks shrink the denominator: they package the loans, securitise some, and sell them, sometimes “at pennies on the dollar”, to private equity funds, and use the cash to pay down short-term borrowing or borrowing from the ECB [► L6.1 @ 1:52].

European bank		Private equity fund	
–Property loans	then:	+Property loans	
+Cash, then	–Short-term borrowing	–Cash	
–Cash			

This is *deleveraging*: shrinking the balance sheet on both sides to reach the desired capital ratio [► L6.1 @ 2:56]. The loans are sold at fire-sale prices, and the banks make no new loans: a credit crunch in Europe. The article is about the connection between banks and other financial institutions, and between banking and the real economy [► L6.1 @ 3:16].

Definition 6.1 (New concepts: capital requirement, securitisation). A *capital requirement* sets a minimum ratio of a bank’s capital (net worth) to its (risk-weighted) assets. *Securitisation* is the pooling of loans into a package whose cash flows back securities that can be sold to investors (the loans are turned into “securities”).

6.2 What are Fed funds?

In the flow of funds matrix for 2011 (here in stock form, assets and liabilities), line 12, “Fed funds and security repos”, shows for the domestic financial sectors \$823 billion of assets and \$1.1 trillion of liabilities: an *interbank* market, in which the financial sector borrows from and lends to itself; the numbers differ because some non-banks participate too [► L6.2 @ 0:07].

Definition 6.2 (New concepts: Fed funds). *Federal funds* are unsecured loans, typically overnight, of reserves (deposits at the Federal Reserve) between banks: a promise to pay reserves tomorrow morning in exchange for reserves today [► L6.5 @ 5:17]. Fed funds are *not* a liability or an asset of the Fed, despite the name: they are assets and liabilities of banks [► L6.2 @ 2:34].

In the hierarchy, from the banks' point of view reserves are money and Fed funds are credit: interbank promises to pay money, a credit expansion on the base of the money on the Fed's balance sheet [► L6.2 @ 1:53]. "If you get off on the wrong foot there, you can be halfway through the lecture before realizing where you are" [► L6.2 @ 2:34].

The rate. The Fed funds effective rate (FRED, St. Louis Fed) fluctuates with the business cycle; in the crisis the Fed lowered its target from 5% to 2% and then to nearly zero, where it has stayed, in a range of 0 to 0.25%, with the effective rate at 0.16% [► L6.2 @ 2:54]. The Fed used to steer the economy by moving this rate; at the zero bound it uses instead the *size* of its balance sheet, quantitative easing [► L6.2 @ 3:57]. Reserves used to be about \$50 billion; now there are over a trillion, excess reserves: reserves are not scarce any more, and yet the Fed funds market is still active, because it serves the payment system [► L6.2 @ 4:38].

IN THE MARKETS: The two jobs of a Fed funds desk

Stigum: the primary job of the manager of a bank's Fed funds desk is to ensure (i) that the bank settles with the Fed, every day, and (ii) that in doing so it holds no more excess reserves than it can carry into the next week [► L6.3 @ 0:09]. The second job, tied to reserve requirements, is now basically defunct: with so many reserves, missing a reserve target is "ridiculous". Only the first, the payment job, matters [► L6.3 @ 0:30].

6.3 Intraday elasticity: daylight overdrafts and CHIPS

The survival constraint, in Stigum's words that the bank must settle with the Fed, takes the precise institutional form of a requirement to *end the day with a non-negative balance at the Fed* [► L6.4 @ 0:08]. It does not bind *during* the day [► L6.4 @ 0:29].

READING: Stigum on the Fed funds desk

Stigum (p. 507) describes the job of a bank's Fed funds desk as ensuring that the bank settles with the Fed and that in doing so it holds no more excess reserves than it can carry into the next period. In the course's terms: (1) satisfy the daily survival constraint, and (2) satisfy the legal reserve requirement (the special Wednesday settlement). The first is the more interesting, especially now that all banks hold massive excess reserves and the requirement does not bind [Notes 6 p.1].

Daylight overdrafts at the Fed. Mehrling spent the summer of 1986 at the Board of Governors as a graduate student, when the Fed was worried about its credit exposure from daylight overdrafts [► L6.4 @ 0:49]. Bank A must pay bank B but has no reserves: it simply runs an overdraft, which is a loan from the Fed; the Fed creates the reserves and credits them to B, which feels paid and knows nothing of the arrangement [► L6.4 @ 1:33].

Fed		Bank A	
+Overdraft of A	+Reserves of B		–Deposits +Overdraft at Fed
Bank B			
	+Reserves		+Deposits

This is “the credit grease that makes the payment system go during the day” [► L6.4 @ 2:54]. The overdraft is a promise to come up with the reserves by the end of the day; if A does not, the Fed is forced to lend overnight, but at a penalty, 100 basis points over the Fed funds rate as said in class, to give A an incentive to find someone who will lend cheaper. The Fed cannot say “pay me or you’re dead”, because it is protecting the payment system; it says “pay me or you owe me more than you want to”. Sometimes a bank cannot find a counterparty and pays the penalty [► L6.4 @ 3:15]. Credit expands during the day and collapses at the end of it: the intraday version of the pattern at every level and every time scale [► L6.4 @ 4:36]. The violation of the survival constraint during the day is essential: it lets banks be “first movers”, paying before they are paid. At its peak the daylight expansion of reserves typically exceeds the (rather small) required reserves, so “it is not stretching the truth much to say that these daylight temporary reserves, not the overnight required reserves, are the actual ultimate means of payment”: the payment system is much closer to an ideal credit system than to an ideal money system [Notes 6 p.1].

CHIPS. The same happens on the balance sheet of a private clearinghouse, CHIPS, a mutual organisation of banks that collects dues to and from during the day and settles at the end of the day on Fedwire [► L6.4 @ 4:57]. Every morning members deposit collateral; during the day A can owe CHIPS while B is owed by CHIPS [► L6.4 @ 6:05]:

CHIPS		Bank A	
Due from A	Due to B		Due to CHIPS
Bank B			
	Due from CHIPS		

Two layers of intraday credit expansion, a public one on the Fed’s balance sheet and a private one on CHIPS’s [► L6.4 @ 6:05]. If A fails, all members of CHIPS are supposed to pay B: B holds an obligation of the clearinghouse, better than one of A, though not as good as one of the Fed [► L6.4 @ 7:28].

Definition 6.3 (New concepts: *real-time gross settlement*). A *real-time gross settlement* (RTGS) system settles each payment individually (gross) and immediately (real time) in central bank money: once received, the reserves are money *now*, not to be netted at the end of the day. Fedwire is RTGS; CHIPS nets during the day and settles at the end, so a CHIPS credit is money only “hopefully”, after settlement on Fedwire [► L6.4 @ 8:09].

Exercise 6.1 (A day at the Fed). Bank A starts the day with 0 reserves, B with 50. During the day A pays B 80 and B pays A 30.

- (a) Track A’s balance at the Fed; what is its maximum daylight overdraft?
- (b) What must A do by the end of the day? Write the T-accounts if it borrows from B in the Fed funds market.
- (c) Repeat when B is unwilling to lend to A, and a dealer D borrows from B and lends to A.

COURSE NOTE: Daylight overdraft policy

The Fed and CHIPS have tightened up over time: the size of overdrafts allowed depends on a bank’s capital, and the Fed charges for them, on the average overdraft during the day, so a trillion-dollar overdraft for a few microseconds costs essentially nothing, but still disciplines the bank, which must know where the money is coming from [► L6.5 @ 2:52]. (Fees on daylight overdrafts were introduced in 1994, with caps related to capital; the notes give the fee as 36 basis points, and add that banks may borrow an amount proportional to capital without paying interest, and that the penalty for overnight borrowing at the discount window, 100 basis points over the Fed funds target, had been lowered by Bernanke to 50, “probably just temporary” [Notes 6 p.2][Notes 6 p.6]. The source cited there is S. Coleman, “The Evolution of the Federal Reserve’s Intraday Credit Policies”, *Federal Reserve Bulletin*, February 2002.) Large security and international transactions cluster at the end of the day, when CHIPS settles through Fedwire [► L6.5 @ 3:53].

6.4 The function of the Fed funds market

At the end of the day, in a closed system, the overdrafts and surpluses sum to zero: if I owe the Fed, somebody out there has excess reserves, and if we find each other we can both end flat and neither needs the Fed; the same for CHIPS [► L6.5 @ 0:48]. *Interbank markets exist to let banks find each other* [► L6.5 @ 1:51]. For CHIPS the relevant interbank market is the Eurodollar market: anyone can be a member of CHIPS, or have a correspondent that is, including many foreign banks; Fedwire is a much more exclusive club, and its members borrow and lend in the Fed funds market. Roughly, Fed funds are domestic, public, “Fed” dollars; Eurodollars international, private dollars; both serve the same function [► L6.5 @ 2:11]. The Fed funds market creates bilateral overnight borrowing to substitute for the centralised overnight lending the Fed would otherwise be forced to do [► L6.5 @ 4:14].

A Fed funds trade. Bank A, short at the end of the day, borrows from bank B, long: it promises reserves tomorrow morning and gets reserves today, which pay off its overdraft; the Fed’s balance sheet expands during the day and contracts back [► L6.5 @ 4:35].

Bank A		Bank B	
+Reserves (repays overdraft)	+Fed funds borrowed	−Reserves	
		+Fed funds lent	

Banks do not wait for the end of the day: a desk anticipates needs and trades all day long [► L6.5 @ 6:40]. Without any overdraft: A borrows reserves from B and then uses them to pay B; the reserves go back and forth and what is left is a Fed funds loan [► L6.5 @ 7:21]:

Bank A		Bank B	
	−Deposits of payer	+Fed funds lent	+Deposits of payee
	+Fed funds borrowed		

Proposition 6.4 (Paying today by promising tomorrow). *A Fed funds loan pushes final settlement into the future: “I’m going to pay you today by promising to pay you tomorrow”* [► L6.5 @ 8:50].

Natural buyers and sellers. Stigum notes that some banks are natural sellers of funds and others natural buyers: small banks in isolated areas, with little demand for loans, have daily inflows exceeding outflows; large city banks lend out all their deposits and more. So the Fed funds market channels funds from country banks to city banks, like the older correspondent banking it replaced, and the flow is still largely intra-regional, as the location of the twelve Reserve Banks reflects [Notes 6 p.3]. Unlike a clearinghouse, a Fed funds trade moves reserves at once, with no netting; once A has bought the money it can sell it again, so by the end of the day outstanding Fed funds credit can be several times the deposits at the Fed [Notes 6 p.3].

6.5 Payment versus funding: an example

Mehrling buys your apartment in Manhattan for \$1 million, with a mortgage from his bank, Citibank; you bank at Chase. Credit checks are “interesting but not monetary economics” [► L6.6 @ 0:08]. Step by step:

- 1. **The loan.** Citibank grants the mortgage by crediting a deposit: a swap of IOUs, but with different timing and different signers: Mehrling owes \$1 million over 30 years, Citibank owes \$1 million on demand; when Citibank signs an IOU it is money, when Mehrling does it is credit [► L6.6 @ 1:09].
- 2. **Funding the payment.** The deposit must go to Chase, and Citibank does not keep \$1 million of idle reserves. Its credit department knows the closing is coming, so it borrows \$1 million in the Fed funds market, not from Chase but from a third party, HSBC, which lends reserves it has (or runs an overdraft) [► L6.6 @ 2:30]. Eventually the mortgage will be packaged and securitised, but that takes weeks; the payment must happen now [► L6.6 @ 3:11].
- 3. **The payment.** Citibank pays Chase in reserves and Chase credits your deposit: it is happy to, because it is settled, it has money [► L6.6 @ 4:14].
- 4. **Chase lends on.** Reserves pay nothing, so Chase lends them in the Fed funds market, to HSBC [► L6.6 @ 5:38].

Citibank		HSBC		Chase	
+Mortgage	+Fed funds (HSBC)	+Fed funds (Citi)	+Fed funds (Chase)	+Fed funds (HSBC)	+Deposit (you)
Mehrling		You			
+Apartment	+Mortgage	-Apartment	+Deposit		

Who lent the money? Follow the chain: Citibank borrowed from HSBC, HSBC from Chase, Chase from you. “In a certain sense I bought it from you and you lent me the money for it”, though Mehrling owes the intermediaries, not you [► L6.6 @ 6:19]. Until the mortgage is financed by some pension fund, it has been funded through the *payment system*, by an expansion of balance sheets: nobody saved extra money so that Mehrling could borrow; you transformed a house into a deposit, he acquired a house and a mortgage [► L6.6 @ 7:00].

HSBC as a dealer. HSBC lends and borrows Fed funds at the same time, flat in exposure: it does so only because it lends at a higher rate than it borrows. It is a *dealer* in Fed funds, quoting rates at which it will lend and borrow and setting them so that over the day inflows match outflows; when they do not, it must find funds itself, or place the surplus [► L6.6 @ 7:41].

Definition 6.5 (New concepts: [matched book](#)). A dealer runs a *matched book* when its borrowing and its lending in a market have the same size (and maturity), so that it earns the spread without net exposure to the price [► L6.6 @ 8:44].

An internal drain. If you withdraw the \$1 million in cash (“1MM”, in Stigum’s notation), Chase must get currency: it asks the Fed to convert reserves into cash, which is physically “downtown at 33 Liberty Street”. The Fed’s liabilities change form, one million more of currency and one million less of reserves; an internal drain, which causes no monetary problem, since both are liabilities of the Fed [► L6.6 @ 9:25].

Exercise 6.2 (The apartment). Redo the apartment example when Chase itself lends Citibank the reserves in the Fed funds market. Then suppose the buyer’s bank and the seller’s bank are the same: what happens to reserves and to the Fed funds market?

6.6 Brokers versus dealers

A *broker* puts buyer and seller together and takes a fee, like a real-estate broker, who does not buy your house and resell it. In a brokered Fed funds trade the loan sits on the balance sheets of A and B only [► L6.7 @ 0:07]. A *dealer* like HSBC stands in between, and the loans sit on its balance sheet: Chase and Citibank probably do not even know that they are, in effect, lending to and borrowing from each other [► L6.7 @ 1:08]. The dealer is “the source of liquidity in these markets”, a theme for the rest of the course [► L6.7 @ 1:50]. (Brokers and dealers were defined in [definition 2.4](#).) The notes draw the difference: brokering leaves the broker’s balance sheet untouched, dealing expands it [Notes 6 p.5]:

Brokered:	Bank A		Broker		Bank B	
	+Reserves	+FF loan			−Reserves	
					+FF loan	
Dealt:	Bank A		Dealer		Bank B	
	+Reserves	+FF loan	+FF loan (to A)	+FF loan (from B)	−Reserves	
					+FF loan	

Brokers keep sheets of bids and offers and look for matches, fine-tuning quotes for credit lines, since the seller is also the lender (Stigum’s “line problems”); the effective rate is discovered at brokers and dealers. Dealers buy funds low and sell high, bridging segments (small and large, unsophisticated and sophisticated, domestic and offshore) and time (intraday, across days): arbitrage, “picking up nickels”, and market making [Notes 6 p.3][Notes 6 p.4][Notes 6 p.5].

Exercise 6.3 (Dealer pricing). A Fed funds dealer quotes 0.15% to borrow and 0.20% to lend. During the morning it receives requests to borrow 300 and offers to lend 100.

- (a) What is its net position if it accepts everything?
- (b) Which way should it move its quotes, and why does this move the effective rate?

6.7 Payment imbalances and the Fed funds rate

During the day the Fed and CHIPS expand to make payments possible; at the end both shrink back, and the Fed funds market is one way to do it. But the credit does not vanish: it moves to the bilateral balance sheets of borrowers and lenders and onto the balance sheets of the Fed funds dealers [► L6.8 @ 0:07]. The centre protects itself (“we won’t extend this overnight, you do it”); daily payment imbalances that could not be cleared show up as expansion of Fed funds credit and of dealer balance sheets, pushed into the future in the hope that tomorrow the flow reverses [► L6.8 @ 1:11]. These quantities are highly sensitive measures of payment patterns, and so are the prices: a dealer with many more borrowers than lenders raises its rate to attract lenders, and lowers it in the opposite case. The Fed funds rate is a *market rate*, and it fluctuates within the day too [► L6.8 @ 1:52].

How does the Fed “set” the rate? Not as in intermediate macro: the Fed does not participate in the Fed funds market at all, neither borrowing nor lending [► L6.8 @ 3:16]. It has a *target*, and it influences the rate by changing the quantity of reserves, by entering other markets, in particular the repo market [► L6.8 @ 3:57]. Almost every day it anticipates payment imbalances that would push the rate away from target, and adjusts the quantity of reserves: it changes money in order to influence credit [► L6.8 @ 4:18]. To add reserves it makes an overnight repurchase agreement loan, for example to a security dealer funding its securities, which ends up with reserves it does not want and lends them in the Fed funds market, putting downward pressure on the rate [► L6.8 @ 5:20]:

Fed		Security dealer	
+Repo loan	+Reserves	+Reserves, then −Reserves	+Repo from Fed
		+Fed funds lent	

The *Fed funds effective* rate is the average of the rates at which funds trade during the day; good borrowers pay less, bad ones more [► L6.8 @ 6:45]. When the Fed adds reserves, the daylight expansion of credit on Fedwire need not collapse to zero at the end of the day: the Fed takes some of it onto its own balance sheet overnight or longer [Notes 6 p.5]. As the former New York Fed desk manager Peter Sternlight put it (quoted by Stigum, p. 506), “if you want to hang your hat on something, hang it on the average effective funds rate over the settlement period” [Notes 6 p.6].

Definition 6.6 (New concepts: target rate, effective rate). The *Fed funds target* is the rate (since 2008 a range) that the Federal Open Market Committee (FOMC) aims at; the *effective* Fed funds rate is the volume-weighted average of the rates actually paid in the market on a given day.

6.8 Secured versus unsecured interbank credit

Fed funds are *unsecured* overnight loans: no collateral. If Citibank does not pay HSBC, HSBC can sue, but there is no asset to seize; a mortgage, by contrast, is secured by the house [► L6.9 @ 0:08]. Eurodollars too are unsecured interbank loans [► L6.9 @ 1:10]. Why lend a million dollars overnight on a promise? Because banks are in a network, it is their business, and tomorrow they will be on the other side; they watch their exposure to each counterparty very closely, with *credit lines*: “we’re already in them for \$10 million, tell them your credit line to us is full”, and they diversify their lending [► L6.9 @ 1:30].

The market, especially the Eurodollar market, is under great stress: with many insolvent banks in Europe, nobody wants to lend unsecured, and even a dealer in the middle worries, because its business is highly leveraged on both sides: if Citibank does not pay HSBC, HSBC must still pay Chase; it is not a pass-through, it faces its own survival constraint [► L6.9 @ 2:53]. Repurchase agreements, next lecture, are *secured*, by a Treasury bill for example [► L6.9 @ 3:54]. This is where large amounts are transferred, where risk is taken and liquidity created: the Fed funds market is the market for the best money in the world, Eurodollars another unsecured market, repo the secured one; understanding them puts us at the very core of the international payment system [► L6.9 @ 4:14].

Definition 6.7 (New concepts: *secured and unsecured credit, credit line*). A *secured* loan is backed by collateral that the lender can take if the borrower defaults; an *unsecured* loan rests only on the borrower’s general promise. A *credit line* is the maximum exposure a lender is willing to have to a given counterparty.

6.9 Required reserves, redux

“The least important part of this lecture”: reserve requirements [► L6.10 @ 0:07]. Stigum describes two-week *maintenance periods* over which a bank must hold an average quantity of reserves, so a bank had two tasks: settle every day (easy, with the Fed funds market) and hit the average without overshooting [► L6.10 @ 0:27].

Monetarism and reserve accounting. Under Volcker, monetarists wanted to use reserve requirements to control the money supply: if deposits require reserves, a shortage of reserves limits the expansion of deposits and of bank credit [► L6.10 @ 1:08].

- With *lagged reserve accounting*, required reserves depended on deposits of two weeks earlier: the loan and the deposit are already there, so the Fed has to supply the reserves, or somebody will miss the requirement; the Fed only accommodates demand [► L6.10 @ 1:50].
- With *contemporaneous reserve accounting*, required reserves depended on deposits in the same period: harder to hit, with both deposits and reserves moving; it caused a lot of chaos in the Fed funds market and was abandoned [► L6.10 @ 2:53].

It was an attempt to impose discipline on a system that seemed out of control, in a period of double-digit inflation; according to Stigum it did not really work [► L6.10 @ 3:36]. Today, with trillions of reserves, requirements are a dead letter, yet the Fed funds market exists for payments [► L6.10 @ 4:17].

Were reserve requirements ever important? Asked whether they will matter again after the crisis, Mehrling answers that a trillion of excess reserves will not go away soon, but the

deeper question is whether the quantity of reserves ever significantly restricted credit or money [► L6.10 @ 4:38]. A parallel banking system, shadow banking, creates close substitutes for money and credit with no reserve or capital requirement, possibly offshore; marginal credit can flow there. Squeezing banks, as with the European capital requirements of the FT article, does not mean there is no credit, only that it does not come from banks: “like a balloon, you punch it here and it blows out there” [► L6.10 @ 5:39]. Mehrling sides with those who think the reserve constraint is ineffective in modern financial markets; it probably was not in 1950, when it was invented and the textbooks were written, because there were no alternatives [► L6.10 @ 6:40].

Exercise 6.4 (Reserve accounting). A bank with deposits of 1000 and a 10% reserve requirement grants a loan of 100 that is spent within the bank. Compare what it must do under lagged and under contemporaneous reserve accounting, and explain why the Fed must accommodate in the first case.

Key formulas and ideas

KEY FORMULAS	
Fed funds	unsecured overnight interbank loans of reserves; assets/liabilities of banks, not of the Fed
Survival constraint	end the day with a non-negative balance at the Fed
Intraday	daylight overdrafts at the Fed (RTGS) and at CHIPS (net): credit expansion during the day
End of day	surplus and deficit banks find each other in the Fed funds (or Eurodollar) market; else penalty borrowing from the Fed
Payment vs funding	a payment funded by interbank borrowing: paying today by promising tomorrow
Dealer	borrow and lends at a spread, matched book; source of market liquidity
Fed funds rate	a market rate; the Fed targets it by adjusting reserves (repo), not by trading Fed funds
Reserve requirements	lagged vs contemporaneous accounting; ineffective with shadow banking

ASIDE: The Fed’s worry about daylight credit

From his summer of 1986 at the Board, Mehrling’s notes recall why the Fed worried: Fedwire is real-time gross settlement, so banks made more payments than they had reserves, planning to receive payments later, an intraday expansion of reserves out of the Fed’s control; the Fed thinks of reserves as money and wants credit to happen elsewhere. It set credit limits, then (1994) charged for average overdrafts, which did little to reduce peak overdrafts, late in the day and driven by securities transactions. Raising the price might push overdrafts onto other systems that charge less, even zero. CHIPS, where payments are good only after end-of-day net settlement, was required on 1 October 1981 to move to same-day settlement, and its collateral requirements were periodically tightened: the idea was to push the credit element out of the payment system into the repo market [Notes 6 p. 6][Notes 6 p. 7].

COURSE NOTE: Dates

The Fed’s “monetarist experiment” of targeting reserves ran from October 1979 to 1982. Contemporaneous reserve accounting was introduced in February 1984 and replaced by lagged accounting again in 1998. (In March 2020, after these lectures, the Fed set reserve requirement ratios to zero.)

Chapter 7

Repos, Postponing Settlement

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L7.1 FT: The Impact of QE3	2:45	Second thoughts on QE3: buying time, and hedge funds betting against it.
L7.2 Money Market Interest Rate Patterns	3:34	LIBOR, Fed funds, repo: the usual order, and its reversal.
L7.3 What is Repo?	3:19	A collateralized loan organised as a sale and repurchase.
L7.4 Repo in Balance Sheets	7:34	Repo and reverse repo around a security dealer; the confusing language.
L7.5 Comparison with Fed Funds	5:09	Repo as an open, dealer-made market.
L7.6 Legal Construction of Repo	9:46	Symmetry, margin, haircut, repo interest (Stigum's example).
L7.7 Security Dealers Balance Sheet	11:07	Primary dealer statistics; dealers as banks; haircuts and liquidity spirals.
L7.8 Repo, Modern Finance, and the Fed	8:49	Repo as money; Fed open market operations as repos with dealers.
L7.9 Interest Rate Spreads: Before the Crisis	5:33	Why repo < Fed funds: Bagehot's market rate and bank rate.
L7.10 Interest Rate Spreads: After the Crisis	8:07	Repo > Fed funds: stress in dollar funding; reading spreads as clues.

Files. Transcripts `materials/transcripts/L07-P*.txt`. Segment 5 has no YouTube captions: its transcript `L07-P05.txt` is extracted from the original `transcript-lecture-7to12.txt`, so its times may be off by a few seconds. Mehrling's notes [Notes 7 p.1]. **Reading.** Stigum, *The Money Market*, chapter on repo (pages 533–534 are quoted in class).

Motivation

The Fed funds market (lecture 6) lets banks meet at the end of the day and push settlement to tomorrow. The repo market does the same, but for almost everybody and against collateral: it is the core funding market of the security dealers, and through them of modern finance. The lecture explains what repo is ([section 7.3](#)), how it appears in balance sheets ([section 7.4](#)), how it is priced ([section 7.5](#)), why dealers that fund themselves this way are banks ([section 7.6](#)), how the Fed uses it ([section 7.7](#)), and what the spread between repo and Fed funds tells us about the system ([section 7.8](#)).

7.1 FT: the impact of QE3

CASE STUDY: FT, September 2012: second thoughts on QE3

John Plender, “Central bank firepower risks creating false sense of security”: QE3 puts a floor under certain security prices but does nothing fundamental; if the problem is fiscal, central banks only buy time, and if nobody uses the time, nothing has been done. His last paragraph: “throughout history monetary experiments have shown a nasty tendency to blow up. . . this experiment is vast; it must not breed complacency” [► L7.1 @ 0:06]. On the same page, “Hedge fund sceptics take on the bulls over QE infinity”: some funds argue that QE1 worked well, QE2 less, QE3 may not even lift asset prices, and trade against those who expect a boom [► L7.1 @ 1:08].

If private investors trade against it, QE buys even less time [► L7.1 @ 2:09]. Central banks act not because they are optimistic, but because nobody else is doing anything; and in repo rates we can see symptoms of the stresses in the system [► L7.1 @ 2:09].

7.2 Money market interest rate patterns

In the FT: US dollar LIBOR overnight 0.1509%, US overnight repo 0.31%, Fed funds effective 0.16% [► L7.2 @ 0:07]. Before the crisis the typical order was

$$\text{repo} < \text{Fed funds} < \text{LIBOR (Eurodollar)},$$

and Mehrling had a whole story for it, from a paper he wrote before the crisis [► L7.2 @ 0:56]. Now repo is *above* Fed funds: odd, since repo is secured and Fed funds are not [► L7.2 @ 1:38].

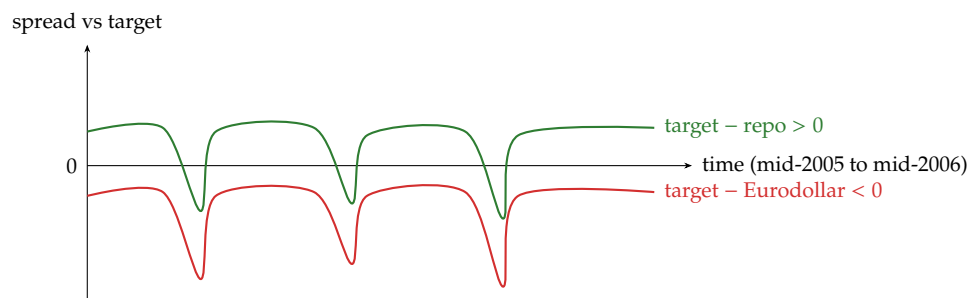


Figure 7.1: Schematic of the pre-crisis chart shown in class [► L7.2 @ 1:58]: Fed funds *target* minus repo was typically positive, target minus Eurodollar typically negative. The downward spikes occur before each step of the 2005–2006 tightening, as market rates rise in anticipation [► L7.2 @ 2:39]. (Shape illustrative.)

7.3 What is repo?

Definition 7.1 (New concepts: repurchase agreement). A repurchase agreement (“repo”, “RP”) is a loan, overnight or for a term, collateralized by a security, organised as a *sale and repurchase*: the borrower of money sells a security today and promises to buy it back tomorrow at a fixed price; the securities go from the borrower to the lender and

back, the money from the lender to the borrower and back [► L7.3 @ 0:08]. (Stigum draws the “first leg” and the “second leg” of the repo [► L7.3 @ 1:34].)

The securities are collateral in a strong sense: if the borrower does not repay, the lender *already holds* the securities and can sell them; there is no need to depend on the borrower’s credit [► L7.3 @ 1:55]. And symmetrically: if the lender does not return the securities, the borrower need not repay the money. Such a *fail* turns into a free overnight continuation of the loan until the securities are delivered [► L7.3 @ 2:37]. The money collateralizes the loan of securities, and the securities collateralize the loan of money.

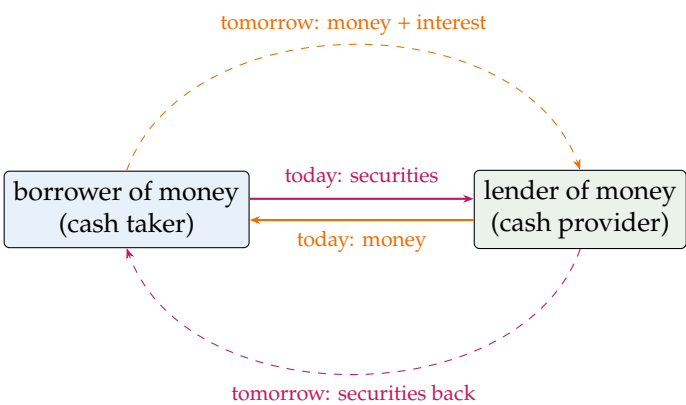


Figure 7.2: The two legs of a repo [► L7.3 @ 1:14]: a sale of securities today and their repurchase tomorrow at a slightly higher price.

7.4 Repo in balance sheets

There are trillions of these overnight loans [► L7.4 @ 0:07]. The language is not standardised: “doing repo”, “reversing in securities”, “repoing securities”; and, Mehrling warns, the same phrase can mean opposite sides of the trade in New York and in London. Balance sheets keep you straight [► L7.4 @ 0:27]. Two rules:

- 1. almost every repo has a *security dealer* on one side [► L7.4 @ 0:48];
- 2. “follow the money”: a repo booked as a liability means money borrowed, as an asset money lent; money is better than securities, so its side tells you where to book [► L7.4 @ 1:54].

Pension fund		Security dealer		Bank	
Repo loan to dealer		Reverse repo (loan to bank)	Repo (from pension fund)	Securities	Repo (from dealer)

- **Pension fund:** lends money overnight, against a Treasury, for interest. For it this is practically a deposit, a way to hold cash, and better than a deposit, because large deposits are not insured: if the bank fails the cash is gone, while in repo it holds a Treasury [► L7.4 @ 2:15].

- **Dealer lending money:** takes in securities; Stigum calls it a *reverse* [► L7.4 @ 3:16]. Securities move one way across the diagram, money the other [► L7.4 @ 3:57].
- **Bank:** holds securities and funds them by repoing them out overnight, instead of with deposits: repo as the liability that funds an asset [► L7.4 @ 4:17].

The dealer's "repo" and "reverse" are the same instrument seen from the two sides; they get different names because the dealer is in the middle, where one is a liability and the other an asset [► L7.4 @ 5:40]. To add to the confusion, the Fed uses the words the other way round: when it borrows from dealers it calls it a "reverse repo", when it lends (open market operations) a "repo" [► L7.4 @ 6:21]. Always translate into balance sheets: who is borrowing money? [► L7.4 @ 7:23] (The notes add the reason for the convention: the borrower of money is also lending securities and could be said to acquire an asset, but the course does not book it that way, following the money, because of the hierarchy of money and credit [Notes 7 p.1].)

Comparison with Fed funds. The repo market is like the Fed funds market: it lets deficit and surplus agents meet and push their liabilities and assets to another day, with a dealer in the middle [► L7.5 @ 0:03]. Two differences [► L7.5 @ 1:25]:

1. **Openness:** anyone with a Treasury bond can do repo, without being a bank or a member of the Federal Reserve System: it is "much more democratic" [► L7.5 @ 1:33];
2. **Dealing versus brokering:** in Fed funds there is little dealing, it is mostly brokered; in repo it is almost all dealing, on the balance sheets of security dealers [► L7.5 @ 0:52].

Why a dealer market? Repo is heterogeneous: any security can be repoed (corporate bonds, mortgage-backed securities, . . .) and anyone can participate. The *name of the dealer*, a primary dealer, a big bank, on the other side of every trade, is what makes the market homogeneous: you only worry about the dealer [► L7.5 @ 2:12]. There is even *general collateral* repo, in which the dealer may substitute one security for another of the same class, to keep things smooth; the dealer is a liability on both sides, to return securities and to return money [► L7.5 @ 3:53]. Repo defaults usually happen not because someone is insolvent but because the securities were sold on and not returned [► L7.5 @ 4:42].

Exercise 7.1 (Who is doing repo?). For each statement, write the T-account entry of every party: (a) "the Fed did \$5 billion of reverse repos with primary dealers"; (b) "the money fund reversed in \$1 billion of Treasuries from a dealer"; (c) "the dealer repoed out its inventory of corporate bonds".

7.5 The legal construction of repo: haircut and interest

Repo is like a collateralized loan, but much more symmetric [► L7.6 @ 0:07]. With a mortgage, the bank has a claim on the house, but foreclosure takes months and legal fees, and nobody intends to use it [► L7.6 @ 0:28]. A repo is an overnight loan of \$10 million: the lender *owns* the security, can sell it or re-pledge it (*rehypothecate*) and trade with it all day, and is only liable for returning it the next day; the borrower can spend the money as it likes [► L7.6 @ 0:48].

Who gives margin to whom? The lender of money fears that the security falls below the value of the loan, which gives the borrower an incentive not to repay; the lender of securities fears the opposite [► L7.6 @ 1:51]. Stigum's answer: it is always the *lender of money*

HISTORY: Repo before the Fed

Repo is "very ancient": in the United States it existed before the Fed, as the way banks traded with each other before the Fed funds market. With a correspondent you could rely on the relationship; with a stranger you wanted collateral: lend against a bond, and you only need to check the bond, not the borrower. Mehrling links the rise of bond ratings (Moody's) to this need to know how much to lend against which bond [► L7.4 @ 4:38]. (John Moody published his first bond ratings in 1909.)
References: R. Sylla, "An historical primer on the business of credit rating", in R. Levich et al. (eds.), *Ratings, Rating Agencies and the Global Financial System*, Kluwer, 2002.

who gets margin [► L7.6 @ 2:31]. “This is part of the hierarchy of money and credit”: money is the better asset, and its holder can demand extra security; and the demand for liquidity is the short side of the market, people pay up to get money, not to get collateral [► L7.6 @ 2:31]. (In swaps, by contrast, typically both sides post margin [► L7.6 @ 9:09].)

Definition 7.2 (New concepts: haircut, repo rate). The *haircut* is the percentage by which the amount lent is smaller than the market value of the collateral; it protects the lender of money against a fall in the collateral’s price. The *repo rate* is the interest rate on the loan, quoted on a simple-interest, actual/360 basis.

Stigum’s example [► L7.6 @ 2:52]–[► L7.6 @ 7:47]. A 10-year Treasury bond trades a bit above par (its coupon is higher than the current 10-year rate). It is offered as collateral for an overnight loan with a haircut of 2%, so the amount lent per 100 of face value is 99.228. The repo rate is 4.92% (annualised). The next day the borrower repays principal plus simple interest for one day on a 360-day year:

KEY FORMULAS

$$\text{repayment} = L \left(1 + r_{\text{repo}} \frac{d}{360} \right), \quad L = (1 - h) P,$$

with P the market value of the collateral, h the haircut, d the number of days (1 overnight, 3 over a weekend, 7 for a week).

The conventions (simple, not compound interest; a 360-day year) date “from time immemorial”: the market prices them in, and overnight they change the result only at the fourth decimal, but traders must get them right [► L7.6 @ 6:04]. On a \$1 million bond the overnight interest is about \$137: the cost of borrowing a million dollars overnight [► L7.6 @ 7:26].

Haircuts make collateral homogeneous. Different securities get different haircuts (2% for a Treasury, 5, 10 or 50% for a mortgage-backed security, which in a crisis may not be accepted at all) and then the same repo rate [► L7.6 @ 4:38]. The haircut is negotiated first, according to the volatility of the collateral’s price, then the rate [► L7.6 @ 4:59].

Exercise 7.2 (Repo arithmetic). A Treasury note with market value 102.50 per 100 face is repoed for 3 days (a weekend), haircut 2%, repo rate 0.30%.

- How much is lent on \$10 million face value, and how much is repaid?
- What is the same interest with compound interest on a 365-day year? Compare.

7.6 The security dealer’s balance sheet

The New York Fed publishes weekly statistics on the primary dealers [► L7.7 @ 0:07]:

- Transactions** by type of security: Treasury coupon securities, agencies (Fannie Mae, Freddie Mac), mortgage-backed, corporate [► L7.7 @ 0:48].
- Net positions** (inventory, hence risk exposure): about \$26 billion of Treasury bills, \$68 billion of coupon securities due in three years or less, negligible longer bonds [► L7.7 @ 1:08]. This is unusual: normally dealers are *short* bills and *long* bonds, picking up the term spread. With Operation Twist the Fed buys bonds from the dealers and

COURSE NOTE: Checking the numbers

The price of the bond is garbled in the captions (“129 30 seconds”). Mehrling’s notes give it: $100\frac{29}{32}$ plus accrued interest [Notes 7 p.2]. Stigum’s Figure 13.1 (a Bloomberg “repo/reverse repo analysis” screen, among the lecture’s readings) gives every number: the Treasury $5\frac{1}{8}\%$ of 15 May 2016, market price 100-29 (yield 5.008%), accrued interest 0.3064 for 22 days, full price 101.2126; collateral set at 102% of the money lent, so the settlement price is $101.2126/1.02 = 99.2281$ (the “2% haircut”); on \$1 million face the money wired on 6 June 2006 is \$992,280.74; at 4.92% for one day on a 360-day basis the repo interest is \$135.61 and the termination money \$992,416.36 on 7 June. (“137” in class is the interest on exactly \$1 million, \$136.67.) Bond prices are quoted in 32nds of a point (one point = 1% of face value).

sells them bills, sucking out their bond inventory and wiping out their short in bills; the dealers absorb the mismatch until they can distribute it [► L7.7 @ 1:49]. Analysts from Barclays, in a Reuters article, explained the high repo rate this way; Mehrling wonders why it should affect the repo rate [► L7.7 @ 2:30]. (Net positions say nothing about hedges with derivatives [► L7.7 @ 4:14].)

- **Financing** (gross): securities in, about \$1.9 trillion, financed overnight or at term, and securities out (short positions) [► L7.7 @ 4:35].

The memorandum items give the repo book, in billions [► L7.7 @ 5:38]:

Primary dealers (\$ billion)			
Reverse repo, overnight	878	Repo, overnight	1,842
Reverse repo, term	1,316	Repo, term	931

Dealers fund their long positions in the repo market and their short positions in the reverse market [► L7.7 @ 6:44]. And look at the maturities: the big number on the asset side is *term*, on the liability side *overnight*. Dealers borrow short and lend long, “just like a bank” (term may be a week, but it is more than a day) [► L7.7 @ 7:25].

Proposition 7.3 (Security dealers are banks). *Overnight repo to dealers is, for institutional investors, a deposit, even better than one because secured; at \$1.8 trillion it is larger than M1. Dealers, trading securities for profit, create an enormous volume of monetary assets: the wholesale money supply [► L7.7 @ 7:04].*

Haircuts and liquidity spirals. Repo must be rolled over, and everything is “up for grabs” at each renewal: haircut and rate [► L7.7 @ 9:28]. If the haircut on a security you finance with repo jumps overnight from 2% to 10%, you have a *funding gap*: yesterday you could borrow 98 cents on the dollar, today 90; if you cannot find the other eight, you must sell the security at a fire-sale price, which pushes prices down further. In the crisis, haircuts on “triple-A” mortgage-backed securities went from 2% to 10% to 50%, and the scramble for funding distorted every other price in the world [► L7.7 @ 9:48].

INTUITION: Leverage and the haircut

With haircut h , a dealer with capital E can hold securities worth at most E/h : leverage $1/h$ (50 at $h = 2\%$, 10 at 10%, 2 at 50%). A rise of the haircut from 2% to 10% forces assets to fall by a factor of five for the same capital: a positive-feedback loop if the sales lower prices and raise haircuts further.

Exercise 7.3 (Funding gap). A dealer has capital 5 and holds MBS worth 100, financed by repo with a 5% haircut.

(a) Write its balance sheet.

(b) The haircut rises to 20%. How much new funding does it need? If it must sell MBS at 90% of value, what is its capital after selling enough to restore the haircut constraint?

COURSE NOTE: The dealer statistics file on the course page

The lecture page links the New York Fed’s weekly report `deal.pdf`, which is overwritten every week. The only copy recovered (`materials/readings/L07-deal.pdf`) is the release of 3 September 2015, not the 2012 one shown in class, so the class numbers above cannot be checked against it. It also uses a later format: financing is broken down by type of security, with no grand total and no “memorandum” lines.

For US Treasuries alone (excluding TIPS), on 26 August 2015, in \$ billion (Table IV):

- **Repo** (securities out): 1,319 in all. Of this, 913 was overnight and continuing, 224 term under 30 days, and 183 term of 30 days or more.
- **Reverse repo** (securities in): 1,193 in all. Of this, 630 was overnight and continuing, 285 term under 30 days, and 277 term of 30 days or more.

The pattern of the lecture survives: dealers fund mostly overnight and lend relatively more at term, borrowing short and lending long.

7.7 Repo, modern finance and the Fed

Most money and banking textbooks do not mention repo: they are about retail money and banks, and a security dealer “is not part of our subject”. This is where modern finance changed the world: not being called a bank does not mean not operating like one, and producing liquid liabilities is producing money supply for someone [► L7.8 @ 0:07]. The lender in repo may be a money market mutual fund, which holds repo and issues shares that function as deposits, to corporations or even foreign central banks: money, directly or passed through a fund [► L7.8 @ 0:49].

Open market operations are repos with dealers. The textbook says the Fed buys or sells Treasury securities to change the money supply; in reality it trades with the primary dealers, by auction: it asks them to bid [► L7.8 @ 1:10]. The New York Fed’s web page lists the last temporary open market operations [► L7.8 @ 2:10]. Before the crisis the Fed was in the repo market almost every day; with excess reserves, rarely [► L7.8 @ 2:31]:

- the latest is a *reverse repo*: the Fed borrows money from dealers against Treasuries, agencies or MBS; dealers submitted \$4.3 billion, the Fed took \$2.25 billion, from Thursday the 13th to Monday the 17th, “part of an operational readiness program”: the Fed is practising a way to drain its excess reserves [► L7.8 @ 2:51];
- for an actual *repo*, which adds reserves, one must go back to August [► L7.8 @ 5:37].

For a long time the Fed refused to call its borrowing “reverse repos” and called them *matched sale–purchases* (MSPs): “I’m the Fed, I print this stuff, why do I need to provide collateral?” Now it calls them reverses, acting like a bank [► L7.8 @ 4:56].

Daily elasticity. Outright purchases of Treasuries happen a couple of times a year, secularly to supply the reserves of a growing economy; on a daily basis the Fed provides elasticity, soaking up excess reserves and meeting temporary demand caused by “operating factors”: changes in the Treasury’s balance at the Fed (a rise drains reserves from banks) and in the public’s cash holdings (a cash drain) [Notes 7 p.4]. Before the crisis the Desk arranged short-term repos on all but 8 business days of 2006, large relative to total bank reserves [Notes 7 p.4][Notes 7 p.5]. The reverses that were on the Fed’s balance sheet were almost all its repo pool for foreign central banks, which thus earned the repo rate on balances at the Fed while ordinary reserves earned nothing [Notes 7 p.4]. The new reverse repos shift the Fed’s liabilities from reserves, high-powered money, to repo, less high-powered (especially at term), and shift its counterparty from banks to dealers [Notes 7 p.5]:

Bank		Dealer		Fed	
–Reserves	–Deposit of dealer	+Reverse repo		+Reverse repo	–Reserves
		–Deposit			

One more level. A Fed repo lends reserves to a dealer against a Treasury bill; the dealer has no account at the Fed, so the reserves end up at the dealer’s bank, credited to its deposit (or repaying a bank loan); if they are excess, the bank lends them in the Fed funds market. This is how open market operations influence the Fed funds rate [► L7.8 @ 6:42]:

Fed		Dealer		Dealer's bank	
+Repo loan to dealer	+Reserves of bank	+Deposit at bank	+Repo from Fed	+Reserves	+Deposit of dealer

7.8 Interest rate spreads

7.8.1 Before the crisis: repo < Fed funds

Stigum puzzles over why Fed funds traded above repo, and gives two reasons [► L7.9 @ 0:30]:

- 1. **a credit spread:** repo is secured, Fed funds are not. Mehrling was never convinced, and the present reversal disproves it: nobody lends unsecured with any thought of default; if you fear default you cut the credit line, you do not ask a higher rate. “Who risks a million dollars to make 137 of interest overnight?” And the borrowers are banks with access to the Fed [► L7.9 @ 1:10];
- 2. **segmentation:** some must borrow Fed funds and have no access to repo. But many can arbitrage, certainly any bank, by borrowing in repo and lending Fed funds [► L7.9 @ 2:32].

Mehrling’s explanation: Bagehot’s two rates. In Bagehot’s world (lecture 9) there was a *market rate* of interest, made by bill brokers in the London money market, and the official *bank rate*, the Bank of England’s discount rate, normally above it; nobody came to the Bank until stress pushed the market rate up to bank rate, and then everybody flooded to it. That was monetary policy then [► L7.9 @ 2:53]. Before the crisis the US worked the same way: the analogue of bank rate was the Fed funds target, the analogue of the market rate was repo [► L7.9 @ 4:19]. Repo below Fed funds is *discipline*: repo is the cheapest funding, so dealers repo out all their securities first; if they need more, they must go to more expensive money, borrowing from their bank not at the Fed funds rate but at Fed funds plus 50 basis points [► L7.9 @ 4:40].

INTUITION: The morning auction

The notes make the mechanism concrete [Notes 7 p.5][Notes 7 p.6]. Dealers expand on thin capital with almost no cash reserves, relying on repo; if short of collateral, they borrow from their clearing bank at about Fed funds +50 bp. At the Fed’s morning repo auction dealers bid for the money; the Fed, wanting to supply needed reserves but keep them scarce, accepts bids at or above the Fed funds target; dealers bid above the market repo rate because any chance of needing the clearing bank’s loan makes the Fed’s money worth it. Result: Fed funds above repo, and overnight money above three-month bills. The premium on the best money gives everyone an incentive to settle at the clearing rather than roll over: “the answer to the otherwise puzzling pattern of interest rates in the money market is nothing more than the natural hierarchy of money and credit”. The argument is developed in Mehrling, “Monetary Policy Implementation: A Microstructure Approach” (2006), and resembles the 19th-century British system of Bagehot and Sayers [Notes 7 p.5].

7.8.2 After the crisis: repo > Fed funds

Since January repo has moved in a band of about 16 to 30 basis points, always above Fed funds effective, sometimes by a lot [► L7.10 @ 0:09]. With the same Bagehot logic: a market rate above the official rate means the Fed is trying to be very elastic, to pull the market rate down by pegging Fed funds below it. That offers an arbitrage: borrow Fed funds, lend in repo, pick up about 15 basis points, and only banks (mostly) can do it [► L7.10 @ 0:51]. That repo *stays* above Fed funds despite the incentive is worrying: there are people who need liquidity and keep bidding for it, and the arbitrage is not enough to meet the needs of the dollar funding markets, which are under stress [► L7.10 @ 1:54]. In the notes: with repo < Fed funds the Fed keeps the better money at a premium, for discipline; with repo > Fed funds it keeps it at a discount, for elasticity: the market rate is repo, the official rate Fed funds, and the Fed wants banks to borrow at the official rate and lend at the market rate, supporting short-term credit markets. A 15 basis point gap suggests this is not working very well, “a large gap in money market terms” [Notes 7 p.6]. This is what the Fed would have been watching when deciding on another QE: pegging Fed funds low was not enough, so now it buys MBS outright, \$40 billion a month, injecting reserves directly and not temporarily [► L7.10 @ 2:36]. Will it work? Keynes said monetary policy is better at slowing the economy than at speeding it up: “pushing on a string” (section 2.7) [► L7.10 @ 3:37].

IN THE MARKETS: Reading spreads as clues

The method: look at patterns of interest rates and of quantities and balance sheets, and ask what must be happening behind the scenes; spreads are “detective clues” to where the strains are [► L7.10 @ 4:37]. In the fall of 2007 the spread of LIBOR over Fed funds reached 100 basis points, never seen in 15 years of teaching; knowing the two markets, it pointed to Europe [► L7.10 @ 4:58]. Prices are visible every day, balance sheets only later; “the spreads are the key”, while the intermediate macro textbook has just one interest rate [► L7.10 @ 5:39]. A 15 basis point reversal looks tiny, but when spreads reverse the incentives and the flows reverse: big information in small numbers [► L7.10 @ 6:42]. Mehrling’s hypothesis was formed in real time, from the FT alone, to be tested against what happens next [► L7.10 @ 7:23].

Definition 7.4 (New concepts: LIBOR). *LIBOR* (London Interbank Offered Rate) was the average rate at which a panel of banks said they could borrow unsecured from other banks in London, for each currency and maturity; US dollar LIBOR is the Eurodollar rate (lecture 8). (After the manipulation scandal that broke in 2012, LIBOR was phased out; US dollar LIBOR ceased in 2023, replaced by SOFR, a rate computed from overnight Treasury repo transactions.)

Exercise 7.4 (The arbitrage). Repo trades at 0.31%, Fed funds at 0.16%. Describe the arbitrage a bank could do, write its T-accounts, and give two reasons why it might not close the spread.

Key formulas and ideas

KEY FORMULAS

Repo	collateralized loan organised as sale and repurchase; fails continue the loan for free
Balance sheets	follow the money; dealer: reverse repo (asset), repo (liability); the Fed's language is the reverse
Margin	always to the lender of money (hierarchy): haircut h , amount lent $L = (1 - h)P$
Repo interest	$L r d/360$, simple interest, 360-day year
Dealers	borrow overnight (repo), lend term (reverse): banks; overnight repo $\approx$ \$1.8 tn, larger than M1
Liquidity spiral	haircuts up $\Rightarrow$ funding gap $\Rightarrow$ fire sales $\Rightarrow$ prices down, haircuts up
Fed	open market operations = repos/reverse repos with primary dealers
Spreads	normal: repo < FF < LIBOR (repo as Bagehot's market rate below bank rate); repo > FF signals funding stress

Chapter 8

Eurodollars, Parallel Settlement

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L8.1 FT: Ring-fencing and the Volcker Rule	9:45	Ring-fencing retail banking; matched book versus proprietary trading.
L8.2 The Eurodollar Market in Crisis	4:51	LIBOR–OIS in 2007–2009.
L8.3 What are Eurodollars?	7:11	Stigum’s Exxon example; dollar deposits outside the US with reserves at a New York correspondent.
L8.4 Why is there a Eurodollar Market?	4:43	Demand for dollars abroad; capital controls; tail and dog.
L8.5 Eurodollar as Global Funding Market	11:33	Eurobanks as money dealers; the interbank market and LIBOR.
L8.6 Liquidity Challenge of Eurodollar Banks	10:59	Lining up cash flows in time; a forward deposit as a swap of IOUs.
L8.7 FRA as Implicit Swap of IOUs	4:34	Forward forwards and forward rate agreements, off balance sheet.
L8.8 Forward Parity, Interest Rates, EH	8:58	Forward interest parity; covered interest parity via a swap of deposits.
L8.9 Forward Parity, Exchange Rates, UIP	2:48	Forward rates are not expected spot rates: EH and UIP fail.
L8.10 Forward Rates are NOT Expected Spot Rates	2:49	Same recording as L8.9 (see the note at the end of the chapter).

Files. Transcripts `materials/transcripts/L08-P*.txt`; Mehrling’s notes [Notes 8 p.1]. **Reading.** Stigum, *The Money Market*, chapter on the Eurodollar market (from p. 212), and on FRAs.

Motivation

The third of the three wholesale dollar money markets, after Fed funds and repo [► L8.2 @ 0:07]. Eurodollars are dollar deposits at banks outside the United States, which hold their dollar reserves not at the Fed but at a correspondent bank in New York: a *parallel* settlement system, one level below the Fed funds market in the hierarchy. The lecture explains what Eurodollars are and why they exist, how Eurobanks act as dealers in money and line up their cash flows in time, how this leads to the first derivatives of the course, forward rate agreements, understood as implicit swaps of IOUs, and to two parity conditions, one of which fails in practice.

8.1 FT: ring-fencing and the Volcker rule

CASE STUDY: FT, autumn 2012: “Banks braced for proposals to ring-fence trading”

A full-page FT feature, “Leaner and meaner”, speculates on the business model of banking after the crisis: banks have learned some lessons, and the regulatory landscape has yet to firm up [▶ L8.1 @ 0:07]. The article on page five refers to two ideas: the UK Vickers Commission’s recommendation that banks’ retail operations be *ring-fenced*, and the US *Volcker rule* limiting proprietary trading [▶ L8.1 @ 0:48].

Ring-fencing. Take the retail side of a bank, retail deposits and retail loans (households, and businesses: the distinction is retail versus *wholesale*, such as Eurodollar borrowing or derivatives), allocate to it part of the bank’s capital and reserves, and fence it off. Everything else, trading assets, wholesale borrowing, must have its own capital and reserves, so that if it loses a lot of money it can fail while the retail part is carved off and kept safe [▶ L8.1 @ 1:29].

The Volcker rule. A bank may not do *proprietary trading*, i.e. speculate on the direction of a price; it may do *market making*, a matched book [▶ L8.1 @ 4:20]. Example: a pension fund holds a risky asset and wants to hedge it with a *credit default swap*, a kind of insurance, though not regulated as insurance and not a large-numbers game, that pays it if the asset fails [▶ L8.1 @ 5:01]. The bank may sell it, but then must find another client and buy an offsetting CDS: on both sides, earning the spread, like the repo dealer of lecture 7 but with a derivative [▶ L8.1 @ 6:26]. What is outlawed is the naked position, betting that the assets will fail [▶ L8.1 @ 7:28]. Must the hedge be the same instrument? What matters is the same risk exposure; but hedging with something only correlated leaves *basis risk*: is that proprietary trading? If it is outlawed, banks cannot hedge, and cannot make the market at all. “The devil is in the details” [▶ L8.1 @ 8:30].

Client 1 (pension fund)		Bank (matched book)		Client 2	
Risky asset		CDS	CDS sold		CDS sold
CDS		bought	to client 1		
bought		from client			
		2			

8.2 The Eurodollar market in crisis

The chart shown plots the Eurodollar rate (LIBOR) minus OIS, the *overnight index swap* rate, “a sort of term Fed funds”, at one and three months, through the crisis [▶ L8.2 @ 0:28]. Normally the spread is a few basis points, and one expects it: banks can borrow and lend in either market, and any gap is an arbitrage [▶ L8.2 @ 1:50]. In August 2007 it jumped from almost zero to almost 100 basis points: a measure of the disorganisation of the world money market [▶ L8.2 @ 2:31]. Mehrling was teaching the course that fall, and told the class something very bad was happening: banks without access to the Fed funds market needed to roll their one- or three-month funding, and banks with access would not lend to them; the rate rose to offer a huge incentive to borrow Fed funds and lend at LIBOR [▶ L8.2 @ 3:12]. After Bear Stearns the Fed seemed to stabilise the spread, high but steady; it lost control at Lehman, regained it, and the crisis subsided [▶ L8.2 @ 3:53].

HISTORY: Vickers and Volcker

The UK Independent Commission on Banking, chaired by Sir John Vickers, published its final report in September 2011, recommending ring-fencing of retail banking; it was enacted in the Financial Services (Banking Reform) Act 2013 and in force from 2019. The Volcker rule, proposed by former Fed chairman Paul Volcker, is section 619 of the Dodd–Frank Act (2010); the final implementing rule was adopted in December 2013.

References: Independent Commission on Banking, *Final Report*, Sept 2011; Dodd–Frank Act, sec. 619.

Definition 8.1 (New concepts: basis point, overnight index swap). A *basis point* is one hundredth of a percentage point ($2\% + 1 \text{ bp} = 2.01\%$); in money markets, with sums of millions, even the fourth decimal matters [► L8.2 @ 1:08]. An *overnight index swap* (OIS) exchanges a fixed rate for the average of the overnight rate (in dollars, Fed funds effective) over a term; its fixed rate is the market’s term version of Fed funds. LIBOR – OIS measures the premium for term unsecured interbank funding.

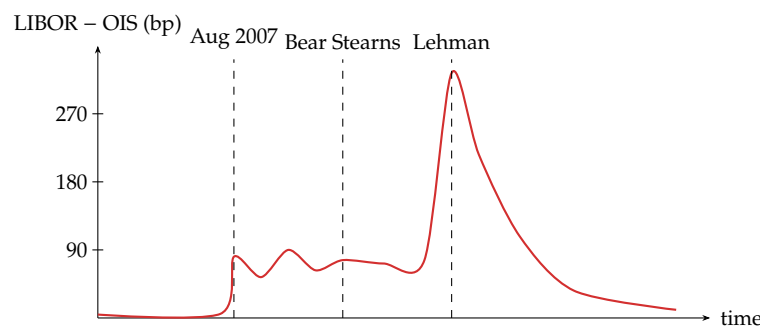


Figure 8.1: Schematic of the LIBOR–OIS spread through the crisis [► L8.2 @ 0:28]: near zero before August 2007, around 100 basis points after it, steadied after Bear Stearns, a peak at Lehman, then back to normal. (Shape illustrative; the three-month spread peaked at about 365 bp in October 2008.)

8.3 What are Eurodollars?

Stigum’s chapter is a recent addition and “still doesn’t go far enough” [► L8.3 @ 0:07]. Her example (p. 212): Exxon moves money from its account at Chase New York to Citibank London, a branch outside the country [► L8.3 @ 0:28]. Behind the scenes Chase transfers reserves at the New York Fed to Citi New York, which acts as correspondent of Citi London [► L8.3 @ 1:55]:

Exxon		Chase NY	
–Deposit at Chase		–Reserves	–Deposit of Exxon
+Deposit at Citi London			
Citi NY		Citi London	
+Reserves	+Deposit of Citi London	+Deposit at Citi NY	+Deposit of Exxon

Stigum stresses that “no money ever leaves the United States”: reserves at the New York Fed only change owner, from Chase to Citi [► L8.3 @ 3:12]. True, but a *deposit* leaves the country, and that is money; and the deposit of Citi London at Citi New York is, for Citi London, its *reserves*: it cannot hold an account at the Fed, but it can hold one at a bank that does [► L8.3 @ 3:53].

A clearer example. A branch makes it look like internal accounting of one bank. Better: Exxon moves its money to Crédit Lyonnais in Paris, which holds its dollar reserves as a deposit at Citi New York, a different bank: a correspondent relationship, and Citi New York certainly considers its reserves to back *its own* deposits [► L8.3 @ 4:55]. Crédit Lyonnais can now go into the dollar business: make dollar loans by creating dollar deposits, and pay withdrawals with its deposit in New York, even though it is in France [► L8.3 @ 6:18]. In the notes' words: inside the US, Citi New York's balance sheet expands and Chase's contracts by exactly as much; outside the US Crédit Lyonnais expands with no corresponding contraction: "from a global perspective, there has been an expansion of credit" [Notes 8 p.1].

Definition 8.2 (New concepts: Eurodollar). *Eurodollars* are dollar-denominated deposits (and loans) at banks outside the United States, including foreign branches of US banks. Their reserves are deposits at banks in the United States, not at the Fed. ("Euro" because the market started in Europe; the name is used for offshore dollars anywhere, and has nothing to do with the euro currency.)

Exercise 8.1 (Exxon's transfer). Redo Stigum's example with Crédit Lyonnais in place of Citi London, and then let Crédit Lyonnais lend \$5 million to a Korean firm that pays a US supplier banking at Chase. Write all the T-accounts, including the Fed.

8.4 Why is there a Eurodollar market?

There is demand: the dollar is the world reserve currency, and traders in Japan and Argentina may invoice and pay in dollars without any office in the US [► L8.4 @ 0:07]. They could do it through accounts in New York, but the Fed prefers that this happens offshore: it wants to focus on domestic credit [► L8.4 @ 0:49]. And a payment system is a credit system: deficit agents must be able to borrow from surplus agents, so people want not only dollar deposits but dollar loans [► L8.4 @ 1:09]. A bank like Crédit Lyonnais wants a matched book in dollars, as many dollar assets as dollar liabilities, so that it has no currency exposure in its euro accounts [► L8.4 @ 1:50].

History and size. The market arose because of controls on international capital flows after the Second World War: large corporations did their dollar business with each other in London, evading US controls, and the US let them, even encouraged them; the controls broke down, then Bretton Woods, and the market kept thriving, growing an order of magnitude larger than the Fed funds market [► L8.4 @ 2:13].

Tail and dog. Before the crisis it was unclear which was the market for dollars, Fed funds or Eurodollars: they traded at the same rate. People at the New York Fed, and Mehrling too, worried that with so much volume offshore, and big banks able to borrow in either market, the Fed could lose control of the rate [► L8.4 @ 3:14]. The crisis settled it: the Eurodollar market froze, the Fed funds market did not, because the Fed supported it. Fed funds are the "real" dollars, Eurodollars a credit extension of them, below Fed funds in the hierarchy [► L8.4 @ 3:34].

The notes add two points [Notes 8 p.2]. The market grew outside the control of US authorities, escaping capital controls and regulatory costs on dollars held at home (reserve requirements, deposit insurance premiums); most barriers have been lifted, but the market survives as a separate entity. And monetary policy: the Fed focuses on domestic money and rates, leaving out the international dollar on the idea that its holders are foreigners,

HISTORY: Origins of the Eurodollar market

The market grew in London in the late 1950s. Commonly cited causes: dollar balances held in Europe by Soviet-bloc banks wary of keeping them in the US (the Paris bank BCEN, whose telex address was "Eurobank", is often credited with the name); the 1957 sterling crisis, when the UK restricted sterling credit to non-residents and London banks turned to dollars; US Regulation Q ceilings on deposit rates; and later US capital controls (the Interest Equalization Tax of 1963, the voluntary credit restraints of 1965).

References: C. R. Schenk, "The origins of the Eurodollar market in London: 1955–1963", *Explorations in Economic History* 35 (1998).

not domestic consumers and businesses, “increasingly a difficult abstraction to defend, since important entities are global”.

8.5 The Eurodollar market as global funding market

The Eurodollar market is not only a payment system: it is *the* world funding market, and the world funding market is a dollar market [► L8.5 @ 0:08]. Ultimate lenders and borrowers anywhere want to lend and borrow in dollars, for periods of time, and the Eurodollar system intermediates the flow: an overnight deposit (a money balance, perhaps petrodollars) funds a six-month term loan, perhaps to a Korean manufacturer [► L8.5 @ 0:50].

Eurodollar system	
Term loan (6 months)	Overnight deposit

This is the repo dealer’s balance sheet of lecture 7, overnight funding against term lending: here a bank acts as a dealer, quoting a deposit rate and a loan rate, a bid–ask spread. “Banks are dealers in money” [► L8.5 @ 2:14]. Unlike repo, Eurodollars are *unsecured*: the security is the whole balance sheet of the bank, its capital, and its backstop, the national government; unlike Fed funds, a brokered market, Eurodollar funding shows up on the balance sheets of the Eurobanks [► L8.5 @ 3:15]. A small country borrowing large amounts is most likely funded, ultimately, in the world dollar market; borrowers and depositors outside the US face exchange risk, a topic for later [► L8.5 @ 4:17].

Outside US regulation. By the home-country rule a bank is supervised where it resides: a French bank dealing in dollars is supervised by the Banque de France and to some extent the ECB, not by the Fed. It can expand its balance sheet in dollars and create what is essentially money, not counted in the US money supply, since it is neither the liability of an onshore bank nor the asset of a US resident. Yet for Exxon a deposit in Europe is “more or less the same thing” as one in New York: they trade at par [► L8.5 @ 5:19]. The crisis chart shows the stress on that par: to keep it, Eurodollars had to pay a full percentage point more, to keep people from “breaking the buck” [► L8.5 @ 6:20].

The interbank market. Individual banks are not matched. Say Crédit Lyonnais has many customers who want dollar deposits, more than it can lend: excess dollar deposits; Citi London can lend any amount but has a thin European deposit base: excess dollar loans. They do business with each other through an interbank loan [► L8.5 @ 7:01]:

Crédit Lyonnais		Citi London	
Interbank loan to Citi London	Dollar deposits	Dollar loans	Interbank deposit from Crédit Lyonnais

LIBOR, the London interbank offered rate, is the rate on this interbank lending [► L8.5 @ 9:47]. People ask why banks do all this lending to each other instead of lending to “real productive activity”: this is why, to move money from where it is to where it is needed [► L8.5 @ 10:28].

It is this market that is breaking down in Europe now: the loans are unsecured, and a bank that is not sure it can lend on may not want to take the deposit in the first place [► L8.5 @ 11:11].

8.6 The liquidity challenge of Eurodollar banks

Eurobanks hold their dollar reserves at a New York bank, not at the Fed: they cannot borrow Fed funds, nor borrow from the Fed; their backstop is their own central bank, and the ECB “can’t print dollars”. So they face more discipline than onshore banks [► L8.6 @ 0:07]. They respond by lining up their cash flows in time much more carefully: deposits are *term* deposits to a specific date (one month, three months), loans run to specific dates (six months), and the banks match not only the amounts but the *dates*, so that when they must pay, someone is paying them [► L8.6 @ 1:09]. Payments do not always come as promised, so reserves are still needed, but one can try; and this is the motivation for forward rate agreements and foreign exchange agreements: instruments to line up cash inflows and outflows in time [► L8.6 @ 2:30].

A forward deposit as a swap of IOUs. Bank X expects to make a three-month loan two months from now (a corporate customer has announced its needs) and wants to lock in the funding now; bank Y expects a deposit inflow in two months and wants to lock in a use for it [► L8.6 @ 4:55]. “All banking is a swap of IOUs”: X makes a two-month deposit at Y of \$1 million, Y makes a five-month deposit at X of \$1 million [► L8.6 @ 6:37]:

$t = 0:$	Bank X		Bank Y	
	Deposit at Y (2 months)	Deposit of Y (5 months)	Deposit at X (5 months)	Deposit of X (2 months)
$t = 2:$	Bank X		Bank Y	
	–Deposit at Y +3-month loan	Deposit of Y (3 months left)	Deposit at X (3 months left)	–Deposit of X +Customer deposit

No cash moves at time 0 [► L8.6 @ 7:23]. At two months X’s deposit at Y matures: Y pays X, which uses the money to make its loan; Y pays with the deposit inflow it expected [► L8.6 @ 7:44]. The swap has locked in a flow of funds from Y to X in two months, for three months; Eurodollar deposits exist to any date, so the timing can be fine-tuned with balance sheet entries alone [► L8.6 @ 9:08]. If the two deposits pay the same rate over the last three months, call it f , X has locked in a funding rate and Y an investment rate, even if three-month LIBOR is much lower by then [► L8.6 @ 9:49].

8.7 FRAs as implicit swaps of IOUs

The literal swap takes up balance sheet space, against which regulators require capital and reserves; so it is done off balance sheet [► L8.7 @ 0:08]. Stigum describes the evolution [► L8.7 @ 0:28]:

1. the *forward forward*: Y promises to make a deposit at X at rate f two months from now;

2. the *forward rate agreement*: Y promises nothing of the sort; X funds its loan at three-month LIBOR when the time comes, and the two banks act as if there were a three-month deposit paying $f - \text{LIBOR}$ between them [► L8.7 @ 1:10].

Definition 8.3 (New concepts: forward rate agreement). A *forward rate agreement* (FRA) between two banks fixes today a rate f for a future period (here from month 2 to month 5); at the start of the period they settle the difference between the reference rate then observed (LIBOR) and f , on a notional amount: if $\text{LIBOR} > f$ the forward lender pays the forward borrower, if $\text{LIBOR} < f$ the forward borrower pays [► L8.6 @ 3:31] [► L8.7 @ 1:55].

Market people describe it as a bet on LIBOR, and it can be used to speculate; but that is not why it exists [► L8.6 @ 4:13]. X funds at LIBOR plus or minus the FRA payment, i.e. at f whatever happens: the FRA implements exactly what the swap of IOUs did, locking in funding, without balance sheet space. “It looks like a side bet on LIBOR, but it’s used to line up payments in time” [► L8.7 @ 2:37]. The notes underline the hidden assumption: X does not care how high LIBOR goes, and Y how low, only so long as the *liquidity is there* to raise and invest the principal when needed: “a key assumption, almost always satisfied but likely to fail when you need it most” [Notes 8 p.4].

Proposition 8.4 (Derivatives as implicit swaps of IOUs). *All the derivatives of the course, interest rate swaps, foreign exchange swaps, credit default swaps, can be viewed as implicit, behind-the-scenes swaps of IOUs: they are integrated in the balance sheet structure, and are “not really off balance sheet”. The method: ask how one would build the same payoff with a swap of IOUs on bank balance sheets; that tells what the instrument is for* [► L8.7 @ 3:18].

Exercise 8.2 (FRA settlement). Bank X buys an FRA at $f = 0.60\%$ on \$10 million for the period from month 2 to month 5 (90 days). At month 2 three-month LIBOR is 0.90% .

- Who pays whom, and how much (paid at the end of the period)?
- Show that X’s all-in funding cost for its loan is 0.60% .

8.8 Forward parity

Forward interest parity. Y would like f high, X low; but they do not haggle: f is fixed by an arbitrage condition [► L8.8 @ 0:07]. There are market rates for two-month and five-month Eurodollar deposits, $R_{0,2}$ and $R_{0,5}$. If the swapped deposits were real deposits at market rates, X would have manufactured a three-month funding rate and Y an investment rate; the only rate both accept is the one implied by the market rates [► L8.8 @ 0:48]:

KEY FORMULAS

Forward interest parity [► L8.8 @ 1:37]:

$$(1 + R_{0,2})(1 + f_{2,5}) = 1 + R_{0,5}$$

(with each rate expressed per its own period; in money-market convention $(1 + R_{0,2} \frac{60}{360})(1 + f_{2,5} \frac{90}{360}) = 1 + R_{0,5} \frac{150}{360}$).

Covered interest parity [► L8.8 @ 7:52]: with S the spot and F_6 the six-month forward exchange rate, both in yen per dollar, and R, R^* the six-month dollar and yen deposit rates,

$$F_6 = S \frac{1 + R^*}{1 + R}.$$

FRAs use this f [► L8.8 @ 2:58].

Covered interest parity. A bank is asked for a yen loan but has only dollar funding, six-month Eurodollars: matched in time, not in currency [► L8.8 @ 2:58]. Forward exchange contracts exist for this; but with a swap of IOUs, it swaps a six-month dollar deposit for a six-month yen deposit with another bank, at today’s spot rate [► L8.8 @ 4:02]. For one dollar: it receives S yen (say 100) to lend; after six months it owes the counterparty $S(1 + R^*)$ yen and is owed $1 + R$ dollars [► L8.8 @ 5:25]. Paying yen and receiving dollars at a ratio fixed today is an implicit forward exchange rate, which must equal the market forward F_6 : covered interest parity [► L8.8 @ 7:52]. Both parities are almost definitions: since anyone can “roll their own” forward from existing instruments, the outright forward must have the same price, or there is an arbitrage [► L8.8 @ 8:33]. (In the yen–dollar example the dollar has no star, “because the dollar is special”, the world funding currency [► L8.8 @ 5:46].)

Bank (yen lender)		Bank B	
Yen loan S	Dollar	Yen deposit at bank S	Dollar
Dollar deposit at B 1	Eurodollar funding 1		deposit of bank 1
	Yen deposit of B S		

The notes spell out the cash flows of this swap (yen per dollar, so a rise of S is a dollar appreciation) [Notes 8 p.6][Notes 8 p.7]: at time 0 the bank pays \$1 and receives S_0 yen; at T it pays $(1 + R^*)S_0$ yen and receives $1 + R$ dollars. If $S_T = F_T$ the two flows have equal value; if $S_T > F_T$ the dollars received buy more yen and the bank gains on the swap (and loses as much on its yen loan), if $S_T < F_T$ the reverse: the bank does not care what happens to the yen. The off-balance-sheet FX swap reproduces exactly these net flows, swapping yen for dollars at S_0 and back at F_T ; unlike the FRA, the principal cannot be netted, because it is in two currencies [Notes 8 p.7].

Exercise 8.3 (Forward interest parity). Two-month Eurodollar deposits pay 0.30% and five-month deposits 0.45% (annualised, actual/360, 60 and 150 days). Compute the implied three-month forward rate two months ahead. If an FRA is quoted at 0.70%, describe the arbitrage.

8.9 Forward rates are not expected spot rates

Two natural guesses [► L8.9 @ 0:08]:

- the forward interest rate $f_{2,5}$ is an unbiased forecast of three-month LIBOR two months from now: the *expectations hypothesis* of the term structure (definition 2.6), $f = \mathbb{E}[\text{future spot rate}]$;
- the forward exchange rate F_6 is an unbiased forecast of the spot rate six months from now: *uncovered interest parity* (UIP), $F_6 = \mathbb{E}[S_6]$, equivalently $\mathbb{E}[S_6]/S = (1 + R^*)/(1 + R)$.

Both fail empirically, all the time: a puzzle for economics [► L8.9 @ 1:10]. Mehrling’s answer, to be developed, is that it has to do with the hierarchy of money and credit, with liquidity: payments today are worth more than promised payments in the future, and not only because of time preference, in foreign exchange as in the term structure [► L8.9 @ 2:12].

KEY FORMULAS

The two violations, as stated in Mehrling’s notes [Notes 8 p.7][Notes 8 p.8]:

- EH fails as $f_{N,T} > \mathbb{E}[R_{N,T}]$: bank X would typically have done better without the FRA, borrowing spot; it pays for insurance, and it is the *borrower* of money who pays, not the lender: “another symptom of the hierarchy of money and credit”;
- UIP fails as: if $F_T < S_0$ (i.e. $R > R^*$), then $F_T < \mathbb{E}[S_T]$: the low-yielding currency does not appreciate enough to equalize yields; typically it actually *depreciates*, contrary to UIP.

Exercise 8.4 (Covered interest parity). The spot rate is 80 yen per dollar, six-month dollar deposits pay 0.5% and yen deposits 0.1% (for six months).

- (a) Compute the six-month forward rate.
- (b) Under UIP, what is the expected spot rate in six months? What does the failure of UIP mean for a bank that funds in yen and lends in dollars?

COURSE NOTE: Segment 8.10

The video “Forward Rates are NOT Expected Spot Rates” (L8.10), filed in the playlist under lecture 21, is the same recording as L8.9, starting 8 seconds later. It is not a missing segment: in the continuous recording of lectures 7–12 (`materials/transcripts/transcript-lecture-7to12.txt`, at 2:08:36–2:11:08) covered interest parity (end of L8.8) is followed by this one closing passage on EH and UIP, and then directly by lecture 9. So L8.9’s title (“Forward Parity, Exchange Rates, UIP”) only partly fits its content: the FX forward parity was already covered at the end of L8.8.

Key formulas and ideas

KEY FORMULAS

Eurodollars	dollar deposits outside the US; reserves = deposits at a New York correspondent; below Fed funds in the hierarchy
Global funding	Eurobanks as money dealers: overnight/term deposits fund term loans; LIBOR = interbank rate
Volcker rule	matched book (market making) allowed, proprietary positions not; basis risk blurs the line
Liquidity	no Fed access $\Rightarrow$ match amounts <i>and</i> dates of cash flows
FRA	implicit swap of term deposits; locks in a future funding/investment rate f off balance sheet
Forward interest parity	$(1 + R_{0,2})(1 + f_{2,5}) = 1 + R_{0,5}$
Covered interest parity	$F = S(1 + R^*)/(1 + R)$
EH, UIP	$f = \mathbb{E}[\text{future rate}]$, $F = \mathbb{E}[S_T]$: both fail empirically (liquidity)

Chapter 9

The World that Bagehot Knew

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L9.1 FT: Depreciation of Iran's Currency	3:37	Official and black-market rial; sanctions on the payment system.
L9.2 Reading: John Hicks	3:38	Hicks, <i>A Market Theory of Money</i> (1989): banks as dealers.
L9.3 Bagehot's World, Wholesale Money Market	7:57	Bills of exchange and their discount.
L9.4 Economizing on Notes: Deposits, Acceptances	8:28	Discounting for deposits; acceptances as guarantees.
L9.5 Managing Cash Flow: Discount, Rediscount	7:29	The discount rate as a tool of cash management; rediscount between banks.
L9.6 Market Rate of Interest	2:44	Decentralised management produces one market rate, the world rate.
L9.7 Central Bank and Bank Rate	8:13	The Bank of England's balance sheet; bank rate.
L9.8 The Bagehot Rule, Origin of Monetary Policy	4:38	Lend freely at a high rate; from lender of last resort to monetary policy.
L9.9 Limits on Central Banking: Internal vs. External Drain	10:20	Discipline from above: gold, cooperation among central banks.

Files. Transcripts [materials/transcripts/L09-P*.txt](#); Mehrling's notes [[Notes 9 p.1](#)]. **Readings.** Bagehot, *Lombard Street* (1873), two chapters; John Hicks, *A Market Theory of Money* (1989), ch. 5–7 ([materials/readings/L09-Hicks.pdf](#), summarised in [section 9.2](#)); Hicks, "A Suggestion for Simplifying the Theory of Money" (1935).

Motivation

So far the course has dealt with *par*: payments, in which a deposit or reserves are delivered to settle a debt; the interest rate stayed in the background [[► L9.2 @ 1:51](#)]. The next two weeks ask where the interest rate comes from and what the central bank has to do with it; after the midterm the course moves up to the exchange rate and out to asset prices, the price

of risk [► L9.2 @ 2:11]. The shift is from payments to *market making*: banks, central banks and security dealers as dealers who create liquidity by standing ready to buy and sell [► L9.2 @ 2:53]. The lecture starts in the 19th century, where things are simpler: the London money market of Bagehot [► L9.3 @ 0:30].

INTUITION: Review and preview, from Mehrling's notes

Review [Notes 9 p.1][Notes 9 p.2]. The money market is a mechanism for making a decentralized payment system work as close as possible to the ideal of one big bank: the central bank as a clearinghouse, Fed funds for banks, repo for dealers, Eurodollars offshore (where foreign banks without the discount window match flows in advance with FRAs and FX swaps), each with a backstop one level up (the discount window for Fed funds, clearing-bank credit lines for repo). The key distinction was money (final settlement) versus credit (postponed settlement): "the difference between money and credit is all about time". Real-time gross settlement and overdraft limits on Fedwire, net settlement on CHIPS, looser limits in repo and Eurodollars protect the core of the payment system from excess elasticity and the core of the credit system from excess discipline. Of the four prices only par has been in focus.

Preview [Notes 9 p.2][Notes 9 p.3]. Credit instruments are now seen as securities with a price in money, and the focus moves to the institutions that make liquid markets in them. Stigum says dealers are like banks, intermediaries (p. 432); Mehrling reverses it: *banks are like dealers*, market makers in money, and so is the central bank. These layers of market making knit the hierarchy together and make the difference between instruments look like one of price (quantity) rather than of moneyness (quality); the hierarchy "occasionally springs out onto the stage during crises".

9.1 FT: depreciation of Iran's currency

CASE STUDY: FT, October 2012: "Iran uses force to strengthen rial"

The rial has depreciated dramatically against the dollar in Tehran's bazaar, where private currency dealers trade. The government steps in and buys dollars at 28,000 rials; on the illegal black market a dollar costs 30,000, a depreciation of about 6% [► L9.1 @ 0:23]. The FT attributes it to Western sanctions: EU oil sanctions shrink Iran's oil revenue, and US financial sanctions make it "slow and expensive" for Iran to receive payments, which are in dollars; investors stay away from the rial and citizens dump it for foreign notes [► L9.1 @ 2:07].

Two points for the course [► L9.1 @ 1:46]:

1. the *foreign exchange dealer*, a first example of the dealers studied from now on (foreign exchange itself comes after the midterm);
2. the cause: an attempt to keep Iran out of the world *payment system*, so that it cannot get dollars.

9.2 Reading: John Hicks

READING: Hicks, *A Market Theory of Money* (1989)

Hicks's last book, written as an old man, influenced Mehrling deeply: he read it as a young professor (he started teaching at Barnard in the fall of 1987) and has had it on the reading list since he first taught money and banking in 1996 [► L9.2 @ 0:07]. As a young man, Hicks had sent monetary economics down what Mehrling sees as a misleading path, with an article of 1933; in 1989 he proposed another way, a market theory of money in which banks are *dealers*, which is exactly what the course does in the next two weeks. "I think of him as the Moses of modern economics": he saw the promised land but did not live to enter it [► L9.2 @ 0:49].

READING: What the excerpt says: Hicks (1989), ch. 5–7, pp. 41–63

Ch. 5, "The Nature of Money".

- A transaction has three parts: the agreement, the delivery of the goods, and the payment.
- Money serves as a standard (unit of account) and as a means of payment. Hicks does not treat "store of value" as a defining function.
- Coinage, from about the 7th century BC, carries a state guarantee of the metal's weight and fineness.

Ch. 6, "The Market Makes its Money".

- Trade credit circulates inside "circles of confidence". A bill that is *accepted* by a well-known house can circulate further.
- Discounting dealers stand between the bill market and outsiders, much as foreign-exchange dealers stand between two currencies.
- The Bank of England (1694) was founded on a loan to the government: the National Debt.

Ch. 7, "Banks and Bank Money".

- Banking grows out of joining two businesses: lending on the bill market, and safe custody of deposits. Custody supplies the funds that lending needs (p. 56).
- Hicks calls a firm a full bank when it does three things: it accepts deposits, discounts bills and makes advances to customers. The fourth function, supplying a medium of payment, follows from the other three.
- Bank notes began as transferable deposit receipts. The cheque replaced them, because it moves the deposit inside the bank and so nothing can be stolen on the way (pp. 57–58).
- The bank's own liabilities are usable as money, even by the bank itself. A loan therefore expands both sides of the balance sheet: from the point of view of the rest of the economy, the bank has "created" money (p. 58).
- Deposits are withdrawable on demand while advances are not, so a bank is exposed to a run. It has three protections (pp. 59–60):
 - (i) the law of large numbers, which fails when risks are correlated;
 - (ii) holding cash, bills and securities that can be sold;

(iii) borrowing from other banks, which leads towards a central bank.

Only the second is under the bank’s own control at any moment, so its management is central to banking.

- *Liquidity* is a quality of an asset held *as a reserve*. Cash used to cover day-to-day flows is a “running asset”, not a reserve. Liquidity therefore cannot be reduced to an arithmetical ratio of reserves to deposits (pp. 61–62).
- Hicks takes the idea from Keynes’s *Treatise on Money* (1930, vol. 2, p. 67). There Keynes ranks three kinds of bank assets by liquidity: first bills and call loans, then investments, then advances.
- In the *General Theory*, by contrast, Keynes treated bills as money. Hicks argues this was natural when the bill rate was below one-half per cent (1932–38), but it was an aberration. Once short rates are significant, no line can be drawn between money and bonds (“ M_1 , M_3 ”). This is exactly the money–credit continuum of the course (p. 63).

9.3 Bagehot’s world: bills of exchange

Firms in a production chain (in Britain, say, a wool producer and a clothing firm) buy and sell intermediate goods from each other: business to business, the wholesale money market [► L9.3 @ 0:51]. Firm A buys goods from firm B on credit, issuing a *bill of exchange*: a promise to pay in 90 days that refers specifically to these goods, which will have been sold by then [► L9.3 @ 2:25]. Business-to-business trade still works this way [► L9.3 @ 3:28].

COURSE NOTE: Which article?

The early article is Hicks’s “A Suggestion for Simplifying the Theory of Money” (*Economica*, 1935), which treated the demand for money as a choice among assets, an approach that led to portfolio theories of money; in class the date is given as 1933 [► L9.2 @ 0:49]. The 1935 paper is the one on the course’s reading list next to the 1989 book. Hicks received the Nobel prize in 1972.

Definition 9.1 (New concepts: bill of exchange, discount). A *bill of exchange* is a written order by which a debtor (here firm A) promises to pay a sum on a given date, typically 90 days, for goods received: a form of trade credit, transferable, “self-liquidating” because the sale of the goods provides the means to pay. To *discount* a bill is to buy it before maturity for less than its face value; the difference is the interest (section 3.6).

The four steps [► L9.3 @ 3:49]–[► L9.3 @ 6:23]:

1. A buys goods from B with a bill;
2. B, needing cash, discounts the bill at a bank, which pays in bank notes, say 95 for a bill of 100 (an exaggeration: it would be closer to 100);
3. in 90 days A sells the goods to its customers for notes;
4. A uses the notes to redeem the bill at the bank.

(1)	Firm A		Firm B	
	+Goods	+Bill	−Goods	
			+Bill	
(2)	Firm B		Bank	
	−Bill		+Bill	
	+Notes		−Notes	

(3-4)		Firm A		Bank	
		-Goods	-Bill	-Bill	
		+Notes,		+Notes	
		then			
		-Notes			

After step 2, B has been paid and is out; A owes the bank, not B [► L9.3 @ 4:31]. This is the basic discount mechanism by which firms pay each other with the help of a bank [► L9.3 @ 6:23].

The bank’s opportunity and danger. The bank gives up notes, which pay no interest, for a bill, which does: good. But it gives up a means of payment for an asset that turns into cash only in 90 days, hopefully: a liquidity risk. Its notes are its reserves against its deposits, which can be withdrawn or transferred at any time [► L9.3 @ 7:04].

9.4 Economizing on notes: deposits and acceptances

Two alternatives to step 2 save notes [► L9.4 @ 0:09]:

- (a) **Discount for deposits.** If B accepts a deposit instead of notes, the bank expands its balance sheet and keeps its notes. Still a liquidity risk: B discounted because it wants to spend, and if it pays a customer of the same bank, fine, but if it pays someone in France the bank must find notes, and behind them gold [► L9.4 @ 0:53].
- (b) **Acceptance.** B may prefer to hold the bill to maturity and earn the full 100 rather than 95, but wants to be able to sell it if needed. So it has the bank *endorse* the bill: the bank promises to pay if A does not. Then the bill is “a 90-day bill on this bank”, almost a time deposit, and easy to sell [► L9.4 @ 2:57].

(a)		Firm B		Bank	
		-Bill		+Bill	+Deposit of B
		+Deposit			

(b)		Firm B		Bank	
		Bill			+Acceptance
		+Acceptance			(contingent)
		(contingent)			

Definition 9.2 (New concepts: acceptance, contingent liability). A *banker’s acceptance* is a bank’s guarantee written on a bill (physically, the word “accepted” and the banker’s signature across it): the bank will pay if the drawee does not. It is a *contingent* liability of the bank and a contingent asset of the holder: it becomes an actual payment only if the event (the drawee’s failure to pay) occurs [► L9.4 @ 4:22].

Mehrling writes the acceptance as a separate balance sheet entry, because in the modern world it is a separate instrument: “an early form of a credit default swap: if they don’t pay,

I'll pay" [► L9.4 @ 5:26]. The bank charges a fee, like an insurance premium, smaller than the discount, say 1% instead of 5%, since it is a guarantee and not a 90-day investment; if A pays, the bank has "money for nothing" [► L9.4 @ 6:50]. Bankers know their firms; the typical problem is not outright default but *delay*: the goods are not sold in time, so payment is late, a liquidity issue [► L9.4 @ 7:52].

9.5 Managing cash flow: discount and rediscount

Focus on the bank: in step 2 bills come in and notes go out, in step 4 notes come in and bills go out. "That's the banking business" [► L9.5 @ 0:07]. The bank holds a *ladder* of bills maturing on known dates (a 90-day bill today is an 89-day bill tomorrow), so it knows the timing of its inflows; on the other side, demand deposits that could all go tomorrow, but whose patterns it knows from experience [► L9.5 @ 0:49]. The banker's business is matching inflows and outflows: use the notes that come in today to discount new bills, hold as little idle cash as possible [► L9.5 @ 1:52]. Two tools:

1. **The discount rate.** Short of notes, a bank gives customers "a bad price": it raises the rate at which it discounts, which is the same as lowering the price at which it buys bills. Discounting slows, maturing bills bring notes back, and later it lowers the rate again. Raising the rate shifts the balance sheet from bills into cash, lowering it from cash into bills [► L9.5 @ 2:33].
2. **Rediscount at another bank.** Instead of sending the customer down the street, the bank itself takes bills it already holds to another bank: an interbank money market [► L9.5 @ 5:20].

So each bank quotes two prices, a rate to discount for customers and a rate to rediscount with other banks: a buy–sell spread. Banks are money dealers, and the prices at which they deal are the discount and rediscount rates [► L9.5 @ 6:46].

INTUITION: Price as a valve on flows

Raising the discount rate does not change the bank's stock of notes at once; it changes the *rate of flow*: fewer bills in, the same maturities out, so the stock of notes recovers over the following days. The interest rate here is a control on the time derivative of reserves, set by a bank watching its own cash position.

Exercise 9.1 (Discount arithmetic). A 90-day bill of 100 is discounted at a discount rate of 4% a year (bank discount basis, 360 days).

- (a) What does firm B receive?
- (b) What is the bank's yield on the money it lays out, expressed as simple interest on a 365-day year?

Exercise 9.2 (A bank's ladder). A bank holds bills maturing at 10 a day for the next 90 days and notes of 50; its depositors withdraw on net 8 a day.

- (a) How many new bills can it discount each day without running down its notes?
- (b) A large withdrawal of 40 occurs today. Describe how it can use its discount rate and the rediscount market to restore its position.

9.6 The market rate of interest

Each bank looks only at its own cash flow and moves its discount rate accordingly; the effect is *coordination*. Whoever quotes the lowest rate gets all the discounts, so rates line up, some a bit off the market in one direction or the other to influence flows: a *market rate of interest* emerges [► L9.6 @ 0:07]. It moves over time with firms’ demand for discounts and the banks’ decisions on deposits, acceptances and how much cash to hold: all these decisions show up in one money market rate, uniform across the economy [► L9.6 @ 1:10]. Lombard Street was the money market not only for wool merchants but for the world: firms A and B might be in two different countries, discounting their bills in London; the London market rate was the world rate of interest, and payment imbalances anywhere showed up in it [► L9.6 @ 1:51].

9.7 The central bank and bank rate

The balance sheet of the Bank of England of 9 January 1924 (after Bagehot’s time, but with the structure he describes) is divided, since Peel’s Act of 1844, into an Issue Department and a Banking Department [► L9.7 @ 0:07]:

Issue Department		Banking Department	
Gold coin and bullion	Notes issued	Notes (reserve)	Deposits (public, bankers, other)
Government debt (fiduciary issue)		Government securities	
		Other securities (discounts, bills)	

- The **Issue Department** issues the currency, mostly against gold, plus a small fiduciary issue against government debt; at the margin it can issue more notes only against more gold, and must pay gold to foreigners who present notes [► L9.7 @ 0:49].
- The **Banking Department** holds part of the notes as its reserve (the rest circulate), government securities and other securities, discounts of trade bills; its liabilities are deposits, not only of banks but of firms and people: unlike the Fed, the Bank was a bank for everyone [► L9.7 @ 1:30].

READING: The bank return of 9 January 1924, in £ million

The figures of the course file (materials/readings/L09-balance_sheet_of_Bank_of_England.pdf, “Table 36.1, Bank return: January 9, 1924”), rounded:

Issue Department	
Government debt 11.0	Notes issued 146.0
Other securities 8.7	
Gold coin, bullion 126.2	
Banking Department	
Government secs 47.3	Capital 14.6
	Rest 3.3
Other securities 77.0	Public deposits 11.8
Notes 19.7	Other deposits
Gold, silver coin 1.8	116.2
	7-day bills 0.01

Each department totals £145.9–146.0 million. The fiduciary issue (government debt plus other securities) is £19.75 million; the rest of the note issue is backed by gold. The Banking Department's reserve is the £19.7 million of notes, about 15% of its deposits.

The Banking Department is in the same bill market as other banks: it discounts for the market and rediscounts for other banks [► L9.7 @ 2:31]. It quotes *bank rate*, the rate at which a banker who needs notes can get them from the Bank [► L9.7 @ 3:12].

Definition 9.3 (New concepts: *bank rate*). *Bank rate* was the Bank of England's published minimum rate for discounting eligible bills: the central bank's rediscount rate, the price at which the banking system could obtain the best money of the system.

The Bank was privately owned and paid dividends, but it held the reserve of the whole British banking system, which was at the same time the reserve of the international money market [► L9.7 @ 3:53]. If the banking system needs money, it discounts bills at the Bank for notes, or, in a real crisis when the Bank runs out of notes, for deposits [► L9.7 @ 4:56]. It could behave like any bank, raising or lowering its rate to manage its own cash flow; Bagehot says it does not, because as keeper of the reserve it has public duties, whether it admits it or not [► L9.7 @ 6:03]. Usually the market rate is below bank rate and nobody goes to the Bank; when everyone scrambles for cash the market rate rises to bank rate, and then everyone goes to the Bank, which “disgorges” its notes and expands its deposits, which banks treat as cash, although deposits are promises to pay notes and notes promises to pay gold [► L9.7 @ 7:05].

Discipline and elasticity in Peel's Act. The division fixed the note issue in aggregate: discipline. Elasticity came from the Bank holding more notes than it needed and lending them through discount, and from deposits, which other banks used as reserves when notes were scarce [Notes 9 p.6]. A profit-seeking central bank would expand like any bank in a credit boom, buying bills against new deposits; but whenever money is tight it can set the rate as a matter of its own policy: high enough that net rediscounting is minimal, lower to let its balance sheet expand, or so high that rediscounting turns negative and its

balance sheet contracts, using the rate to control the pace. “Bank rate thus moves as a policy variable, while the market rate moves as a market price” [Notes 9 p.6].

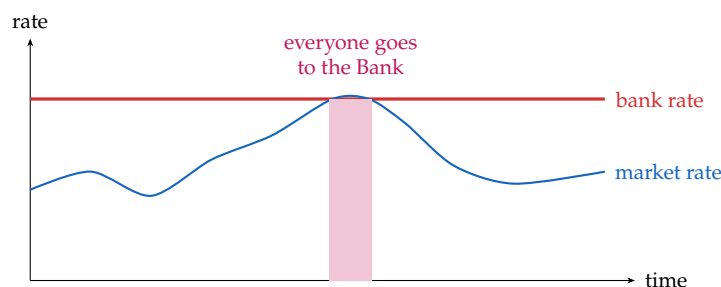


Figure 9.1: Bagehot’s two rates [► L9.7 @ 7:05]: normally the market rate is below bank rate and nobody borrows from the Bank; in a scramble for cash the market rate rises to bank rate, which acts as a ceiling, and the Bank lends freely there.

Exercise 9.3 (The two rates). Explain, with the T-accounts of a bank and of the Bank of England, what happens when the market rate reaches bank rate and a bank rediscounts 20 of bills at the Bank, first when it is paid in notes and then when it is paid in a deposit at the Bank.

9.8 The Bagehot rule and the origin of monetary policy

With bank rate the Bank of England is engaged in monetary policy, whether it knows it or not: “this is the origin of monetary policy” [► L9.8 @ 0:08]. In its minimal form: set a maximum rate, and when the crisis comes act passively, lending to all comers at that high rate [► L9.8 @ 0:28].

Proposition 9.4 (The Bagehot rule). *In a crisis the central bank should lend freely, at a high rate, against good security [► L9.8 @ 0:48]. That is all Bagehot says in Lombard Street; he does not say to move bank rate around to influence the economy [► L9.8 @ 0:48].*

Dealer, not lender, of last resort. Bagehot called this the *lender* of last resort; Mehrling’s notes call it *dealer* of last resort, to fix ideas useful for modern conditions. Narrow profit maximization is not the right policy for a central bank: its ability to relax the survival constraint of banks below it brings responsibility for the balance between discipline and elasticity. In an internal drain its job is to lend its note reserve “to the last farthing”, and even create deposits as a substitute for notes [Notes 9 p.7].

Elasticity and discipline in one phrase. “Lend freely” is elasticity: otherwise borrowers who cannot make their payments default, and liquidity kills you quick. “At a high rate” is discipline: only those who really need it come, and they repay fast, borrowing in the market again as soon as the market rate falls; the lending is self-liquidating [► L9.8 @ 3:13].

From backstop to policy. The market knows the Bank is waiting in the wings and behaves accordingly, taking more risk, lending more; this gives the Bank influence over the economy, and once it realises it, it can move bank rate up and down rather than leave it

at, say, 6%: lower it to encourage credit growth, raise it to discourage it ("I'm not going to help you, you're on your own"). "That's the art of central banking" [► L9.8 @ 1:09]. Sitting at the top of the hierarchy, the central bank can manage it, moving bank rate, and be lender of last resort in trouble; but it can push rates around with its balance sheet only if it lets the market move onto its balance sheet: it can relax tension or cause it [► L9.8 @ 2:31].

9.9 Limits on central banking: internal and external drain

Looking down: internal drain. If notes drain out of the banks into the hands of firms, the Bank can ease the drain by releasing its notes and creating good substitutes for them, or choose not to: a policy choice, without constraint [► L9.9 @ 0:09].

Looking up: external drain. If foreigners want notes, and ultimately gold, the Bank must give them gold. London may be the centre of the world money market, but the Bank of England is one central bank among others, and if they will not accept deposits at the Bank as equivalent to notes and demand gold, it must pay: that is the gold standard [► L9.9 @ 1:16]. The Bank faces the same constraint as any bank, measured in gold in and gold out for the whole country [► L9.9 @ 2:18]. And it uses the same tool: it raises its discount rate to say "don't come to me to discount and then take the gold to France" [► L9.9 @ 2:38]. If it does not, it can lose all its gold and have to suspend payments [► L9.9 @ 3:20].

IN THE MARKETS: Drawing gold from the moon

A saying of the time held that a high enough bank rate would "draw gold from the moon". The mechanism is a contraction of credit: at a very high rate no new bills come for discount, maturing bills are repaid, in gold, and no gold flows out again. The Bank secures its own position and maintains the gold parity by contracting credit [► L9.9 @ 4:23].

So the Bank of England is a source of elasticity looking down, and faces discipline looking up [► L9.9 @ 5:24]. Young stresses how little gold ran the whole world monetary system: only the gold in the Banking Department moved, the rest was locked behind the notes; this was possible because of the sophistication of the money market, where gold could be borrowed, also from other central banks [► L9.9 @ 5:48].

INTUITION: Cooperation and its limits

Central banks willing to borrow and lend gold among themselves could jointly relax their survival constraint, as banks do in the Fed funds market. But central bank cooperation, "easier said than done", involves national interests, politicians and finance ministers. Without it the survival constraint binds on central banks, which then have no choice but to raise rates, even if that crunches all the banks below them: discipline is transmitted down the hierarchy. "Does this sound familiar?" [► L9.9 @ 6:49]. The notes push the point: from the world's perspective gold flows between countries change the world gold stock no more than internal flows change the domestic one; if we accommodate the latter, why not the former? Only because it needs cooperation (the French central bank lending back the gold flowing out of England): a practical problem, not a necessity. The system does, however, need some overall disciplining factor [Notes 9 p.8].

The money market as coordinator. The central bank is in between, not at the absolute top: it can relax its own constraint with other central banks, and impose discipline below only if it has discipline imposed on itself, or chooses to [► L9.9 @ 8:13]. This is the world Bagehot knew, “not so different from our own”, and the origin of all later thinking about central banks, before IS/LM [► L9.9 @ 8:34]. The money market rate was a reliable indicator of stress, watched by everyone, coordinating the economy and, because London was the world money market, the world: “enough to make you think the money market is god”. Hence *Lombard Street* was a bestseller [► L9.9 @ 9:14].

Exercise 9.4 (External drain). Foreign central banks present notes worth 10 to the Issue Department for gold. Write the T-accounts of both departments. What can the Banking Department do with bank rate, and with what consequences for British banks?

Key formulas and ideas

KEY FORMULAS	
Bill of exchange	90-day trade credit tied to goods; discounted at a bank for notes: bank gains interest, takes liquidity risk
Economizing on notes	discount for deposits; acceptance = contingent guarantee (early CDS)
Discount rate	raising it slows inflow of bills, notes accumulate: a cash-flow management tool
Banks as money dealers	discount rate (to customers) and rediscount rate (between banks): a bid–ask spread
Market rate	emerges from decentralised cash management; London rate = world rate
Bank rate	central bank’s rediscount rate; normally above the market rate, a ceiling in crises
Bagehot rule	lend freely (elasticity) at a high rate (discipline) against good security
Drains	internal: policy choice; external (gold): discipline from above, relaxed only by central bank cooperation

HISTORY: Bank rate in the 19th century

Bank rate was the Bank of England’s minimum discount rate; from the 1850s, as the Bank came to manage the gold reserve actively, it was changed often, and rates of 6–10% were used in crises (10% in 1857 and 1866, 9% in 1873). Peel’s Act of 1844 was suspended in 1847, 1857 and 1866, allowing the Bank to issue notes beyond the legal limit. Bagehot’s *Lombard Street* (1873) appeared in the year of a further crisis.
References: R. S. Sayers, *The Bank of England 1891–1944*, Cambridge UP, 1976; Bagehot, *Lombard Street*, 1873.

Chapter 10

Dealers and Liquid Security Markets

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L10.1 FT: Asymmetric Credit Growth in Europe	6:33	IMF: European banks deleveraging; deposits flee from periphery to core.
L10.2 Market Liquidity, Dealers, and Inventories	7:10	Market versus funding liquidity; the supermarket and its inventories.
L10.3 Two-sided Dealer Basics	6:31	Two inventories; dealers as smoothers in price and time.
L10.4 Economics of the Dealer Function: the Treynor Model	11:43	Position limits, outside spread, inside spread.
L10.5 Leveraged Dealer Basics	7:22	Dealing with borrowed money; the Volcker rule and matched book.
L10.6 Real World Dealers	7:56	Gross versus net exposure; inventories “out in the world”.
L10.7 Arbitrage and the Assumption of Perfect Liquidity	9:41	Dealers move liquidity across markets; perfect liquidity as a limit case; Fischer Black’s “Noise”.

Files. Transcripts [materials/transcripts/L10-P*.txt](#); Mehrling’s notes [[Notes 10 p.1](#)]. **Reading.** Jack Treynor, “The Economics of the Dealer Function”, *Financial Analysts Journal* (1987) (not among the course files); Fischer Black, “Noise” (1986) ([materials/readings/L10-Noise.pdf](#)); IMF GFSR Box 2.3 and the FT article ([materials/readings/L10-*.pdf](#)). Recommended: Larry Harris, *Trading and Exchanges: Market Microstructure for Practitioners* (2003).

Motivation

Lecture 9 showed banks as dealers in money in Bagehot’s world. This lecture builds the general model of a dealer, Treynor’s, which the course uses from now on: for security dealers here, for banks as dealers in money in lecture 11, for foreign exchange and swaps later. The question it answers: where does *market liquidity* come from? Answer: from dealers, who supply it for a profit, and in doing so move prices away from fundamental values. Perfect liquidity, the assumption behind most of finance and general equilibrium theory, is shown to be a logically impossible limit.

10.1 FT: asymmetric credit growth in Europe

CASE STUDY: FT, 9 October 2012: “IMF warns eurozone on capital flight”

The IMF’s *Global Financial Stability Report* warns that unless eurozone officials step up their response, European banks would dump \$2.8 trillion of assets, more than 7% of their balance sheets, by the end of next year; banks in the periphery about 10% [► L10.1 @ 0:07]. Mehrling notes that the FT mislabelled one of the IMF’s charts, “an F for accuracy”: pay attention to details [► L10.1 @ 0:48]. The article (by Claire Jones and Michael Steen) adds three points:

- The IMF calls the result “extreme fragmentation” of euro-area funding markets. Its estimates are based on the 58 largest EU banks.
- With additional national measures, deleveraging would be \$2.3 trillion, 6% of balance sheets.
- Draghi: “You can’t have a union when you have certain countries that are permanent creditors and a set of countries that are permanent debtors.”

The chart is from Box 2.3 of the IMF *Global Financial Stability Report*, October 2012. Its Figure 2.3.3 shows euro-area bank credit to the non-bank private sector (Dec 2009 = 100) rising in the core and falling in the periphery.

Depositors move money from banks in the periphery (Spain, Italy, Portugal, Ireland) to banks in the core, fearing bankruptcy or a break-up of the euro, in which they would end up with a liability of, say, the Irish rather than the German central bank [► L10.1 @ 1:28]. This involves movements of reserves among central banks (TARGET2 balances, left for later) [► L10.1 @ 2:29]. For the periphery banks, losing deposits means they must find other funding, impossible, or shrink assets: they shed securities and loans and try to raise capital, and they make no new loans [► L10.1 @ 2:50].

Periphery bank		Core bank	
–Securities	–Deposits	+Reserves / securities	+Deposits
–Loans	+Capital		

For European banks as a whole the leverage ratio, total liabilities to capital, fell from about 25 to 23 (banks have 3–4% capital) [► L10.1 @ 3:32]. The average hides the asymmetry: bank credit to the private sector shrinks in the periphery and keeps growing in the core [► L10.1 @ 4:37]. The IMF adds that the ECB’s pledge to buy government debt lowered sovereign yields, but it is too early to tell whether it relieves deleveraging pressure [► L10.1 @ 6:01]. The question of the lecture: how can a mere *pledge* to buy lower bond yields? [► L10.1 @ 6:01]

10.2 Market liquidity and inventories

Definition 10.1 (New concepts: funding liquidity, market liquidity). *Funding liquidity* is the ability to raise money, to replace funds, the problem of the periphery banks [► L10.2 @ 0:28]. *Market liquidity* is the ability to buy or sell an asset quickly, in volume, without moving the price much; it is a property of the market for a particular asset [► L10.2 @ 0:49]. In Hicks’s terms, a liquid market has a *continuous* price: it may fluctuate,

but without jumps or “air pockets”; an illiquid market trades in fits and starts, at discontinuous prices [► L10.2 @ 1:51].

Supply and demand say nothing about the market structure behind such continuity: the price simply moves from one equilibrium to another, an abstraction from the institutions that make it continuous, the dealers [► L10.2 @ 3:01].

The supermarket. Intuition from a dealer we know, the West Side Market: in the early morning it is empty and the shelves full; in the evening “all of New York” is in there and the shelves are depleted; the prices are the same morning and evening [► L10.2 @ 3:42]. Supply and demand are not lined up in time, yet the price does not move. The secret is *inventories*: the shelves absorb the flow of demand and are restocked at night, from inventories in New Jersey, the wholesale market, which in turn are restocked from farms: “dare I say a hierarchy of inventories” [► L10.2 @ 4:43]. The supermarket is a *one-sided* dealer: it sells retail but does not buy from you (come in with a box of mangoes and they will ask whether you stole it); it buys wholesale. An investment bank underwriting an initial public offering is a one-sided security dealer: it buys the whole issue and distributes it, holding inventory meanwhile [► L10.2 @ 6:07]. Even Japanese “just-in-time” production does not eliminate inventories, the notes observe: it pushes them back before production, into the parts and components held by suppliers [Notes 10 p.2].

10.3 Two-sided dealers

A two-sided dealer trades a security for money and money for the security, either way; so it needs two inventories, of cash and of securities. A simple (hypothetical) dealer with fixed capital allocates it between the two and lets them fluctuate: when the public sells securities to it, its cash falls and its securities rise, and vice versa [► L10.3 @ 0:28].

Unleveraged dealer	
Cash	Capital
Securities	

With large enough inventories the price would not move at all [► L10.3 @ 2:12].

Life without a dealer. A buyer must find a seller: either raise the price offered (“I know it’s worth 10, but I’ll pay 12”) or wait. Time and price are the two costs [► L10.3 @ 2:33]. The housing market is the example: a broker shows you houses, and you wait; knocking on the door of an owner who is not ready to sell, you overpay; needing to sell quickly, you cut the price [► L10.3 @ 3:56]. Dealers are *smoothers*: they can flatten the price and reduce the waiting time to nothing, because they are always ready to take the other side [► L10.3 @ 3:36]. They do it to make money, so we need the economics of the dealer function [► L10.3 @ 4:58].

INTUITION: A missing piece of microeconomics

Price theory, and general equilibrium, assume perfect liquidity: a market-clearing price at every moment. This assumption must be false: dealers supply market liquidity, and not for free; if liquidity were a free good, infinite in quantity, no dealer would supply it. “The abstraction we have in price theory is an abstraction that’s logically

impossible for liquid markets" [► L10.3 @ 5:19].

10.4 The economics of the dealer function: Treynor's model

Treynor's diagram has the dealer's *inventory* on the horizontal axis and the *price* on the vertical axis (fig. 10.1). It has three elements.

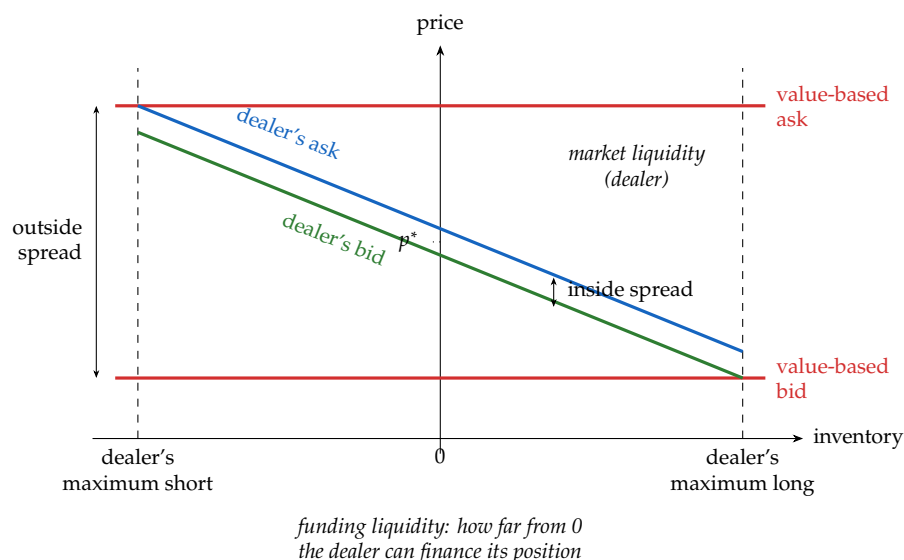


Figure 10.1: Treynor's model of a dealer [► L10.4 @ 0:13]. The dealer quotes an ask and a bid (inside spread) that fall as its inventory grows; the value-based traders' prices (outside spread) bound the market; the position limits come from the dealer's funding. p^* , the midpoint at zero inventory, can be read as fundamental value [► L10.7 @ 4:07].

1. **Position limits.** The dealer's maximum short and maximum long positions. A long position gains if the price rises, a short position gains if it falls; at zero inventory the dealer is not exposed [► L10.4 @ 0:36]. The limits come from capital, but more generally from the ability to *borrow* to fund the inventory: a banker who will lend up to some amount and no more, or the dealer's own risk appetite. They are soft in reality, hard in the model, and symmetric only for convenience [► L10.4 @ 1:21]. This is *funding liquidity*.
2. **The outside spread.** Behind the scenes are the fundamental forces of supply and demand and deep pockets: the *value-based traders*, "Warren Buffett in the wings", who buy when the price is low enough (the value-based bid) and sell or short when it is high enough (the value-based ask) [► L10.4 @ 3:07]. They are the dealer's "liquidity provider of last resort": a dealer near its long limit can unload its inventory to them. Knowing they are there, the dealer is willing to let inventory build up, and so to supply liquidity [► L10.4 @ 4:08]. In a crisis the value-based bid goes to zero; dealers stop making markets and prices are in free fall [► L10.4 @ 5:11].
3. **The inside spread.** The dealer's own quotes, bid and ask, are what you see in the market when your broker quotes you Apple stock; the outside spread you see only if inventories push the price to it [► L10.4 @ 5:51]. Two things determine it [► L10.4 @ 6:34]:

- (a) its *width*: the volatility of the price; holding inventory of a volatile asset is risky, and a wider spread protects the dealer [► L10.4 @ 6:55];
- (b) its *slope*: adverse selection; the dealer trades with anyone, and the counterparty may know more. A growing inventory means many people are selling to it: the order flow is information, “maybe they know something I don’t know”, so the dealer buys more only at a better price, and lowers both quotes [► L10.4 @ 7:38].

Definition 10.2 (New concepts: inside and outside spread, adverse selection). The *inside* (dealer) spread is the difference between the dealer’s ask and bid; the *outside* spread is the difference between the prices at which value-based investors will sell and buy. *Adverse selection* is the risk of trading with a better-informed counterparty: the trades that happen are disproportionately those that are bad for you.

Proposition 10.3 (Dealers turn funding liquidity into market liquidity). *Security dealers demand funding liquidity, borrowing to finance their inventories, and supply market liquidity, quoting prices at which anyone can trade* [► L10.4 @ 10:02].

The notes describe the liquidity the dealer supplies as an *option to trade*: at the quoted prices, either way, if you want to, but it is up to you. Anyone who places a limit order supplies liquidity in this sense, and amateur day traders often do, “until they run out of money” (the concept is developed in Larry Harris, *Trading and Exchanges*, 2003) [Notes 10 p.4]. In the notes’ Treynor model the value-based traders’ outside spread may be “maybe 20% below and 20% above” their estimate of fundamental value; when the dealer hits his position limits, the value-based trader becomes market maker of last resort [Notes 10 p.4].

KEY FORMULAS

A linear Treynor dealer (a simple parametrisation of fig. 10.1): with inventory $I \in [-I_{\max}, I_{\max}]$,

$$\text{ask}(I) = p^* + \frac{s}{2} - \lambda I, \quad \text{bid}(I) = p^* - \frac{s}{2} - \lambda I,$$

with inside spread s (rising with volatility) and slope λ (rising with adverse selection), chosen so that the quotes reach the value-based ask and bid at the position limits: $\lambda I_{\max} + s/2 = \text{half the outside spread}$.

Exercise 10.1 (The pledge). Use the Treynor diagram to explain how a central bank’s credible pledge to buy a government bond at a given price can lower its yield even before any purchase: which element of the diagram does the pledge change?

10.5 Leveraged dealers and the Volcker rule

Dealing is a very competitive business, with pressure to quote narrow spreads; one way to beat the competition is to trade on less capital, with borrowed money, so that the profit on capital is multiplied by leverage [► L10.5 @ 0:28]. Leverage is the ratio of debt to equity; deleveraging, as in the FT story, reduces it [► L10.5 @ 1:10]. Capital-funded dealers exist, but they are the value-based traders: ask Warren Buffett what his balance sheet looks like, “billions of dollars of cash waiting to buy a railroad”; the inside spread is made by leveraged dealers [► L10.5 @ 1:52]. In the extreme, fully borrowed: the “inventory of cash” is the banker who lends [► L10.5 @ 2:33].

<i>long:</i>	Leveraged dealer		<i>short:</i>	Leveraged dealer	
	Securities	Loan from bank		Cash	Securities (owed)

Making markets now means expanding and contracting the balance sheet itself, with only a thin layer of capital to absorb price changes [► L10.5 @ 3:14]. Mehrling knows of no regulatory leverage limit for dealers; they are usually part of larger organisations with more capital [► L10.5 @ 4:16].

The Volcker rule again. It says that banks cannot have subsidiaries that take positions: they may deal, but with a *matched book*, net inventory zero. The balance sheet can be very large, long and short positions of equal size, but there is no exposure to the price [► L10.5 @ 4:57]. The rule comes from the view that banks, or their dealer subsidiaries, took proprietary risk that, with FDIC insurance, fell on the taxpayer [► L10.5 @ 6:18]. It is about exposure to price risk, not leverage: “a matched-book dealer can be infinitely leveraged” without price risk, and will try to persuade regulators it needs no capital. But there is no perfect hedge, and matched books carry other risks, liquidity risk among them [► L10.5 @ 5:58].

10.6 Real-world dealers

Real dealers hedge: long some securities and short others that are correlated [► L10.6 @ 0:07]. Their balance sheet is the repo dealer’s of lecture 7: reverse repo brings securities in (and they may be sold short), repo sends them out as collateral for borrowing cash [► L10.6 @ 0:27].

Definition 10.4 (New concepts: gross and net exposure). *Gross exposure* is the size of the balance sheet; *net exposure* is long positions minus short positions. A matched book can have a huge gross and zero net exposure; a small one-sided book can have a small gross and a large net exposure [► L10.6 @ 1:50].

Typically (apart from the present distortions from QE3) dealers’ net exposure is to the *term spread*: long long-term bonds, short short-term ones, borrowing short and lending long like a bank; they profit from the failure of the expectations hypothesis [► L10.6 @ 2:57]. The dealer business is building up gross exposure while controlling net exposure [► L10.6 @ 3:59]. The Treynor model is about net exposure, price risk; gross exposure is liquidity risk, since the positions must be funded and counterparties may fail, a topic for the banking lectures [► L10.6 @ 3:59]. Managing price risk well allows tighter spreads, which is how dealers compete [► L10.6 @ 4:41].

How dealers make money [► L10.6 @ 5:22]–[► L10.6 @ 7:48]:

1. **Access to inventories:** knowing where securities are (which long-only pension funds hold what) and having a good relationship with a banker, perhaps the parent bank, for cash. The inventories are not on the dealer’s balance sheet, where they would need funding, but “out in the world”, like the mangoes in New Jersey;
2. **watching the net position** and moving prices so as not to be “bagged” by better-informed traders, widening the spread when volatility rises, minute by minute;
3. **offsetting** long with short exposures, buying at the bid and selling at the ask: the more both-way business, the more spread earned.

INTUITION: A just-in-time inventory system that straddles the hierarchy

Through repo the dealer reaches the inventory of cash, through reverse the inventory of securities: it behaves as if it had inventories that are actually out in the market, a just-in-time system. In a crisis it matters where the inventories are: cash comes from higher in the hierarchy, from banks, securities from lower, from security holders; the dealer straddles the layers. Troubles come when it must produce securities it reversed in and sold, and more seriously when it must produce money it repoed in and spent [Notes 10 p.5][Notes 10 p.6].

Exercise 10.2 (Gross and net). A dealer is long \$5 billion of 10-year Treasuries and short \$4.8 billion of 2-year Treasuries, all financed by repo and reverse repo.

- Compute gross and net exposure in dollar terms.
- Why is the net exposure in dollars a poor measure of its price risk? What would you use instead?

10.7 Arbitrage and the assumption of perfect liquidity

A dealer long Treasury bonds and short Treasury bills is *transporting liquidity* from one market to another: if every security needed its own dealer with its own capital, markets would be much less liquid [► L10.7 @ 0:11]. Arbitrage links all markets: liquidity is a *macroeconomic* concept, not market by market [► L10.7 @ 0:59]. Dealers choose where to deploy their scarce resources, seeking wide spreads and little competition; so some markets have wide spreads, some steep bid–ask curves (trade in size and the price moves), some, like Treasury bills, are nearly flat. Beyond this cross-section, macroeconomic fluctuations of liquidity move all prices together [► L10.7 @ 1:41].

Prices away from value. In Treynor’s model the price moves with the dealer’s *inventory*, not with the fundamental value: the dealer lowers its price not because it knows the asset is worth less but because its inventory is large [► L10.7 @ 3:05]. If fundamental value is the midpoint at zero inventory, then whenever the dealer holds inventory, “the price is wrong” [► L10.7 @ 4:07].

Proposition 10.5 (Perfect liquidity is a limit case). *Most asset pricing, and general equilibrium theory, assume perfect arbitrage: the outside spread collapses onto the fundamental value, value-based traders buy at any price a tiny bit below it and sell a tiny bit above, the inside spread is squeezed to zero, and liquidity is a free good [► L10.7 @ 4:49]. In Treynor’s world your ability to trade comes from trading at a price different from fundamental value: liquidity is provided for profit, and in fluctuating quantity [► L10.7 @ 6:36].*

The notes add a “philosophical question” [Notes 10 p.6][Notes 10 p.7]: finance assumes that arbitrage creates perfect liquidity by entering at the smallest deviation; but in a fully efficient market arbitrage would not be profitable, all bets being fair, so position takers would not make money, would compete less for market making, and markets would be less liquid, with wider spreads and more volatile prices: “in practice it seems that we must expect liquidity to enter into asset prices”. Arbitrageurs, by taking positions against temporary distortions, spread their impact into other markets and so counteract them: “porters” of liquidity, producing a result “no part of their intention”. Arbitrage and liquidity are two sides of the same coin [Notes 10 p.6]. Some markets, and some times, are

COURSE NOTE

The notes give the crisis examples: “fails” (collateral not returned) became so widespread that the Fed lent its own Treasuries to dealers through the Term Securities Lending Facility, and when the repo market collapsed it opened the Primary Dealer Credit Facility, a lender of last resort for dealers [Notes 10 p.6]. (Both were created in March 2008, around the collapse of Bear Stearns.)

close to the frictionless limit; others are not [Notes 10 p.7].

READING: Black, "Noise" (1986), section I "Finance", pp. 529–533

- Black contrasts trading on *information* with trading on *noise* as if it were information.
- Without noise traders there would be almost no trading in individual assets. An informed trader knows that only another informed trader takes the other side, and so declines to trade. Without trading in individual shares, portfolios, index futures and options could not be priced either.
- Noise trading makes markets liquid and makes it pay to collect costly information. But it also puts noise into prices: "What's needed for a liquid market causes prices to be less efficient."
- Informed traders do not eliminate the noise, because larger positions mean more risk and they can never be sure their information is not already in the price.
- Price wanders away from value like a drunk, and is pulled back the harder the further it strays. Hence the definition of efficiency: price within a factor of 2 of value.

The connection with Treynor: Black's noise traders are the flow the dealer absorbs, and his informed traders are Treynor's value-based traders. The band within which price may wander is the outside spread.

"Money and banking matters": it is not plumbing behind the scenes to be abstracted from in theories of value; it matters for asset pricing, income, employment and business cycles, as the crisis showed [► L10.7 @ 9:03].

Key formulas and ideas

KEY FORMULAS

Market vs funding liquidity	trade in volume without moving the price vs ability to raise funds
Inventories	the secret of continuous prices (supermarket, hierarchy of inventories)
Treynor	position limits (funding), outside spread (value-based traders), inside spread (width ~ volatility, slope ~ adverse selection)
Dealers	demand funding liquidity, supply market liquidity
Leverage	dealers trade on borrowed money; matched book: no price risk, but liquidity risk
Gross vs net	size of balance sheet vs longs – shorts; dealers build gross, control net
Arbitrage	transports liquidity across markets: liquidity is macroeconomic
Perfect liquidity	outside spread → value, inside spread → 0: a logically impossible limit; Black: price within a factor 2 of value

HISTORY: Treynor, Black and "Noise"

Jack Treynor, a friend of Fischer Black, published "The Economics of the Dealer Function" in the *Financial Analysts Journal* (1987); in 1971 he had already written, under the pseudonym "Walter Bagehot", "The Only Game in Town" on market makers and informed traders. Black, by then a partner at Goldman Sachs, gave his presidential address to the American Finance Association, "Noise", in New York in December 1985 (*Journal of Finance*, July 1986). There he proposed to call a market efficient if price is within a factor of 2 of value, more than half and less than twice value, and judged that by this definition almost all markets are efficient almost all of the time, "almost all" meaning at least 90%. Mehrling reads the factor of 2 as Treynor's outside spread [► L10.7 @ 6:58]. (In class the order of the two works is given the other way round.)

References: J. L. Treynor, *Financial Analysts Journal* 43(6), 1987; W. Bagehot (pseud.), *FAJ* 27(2), 1971; F. Black, "Noise", *Journal of Finance* 41(3), 1986.

Exercises

Standalone exercises follow the section they practise; connected or integrative ones are at the end.

Exercise 10.3 (A linear dealer). A dealer has $p^* = 100$, inside spread $s = 0.2$, position limits ± 1000 , and value-based bid and ask at 95 and 105.

- (a) Compute λ and the quotes at inventories $-1000, 0, 500, 1000$.
- (b) The public sells the dealer 100 units, one at a time. What is the average price received, and how does it compare with p^* ?

Exercise 10.4 (Crisis in the model). In the model of the previous exercise the value-based bid falls to 70. Redraw the diagram. What happens to the dealer's willingness to absorb sales, and to the price path when the public sells 1500 units?

Chapter 11

Banks and the Market for Liquidity

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L11.1 FT: Money Market Mutual Funds	6:46	The Reserve Primary Fund in court: a run on a deposit substitute.
L11.2 Banks as Money Dealers, a Puzzle	4:15	Banks make markets at par, with no spread: how, and why profitably?
L11.3 Security Dealers as Money Dealers	11:19	Matched book and speculative book; the dealers' repo book rearranged.
L11.4 Adapting Treynor to Liquidity Risk	6:33	The Treynor model for term funding: dealers in liquidity.
L11.5 Digression: Evolution of American Banking	11:30	From bank intermediation to parallel and shadow banking.
L11.6 The Fed in the Fed Funds Market	12:48	The Fed's outside spread (IOER, discount rate) and daily stabilisation.
L11.7 Return to the Initial Puzzle	2:14	Payments and money dealing as joint businesses.

Files. Transcripts `materials/transcripts/L11-P*.txt`; Mehrling's notes [Notes 11 p.1]. **Reading.** Treynor (lecture 10); Stigum on banks.

Motivation

Lecture 10 built the Treynor model for security dealers, who turn funding liquidity into market liquidity. This lecture climbs one step up the hierarchy, to banks [► L11.2 @ 1:09]. Banks make a market between currency and deposits at par: a fixed price, with no bid–ask spread. How can that be possible, and profitable? [► L11.2 @ 1:49] The route is indirect: first see security dealers as dealers in *money*, apply Treynor's model to liquidity risk, then bring in the Fed as the maker of the outside spread in the money market; the answer comes at the end.

11.1 FT: money market mutual funds

CASE STUDY: FT, October 2012: “Court drama puts focus on money funds”

Anyone wondering why Mary Schapiro, chairman of the SEC, pushes so hard to reform money market funds should sit in courtroom 6B of Manhattan’s federal courthouse, where the SEC is reliving 2008: the oldest money market fund, the Reserve Primary Fund, suffered a run when investors discovered it was owed money by bankrupt Lehman Brothers [► L11.1 @ 0:07]. The fund was started by Bruce Bent, and the case involves him and his son [► L11.1 @ 1:08].

Definition 11.1 (New concepts: money market mutual fund, net asset value). A *money market mutual fund* (MMMF) holds money market assets (time deposits, Eurodollars, commercial paper, repo) and issues shares meant to look and feel like deposits [► L11.1 @ 1:28]. Like any mutual fund, the value of its shares is the value of its assets: the *net asset value* (NAV) per share [► L11.1 @ 2:30]. Money funds use accounting that keeps the NAV at \$1, absorbing fluctuations in the interest they pay, so that a share can be treated as a dollar deposit [► L11.1 @ 3:11]. “Breaking the buck” means the NAV falling below \$1.

Money market fund	
Commercial paper (incl. Lehman)	Shares (NAV kept at \$1)
Eurodollar time deposits	
Repo	

Money funds were a response to a regulatory constraint, *Regulation Q*, which forbade banks to pay interest on (demand) deposits: Bent offered interest on something very like a deposit, and “the world beat a path to his door” [► L11.1 @ 3:52]. In September 2008 the fund’s Lehman commercial paper became worthless; the loss could not be absorbed in the interest rate (it would have needed a negative rate), holders ran, and shares were redeemed below a dollar. The fund was diversified, so it lost only its Lehman paper, a few cents per share [► L11.1 @ 4:33]. The SEC claims the Bents told investors they intended to support the fund with their own money when they had no such intention; Bent replies that it was a run, a liquidity problem that no private wealth could stop [► L11.1 @ 5:35]. The run stopped when the Treasury in effect created deposit insurance for money funds [► L11.1 @ 5:55].

Exercise 11.1 (A run on a money fund). A money fund has \$100 of assets, of which \$3 of commercial paper of a bank that fails (worth zero), and 100 shares.

- What is the NAV? What happens if it keeps paying \$1 per share to the first 50% of holders who redeem?
- Why is it rational to run? Which institution’s guarantee stops the run, and why?

11.2 The puzzle: banks as money dealers

A bank makes a market between two layers of the hierarchy, currency and deposits, at par (section 2.6) [► L11.2 @ 1:49]. This seems to contradict the Treynor model on two counts [► L11.2 @ 2:30]:

1. the price is *fixed*: it cannot fall as inventories grow, which is how a dealer protects itself;
2. there is *no bid-ask spread*: currency goes into deposits and deposits into currency at the same price.

So: how is it possible, and why is it profitable? “When you confront a puzzle like this, it’s better not to hit it head on”: back up to security dealers [► L11.2 @ 3:34].

11.3 Security dealers as money dealers

Two books. Split a security dealer’s book in two [► L11.3 @ 0:21]:

- a *matched book*: securities in 100, securities out 100, the same securities, long and short, netting to zero;
- a *speculative book* (Treynor’s book): a net long position of 10 in bonds financed by borrowing of 10, exposed to price risk.

In real dealers’ balance sheets most of the book is matched, gross exposure, and only a little is net; they make a lot of money on the matched book too, which Treynor’s model ignores [► L11.3 @ 1:27].

The gestalt switch. Now look at the security dealer as a dealer in *money*: the securities are just collateral for borrowing and lending money [► L11.3 @ 3:11]. From the New York Fed’s primary dealer statistics (the memorandum items, in billions) [► L11.3 @ 4:35]:

Primary dealers as money dealers			
Overnight reverse	854	Overnight repo	1,796
Term reverse	1,253	Term repo	826

Reverse and repo are the same instrument from opposite sides, so there is a matched book underneath [► L11.3 @ 4:58]. Assume, for the sake of the concept, that the term instruments have the same term [► L11.3 @ 5:39]. Carve out the matched parts [► L11.3 @ 6:00]:

Matched book				The rest	
Overnight reverse 854		Overnight repo 854		Term reverse Net financing (bonds)	427 515
Term reverse 826		Term repo 826		Overnight repo 942	

The numbers are the same, only rearranged; the “net financing” of 515 is bonds financed in the overnight repo market. One more step: finance the bonds with term repo instead, adding 515 of term repo on both sides [► L11.3 @ 8:29]. The rest becomes two separate exposures, read line by line:

COURSE NOTE: Facts about the Reserve Primary Fund

In class: the first money fund, started in 1969, redeemed at about 98 cents [► L11.1 @ 1:08]. The Reserve Fund was launched by Bruce Bent and Henry Brown in 1971–1972; its NAV fell to \$0.97 on 16 September 2008. On 19 September the Treasury announced a temporary guarantee program for money market funds. Regulation Q prohibited interest on demand deposits and capped the rates on time and savings deposits.

Two exposures			
Term reverse	942	Overnight repo	
Bonds	515	942	
		Term repo	515

1. **Bonds 515 against term repo 515:** Treynor's *price risk* [► L11.3 @ 9:10];
2. **Term reverse 942 against overnight repo 942:** borrowing overnight, lending for a month, both secured. "You're basically a bank": the overnight repo is like a demand deposit, the term reverse like a one-month loan. This is *liquidity risk*: the funding must be rolled every day, and could run, as the Reserve Primary Fund did [► L11.3 @ 9:31].

Definition 11.2 (New concepts: *price risk*, *liquidity risk* (of a dealer)). A dealer's *price risk* is its exposure to changes in the prices of the securities it holds net (its net position). Its *liquidity risk* is its exposure to the need to refinance short-term borrowing that funds longer-term lending, i.e. to the market for funding.

Exercise 11.2 (Rearranging a repo book). A dealer has overnight reverse 300, term reverse 700, overnight repo 800, term repo 250 (assume equal terms).

- (a) Separate the matched book and the rest.
- (b) What is its net bond position? Rewrite the rest as a price-risk exposure and a liquidity-risk exposure.

11.4 Adapting Treynor to liquidity risk

Put *liquidity risk* on the horizontal axis of Treynor's diagram: the scale of the maturity mismatch, up to some maximum [► L11.4 @ 0:07]. There are two prices, the overnight rate and the term rate; what matters is the spread between them, the dealer's profit [► L11.4 @ 0:49]. Take the overnight rate as fixed by the Fed (Fed funds, and by arbitrage repo) for the moment, and focus on the *term* rate [► L11.4 @ 1:09]. Money markets quote yields, not prices; a yield is an inverse price, so the curves slope *upward* and bid and offer swap places [► L11.4 @ 1:52].

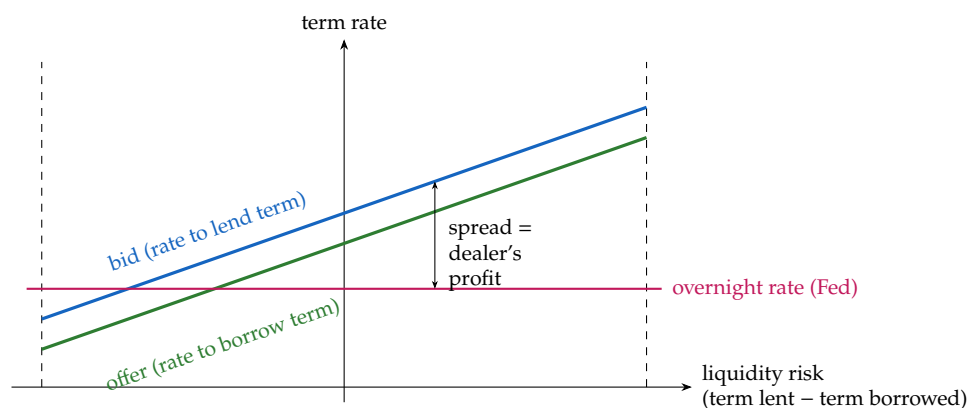


Figure 11.1: Treynor's model applied to liquidity [► L11.4 @ 1:52]: quotes are yields, so the curves slope upward. To take on more liquidity risk (lend more at term against overnight

funding) the dealer must be paid a larger spread of the term rate over the overnight rate.

Proposition 11.3 (The price of liquidity risk). *For dealers to take on more liquidity risk they must be paid, and what they are paid is the spread between the term rate and the overnight rate. With a big enough spread, “give it a whirl, life is short”: they expand their balance sheets, although a run is always possible [► L11.4 @ 2:57].*

Two margins. The bond price is driven by the dealer’s net bond position (515), the term rate by its liquidity position (942); there is no reason for them to be equal. A dealer can change its bond position without changing its liquidity risk, by expanding both sides, and vice versa: a profit-seeking dealer adjusts both margins, dealing in bonds and in money separately [► L11.4 @ 3:41]. The same diagram describes a bank, with deposits in place of overnight repo and term loans to companies in place of term reverse: a bank is a dealer whose “security” is money [► L11.4 @ 5:24]. The other side of a bank, the payment system, is what puzzled us [► L11.4 @ 6:06].

Exercise 11.3 (Pricing liquidity). A dealer can borrow overnight at 0.30% and lends term (one month) at a rate that rises by 1 basis point for each \$10 billion of term lending it does. It must earn at least 5 basis points over the overnight rate on its marginal lending. What is its maximum term lending if the market term rate is 0.40% at zero dealer position?

11.5 Digression: the evolution of American banking

Why this way of thinking is now necessary, and why textbooks, by a kind of hysteresis, still do not use it; it also helps read Stigum, who belongs to an earlier generation [► L11.5 @ 0:07].

1. **Banks as intermediaries** (“*It’s a Wonderful Life*” banking of the 1950s): households save in bank deposits, banks lend to businesses for capital investment; Janus-faced, giving households the short-term liquid asset they want and firms the long-term liability they need. Banks also hold government securities and borrow wholesale, which lets loans and deposits move separately; capital absorbs losses, reserves at the Fed serve payments [► L11.5 @ 1:12].
2. **The death of loans:** corporations found the commercial paper market cheaper than bank loans [► L11.5 @ 4:19].
3. **The death of deposits:** households and businesses moved money to money market funds, which paid interest [► L11.5 @ 4:39].
4. **A parallel banking system:** finance companies (General Motors’ financing arm making auto loans) issue commercial paper, money funds buy it and issue deposit substitutes; intermediation bypasses banks [► L11.5 @ 5:20].
5. **The shadow banking system:** households issue mortgages, which are packaged into residential mortgage-backed securities, sliced into AAA “quasi Treasury bills” held by a shadow bank (on the balance sheet of a European bank, or a structured investment vehicle of Citibank), which uses them as collateral to borrow in the wholesale money market, through asset-backed commercial paper and repo, bought by money funds that issue shares [► L11.5 @ 6:43]. None of it passes through a bank

with an account at the Fed, and much happened offshore: UBS or Deutsche Bank borrowing in the dollar money market to invest in dollar RMBS [► L11.5 @ 9:12].

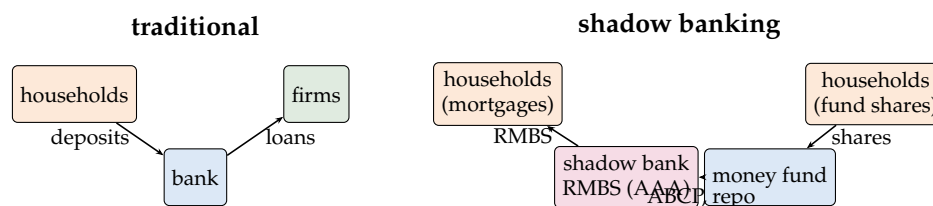


Figure 11.2: From bank intermediation to market-based credit [► L11.5 @ 1:33]: the shares of money funds fund mortgages through the wholesale money market, without passing through a bank. (Arrows point from the funder to the funded asset.)

The notes add what happened in between [Notes 11 p.4][Notes 11 p.5]. Banks hit by the death of loans adjusted by providing backup lines of credit to commercial paper issuers. The death of deposits came as Regulation Q ceilings eroded and money market funds competed: “disintermediation”, ultimate borrowers and lenders taking interest rate risk without a bank in between. As banks lost their core business they were kept out of others (the Securities Industry Association resisted bank underwriting) and subjected to capital requirements others did not face, partly for the safety of the payment system and because of their privileged access to the Fed: this pushed them to strip the balance sheet, FRAs for deposits, futures for actuals. The shadow banking system faces the liquidity and solvency risks of traditional banking without the government backstops, relying on the wholesale money market and the CDS market instead; its lender of last resort is the traditional banking system, through lines of credit and liquidity commitments. (On the older practice of managing bank interest rate and liquidity risk the notes cite M. Stigum and R. Branch, *Managing Bank Assets and Liabilities*, 1983.)

INTUITION: Two prices, both made by dealers

A shadow bank makes money on two prices: the price of its assets, essentially a bond price, and the price of its funding, a term interest rate. Both are made in dealer markets, by the two versions of Treynor’s model of this and the previous lecture. “The modern banking system is a capital-market-based credit system, not a bank-loan-based system”: hence the central place of the economics of the dealer function in the course [► L11.5 @ 10:16].

11.6 The Fed in the Fed funds market

Now unbracket the overnight rate [► L11.6 @ 0:07]. The Fed does two things in the Fed funds market [► L11.6 @ 1:08]:

1. it makes the *outside spread*;
2. in normal times it *stabilises* the rate within it.

The outside spread today. Since the crisis the Fed pays *interest on excess reserves* (IOER), 0.25%, where before reserves earned nothing, like cash: a floor [► L11.6 @ 1:49]. (Fed funds effective at 0.16% is below the floor because some lenders cannot hold accounts at the Fed

and lend to banks that can, which pocket the spread by depositing at 0.25% [► L11.6 @ 2:51].) The ceiling is the discount rate, 0.75%: the Fed lends to banks at that rate, so Fed funds cannot really go above it [► L11.6 @ 3:32]. With a trillion dollars of excess reserves, the market is “smack up against” the floor: the Fed is flooding the market with liquidity. This is unusual, maybe pathological, and the Fed is experimenting with ways to soak up reserves: reverse repos, and last month term deposits [► L11.6 @ 4:13]. The notes add: the Fed’s communication says it expects to widen this 50 basis point outside spread, with IOER some 75 basis points below the target; in a way the effective rate should be thought of as 25, not 16, which helps explain why repo trades above Fed funds, “a lot of distortion in the system right now”; and the first offering of the Fed’s own term deposits took place on 10 September 2012 [Notes 11 p.5][Notes 11 p.6][Notes 11 p.7].

Before the crisis. The floor was zero; the discount rate was set 100 basis points above the Fed funds *target*: an outside spread that never bit, because the Fed stabilised the rate around the target every day, anticipating fluctuations in the demand for reserves and supplying it, sometimes with repos twice a day; whole books were written on forecasting the need for reserves [► L11.6 @ 5:34].



Figure 11.3: The Fed’s outside spread in the Fed funds market [► L11.6 @ 2:09]: before the crisis a wide spread with the rate stabilised at the target inside it; in 2012 a narrow spread with the market pressed against the floor by a trillion dollars of excess reserves. (Schematic.)

How the Fed stabilised. If the market pushed Fed funds above the target, the Fed added reserves with a *temporary open market operation*: a repo with a dealer, a purchase and sale of a Treasury for a fixed period, overnight, three days, a week (an outright purchase would be permanent) [► L11.6 @ 8:40]. The dealer, which has no Fed account, uses the cash to repay a loan from its bank, borrowed at Fed funds plus 50 basis points: it replaces expensive borrowing with cheap repo. The bank’s reserves rise, and it lends them in the Fed funds market, pushing the rate down [► L11.6 @ 10:08].

Fed		Dealer		Bank	
+Repo (dealer)	+Reserves (bank)		+Repo from Fed –Loan from bank	+Reserves –Loan to dealer	

In doing so the Fed moves dealer inventories and so their prices; and it puts part of the Fed funds market on its own balance sheet, borrowing from a bank and lending to a dealer: a “dealer of first resort”, trading not at the outside spread but inside the inside spread, since it asks the dealers to bid and takes the best bids [► L11.6 @ 11:32].

Definition 11.4 (New concepts: interest on excess reserves, corridor). *Interest on excess reserves* (IOER), paid by the Fed since October 2008, is the rate on reserve balances above the requirement; it sets a floor under the overnight rate for institutions that can hold reserves. Together with the discount rate as ceiling, it forms a *corridor* within which the policy rate moves.

Exercise 11.4 (The corridor). With IOER at 0.25% and the discount rate at 0.75%:

- (a) Explain why Fed funds can trade below 0.25%.
- (b) Describe what the Fed must do to raise the effective rate to 0.50% with \$1 trillion of excess reserves outstanding, using the tools mentioned in class (reverse repos, term deposits).

11.7 Return to the initial puzzle

Banks, like security dealers, have two faces: on one side they are money dealers, in the money market; on the other they operate the payment system [► L11.7 @ 0:07].

Proposition 11.5 (How banks can make markets at par).

- (i) Possible: *all the slack is taken up in the money market: the ability to expand and contract the balance sheet and to get reserves when needed is what lets a bank hold par in the payment system* [► L11.7 @ 0:49].
- (ii) Profitable: *the money-market side earns a spread; the payment side may even lose money, but the two are on the same balance sheet, complementary, and the joint business is profitable* [► L11.7 @ 1:10].
- (iii) *As for dealers, the size of the payment business and of the funding exposure are separate margins the bank can move independently* [► L11.7 @ 1:50].

Key formulas and ideas

KEY FORMULAS

Money funds	deposit substitutes, NAV kept at \$1; vulnerable to runs (Reserve Primary Fund, 2008)
Dealer's books	matched book (gross) + speculative book (net, Treynor)
Rearrangement	repo book = matched book + price risk (bonds vs term repo) + liquidity risk (term reverse vs overnight repo)
Liquidity Treynor	term rate on vertical axis (upward-sloping quotes); dealers paid the term-overnight spread to bear liquidity risk
Evolution	intermediation → death of loans and deposits → parallel banking → shadow banking (capital-market-based credit)
Fed	outside spread: IOER (floor), discount rate (ceiling); stabilisation by temporary repos with dealers
Banks at par	possible thanks to the money market; profitable as a joint business with it

Chapter 12

Lender/Dealer of Last Resort

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L12.1 FT: Citibank and the SIVs	5:10	Pandit resigns; Citi's structured investment vehicles and the collapse of ABCP.
L12.2 The Art of Central Banking	3:42	Hawtrey (1932); the Fed's crisis facilities.
L12.3 Evolution of Monetary Policy: 1951–1987	7:06	Quantity and price views; operating targets from free reserves to Fed funds.
L12.4 The Taylor Rule: 1987–2007	7:04	The rule, the Fisher effect, inflation targeting.
L12.5 Monetary Transmission Mechanism	10:00	From Fed funds to term rates to asset prices, through three dealer markets.
L12.6 Anatomy of a Normal Crisis	8:18	Dealers and banks absorb an imbalance, for a price.
L12.7 Anatomy of a Serious Crisis	4:32	The Fed as backstop of the banks that backstop the dealers.
L12.8 Should the Fed Intervene or Not?	8:40	Elasticity against spirals, discipline against accumulated mistakes.
L12.9 The Fed as Dealer of Last Resort: 2007–2009	15:50	The stages of the crisis in balance sheets.

Files. Transcripts `materials/transcripts/L12-P*.txt`; Mehrling's notes [[Notes 12 p.1](#)]. **Readings.** Stigum, chapter 9 on monetary policy; R. G. Hawtrey, *The Art of Central Banking* (1932), chapter 4 (quoted); FRBNY, "Forms of Federal Reserve Lending" (`materials/readings/L12-Forms_of_Fed_Lending.pdf`).

Motivation

The last lecture before the midterm puts the pieces together: the three dealer markets of lectures 6–11 (Fed funds, term funding, bonds) form the *transmission mechanism* of monetary policy; normal fluctuations are absorbed by dealers and banks; serious crises need the Fed as lender of last resort to the banks; and in 2007–2009 the Fed went further, becoming the money market itself and finally a *dealer of last resort*. The lecture is mostly about lender of last resort in the money market; dealer of last resort in capital markets needs derivatives and comes after the midterm [[► L12.2 @ 0:07](#)].

12.1 FT: Citibank and the SIVs

CASE STUDY: FT, October 2012: Vikram Pandit resigns

Citigroup's CEO Vikram Pandit resigned under board pressure and was replaced by Michael Corbat, who had run Citi Holdings, where the toxic assets were put to be worked out [► L12.1 @ 0:07]. Citi, built by Sandy Weill as a "state-of-the-art financial supermarket", came close to nationalisation in the crisis [► L12.1 @ 0:49]. The FT's timeline omits an episode Mehrling would add: in November 2007, as Citi announced losses that grew every few days (5, then 8, then 11 billion), it received a \$7.5 billion capital infusion from the Abu Dhabi Investment Authority: professional investors still thought it was a minor blip, as many economists thought subprime too small to matter [► L12.1 @ 1:30]. It was the beginning of the global financial crisis [► L12.1 @ 3:11].

The SIVs. Citi was a famous part of the shadow banking system: it set up off-balance-sheet *structured investment vehicles* that held AAA tranches of residential mortgage-backed securities, funded with asset-backed commercial paper [► L12.1 @ 3:31]. The first thing that happened in the crisis was the collapse of the SIVs and of the ABCP market: 90-day paper funding 30-year mortgages; at maturity the holders, money funds, declined to roll, the SIV found no other funding and borrowed from its host, Citibank, which ended up with the assets on its own balance sheet [► L12.1 @ 4:33].

Definition 12.1 (New concepts: *structured investment vehicle*, *ABCP*). A *structured investment vehicle* (SIV) is an off-balance-sheet entity sponsored by a bank that holds long-term assets (here AAA mortgage-backed securities) funded by short-term debt. *Asset-backed commercial paper* (ABCP) is commercial paper secured by a pool of assets, typically 30 to 90 days.

12.2 The art of central banking

Ralph Hawtrey, *The Art of Central Banking* (London, 1932, "an auspicious year"), chapter 4, subtitled "the lender of last resort": "a central bank is a bankers' bank", and the central bank is the lender of last resort; Hawtrey, in the line of Bagehot and up through Keynes, Sayers and Goodhart, was an inspiration for Mehrling [► L12.2 @ 0:48]. (His copy, withdrawn from the Kirkland House library at Harvard, cost three dollars [► L12.2 @ 1:50].)

The crisis facilities. The New York Fed lists the Fed's lending facilities: regular open market operations and the discount window, and then a long list that was not there before the crisis [► L12.2 @ 2:10]: the term discount window program, the Term Auction Facility, the Primary Dealer Credit Facility, transitional credit extensions, reciprocal currency arrangements (the liquidity swaps), securities lending and the Term Securities Lending Facility, the ABCP Money Market Fund Liquidity Facility, the Commercial Paper Funding Facility, the Money Market Investing Funding Facility, the Term Asset-Backed Securities Loan Facility [► L12.2 @ 2:51]. "This is what put a floor under it" [► L12.2 @ 3:33]. The box below summarises the New York Fed's chart.

READING: FRBNY, “Forms of Federal Reserve Lending” (July 2009)

Facility	Announced	Who	Collateral	Term
Regular OMOs	–	primary dealers	Treasuries, agencies, agency MBS	o/n–14 days
Discount window	–	depository institutions	full DW range	overnight ^a
Term DW program	17 Aug 2007	primary-credit banks	full DW range	up to 90 days
TAF	12 Dec 2007	primary-credit banks	full DW range	28 or 84 days
Liquidity swaps	12 Dec 2007	selected central banks	foreign currency	o/n–3 months
TSLF	11 Mar 2008	primary dealers	Treasuries, agencies, MBS, IG debt (lends Treasuries)	28 days
PDCF	16 Mar 2008	primary dealers	tri-party repo collateral	overnight
AMLF	19 Sep 2008	banks, BHCs, foreign-bank branches	first-tier ABCP	ABCP maturity
Transitional credit	21 Sep 2008	GS, MS, Merrill broker-dealers	DW and tri-party collateral	overnight
CPFF	7 Oct 2008	CP issuers (via an SPV)	new 3-month CP	3 months
MMIFF	21 Oct 2008	money market funds (via SPVs)	CDs, bank notes, CP	–
TALF	25 Nov 2008	all US persons	AAA ABS, CMBS	3 or 5 years

^a Up to several weeks for small banks. Everything except securities lending (bond for bond) adds reserves. The chart also lists ordinary securities lending: Treasuries for Treasuries, daily, to primary dealers. Most facilities are run by the New York Fed; the AMLF was run by the Boston Fed, and the discount window programs by all twelve Reserve Banks.

12.3 Normal monetary policy

12.3.1 Quantity or price?

Once central banks accepted Bagehot’s responsibility, they started to think how to avoid crises in the first place, and invented monetary policy, which evolved from 1873 to what it was before the crisis [► L12.3 @ 0:07]. Stigum shows an ambivalence about what the Fed controls: a quantity (money, reserves, a balance sheet measure) or a price (the Fed funds rate)? [► L12.3 @ 0:48] Those who emphasise quantities have the equation of exchange $MV = PT$ in mind; those who emphasise rates, the money market view [► L12.3 @ 1:30]. The two can be translated into each other:

- **quantity:** the Fed sits on top of the hierarchy (the pyramid of lecture 2) and controls

the best money of all, its own liability, high-powered money, which it uses to influence the balance between discipline and elasticity [► L12.3 @ 2:12];

- **price:** the Fed pegs the shortest end of the term structure, an administered policy rate, while the market determines the long end, security prices [► L12.3 @ 3:38].

Either way, monetary policy is about the *transmission mechanism* from what the Fed controls, its own balance sheet, to the financial system and the economy [► L12.3 @ 4:20].

Operating targets (Stigum) [► L12.3 @ 4:41]:

- 1951: free reserves;
- 1979: non-borrowed reserves (Volcker's "Saturday night special");
- 1983: borrowed reserves;
- 1987: the Fed funds rate, ever since.

Mehrling is not sure the Fed changed much what it actually did, rather how it explained it [► L12.3 @ 5:45].

KEY FORMULAS

Reserve accounting [Notes 12 p.2]:

$$TR = RR + ER = NBR + BR, \quad 0 = (RR - NBR) + (ER - BR),$$

total reserves split into required and excess, or into non-borrowed and borrowed; $ER - BR$ is *free reserves*.

From these identities, targeting excess reserves means accommodating changes in required reserves, and targeting borrowed reserves means accommodating changes in non-borrowed reserves: not the strict quantity control of the Currency School and the monetarists. The quantity target put pressure on the market (banks dislike borrowing from the Fed) while letting the market set the rate; even the non-borrowed reserves regime can be read as a stealth interest rate target [Notes 12 p.2][Notes 12 p.3]. Throughout, the constant is the Fed's idea that its job is to balance elasticity and discipline to control the flow of credit [Notes 12 p.2]. Stigum (3rd edition) is non-monetarist: Fed technicians "knew that the Fed could not control money supply with the precision envisioned in textbooks", and no macroeconomic theory for credit aggregates comparable to the quantity theory yet exists [Notes 12 p.1][Notes 12 p.2]. In 1979, facing double-digit inflation, Volcker presented himself as a convert to monetarism; most people think it was a convenient language: saying "I'm going to jack the Fed funds rate up to 15%" might have cost him his confirmation, saying "I'm a monetarist" had the same consequence without saying it, and he stopped inflation in its tracks [► L12.3 @ 6:05].

12.3.2 The Taylor rule (1987–2007)

1987 is when Mehrling got his job; by the eve of the crisis almost all economists agreed on what the Fed was doing and should do: the *Taylor rule* [► L12.4 @ 0:07].

HISTORY: The Saturday night special

On Saturday 6 October 1979 the FOMC, chaired by Paul Volcker since August, announced that it would control the supply of bank reserves (non-borrowed reserves) instead of the Fed funds rate; the rate rose to around 20% in 1980–1981. The approach was relaxed in 1982.
References: Federal Reserve press release, 6 Oct 1979; A. H. Meltzer, *A History of the Federal Reserve*, vol. 2, Chicago UP, 2009.

KEY FORMULAS

Taylor rule (in the form given in class) [► L12.4 @ 0:49]:

$$r = \rho + \pi^e + \alpha (\pi^e - \pi^*) + \beta (y - y_f),$$

with r the target (Fed funds) rate, ρ the natural real rate of interest (the return on real capital), π^e expected inflation, π^* the inflation target, $y - y_f$ the gap between output and its full-employment level [► L12.4 @ 1:09].

Fisher effect: $r = \rho + \pi^e$, nominal rate = real rate + expected inflation [► L12.4 @ 2:31].

- As a regression, the rule fits the behaviour of central banks around the world well in normal times (not in the crisis): a positive description; a large literature then made it normative [► L12.4 @ 1:51].
- The first part, $\rho + \pi^e$, is the *Fisher effect* (Irving Fisher, Yale): what markets do by themselves, compensating creditors for expected inflation. Fisher himself thought people often got inflation wrong, causing fluctuations; modern economics, with rational expectations, assumes π^e is unbiased [► L12.4 @ 2:56].
- The rest is the Fed leaning against the wind: if expected inflation exceeds the target, raise the rate by more than the difference (more discipline); if output is below full employment, lower it (more elasticity) [► L12.4 @ 4:18].

Careers were made on the optimal α and β , on whether to add a financial-stability term: “inflation targeting” spread all over the world. Economics came to believe it had mastered the economy to the point of arguing whether a coefficient should be 1.1 or 1.2; the Great Moderation, then the crisis. “Nobody believes this anymore” [► L12.4 @ 4:59]. Mehrling, who was reading Hawtrey instead, was better prepared: “in the land of the blind the one-eyed man is king” [► L12.4 @ 6:22].

In the notes’ terms, the Fed imposes discipline when prices rise too fast and adds elasticity when they do not rise fast enough; modern money management, no longer bound to the gold standard, watches the price level instead of gold flows as the sign of imbalance between cash flows and commitments. Inflation targeting is now in question, since the central bank may have fed the bubble by watching inflation rather than asset prices: “maybe we will be seeing a return to Hawtrey”, the inherent instability of credit working through lending on collateral [Notes 12 p.3][Notes 12 p.4].

Exercise 12.1 (Taylor rule). With $\rho = 2\%$, $\pi^* = 2\%$, $\alpha = 0.5$, $\beta = 0.5$:

- compute the prescribed rate for $\pi^e = 3\%$ and an output gap of -2% ;
- what rate does the rule prescribe for $\pi^e = 1\%$ and a gap of -6% ? What happens at the zero lower bound?

12.4 The monetary transmission mechanism

The three dealer diagrams of lectures 10–11, in reverse order, form the transmission mechanism [► L12.5 @ 0:08]:

- Fed funds** (overnight): the Fed makes the outside spread (IOER, discount rate); the horizontal axis is *settlement risk*, the risk of pushing payments to tomorrow [► L12.5 @ 1:11];
- term funding** (three months): *liquidity risk*, borrowing short and lending long;

COURSE NOTE: Taylor’s original rule

John Taylor (“Discretion versus policy rules in practice”, *Carnegie–Rochester Conference Series*, 1993) wrote $i = \pi + 0.5y + 0.5(\pi - 2) + 2$, with π inflation over the previous four quarters, y the percentage output gap and 2% for both the real rate and the inflation target. The total response of i to inflation is $1 + 0.5 = 1.5 > 1$ (the “Taylor principle”). In the form above the coefficient on π^e is $1 + \alpha$, so the principle requires $\alpha > 0$; in class Mehrling says “ α is greater than one” [► L12.4 @ 4:18].

upward-sloping quotes in yields [[▶ L12.5 @ 2:37](#)];

3. **bonds**: asset prices, downward-sloping quotes; *price risk* [[▶ L12.5 @ 2:58](#)].

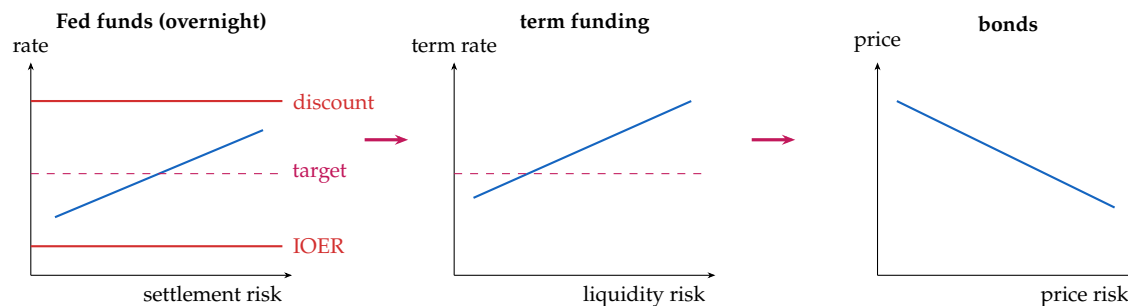


Figure 12.1: The transmission mechanism as three dealer markets [[▶ L12.5 @ 0:28](#)]: the Fed sets the overnight rate within its outside spread; the term rate must exceed it by enough to pay dealers for liquidity risk; bond prices depend on the cost of funding dealers' inventories. Raising the Fed funds target raises term rates and lowers bond prices.

The story [[▶ L12.5 @ 4:04](#)]:

1. the Fed sets a target Fed funds rate and enforces it by trading (temporary open market operations); the level of the target is a choice between leaning toward discipline and leaning toward elasticity, and the same rate can mean elasticity in some circumstances and discipline in others [[▶ L12.5 @ 4:25](#)];
2. raising the target (and with it the discount rate) narrows the gap between Fed funds and the term rate, which dealers do not like: they raise the term rate;
3. funding becomes more expensive, so bond dealers want smaller inventories: downward pressure on bond prices;
4. asset prices, mortgage-backed securities among them, affect the price of loans and so the real economy.

This works through the price of money and the price of capital, and instantly, through arbitrage in securities markets; it does not need banks lending multiples of their reserves (the money multiplier) or a bank-lending channel: institutions that no longer exist [[▶ L12.5 @ 7:31](#)]. The Fed may not move prices exactly as it wants, because the market watches and anticipates it; but “monetary policy does matter” in a deregulated system. What it *should* do has to be rethought [[▶ L12.5 @ 8:53](#)].

Exercise 12.2 (Transmission). Using the three diagrams of [fig. 12.1](#), explain step by step what happens to the three-month rate and to bond prices when the Fed raises its target by 25 basis points, and why the effect may be smaller if the market had already anticipated it.

12.5 Anatomy of a normal crisis

A “normal crisis” is the kind of imbalance that can happen every day [[▶ L12.6 @ 0:07](#)]. Households (non-banks) decide, for whatever reason, maybe only for a day, that they

want to hold money instead of securities [► L12.6 @ 0:28]. The dealer system exists to absorb imbalances in order flow [► L12.6 @ 1:09]:

Households		Dealers		Banks	
–Securities		+Securities	+Repo (from bank)	+Repo (to dealer)	+Deposits
+Deposits					

The household sector sells securities on net; dealers buy them and finance them with repo from banks, which lend by expanding their balance sheets [► L12.6 @ 1:32]. “Sort of magical”: households think they sold to other households, but the securities were absorbed by the dealer and banking system, and their deposits are *new* deposits, indistinguishable from old ones [► L12.6 @ 2:55].

The price of absorption. Dealers and banks do it to make money [► L12.6 @ 3:36]. The dealer, seeing the imbalance in its order book, buys at a lower price; when flows reverse, it sells higher. The bank lends at a slightly higher rate, and later liquidates at a lower one [► L12.6 @ 3:57]. The households pay without seeing it, through a lower price for their bonds [► L12.6 @ 5:40]. These price distortions away from fundamental value are possibly *equilibrating*: low prices attract other buyers, the positions are liquidated [► L12.6 @ 6:43]. Even a seller with no information, who just needs cash for a payment, moves the price, because the buyer fears he knows something; it comes back once the market realises he does not [► L12.6 @ 7:23].

The notes name the two things that happen in such a crisis [Notes 12 p.5]: the dealer system *buys time*, letting the private sector think accounts are settling (cash today instead of a promise) while settlement is in fact delayed, with new cash created for the purpose; and during the delay prices are pushed away from equilibrium, which pressures the system actually to settle: higher rates raise the cost of delay. Dealers and banks do it only if they “see a light at the end of the tunnel”. Many liquidity crises of market makers never become banking crises: “normal crises”, in which the liquidity constraint does its job of keeping the economy coherent [Notes 12 p.5][Notes 12 p.6].

Exercise 12.3 (A normal crisis). Households sell \$100 of bonds. Dealers buy them at a price 1% below fundamental value, financed by repo at the term rate. Three days later the flow reverses and dealers sell at fundamental value. Write the T-accounts at each step and compute the dealers’ profit if repo costs 0.5% a year.

12.6 Anatomy of a serious crisis

In a normal crisis there is no Fed: the private system absorbs it and the Fed funds rate stays put [► L12.7 @ 0:07]. A big shock makes dealers and banks anxious; that is when the Fed comes in [► L12.7 @ 0:28]. As in Bagehot’s world, the central bank sets a rate above the market rate; if the market on its own would push rates above it, the central bank becomes lender of last resort [► L12.7 @ 1:10]:

Fed		Banks		Dealers	
+Loans to banks	+Reserves	+Repo to dealers +Reserves	+Deposits +Borrowing from Fed	+Securities	+Repo

If the Fed backstops the banks, the banks are more willing to backstop the dealers; this keeps a large shock within bounds [► L12.7 @ 1:52].

Proposition 12.2 (Why the central bank can stop a crisis). *In a crisis everyone wants money and nobody wants securities; the securities must go somewhere. The lender of last resort works by letting the unwanted securities be absorbed into the financial sector and creating the cash people want to fund them: expansion of dealers, of banks, ultimately of the Fed. The Fed backstops the banks' short positions in money, i.e. their liquidity risk, and it cannot itself run into trouble with short positions in money, because its liability is the ultimate money, at least domestically: it operates as a money market dealer* [► L12.7 @ 3:04].

12.7 Should the Fed intervene or not?

Private dealers absorb fluctuations for profit; the Fed's motive is different: when should it take the pressure off, and when let it ride? The balance between discipline and elasticity [► L12.8 @ 0:09]. The Fed can always provide elasticity, but that does not mean it should [► L12.8 @ 1:11].

For elasticity. Prices may become so distorted that people make bad decisions; or distortions may stop being equilibrating: falling asset prices make assets unusable as collateral, holders dump them, a *liquidity spiral*, and the system collapses [► L12.8 @ 1:11].

The value investors. A student asks: why would value-based investors not step in? In this picture the Fed backstops banks, not dealers (dealer of last resort in the money market) [► L12.8 @ 2:13]. It may intervene directly in asset markets, as it did, because the value investors did not: they refused, or lost their shirts, or wait for prices of 20 cents on the dollar, which would come only with a great depression; the Fed is not there to make money for value investors, it is concerned with unemployment. If value investors do step in, the Fed need not [► L12.8 @ 2:54].

Three reasons for elasticity (the notes). [Notes 12 p.7][Notes 12 p.8]

1. **government bank:** a liquidity crisis in any market ends as a demand for bank lending, met in the end by banks dumping securities, overwhelmingly government securities; a Fed that let the government debt market go illiquid "would be a Fed soon under direct Congressional control", and by keeping that market liquid it keeps others liquid too;
2. **bankers' bank:** sharp price moves disrupt leveraged balance sheets and force liquidation (lower collateral values, less borrowing), which makes the dislocation bigger; "from time immemorial" the lead banker steps in, even out of self-interest;
3. **stabilization policy:** borrowers raise cash by cutting spending, which is others' income, spreading the problem; liquidating commodities lowers the price level and raises the real burden of debt: the debt deflation of the Great Depression.

The Fed can do it because it backstops the dealers' short cash positions by going short cash itself, and it prints the stuff: no short squeeze on the Fed, domestically; internationally it faces a reserve constraint too [Notes 12 p.7].

For discipline. Why not always be elastic? Minsky: what moves money market rates is the mismatch between cash commitments and cash flows; if rates never moved, nothing would stop agents from pushing commitments into the future forever. Discipline makes mistakes get recognised: at some point you pay or go bankrupt [► L12.8 @ 4:56]. Otherwise mistakes build up like dry tinder, and “Smokey the Bear” produces larger forest fires [► L12.8 @ 6:18]. In stress people are forced to sell at prices they would never accept, and so to recognise their mistakes and start afresh; but too much discipline “kills the patient” [► L12.8 @ 6:59].

The notes quote Minsky’s image: “the cash commitments of each unit depend on the cash commitments of every other unit. The whole web of interlocking commitments is like a bridge we spin collectively out into the unknown future toward shores not yet visible.” If the Fed always meets the demand for liquidity, balance sheets are never adjusted and the imbalance shows up elsewhere, in inflation or an exchange crisis: masking the symptoms does not resolve the problem. Too ready, it prevents adjustment; not ready enough, adjustment may be fatal: “it is a fine line” [Notes 12 p.8][Notes 12 p.9].

INTUITION: The art of central banking

Central bankers “feel this in their bones”: not every hiccup, not even every significant stress, deserves elasticity. In this crisis the Fed did not flood the market at once: it let some bad mortgages “walk” while preventing collapse; then things got worse than expected and it did more, and more [► L12.8 @ 6:38].

12.8 The Fed as dealer of last resort, 2007–2009

The Fed’s balance sheet in the crisis (lecture 3) shows the stages [► L12.9 @ 0:07]:

1. **Interest rates** (August 2007 to Bear Stearns): Fed funds from 5% to 2%, to make it more profitable for banks to take on liquidity risk and support asset prices indirectly [► L12.9 @ 0:49];
2. **composition** (after Bear Stearns): of about a trillion of Treasury bills, \$500 billion sold and the proceeds lent to brokers and dealers [► L12.9 @ 1:09];
3. **expansion** (after Lehman and AIG): about two trillion more of deposits and of lending; the interbank market broke down and “the Fed became the interbank market”: surplus institutions deposited at the Fed, deficit institutions borrowed from it [► L12.9 @ 1:30];
4. **capital market**: purchases of mortgage-backed securities [► L12.9 @ 2:11].

The crisis in balance sheets [► L12.9 @ 2:32]–[► L12.9 @ 10:15]:

1. **Before**: Citi’s SIV holds RMBS funded by ABCP, held by a money fund that issues shares to (international) investors; nothing is on Citi’s or the Fed’s balance sheet.
2. **First stage**, which looked like a normal crisis: the money fund declines to roll the ABCP; Citi replaces the funding with a loan to the SIV, funded on its own balance sheet, say with repo. The fund now holds a liability of Citibank with recourse to its whole balance sheet. Citi did this as a business: take assets off customers who no longer want them, and charge for it [► L12.9 @ 3:56].
3. **Second stage**: the fund prefers Citi’s unsecured commercial paper, based on its whole book (Lehman too issued such paper, held by the Reserve Primary Fund),

or Eurodollar deposits at foreign banks [► L12.9 @ 5:39]. The shadow banking system collapses onto the traditional banking system, backstopped by the Fed and the FDIC [► L12.9 @ 7:23].

4. **Third stage:** European banks, with little access to the Fed, cannot roll their dollar funding when money funds stop renewing Eurodollar deposits [► L12.9 @ 8:04]. The Fed lends dollars to foreign central banks through liquidity swaps (about \$600 billion at the peak), and the Treasury issues bills through a supplementary account at the Fed, so that money funds can hold what they want, Treasury bills [► L12.9 @ 8:48].

Fed + Treasury		ECB		Money fund	
+Swap (loan to ECB)	+Treasury bills	+Dollar loans to European banks	+Swap (dollars owed to Fed)	–Eurodollar deposits	+Treasury bills

The stages in the funding market (the notes). Money funds first refused to roll ABCP, replaced by repo, still secured but shorter (fall 2007); at Bear Stearns (March 2008) repo became suspect too, haircuts rose and funding shifted to unsecured Eurodollars and financial commercial paper; at Lehman these became suspect as well. Each time the trouble showed up as upward pressure on money market funding, ultimately the need to maintain par [Notes 12 p.9].

The correspondent channel. Why did the Fed support foreign banks? To stop them dumping RMBS, and because of the pressure they put on US banks through correspondent banking, visible in the Eurodollar–term Fed funds spread [Notes 12 p.10]. A British bank short of dollars first goes to its New York correspondent, “lender of first resort”, which lends at the Eurodollar rate using a daylight overdraft at the Fed and then seeks reserves in the Fed funds market; the Fed, committed to its target, supplies them by repo. The Eurodollar–OIS spread was the incentive for correspondents to let foreign banks indirectly tap the Fed; the liquidity swaps took the pressure off by channelling the funds through central banks instead [Notes 12 p.10][Notes 12 p.11]:

British bank		NY correspondent		Fed	
–Eurodollar deposit	+Interbank loan	+Daylight overdraft, then +Repo		+Repo	+Reserves
+Interbank loan					

From the best money down the hierarchy. The sequence: first the normal absorption, letting prices move to give the banking system an incentive; then lower Fed funds; then direct entry into the term funding market (liquidity swaps, lending to dealers, the term facilities), backstopping banks; finally, as value investor, the mortgage-backed securities market [► L12.9 @ 10:35]. The Fed moved down the hierarchy, putting more and more of the financial system on its own balance sheet. At no point did it want to: central bankers are conservative, they do not want credit risk or interest rate risk, which is why their money

is the best money; but in a crisis, “lend freely at a high rate against what would be good security in a normal time” [► L12.9 @ 11:57].

INTUITION: A floor under the system

Normally the Fed acts at a distance, setting the overnight rate and letting profit-seeking dealers and arbitrage transmit it [► L12.9 @ 12:39]. When imbalances built up over years are too large, the crisis is not temporary and keeps getting worse, and the Fed’s responsibility as ultimate backstop forces it to do more, until it has put a floor under the system. It can, because its liabilities are the best money: ultimately it could buy every risky security for cash, though that would be the end of the market system [► L12.9 @ 13:19]. It always hopes to do just enough; the ECB is doing that now, hoping the pressure will make politicians act: perhaps “a slow-motion collapse” of the European system, in which long periods of nothing alternate with bursts of excitement [► L12.9 @ 14:41].

Definition 12.3 (New concepts: *dealer of last resort*). A central bank acts as *dealer of last resort* when, instead of only lending to banks against collateral (lender of last resort), it stands ready to buy and sell, or to backstop the prices of, securities that private dealers will no longer absorb, putting a floor under their prices.

Exercise 12.4 (The SIV unwinds). An SIV holds RMBS of 100 funded by ABCP of 100 held by a money fund. Write the T-accounts (SIV, Citi, money fund, Fed) for each stage of section 12.8, ending with the money fund holding Treasury bills.

Key formulas and ideas

KEY FORMULAS	
Quantity vs price	controlling the best money (top of the hierarchy) ≡ pegging the short end of the term structure
Taylor rule	$r = \rho + \pi^e + \alpha(\pi^e - \pi^*) + \beta(y - y_f)$; Fisher effect $r = \rho + \pi^e$
Transmission	Fed funds (settlement risk) → term rate (liquidity risk) → bond prices (price risk) → economy
Normal crisis	dealers and banks absorb order imbalances for a price (temporary distortions)
Serious crisis	Fed backstops banks, banks backstop dealers; the Fed cannot fail on short positions in its own money
Intervene?	elasticity against liquidity spirals; discipline to make mistakes recognised
2007–2009	rates → composition → expansion (Fed = interbank market) → swaps, facilities → MBS: dealer of last resort

Part II

Advanced: foreign exchange and capital markets

Chapter 13

Chartalism, Metallism and Key Currencies

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L13.1 FT: Autonomy of Bank of Japan	2:14	A timid Bank of Japan, a stronger yen.
L13.2 Key Currencies as a Hierarchical System	8:38	Quoting conventions; the hierarchy of currencies.
L13.3 What is Money? Chartalism versus Metallism	8:37	Schumpeter's two traditions; king's money and international money.
L13.4 Chartalism as a Theory of Money	2:10	Currency as zero-interest government funding; taxes.
L13.5 Quantity Theory of Money	4:39	$MV = PT$ as a way to tame the power of the state; velocity and bank money.
L13.6 Purchasing Power Parity	3:02	The chartalist theory of the exchange rate.
L13.7 Metallism as a Theory of Money	5:07	Mint pars, gold points, forward parity.
L13.8 A Money View of International Payments, FX Dealers	11:36	A deficit country pays in dollars it does not have: matched-book and speculative FX dealers.
L13.9 Chartalism, Metallism, and the Money View Compared	4:29	Goods, assets, money: three readings of the exchange rate.
L13.10 Private and Public Money: A Hybrid System	7:30	How war finance merged the king's money and the bankers' money.
L13.11 Hybridity in FX Market-making	5:33	Private dealers and central banks along the currency hierarchy.

Files. Transcripts [materials/transcripts/L13-P*.txt](#); Mehrling's notes [Notes 13 p.1].

Motivation

The second part of the course begins with four lectures on the *exchange rate*, one of the four prices of money ▶ [L13.1 @ 0:07](#). Foreign exchange is the relative price of monies, so it is a

money market, not a capital market [► L13.2 @ 2:05]. To talk about the price of one money in terms of another we must face the question avoided so far, by focusing on credit: what kind of thing is money? [► L13.2 @ 3:07] Two traditions answer it, chartalism and metallism; each yields a theory of the exchange rate; the money view offers a third, and the modern system turns out to be a hybrid of the two kinds of money.

13.1 FT: autonomy of the Bank of Japan

CASE STUDY: FT, October 2012: "Autonomy of BoJ called into question"

The Bank of Japan was less aggressively expansionary than the market expected, and the yen rose about 0.5% against the dollar within 30 minutes of the announcement: bad for a big export economy that has long struggled with deflation [► L13.1 @ 0:29]. "Set alongside the spectacular easing of the US Federal Reserve or the European Central Bank, the more subdued Bank of Japan often struggles to compete"; and if prices keep falling, raising the real cost of borrowing, executives will not lever up [► L13.1 @ 1:31]. The article links central bank balance sheets to the price of money relative to foreign monies [► L13.1 @ 1:53].

13.2 Key currencies as a hierarchical system

Quoting conventions. A FRED chart shows the dollar price of gold, rising from 2003 to about \$1800 an ounce, and the dollar price of a euro, about \$1.30 [► L13.2 @ 0:00]. There are two ways to quote an exchange rate [► L13.2 @ 1:01]:

Definition 13.1 (New concepts: quoting conventions). In the *American convention* the exchange rate is the number of dollars per unit of foreign currency (\$1.30 per euro); in the *European convention* it is the number of units of foreign currency per dollar (0.77 euro per dollar). The theory is the same, but every formula changes into its inverse [► L13.2 @ 1:01]. In this course, so far, the European convention has been used.

The international hierarchy. So far the best money was reserves at the Fed (and currency), then Fed funds, then Eurodollars, all promises to pay reserves and trading at about the same price [► L13.2 @ 2:26]. What lies above? A stylised description of what one sees [► L13.2 @ 4:08]:

1. the **international dollar**, the world reserve currency, in which most trade is denominated and settled [► L13.2 @ 4:28];
2. the **domestic dollar**, at par with it (with difficulty in the crisis, as seen) [► L13.2 @ 4:55];
3. **key currencies**: the yen in Asia (eventually perhaps the renminbi), the pound and the euro in Europe: with the dollar the four most traded [► L13.2 @ 5:15];
4. other **majors**: the Swiss franc, the Canadian and Australian dollars: deep, liquid markets against the dollar, where very large trades do not move the price [► L13.2 @ 5:35];
5. the **minor** currencies of the hundreds of other countries, which do not trade against each other: to trade Hungarian forints against Mexican pesos one goes through dollars [► L13.2 @ 7:20].

In trading volume (as quoted in class), 51% involves only the dollar, euro, yen and sterling, and 85% of trades have the dollar on one side [► L13.2 @ 6:39]. The notes give the BIS figure as 84.9%, and quote a practitioner, “the foreign exchange market is largely the price of the dollar” (D. DeRosa, 2011); but the dollar in question is largely the international *private* dollar, bank money, not state money [Notes 13 p.2].

IN THE MARKETS: The FX market is a money market

Majors are high-volume, liquid pairs with tight spreads, all with the dollar on one leg (EUR/USD, GBP/USD, AUD/USD, USD/JPY, USD/CAD, USD/CHF); “cross” pairs have no dollar leg; minors trade against some major, and minor crosses hardly trade at all. The hierarchy carries over into derivatives. Of about \$4 trillion a day, \$1.5 trillion is spot and \$1.8 trillion FX swaps, with \$0.5 trillion outright forwards, \$0.2 trillion options and exotics, \$43 billion currency swaps: overwhelmingly short term, “fundamentally a money market, not a capital market” [Notes 13 p.3]. (Figures from the BIS Triennial Survey of 2010.)

“It really is a dollar system.” Being at the centre makes it hard for Americans to think about foreign exchange, which Europeans think about all the time [► L13.2 @ 8:00].

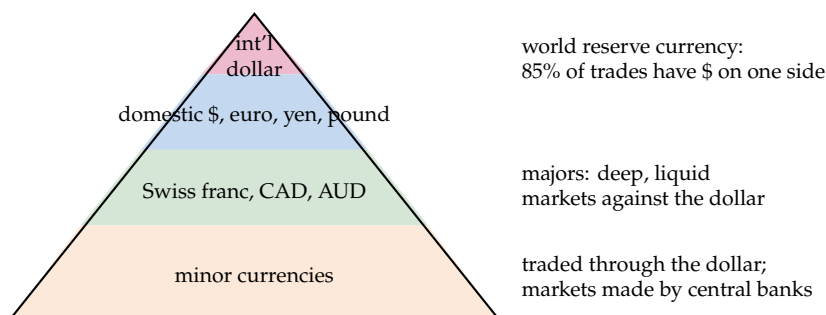


Figure 13.1: The hierarchy of currencies [► L13.2 @ 4:28](a stylised description, not a theory).

Exercise 13.1 (Conventions). The euro trades at \$1.30. Write the rate in the European convention. If the dollar depreciates by 10% against the euro, what are the new quotes in both conventions? Rewrite covered interest parity (lecture 8) in the American convention.

13.3 What is money? Chartalism versus metallism

The two words come from Joseph Schumpeter’s treatise on the history of economic analysis, which classifies monetary theories into chartalist and metallist [► L13.3 @ 0:20].

Definition 13.2 (New concepts: chartalism, metallism). *Chartalism*: a piece of money is an emblem of the power of the state that issues it; the quintessential money is fiat currency, “the king’s money”. Its classic theorist is Georg Friedrich Knapp [► L13.3 @ 0:41]. *Metallism*: the essence of money is a physical, metallic value, gold or silver [► L13.3 @ 1:43]. (Chartalism from Latin *charta*, a token or ticket, the state-stamped token; *fiat*, “let it be done”.)

Both traditions allow a large superstructure of credit, so most of the course is consistent with either; but the debate can get very heated [► L13.3 @ 2:04]. For metallists inconvertible

money is a kind of robbery, unanchored, anxiety-inducing; for chartalists the state is the ultimate authority, and who is gold to tell it what to do? It is discipline versus elasticity again: state money against the discipline of gold and of the market [► L13.3 @ 2:45].

Two moneys in history. When two traditions are at loggerheads forever, look for a higher point of view, or for a historical origin; here both apply [► L13.3 @ 3:45]. Before the modern nation-state there were two parallel monetary systems in the same country [► L13.3 @ 4:06]:

- the **king’s money**: local, domestic, used in retail trade and to pay taxes; the king’s face on it matters if he is your king;
- **international money**: gold, which you simply weigh, used in wholesale and international business.

Both circulated, with an exchange rate between them. Mehrling learned this from Fernand Braudel, *The Wheels of Commerce*, volume 2 of his trilogy on the 15th to 18th centuries, which describes the wheels of trade at the lowest and at the highest level [► L13.3 @ 5:29]. It is no surprise that there are two traditions in monetary theory: there were always two kinds of money [► L13.3 @ 6:10].

King’s money system		International money system	
	King’s money		Gold
	domestic		business
	credit super-		credit super-
	structure		structure

Each money is at the top of its own hierarchy [► L13.3 @ 6:31]. The systems touch: merchants who trade internationally and sell retail move between them, and kings fighting foreign wars need gold to pay soldiers and buy war material abroad [► L13.3 @ 7:13]. Today there are many hierarchies: each currency has its domestic credit hierarchy, and the international dollar a whole international one; the world funding markets are dollar markets [► L13.3 @ 8:15].

13.4 Chartalism: the state, the quantity theory, PPP

Currency as cheap funding. If money is a creature of the state, think of the central bank as a government bank: the Treasury issues bills paying, say, 5%; the central bank buys them and issues currency paying 0%. Looking through the central bank, currency is zero-interest funding for the government, which is why states like to monopolise its issue [► L13.4 @ 0:00].

Treasury		Central bank	
(authority to tax)	Treasury bills (5%)	Treasury bills (5%)	Currency (0%)

What supports the value of such a liability? The government’s authority, especially its taxing authority: it insists that taxes be paid in its currency, and if you do not pay, it throws you in jail [► L13.4 @ 1:26].

HISTORY: Knapp and Schumpeter

Georg Friedrich Knapp’s *Staatliche Theorie des Geldes* (1905; English *The State Theory of Money*, 1924) argued that money is a creature of law, defined by what the state accepts in payment. Schumpeter’s posthumous *History of Economic Analysis* (1954) contrasts “theoretical metallism” with “theoretical cartalism”.

Knapp’s opening sentence is “Money is a creature of law”. In his words, payments are made with “chartal” pieces, whose validity in units of value is fixed by law and is “independent of the contents of the pieces”. The English edition translates only the first three of the four chapters; the fourth, on the currency history of England, France, Germany and Austria, is omitted. Schumpeter defines *theoretical metallism* as the view that it is “logically essential” for money to consist of, or be covered by, a commodity, whose exchange value is then the source of money’s purchasing power. He holds this view untenable. He distinguishes it from *practical metallism*, a policy preference for a metallic standard. The two need not go together: one can reject the theory and still prefer gold out of distrust of politicians.

References: G. F. Knapp, *The State Theory of Money*, Macmillan, 1924, ch. 1 ([materials/readings/L13-Knapp_StateTheoryMoney.pdf](#)); J. A. Schumpeter, *History of Economic Analysis*, Oxford UP, 1954, part II ch. 6, §2 “Fundamentals” ([materials/readings/L13-schumphea.pdf](#)).

The quantity theory. Economists turn this into a proper theory with the quantity theory of money [► L13.5 @ 0:00]:

KEY FORMULAS

Quantity theory [► L13.5 @ 0:21]: $MV = PT$ (money $\times$ velocity = price level $\times$ transactions). With V fixed by institutions and T by the real economy, M determines P [► L13.5 @ 0:43].

Purchasing power parity [► L13.6 @ 0:23]: $EP = P^*$, with E the exchange rate (foreign currency per dollar), P and P^* the domestic and foreign price levels: the exchange rate is the relative price of goods in the two countries.

The quantity theory tells the government: you can say what money is, but not what it is worth; print too much and it is inflated away [► L13.5 @ 1:25]. People get heated about it because of a deep anxiety about the power of the state and about green pieces of paper that could become worthless; the equation is “the way economists regulate their anxiety” [► L13.5 @ 2:06]. But the government is not the only issuer: if quantity causes prices, bank money should worry us too; it seems to worry people less because it looks like credit [► L13.5 @ 2:48]. And since the hierarchy expands and contracts, velocity is not constant: in an expansion private money grows with transactions even if government money does not, which in the equation appears as rising V [► L13.5 @ 3:28].

PPP. If money is fiat and its quantity sets the price level, the natural theory of the exchange rate is *purchasing power parity*: the quantity theory works behind the scenes in each country, and the relative price of monies is just the relative price of goods [► L13.6 @ 0:02]. Two prices of money, the exchange rate and the price level, connected, and no interest rate anywhere [► L13.6 @ 1:27]. It suits economists who look through the veil of money to a barter economy of goods for goods [► L13.6 @ 1:50]. But it does not hold empirically except perhaps in the long run: price levels move slowly, exchange rates “bop all around” [► L13.6 @ 2:30].

In growth rates (the notes’ form) [Notes 13 p.5]:

$$SP = P^* \implies \frac{\Delta S}{S} + \frac{\Delta P}{P} = \frac{\Delta P^*}{P^*},$$

with S pounds per dollar: exchange rate changes mirror inflation differentials, which in turn reflect money growth differentials. The notes add the doubts: in the long run, causation may run from prices to money (most money being credit) and from exchange rates to domestic prices; and the quantity theory leaves out that the public holds less real money when it expects inflation, and that private promises to pay the king’s money serve as close substitutes [Notes 13 p.5]. Mehrling’s summary: PPP sees the exchange rate as a relative price of goods (the economics view), CIP as a relative price of assets (the finance view); neither as a relative price of *money*, the money view [Notes 13 p.5][Notes 13 p.6].

Exercise 13.2 (PPP). A basket costs 100 dollars in the US and 12,000 yen in Japan, and the yen trades at 80 per dollar. What does PPP predict? US inflation is 2% and Japanese inflation 0% for a year, while the yen goes from 80 to 78 per dollar. Has PPP moved closer or further?

13.5 Metallism: mint pars and gold points

If money is gold, the exchange rate depends only on how much gold each money contains [► L13.7 @ 0:03]. If the dollar is x ounces of gold and the pound y ounces, the exchange rate is the ratio of the *mint pars*, x/y [► L13.7 @ 0:45]. Arbitrage enforces it: if the rate deviates,

melt the heavier coin into the lighter one [► L13.7 @ 1:29]. But that means shipping gold, which is costly, so the rate can deviate within a band [► L13.7 @ 2:10].

Definition 13.3 (New concepts: *mint par*, *gold points*). The *mint par* is the quantity of gold a currency unit represents at the mint; the ratio of two mint pars is the *mint parity* exchange rate. The *gold points* are the exchange rates above and below mint parity at which it pays to ship gold, given the costs of shipping and insurance: outside them arbitrage moves gold, inside them it does not [► L13.7 @ 2:54].

Mint parity with gold points is a finance view: the forward parity condition of lecture 8 (in class here, with the spot rate E related to the forward rate by the ratio of the two interest rates) holds under the gold standard too, because the gold points bound the spot rate between a floor and a cap [► L13.7 @ 3:20]. A floor and a cap suggest Treynor's outside spread, with dealers trading inside it and determining the exchange rate: that is where the money view enters [► L13.7 @ 4:24].

Exercise 13.3 (Gold points). Mint parity is \$4.8665 per pound; shipping gold from New York to London costs 0.5% of its value. Between which exchange rates is gold not shipped? In which direction does gold flow if the pound trades at \$4.92 in New York?

13.6 A money view of international payments

A *deficit* country has bought from a *surplus* country and owes \$10, in dollars, the international reserve currency (neither is the US) [► L13.8 @ 0:03]. If it has \$10, no problem; but suppose it does not [► L13.8 @ 1:25]. The point of a payment system is that willing buyers and sellers are not stopped by not having the means of payment at hand; domestically the answer was credit; internationally it is harder [► L13.8 @ 1:47].

The FX dealer. A foreign exchange dealer in the middle buys $10E$ of the deficit country's currency and sells it \$10 spot, which settles the debt: the surplus country has been paid, the deficit country thinks it is off the hook [► L13.8 @ 2:49]. The dealer made the payment, with dollars "created from nothing", promises to pay dollars: a short position in dollars against a long position in foreign exchange, just as a bank makes the payments its customers cannot [► L13.8 @ 5:00].

Hedging in the term market. The dealer is exposed to the exchange rate, and to the surplus country demanding currency for its dollar claim [► L13.8 @ 6:33]. It hedges by trading term deposits with another dealer: a \$10 term deposit asset and a $10E$ foreign-currency term deposit liability; whatever the exchange rate does, it gains on one side what it loses on the other [► L13.8 @ 6:54]. This uses forward interest parity: in effect it hedges its spot position at the forward rate. It is now a *matched-book* FX dealer [► L13.8 @ 7:55]. Someone must take the other side of those term deposits: a *speculative* dealer, exposed to exchange risk and compensated for it [► L13.8 @ 8:58].

HISTORY: The classical gold standard

Under the classical gold standard (roughly 1880–1914) the dollar was defined as 23.22 grains of pure gold and the pound as 113.0016 grains, giving a mint parity of about \$4.8665 per pound; the gold points between New York and London were typically within about 0.5–1% of parity.
References: L. Officer, *Between the Dollar–Sterling Gold Points*, Cambridge UP, 1996.

Deficit country		Matched-book FX dealer		Speculative FX dealer	
–FX $10E$	–Due to surplus \$10	FX $10E$ (spot)	\$10 spot	FX $10E$ term	\$10 term
		\$10 term	FX $10E$ term		

Surplus country	
–Due from \$10	
+\$10 spot	

Foreign exchange trading is about \$4 trillion a day, one of the largest markets [► L13.8 @ 8:37]. Every asset is someone's liability, but the functions differ: the matched-book dealer bears another kind of risk (later), and can be called a forward-interest-parity dealer; the speculative dealer, recognisable from Treynor's model, bears exchange risk and must expect a profit, which is what moves the forward rate away from the expected spot rate: an uncovered-interest-parity dealer [► L13.8 @ 9:39].

Definition 13.4 (New concepts: spot and forward FX, matched-book and speculative FX dealers). A *spot* foreign exchange transaction is settled now (in practice in two business days); a *forward* one at a future date at a rate fixed today. A *matched-book* FX dealer balances its currency positions (spot against term), bearing no exchange-rate risk; a *speculative* FX dealer holds an open position in a currency and bears the exchange risk.

Exercise 13.4 (A payment through FX dealers). Redo the deficit-country example with numbers: $E = 1.25$ units per dollar, six-month dollar rate 1%, foreign rate 3%. Write all the T-accounts, compute the forward rate the matched-book dealer locks in, and state what the speculative dealer gains or loses if the spot rate in six months is 1.30.

13.7 Three readings of the exchange rate

Tradition	The exchange rate is	Theory
Chartalism	relative price of goods	purchasing power parity; trade (current) account
Metallism (finance)	relative price of assets	interest parity; capital account
Money view	price of money in terms of money	dealers' balance sheets

Figure 13.2: What kind of a thing is an exchange rate? [► L13.9 @ 0:04]

The chartalist road leads to the exchange rate as a relative price of tradeable goods (PPP); the metallist road to a relative price of financial assets, interest rates. Some theories stress the trade or current account, others the capital account [► L13.9 @ 0:04]. The money view sees the exchange rate as the price of money in terms of money, on the dealers' balance sheets [► L13.9 @ 1:06]. Economics abstracts from liquidity to price commodities, finance to price assets; the course is about liquidity, so it cannot. PPP and interest parity are not wrong, but they abstract from liquidity, so they cannot be fully right [► L13.9 @ 1:27].

INTUITION: Pushing exchange risk into the future

In the example nobody lends: not the surplus country, not the FX dealer; they swap currencies. The dealer takes risk in the spot market and hedges it in the forward

market, where someone else takes it: exchange rate risk located today is pushed into the future, exactly as the money market pushes a payment one cannot make today to tomorrow or 90 days. The speculative dealer’s position is a *carry trade*, and it is compensated because the payment system needs it [► L13.9 @ 2:08].

13.8 Private and public money: a hybrid system

How did the two parallel systems of the 15th century become today’s? Mostly through war [► L13.10 @ 0:05]. Start with two separate systems [► L13.10 @ 1:34]:

Government bank		Private bankers’ bank	
Treasury bills	King’s money (currency)	Gold Private credit	Deposits (promises to pay gold)

The government bank needs no reserves: the king can always print more pieces of paper with his head on them, a cheap way to fund himself [► L13.10 @ 2:36]. Then comes war, and, as in the Civil War (section 3.4), the Treasury asks the private bank for a loan, which it makes by creating deposits, and then withdraws them in gold [► L13.10 @ 3:17]:

Government		Private bankers’ bank	
+Gold	+Loan from bankers’ bank	+Loan to government –Gold	(+Deposits, then –Deposits withdrawn in gold)

“Here’s hybridity”: the government bank has all the gold, and the bankers’ bank starts using domestic currency as its reserve [► L13.10 @ 4:35]. It happened in the Bank of England, in the US, in most countries: the state used the private and international bank for war finance, twisting its arm [► L13.10 @ 5:17].

Proposition 13.5 (Why hybridity lasted). *After the war both sides kept the arrangement, “a marriage made in heaven”: the state blessed private money, giving bank deposits legal protection as money (political cover for private money issuers); the private market accepted Treasury bills as promises to pay money, giving the state access to capital markets. A government that funds itself by printing money cannot fund much; one that sells Treasury bills to the private sector can fund a huge war. Modern monetary systems are always hybrid [► L13.10 @ 5:37].*

The notes complete the sequence with a postwar step: the government refinances the loan with currency and bills, keeps the gold as international reserve and requires banks to use its currency as domestic reserve [Notes 13 p.8]:

Central bank (postwar)		Private bank	
Gold Treasury bills	Currency	+Currency +Treasury bills	Deposits

In the prewar parallel system domestic credit was tied to the inferior king's money, and repressed; international credit to the superior money, inaccessible to most people. In the US, the Civil War created national banking but no central bank, and the New York Clearinghouse, a private bankers' bank, kept private money on top. The Fed (1913) switched the domestic hierarchy, putting public money on top, useful for the war finance that followed, but it left private money (gold) on top internationally: "a contradiction that would take another hundred years to work out, and it is not completely worked out even today" [Notes 13 p.8][Notes 13 p.9]. On the historical parallel systems the notes cite F. Braudel, *The Wheels of Commerce*, and M.-T. Boyer-Xambeau, G. Deleplace and L. Gillard, *Private Money and Public Currencies: The 16th Century Challenge* (1994) [Notes 13 p.3].

13.9 Hybridity in FX market-making

Are the FX dealers private banks or central banks? "It depends" on the place in the currency hierarchy [► L13.11 @ 0:03]:

- in the dollar and the majors, a huge number of private dealers, highly liquid, with very thin margins;
- in the Hungarian forint, the central bank: there is no deep and liquid private market; the weaker the currency, the more it is all central bank [► L13.11 @ 1:05].

So the theory of exchange rates must be about the relationship between private, profit-seeking FX dealers and the public, welfare-seeking central bank [► L13.11 @ 1:26]. (Not that central banks are the good guys: in many countries the central bank is the agent of a rapacious dictator, and the private dealers who refuse to do business with it are the good guys [► L13.11 @ 2:12].) For some currencies the exchange rate is mostly politics; for others the central bank is a dealer of last resort in the wings [► L13.11 @ 2:33]. Under the gold standard the outside spread was the gold points; today there are none, but central banks, by controlling short-term interest rates, create an outside spread for the FX market, as the next lectures show [► L13.11 @ 3:14].

IN THE MARKETS: Why so much FX trading?

Foreign exchange is more "mind-blowing" than domestic money markets: all of a money market in one country, all of it in another, and a set of dealers in between [► L13.11 @ 3:34]. Economists and finance professors simplify it to a relative price of goods or assets, and wonder why \$4 trillion a day is traded. But behind a \$10 trade there are many trades "below the line" that separate the risk into categories and put each in the right place; today it is electronic and cheap, so there is a lot of positioning without burning real resources [► L13.11 @ 4:16].

Key formulas and ideas

KEY FORMULAS

Conventions	American: \$ per foreign unit; European: foreign units per \$ (inverse formulas)
Currency hierarchy	international \$ > domestic \$, euro, yen, pound > other majors > minors (via \$)
Chartalism	money = state token; currency = zero-interest funding; value from taxes; $MV = PT$; $PPP\ EP = P^*$
Metallism	money = gold; exchange rate = ratio of mint pars within gold points; interest parity
Money view	exchange rate = price of money in money, set by FX dealers (matched-book and speculative)
Hybridity	war finance merged king's money and bankers' money; modern systems always hybrid
FX dealers	private in major currencies, central banks in weak ones; central banks set the outside spread via interest rates

Chapter 14

Money and the State: International

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L14.1 FT: Costs of Japan's Monetary Policy	2:49	Zombie firms and a strong yen.
L14.2 Reading: Robert Mundell	10:27	Mundell's 20th century in the language of the hierarchy: central banks avoiding the discipline of gold.
L14.3 Confrontation	11:48	Act one: the Fed and the gold standard, 1931.
L14.4 Contradiction	14:08	Act two: national management versus fixed rates; Keynes's bancor plan versus the IMF.
L14.5 The Dollar System	7:04	Kindleberger: the US as a bank to the world; the end of Bretton Woods.
L14.6 Flexible exchange	8:28	Act three: floating rates, inflation, inflation targeting, speculative FX markets.
L14.7 Global Financial Crisis	17:58	European banks funding US mortgages in dollars; central bank swap lines; return of home bias.

Files. Transcripts [materials/transcripts/L14-P*.txt](#); Mehrling's notes [[Notes 14 p.1](#)]. **Reading.** R. A. Mundell, "A Reconsideration of the Twentieth Century", *American Economic Review* 90(3), 2000, 327–340.

Motivation

Lecture 13 placed the dollar at the top of a hierarchy of currencies. This lecture tells the history of the international monetary system in the 20th century, following Mundell's Nobel lecture, and translates it into the language of the course: discipline versus elasticity at the top of the hierarchy [► L14.2 @ 3:12]. It ends with the global financial crisis, where the same tension reappears as dollar funding of non-US banks and central bank swap lines [► L14.7 @ 0:03].

14.1 FT: the costs of Japan's monetary policy

CASE STUDY: FT, 2012: John Plender, "Japan counts zombie cost of ultra-loose monetary moves"

About 20 years earlier Japan had a financial crisis; its responses included quantitative easing, the first experiment with this kind of unconventional policy, and fiscal expansion. The economy never really got going again and has had deflation ever since [► L14.1 @ 0:07]. Plender worries about the structural damage of long periods of low rates: uncompetitive firms do not go out of business, because rolling over their loans costs them almost nothing, and by competing they keep the productive firms from making profits [► L14.1 @ 0:50]. The discipline avoided for so long now comes in through the back door: a yen at 79–80 per dollar makes Japanese exports uncompetitive [► L14.1 @ 1:53].

Definition 14.1 (New concepts: **zombie firm**). A firm that is not viable but survives because cheap credit lets it roll over its debts (the living dead of horror films: neither alive nor buried) [► L14.1 @ 1:11].

14.2 Mundell's reconsideration of the twentieth century

Still no global currency. Mundell wrote the article in 1999, published in 2000 when he received the Nobel prize; it ends optimistically: after a bad century we now have stable units, the euro, the dollar and the yen, and perhaps we are on our way to an international monetary system [► L14.2 @ 0:03]. Twelve years later the yen and the euro still fluctuate a lot against the dollar and against each other, there is no global currency, the euro (born in 1999) is in crisis, and after the financial crisis there has been some retreat from financial globalization towards "national capitalism" [► L14.2 @ 0:44].

The good old days of gold. For Mundell the international gold standard is the "good old days" because the payment system disciplined central banks and finance ministries, and, since everyone adjusted to the same discipline, the whole world was integrated and stable [► L14.2 @ 3:32]. There were problems, notably a tendency to long-run deflation when the value of gold changed, but in retrospect the system looks good. It broke down in World War I [► L14.2 @ 4:55].

Central banks avoiding discipline. The 20th-century story is of countries, central banks and finance ministers who found the discipline too harsh and tried to get out from under it [► L14.2 @ 5:36]. In the hierarchy gold > currency > deposits, the discipline sits in the link between gold and currency: as the economy grows, currency expands but gold does not [► L14.2 @ 6:20]. Mundell keeps in mind two clean ways out:

1. **revalue gold:** a higher price of gold is like more gold; but all countries must do it together, and since gold holdings differ it is a wealth transfer, which international politics could not agree on [► L14.2 @ 6:46];
2. **create artificial gold:** increase the quantity instead of the value, as attempted with the SDR at the IMF, but too little too late [► L14.2 @ 7:47].

Neither was found, so there was "a lot of bumbling around". The main culprit is the Fed: a new central bank (1913), with no history, constantly making errors; but the US is dominant,

COURSE NOTE

On the FRED chart shown, the euro is quoted as dollars per euro (American convention) and the yen as yen per dollar (European convention), so a rise of the first line and a fall of the second both mean appreciation against the dollar. With no movement between yen and euro the two lines would move in opposite directions [► L14.2 @ 1:04].

everyone pegs to the dollar, and what the Fed does the world does. Instead of a gold standard, a dollar standard with an immature Fed [► L14.2 @ 8:27]. Then floating, until in 1999 Europe decided it no longer wanted floating rates among its currencies [► L14.2 @ 9:32].

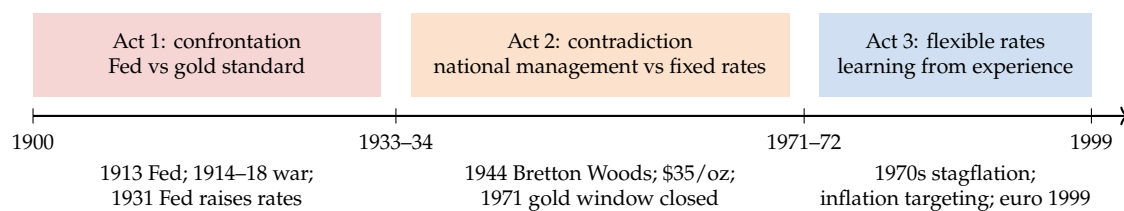


Figure 14.1: Mundell's 20th century in three acts, with the dates used in the lecture [► L14.3 @ 0:03][► L14.4 @ 0:45][► L14.6 @ 1:12].

READING: R. A. Mundell, *A Reconsideration of the Twentieth Century* (2000)

Mundell's Nobel lecture (delivered December 1999). Read with the hierarchy of money in mind: every episode is a central bank, at the top of a national hierarchy, facing an international standard above it [► L14.2 @ 9:52].

Some facts from the article (materials/readings/L14-Mundell.pdf), checked against the scan:

- Cassel had warned in 1925 and 1928 that restoring gold would bring a scarcity of gold and falling prices (p. 329).
- Wholesale prices fell from their 1929 peak to September 1931 by between 22% (Germany) and 40.5% (Japan); the United States figure is 29.5% (p. 329).
- In October 1931 the Fed raised the rediscount rate in two steps from 1½ to 3½%; US wholesale prices fell 35% between 1929 and 1933 (p. 330).
- Bank failures rose from about 500 a year in the 1920s to 1,350 in 1930, 2,293 in 1931 and 1,453 in 1932 (p. 330).
- In 1934 the dollar's gold value was cut by 40.94%, raising the official gold price by 69.33% to \$35 an ounce (p. 331).

Both quotations used below (p. 330 on "pumping liquidity", p. 331 on the counterfactual) are verbatim in the article.

Exercise 14.1 (Two ways to more gold). World gold reserves are fixed while nominal incomes must double. Show the two adjustments Mundell has in mind (price of gold, price level) and explain who gains and who loses from each when gold holdings are unevenly distributed.

14.3 Act one: confrontation (1900–1933)

Mundell calls the first act the confrontation of the Fed and the gold standard; it really begins only after World War I [► L14.3 @ 0:03]. The Fed is created in 1913; almost immediately there is war, and every country goes off gold except the US. That was no feat: gold was flowing to the US to pay for war material [► L14.3 @ 1:05]. So gold did not discipline the US during the war, and there was some wartime inflation; the discipline came back after

the war, with deflation in 1920–21, though not enough to return to pre-war price levels [► L14.3 @ 1:47]. One by one the mark, the pound and the franc returned to gold, not all at the pre-war parities (the French devalued) [► L14.3 @ 2:28].

Gold becomes scarce. Mundell's point: each currency that re-links to gold adds to the demand for gold reserves, so gold becomes increasingly scarce, undervalued [► L14.3 @ 3:13]. A first chance to revalue gold was missed. There are two ways to have "more gold": raise the price of gold, or keep it and lower the price of everything else, which is deflation, bad for the economies and resisted by national central banks [► L14.3 @ 3:54].

The 1931 mistake. The key mistake, for Mundell, was the Fed's in 1931: to defend the gold standard it raised interest rates in the middle of a deflation, from 1.5% to 3.5%, which in deflation are very high real rates [► L14.3 @ 4:40]. Gold flowed to the US, forcing the other countries off gold, eventually the US too, and the depression became worldwide. Defending the parity of the dollar destroyed the gold standard, and at home it sacrificed the US banking system, which collapsed; Roosevelt was elected and the banking legislation of the 1930s followed [► L14.3 @ 5:42].

INTUITION: Two pars, one choice

There are two par relationships: the mint par (dollar to gold) and the deposit par (deposits to currency). Defending the first, the Fed sacrificed the second: people preferred currency to deposits, banks were run and closed. Alternatively it could have stopped paying gold and supported the banks as lender of last resort. "You can't do both": lender of last resort to your banks, or defender of the dollar's value in world markets [► L14.3 @ 8:48].

The notes quote Mundell (p. 330): "Instead of pumping liquidity into the system, it chose to defend the gold standard", compounding the secular deflationary tendency (every return to gold raised the monetary demand for gold, so lowered gold prices of everything else; France, which devalued, got a fresh start) with explicitly deflationary policy. Falling farm prices meant loan defaults, tight policy added liquidity pressure, and bank failures added a further deflationary impulse (on which the notes cite Irving Fisher, "The Debt-Deflation Theory of Great Depressions", 1933) [Notes 14 p.1][Notes 14 p.2].

A monetary theory of history. On p. 331 Mundell writes that had the price of gold been raised in the late 1920s, or had the major central banks pursued price stability instead of adhering to the gold standard, there would have been no Great Depression, no Nazi revolution and no World War II [► L14.3 @ 6:44]. Mehrling finds this overboard but dramatic: the point is that instability of the world polity and of the world monetary system are tied together. As a student of politics he would say that both are symptoms of the same failure: we could not get together to do the right thing [► L14.3 @ 7:45].

Why did the Fed err? A further story: Benjamin Strong, who ran the New York Fed and knew all the world's central bankers, had died, and so had his adviser Allyn Young; the young Fed had no deep bench, and in the crisis the knee-jerk conservative reaction was to defend gold. Some say that had Strong lived the depression might not have happened; "we'll never know" [► L14.3 @ 9:53].

HISTORY: Mundell

Robert A. Mundell (1932–2021), Canadian economist, received the 1999 Nobel memorial prize for his analysis of monetary and fiscal policy under different exchange rate regimes and of optimum currency areas; he is often called an intellectual father of the euro. *References:* NobelPrize.org, "Robert A. Mundell, Facts", 1999; R. A. Mundell, *AER* 90(3), 2000.

HISTORY: The Fed in October 1931

After Britain left gold (21 September 1931), the Federal Reserve Bank of New York raised its discount rate from 1.5% to 2.5% on 9 October and to 3.5% on 16 October 1931 to stop the gold outflow, while US banks were failing in large numbers. *References:* M. Friedman and A. J. Schwartz, *A Monetary History of the United States, 1867–1960*, Princeton UP, 1963, ch. 7; B. Eichengreen, *Golden Fetters*, Oxford UP, 1992, ch. 9.

14.4 Act two: contradiction (1934–1971)

The second act is the confrontation of national macroeconomic management with a fixed exchange rate system, set up at Bretton Woods but starting already in the late 1930s [► L14.4 @ 0:03]. Every finance minister and central bank wants policies good for its own country, and at the same time commits to fixed rates with other countries: this cannot be squared unless someone absorbs all the contradiction, which would have been the US, and it did not [► L14.4 @ 1:06].

COURSE NOTE

In class the date of Strong's death is given as "1929 I think". Benjamin Strong died on 16 October 1928, in New York (he had long suffered from tuberculosis). Allyn Young died in London on 7 March 1929, of pneumonia during an influenza epidemic, as said. The biography mentioned is L. V. Chandler, *Benjamin Strong, Central Banker*, Brookings, 1958 [► L14.3 @ 10:14].

Keynes's bancor plan. Before Bretton Woods (a resort in New Hampshire where the post-war plan was made), Keynes proposed a world central bank and a true global currency, bancor [► L14.4 @ 1:48]. Deficit countries would borrow bancor, created out of thin air by the international bank and credited to the surplus countries; so the quantity of international money expands with the imbalances. It is the classic banking solution, a lender of last resort at the international level, just as inside the US a deficit agent ultimately borrows from a bank that borrows from the central bank [► L14.4 @ 3:19].

Definition 14.2 (New concepts: *bancor*, *asymmetry of adjustment*). *Bancor*: Keynes's proposed international bank money (from French *banque* and *or*, "bank gold"). *Asymmetry of adjustment*: under gold (and under the IMF) the deficit country must adjust, because it has to acquire the international reserve, while the surplus country need not: the survival constraint at the level of nations [► L14.4 @ 6:07].

Deficit country (UK)		International bank	
	+Loan from int'l bank (bancor)	+Loan to deficit country	+Bancor deposit of surplus country
Surplus country (US)			
	+Bancor deposit		

Figure 14.2: The bancor plan: imbalances expand the international bank's balance sheet [► L14.4 @ 3:19].

Why the US said no. The US asked: which of these agents is us? Clearly the surplus country, facing the UK, all of Europe, everyone else. The plan looked like a mechanism forcing the US to lend to other countries, through an intermediary it did not control; true after the war, perhaps not forever [► L14.4 @ 4:44]. Worse, to remove the asymmetry Keynes proposed charging interest on the surplus country's deposit as well as on the loan, to push the US to spend its bancor on European goods: a forced loan at a negative rate [► L14.4 @ 6:07]. It was an attempt to eliminate the survival constraint for the deficit countries, which needed US goods to rebuild [► L14.4 @ 7:29].

What we got: the IMF. The US plan. In stylised form (not the IMF's actual, more complicated accounting, seen in problem set 1), member countries deposit gold and

their own currencies and receive deposit-like claims, now called special drawing rights [► L14.4 @ 7:51]. The Fund ends up with all kinds of currencies, the important one being the dollar [► L14.4 @ 10:44]. Deficit countries pay surplus countries in SDRs: the quantity of international money does not change, it is only transferred, unlike bancor [► L14.4 @ 11:25]. When you hit zero you borrow from the IMF, at a positive interest rate and with policy conditionality: exactly the asymmetry Keynes wanted to avoid [► L14.4 @ 12:06]. Keynes died shortly after; perhaps he accepted the IMF thinking it could be fixed later (“the famous economist death theory of international finance”, Mehrling’s “pet theory”) [► L14.4 @ 12:49].

Definition 14.3 (New concepts: special drawing right (SDR)). An international reserve asset created by the IMF and held by member countries, which can use it to make payments to one another; “paper gold” of a fixed quantity [► L14.4 @ 9:12]. (A *drawing right* is a right to draw, to obtain other currencies, from the Fund.)

Discipline versus elasticity. The IMF is a discipline system, bancor an elasticity system; maybe bancor would have been too elastic and the IMF too disciplined. For a while the shortage of ultimate international reserves was mitigated by the expansion of dollars further down the hierarchy [► L14.4 @ 13:10].

Exercise 14.2 (Bancor accounting). The UK runs a deficit of 100 bancor with the US for three years. Write the balance sheets of the UK, the international bank and the US at the end, with 1% interest charged on both debit and credit balances. Do the same under the stylised IMF, with initial SDR holdings of 150 each.

14.5 The dollar system

Under Bretton Woods the hierarchy was: gold plus SDRs (a fixed quantity of paper gold) at the top; the dollar, pegged to gold at \$35 an ounce; the other currencies (mark, pound, franc) pegged to the dollar [► L14.5 @ 0:03]. The extra reserves that allowed post-war growth came neither from new gold, nor SDRs, nor revaluation, but from more dollars [► L14.5 @ 0:44].

The US as a bank (Kindleberger). Mundell emphasizes trade deficits as the way dollars went abroad. Charles Kindleberger, Mundell’s professor at MIT, stressed a different mechanism, which needs no trade deficit at all: the US lends long term to the rest of the world (financing European reconstruction by buying foreign bonds), while the rest of the world, which wants to grow, accumulates short-term dollar balances as monetary reserves [► L14.5 @ 1:06].

United States (as a bank)		Rest of the world	
Gold	+Dollar	+Dollar reserves	+Bonds issued
+Foreign bonds (long term)	deposits of the rest of the world (short term)		

Figure 14.3: Kindleberger’s view: the US as a bank to the world, borrowing short and lending long, with no net capital or trade flow [► L14.5 @ 2:42].

HISTORY: Bretton Woods and Keynes

The United Nations Monetary and Financial Conference met at Bretton Woods, New Hampshire, 1–22 July 1944; the US plan (Harry Dexter White) prevailed over Keynes’s Clearing Union. Keynes died on 21 April 1946, a month after the inaugural meeting of the IMF and World Bank governors at Savannah. References: B. Steil, *The Battle of Bretton Woods*, Princeton UP, 2013; R. Skidelsky, *John Maynard Keynes*, vol. 3, Macmillan, 2000.

COURSE NOTE

Keynes’s 1942–43 Clearing Union proposals did charge a fee on large *credit* balances as well as on debit balances (at rates rising with the size of the balance, of the order of 1–2% a year); the 3% in class is an illustrative number. Reference: “Proposals for an International Clearing Union”, Cmd. 6437, HMSO, April 1943.

COURSE NOTE

The lecture’s IMF balance sheet is a deliberate simplification. Historically, members paid their quota partly in gold and partly in their own currency and could draw on the Fund up to limits; SDRs were created only later, by the First Amendment to the Articles (agreed 1967, in force 1969), with a first allocation of about SDR 9.3 billion in 1970–72. They are allocated to members, not issued against deposits of gold. Reference: IMF, “Special Drawing Rights (SDR)” factsheet; M. G. de Vries, *The International Monetary Fund 1966–1971*, IMF, 1976.

The US is a bank holding the world's international reserves, as long as the world accepts dollars and does not demand gold: a sort of bancor plan, but on the US balance sheet; and the US earns the spread, because dollar reserves pay little or no interest [► L14.5 @ 3:03]. The IMF refused to expand reserves at the top of the hierarchy; the US expanded them one step down [► L14.5 @ 4:04].

The end of Bretton Woods. Dollars expanded, gold did not, so gold and SDRs became undervalued and stress built up [► L14.5 @ 4:24]. Again: revalue gold or create more SDRs. In 1967 there was an increase in SDRs, about \$10 billion, not enough. Mundell tells, for the first time in Mehrling's reading, that Arthur Burns sounded out European central bankers about revaluing gold and suggested to Nixon to do it right after the 1968 election, which did not happen [► L14.5 @ 4:44]. By 1971 there were too many dollars for the gold, there was a run on the dollar, and instead of paying out the gold Nixon closed the gold window [► L14.5 @ 5:48]. Kindleberger argued all his life that it would have been fine had foreigners accepted dollars as the reserve and not insisted on the "fiction" of gold [► L14.5 @ 6:15].

Exercise 14.3 (The US as a bank). Starting from the T-accounts of figure 14.3, show what happens when foreign central banks convert \$10 of their dollar balances into gold. Why is the system a bank-run problem, and what are the options of the US?

HISTORY: Gold at \$35 and the closing of the gold window

The Gold Reserve Act of January 1934 set the dollar price of gold at \$35 an ounce (from \$20.67). On 15 August 1971 President Nixon suspended the convertibility of dollars held by foreign official institutions into gold. The view of the US as "bank of the world" was set out by E. Despres, C. P. Kindleberger and W. S. Salant, "The Dollar and World Liquidity: A Minority View", *The Economist*, 5 February 1966.
References: Federal Reserve History, "Nixon Ends Convertibility of U.S. Dollars to Gold" and "Gold Reserve Act of 1934".

14.6 Act three: flexible exchange rates (1972–1999)

The contradiction destroyed fixed rates; central bankers and finance ministers were now free to run their economies their own way, and the 1970s were not a good time: stagflation, worldwide [► L14.6 @ 0:03]. Mundell calls 1972–1999 the experiment with flexible exchange rates and learning from experience [► L14.6 @ 1:12].

Definition 14.4 (New concepts: stagflation). Inflation and unemployment (stagnation) at the same time; the word joins *stagnation* and *inflation* [► L14.6 @ 0:23].

What was learned. Flexible rates are very volatile, and without international discipline national policy tends to inflation: the greatest peacetime inflation, because no central bank, least of all the Fed, faced any discipline [► L14.6 @ 1:40]. The stagnation came from uncertainty: with exchange rates moving 100% back and forth you cannot tell where your comparative advantage lies, and you invest in the wrong things [► L14.6 @ 2:44]. Two consequences:

1. **central bank self-discipline:** gold was too much discipline, but some is needed; central banks impose price discipline on themselves, inflation targeting [► L14.6 @ 3:27]. Friedman and the advocates of floating thought this would also stabilize exchange rates (PPP, lecture 13); it did not: inflation differentials were small, yet exchange rates moved all over the place [► L14.6 @ 4:09]. Mundell wants central bank discipline *and* fixed rates, central banks cooperating [► L14.6 @ 4:50];
2. **private speculative markets:** firms selling abroad must hedge their exchange risk, so forward FX markets develop, very sophisticated; but the same markets allow speculation, such as the carry trade, which pushes currencies apart and then, when reversed, snaps them back [► L14.6 @ 4:50]. The dealers take risk and want to be paid for it, which distorts prices (next lecture) [► L14.6 @ 6:34].

Definition 14.5 (New concepts: *carry trade*). Borrowing in a low-interest currency and lending in a high-interest one (borrow yen, lend dollars) to pick up the interest differential, the *carry*, bearing the exchange risk [► L14.6 @ 5:53].

Mundell's hope. In 1999 Mundell foresaw three key currencies, each anchoring a large area (Asian currencies pegged to the yen, the dollar spreading to Latin America, the euro taking over Europe), linked to each other into a unified world system. That is not what happened [► L14.6 @ 7:38].

The notes add [Notes 14 p.4][Notes 14 p.5]: a system of national currencies with floating rates is typical in wartime, when commerce is restricted, “quite anomalous in peace time”. Friedman persuaded himself that market prices beat administered ones and that domestic price stability would bring exchange stability through PPP; Mundell disagrees, valuing fixed rates as par clearing knits together the states of the US. Inflation targeting did bring price stability, but not exchange stability: volatility of dollar, euro and yen continued. A lasting legacy of the period of disorder is the currency futures market for hedging exchange risk (lecture 18). And Mundell's three-currency picture is not symmetric: the world funding market is a dollar market, as the crisis showed. In the 1960s, too, Mundell suggests that more SDRs, substituted for gold in reserves, would have lowered the demand for gold and taken the pressure off; one response to the recent crisis was indeed a new issue of SDRs by the IMF [Notes 14 p.4].

14.7 The global financial crisis

Dollar funding outside the US. In the build-up to the crisis, big European banks (UBS, for example) bought US residential mortgage-backed securities and funded them short term in the international dollar funding markets: shadow banking, though on balance sheet [► L14.7 @ 0:23]. The money came, say, from money market mutual funds whose shares, a deposit substitute, were held by non-US residents, including central banks and countries that prefer to keep dollars offshore, out of reach of the US legal system [► L14.7 @ 1:52]. Dollar borrowing and lending outside the US, by institutions without access to the US banking system or to the Fed [► L14.7 @ 2:54].

The run. After Lehman and AIG, money funds that had lost on Lehman commercial paper wanted only Treasury bills and asked for their money back. The European bank went to the ECB, which could lend euros but had no dollars: “I’m not the Fed” [► L14.7 @ 3:15].

The swap-line operation. The engineered solution [► L14.7 @ 5:20]:

1. the Fed and the ECB swap deposits: the Fed creates dollars, the ECB creates euros (a liquidity swap);
2. the ECB lends the dollars to the European bank, which rolls its funding and need not sell its mortgage securities;
3. the dollars end up as a deposit of the US Treasury at the Fed, which issues additional T-bills to the money funds: the funds wanted T-bills, could not hold deposits at the Fed, and the Treasury acted as the intermediary.

COURSE NOTE

The IMF made a general SDR allocation of about SDR 161 billion (some \$250 billion) in August 2009, plus a special allocation in September 2009. The notes also mention, as an alternative proposal of the 1940s, Benjamin Graham's commodity-reserve currency (*World Commodities and World Currency*, 1944) [Notes 14 p.2]. Reference: IMF, “Special Drawing Right (SDR)” factsheet.

European bank		ECB		Fed	
RMBS	–MMF funding	+Dollar deposit at Fed	+Euro deposit of Fed	+Euro deposit at ECB	+Dollar deposit of ECB
	+Dollar loan from ECB	+Dollar loan to bank	–Dollar deposit (lent)		(then Treasury deposit)
Treasury		Money market fund			
+Deposit at Fed	+T-bills	–Loan to European bank	+T-bills	Shares	

Figure 14.4: The central bank liquidity swap in the crisis, as drawn in class (changes only) [► L14.7 @ 4:37].

Definition 14.6 (New concepts: central bank liquidity swap line). An agreement by which two central banks exchange their own currencies, to be swapped back later at the same exchange rate; the receiving central bank lends the foreign currency on to its own banks [► L14.7 @ 6:27].

Temporary cooperation, then repatriation. This is the central bank cooperation Mundell asks for, but only a holding action, to keep the shadow banks from dumping their securities and collapsing the market; eventually the Fed bought mortgage securities itself [► L14.7 @ 7:54]. The permanent solution was repatriation: bringing the securities back home, undoing the internationalization of finance; a return of *home bias* in investment, because lender-of-last-resort support is national [► L14.7 @ 8:56]. In the notes: the money fund still funds the shadow bank’s MBS, but through a chain, money fund to Treasury to Fed to ECB to shadow bank; afterwards the construction was dismantled by taking the MBS back to the US, where dollar funding was easier to cobble together: “it is no accident that there is now over a trillion MBS on the balance sheet of the Fed” [Notes 14 p.5][Notes 14 p.6]. The dollar is a national currency, the liability of the Fed; there is no global currency [► L14.7 @ 9:58].

Definition 14.7 (New concepts: home bias). The tendency of investors to hold domestic assets more than international diversification would suggest [► L14.7 @ 9:38].

The reversal of globalization worries Mehrling: we saw it at the end of the 1920s, and it is a headwind against which national policies do not work; countries try fiscal or monetary stimulus and only become more indebted and vulnerable [► L14.7 @ 10:40]. Meanwhile the system is held together by international central bank cooperation and by speculative trading [► L14.7 @ 11:42].

Is the Fed taking risk? A student asks; Congress asked Bernanke the same about the roughly \$600 billion of swaps [► L14.7 @ 12:03]. Two answers:

COURSE NOTE

The Treasury deposit is the Supplementary Financing Program, announced in September 2008: the Treasury issued extra bills and left the proceeds on deposit at the Fed. The Fed’s swap lines peaked at about \$580 billion in December 2008 (“600 billion” in class).
References: Federal Reserve, H.4.1 releases 2008; Federal Reserve History, “Central Bank Liquidity Swaps”.

1. the swap is reversed at the same amounts: the Fed gets back its dollars, the ECB its euros. Each books in its own currency, so whatever the exchange rate does nobody books a loss; commercially, with both parties keeping their books in the same currency, one side would show a loss [► L14.7 @ 13:07];
2. it is international lender-of-last-resort action: with private funding gone, the central banks had no real choice; they were doing their job (“try to explain this to Congress”) [► L14.7 @ 14:49].

Standing swap lines. The big central banks (Japan, ECB, Switzerland, Bank of England, Fed) have agreed swap lines in all currencies, of unlimited amount: they are aware of the history and of the stakes, that monetary instability means economic and then social instability [► L14.7 @ 15:50]. Maybe they overestimate their influence: without political will, a lender of last resort cannot prevent social instability; wars happen when people cannot figure out what to do next, and in war the central bank becomes the financing agent of the government. “I’d rather not go down that route” [► L14.7 @ 17:12].

Exercise 14.4 (Swap gains and losses). The Fed swaps \$100 for euros at \$1.25 per euro; three months later the euro trades at \$1.40. Compute the gain or loss of each central bank if both kept their books in dollars, and then if each kept its books in its own currency.

COURSE NOTE

The coordinated action of 30 November 2011 involved six central banks (the five named plus the Bank of Canada). The swap lines became standing, unlimited arrangements among these six only on 31 October 2013, after this lecture. *Reference:* Federal Reserve press releases of 30 November 2011 and 31 October 2013.

Key formulas and ideas

KEY FORMULAS

Mundell’s thesis	integration and stability of the world economy depend on the discipline of an international standard
Escapes from gold	revalue gold (price) or create paper gold (quantity); neither coordinated in time
1931	defending the mint par sacrificed the deposit par: LLR to banks or defend the currency, not both
Bancor vs IMF	elasticity (reserves grow with imbalances) vs discipline (fixed SDRs, deficit country adjusts)
Dollar system	US as a bank: short-term dollar liabilities, long-term foreign assets, gold reserve
Floating	inflation, volatility; answers: inflation targeting, private hedging/speculative FX markets
Crisis	offshore dollar funding; swap lines; repatriation and home bias; no global currency

Chapter 15

Banks and Global Liquidity

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L15.1 FT: European money market funds	2:37	EUR 100 billion moved across currency areas.
L15.2 International transactions	11:42	Bagehot's world, international: bills discounted in London, two FX dealers, the Bank of England.
L15.3 Dealer model for foreign exchange	10:05	Gold points as the outside spread of FX dealers.
L15.4 Central banking, defense of domestic exchange	8:43	The deficit country's central bank as dealer: pay out, sell assets, borrow reserves.
L15.5 Bank of England, defense against external drain	12:08	Bank rate as outside spread; external drain; raising Bank rate; suspension.
L15.6 Toward a theory of exchange	9:37	Without gold both outside spreads vanish; Keynes's <i>Tract</i> (1923) and covered interest parity.

Files. Transcripts `materials/transcripts/L15-P*.txt`; Mehrling's notes [[Notes 15 p.1](#)] (headed "18." in the PDF, from an earlier numbering of the course).

Motivation

Today's world is one of flexible, volatile exchange rates (next lecture). This lecture goes back a hundred years, to the gold standard, to introduce concepts needed next time [[► L15.1 @ 2:10](#)]. It is "the world that Bagehot knew" (lecture [9](#)) again, now with its international piece: firms in different countries trading by bills of exchange discounted in London, foreign exchange dealers, and the Bank of England on top [[► L15.2 @ 0:03](#)]. The tool is Treynor's dealer model (lecture [10](#)), applied twice: to exchange rates and to interest rates.

15.1 FT: European money market funds

CASE STUDY: FT, 2012: “Europe’s money market funds look further afield”

Citing a Fitch Ratings report: rating downgrades, worries over the eurozone and lower bank demand for short-term funding are pushing European money market funds towards Asia and the Middle East. Over two years they moved almost a fifth of their assets, about EUR 100 billion, from the UK, the US and the eurozone periphery to Germany, France, the Netherlands, the Nordics, Asia and the Middle East [► L15.1 @ 0:48]. Moving EUR 100 billion across currency areas has exchange-rate consequences: part of the volatility of flexible rates comes from such flows, and from people trying to avoid currencies that are going to depreciate [► L15.1 @ 1:29].

15.2 International transactions in Bagehot’s world

The bill discounted in London. A surplus firm (or country) sells goods to a deficit firm and receives a bill of exchange, a promise to pay in 90 days; neither is in England [► L15.2 @ 0:44]. The bill is not money, so the seller takes it to a private bank in the City of London, the centre of the world money market in 1873, which discounts it, accepting the bill and paying out sterling notes [► L15.2 @ 1:50]. (Notes rather than deposits, because the intuition is sharper [► L15.2 @ 2:54].) The surplus firm is paid before the deficit firm has paid. In 90 days the deficit firm pays the bill with notes, which flow back to the bank [► L15.2 @ 3:14]. The banks manage the outflow and inflow of notes, holding bills that mature every day: classic 19th-century bill banking [► L15.2 @ 3:56].

The foreign exchange dimension. Sterling notes are high-powered money for international banking, but they cannot pay the surplus firm’s workers; and the deficit firm sells its goods for its own currency and must turn it into sterling to pay the bill [► L15.2 @ 4:38]. So there are two foreign exchange dealers (or banks: in this course a bank is a special kind of dealer and a dealer a special kind of bank) [► L15.2 @ 5:39]:

- the **surplus dealer** buys sterling notes from the surplus firm and pays out domestic currency [► L15.2 @ 6:01];
- the **deficit dealer**, 90 days later, sells sterling notes to the deficit firm for its domestic currency [► L15.2 @ 7:06].

The two domestic currencies may differ; what the dealers have in common is that both exchange against the international currency, sterling [► L15.2 @ 8:10].

The Bank of England. In 1873 sterling notes are liabilities of the Issue Department of the Bank of England, which holds gold and keeps the notes at par with it [► L15.2 @ 8:31]. Both dealers use Bank of England notes as international reserves, and may cash them for gold or vice versa: the Bank makes a two-way market between notes and gold [► L15.2 @ 8:52]. Three kinds of dealers, in a hierarchy: gold at the top, Bank of England notes promising gold, then the other currencies kept at par with those notes [► L15.2 @ 9:58].

ASIDE: Coming attractions

The same day’s FT reported China lifting foreign access to its markets, linked to the week’s reading by McCauley (BIS) on the internationalization of the renminbi, discussed next lecture [► L15.1 @ 0:07].

Surplus firm		City bank	
–Goods		+Bill	–Notes
+Bill, then		(90 days	(paid out)
–Bill +Notes		later: –Bill)	(90 days
			later:
			+Notes)
Deficit firm			
	+Goods	+Bill (due in	
		90 days)	
Surplus dealer		Deficit dealer	
+Sterling	+Domestic	–Sterling	–Domestic
notes	currency	notes	currency
Bank of England (Issue Dept.)			
	±Gold	±Notes	

Figure 15.1: International trade finance in 1873, as drawn in class [► L15.2 @ 1:08].

This is how it worked, and how it works today with the dollar: most world trade is invoiced in dollars, though much of it does not involve the US; back then it was sterling [► L15.2 @ 10:59].

Exercise 15.1 (A bill from Buenos Aires). An Argentine exporter sells wheat to a German importer against a 90-day bill for £1000, discounted in London at 4% a year. Write all the T-accounts on the day of discount and 90 days later, including the two FX dealers. How much sterling does the exporter receive?

15.3 The dealer model for foreign exchange

Recall from lecture 13: if the dollar is x ounces of gold and the pound y ounces, the mint parity exchange rate, quoted in pounds per dollar (the convention of this course), is x/y [► L15.3 @ 0:44]. The actual rate can deviate, because moving gold is costly:

KEY FORMULAS

Gold points [► L15.3 @ 1:46]: with δ set by the cost of shipping gold,

$$\frac{x}{y} - \delta \leq E \leq \frac{x}{y} + \delta \quad (E \text{ in pounds per dollar}).$$

Inside the band there is no arbitrage to snap the rate back. This is the Treynor model hinted at two lectures ago: the gold points are an *outside spread*, and the FX dealers do business inside it [► L15.3 @ 2:26].

The deficit country's dealer. Let the deficit country be the US (deliberately: the world currency is the pound, not the dollar; “God” did not plan the dollar to be the reserve

currency, certainly not in 1873) [► L15.3 @ 0:24]. Its dealers buy dollars and sell sterling: long the domestic currency, short the international one, losing reserves, so long liquidity risk [► L15.3 @ 3:50]. They do it only at a lower price of the dollar: with pounds per dollar, a lower number is a depreciated dollar [► L15.3 @ 5:13]. They take exchange risk only if paid in expectation, hoping the dollar snaps back so they can sell high [► L15.3 @ 6:14].

Why the dealer dares. What if the dollar keeps falling? It cannot fall far: dollars can be taken to the central bank and converted into gold, which can be shipped to England for sterling notes. The gold point is a floor, the outside spread; the dealers' quotes are the inside spread [► L15.3 @ 6:55]. The quotes seen in the market are the dealers'; everyone knows the mint par, and a large volume of business pushes the price to the gold points, where gold flows keep it within bounds as long as central banks can supply the gold [► L15.3 @ 8:37].

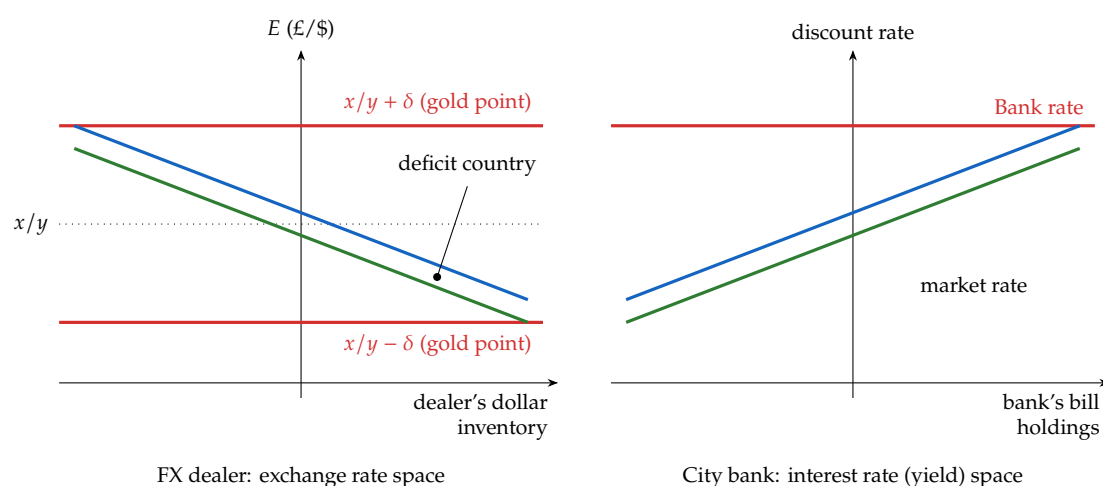


Figure 15.2: Two outside spreads of the gold standard: the gold points for FX dealers [► L15.3 @ 3:28] and Bank rate for the discounting banks [► L15.5 @ 1:26]. In yield space the curves slope up. Schematic.

Exercise 15.2 (Gold points). The dollar is 0.04838 ounces of gold and the pound 0.2354 ounces; shipping gold costs 0.6% of its value. Compute the mint parity in pounds per dollar and the gold points. Draw the FX dealer's diagram when the US is a deficit country.

15.4 The central bank defends the domestic exchange

Suppose the dollar is pushed to the gold point: private dealers, who have as much risk as they want, bring domestic currency to the central bank and ask for international currency, gold or sterling notes [► L15.4 @ 0:04]. Domestic currency is the central bank's own liability, so it is simply cancelled: the central bank's balance sheet shrinks, it loses reserves and high-powered money contracts [► L15.4 @ 1:10]. Textbooks say a currency is defended by contractionary monetary policy; here that follows literally from acting as a dealer, not from an intention [► L15.4 @ 1:51].

Its gold and notes are limited, so the central bank must think ahead. Three logical

COURSE NOTE

The US central bank in this example is hypothetical: in 1873 the US had no central bank (the Fed dates from 1913), as Mehrling remarks [► L15.3 @ 7:36].

options, as in the flow-of-funds accounting [► L15.4 @ 2:33]:

1. **pay out reserves** (dishoarding);
2. **sell an asset** for reserves, for example Treasury bills, which pushes bill prices down and interest rates up [► L15.4 @ 3:17];
3. **borrow reserves** from another central bank, possibly the Bank of England itself [► L15.4 @ 3:58].

Deficit country's central bank	
–Gold or sterling notes	–Domestic currency
(1) –Reserves	
(2) –T-bills, +Reserves	(3) +Borrowed reserves
(3) +Reserves	

Figure 15.3: Defending the domestic exchange at the gold point [► L15.4 @ 0:50].

The central bank is a backstop not so much because it holds large reserves, but because it can replenish them: it has assets (its special relationship with the sovereign) and relationships with other central banks, which a private dealer lacks [► L15.4 @ 4:42].

INTUITION: What “raising rates to defend the currency” really means

The notes read the three channels in balance sheets [Notes 15 p.3][Notes 15 p.4]. Upward pressure on the currency is no problem: issue currency to buy reserves. Downward pressure means buying the currency, the central bank's own liability, so its balance sheet contracts: paying out reserves works only while they last; selling Treasury bills pushes their price down and interest rates up, perhaps a lot if speculators fear devaluation. “We often hear loose talk about central banks defending their currency by raising interest rates”: what really happens is that the central bank offers interest-bearing securities in exchange for currency, hoping the interest is enough to keep holders from demanding international reserves. Borrowing reserves from other central banks (or the IMF, “somewhat like an international central bank”) leaves the balance sheet's size unchanged: part of the monetary base in private hands shifts to other central banks. In every case, contraction of the money supply supports the currency because the central bank buys its own liability. And the deficit entity that must adjust need not be below the Bank of England: it can be the Bank itself [Notes 15 p.4].

INTUITION: What counts as reserves depends on the layer

A student asks what the difference is between domestic currency and reserves. For the US central bank in a sterling world, reserves are gold or Bank of England notes; for a bank inside the US, reserves would be domestic currency. What counts as reserves and what as credit depends on where you are in the hierarchy. Foreign exchange is everything learned so far, one layer higher, with one difference: inside a country the relations are par relations, between countries the rates move between the gold points [► L15.4 @ 5:44].

The abstraction that removes liquidity. Economists often treat the gold points as so narrow that the rate is just the mint par ratio, $\delta = 0$, a horizontal line. That abstracts from the essence of monetary economics: with no price movement no dealer would make a market, and without dealers there is no liquidity. The course puts back the market mechanics that give liquidity [► L15.4 @ 7:28].

Exercise 15.3 (Three ways to defend). A central bank holds 50 of gold, 100 of T-bills, and has 150 of currency outstanding. Dealers bring 60 of currency for gold. Show the balance sheet after each of the three defenses, and say which ones raise domestic interest rates.

15.5 The Bank of England and the external drain

City banks as dealers in yield space. The City banks discount bills and wait 90 days for the money: they lend long and borrow short, take liquidity risk, and so are dealers too, in interest-rate space; in yield space the bid and offer curves slope up, and the outside spread is *Bank rate* [► L15.5 @ 0:03]. A bank losing cash faster than it comes in raises its discount rate, so fewer bills come in and the inflow exceeds the outflow [► L15.5 @ 1:47]. If all banks do it, the market rate of interest rises, eventually to Bank rate, at which the Bank of England, as lender of last resort, rediscounts bills for notes [► L15.5 @ 2:28]. The banks dare to discount because the backstop is there [► L15.5 @ 4:17].

Internal drain. The Banking Department pays out the notes it holds as its reserve, or it expands its balance sheet on both sides, crediting deposits: as long as banks accept deposits as notes it economizes on its note issue and keeps rates from rising above Bank rate (lecture 9) [► L15.5 @ 4:39].

External drain. New today: what if the dollar is bid *up* to the other gold point? That outside spread is the Bank of England's market: surplus dealers with too many notes ask for gold, and the Issue Department pays, –notes, –gold, defending the mint par of sterling [► L15.5 @ 5:40]. Its gold is limited, so like any central bank it seeks more reserves. Its equivalent of selling Treasury bills is to **raise Bank rate**, at a time when everyone is bringing bills for discount: the world rate of interest rises, borrowers repay and do not renew, and gold flows in [► L15.5 @ 7:26]. But a high Bank rate can bankrupt people and cause further problems [► L15.5 @ 8:50].

Definition 15.1 (New concepts: external drain, suspension of specie payments). An *internal drain* is a demand for the central bank's notes from the domestic system; an *external drain* is a demand for gold to pay abroad [► L15.5 @ 5:40]. *Suspending specie payments*: the central bank stops converting its notes into gold ("specie", coined metal, from Latin *in specie*, "in kind") while still issuing notes and deposits [► L15.5 @ 8:50].

The nuclear option. The third option is suspension: "I'm no longer going to pay notes in gold", the system becomes a pure credit system for the time being [► L15.5 @ 9:11]. Other central banks that suspend drop out of the international system; when the Bank of England does it, it *is* the international system: people keep using sterling as reserves, now unattached to gold, and perhaps decades later it returns to gold [► L15.5 @ 9:51].

KEY FORMULAS

The gold standard as discipline [► L15.5 @ 10:53]. Domestically the central bank is at the apex and can always print: no internal run can trouble it. Internationally an external drain forces it to defend its gold by raising Bank rate, at the cost of the domestic economy, because foreigners did their banking business in London; the only alternative is suspension.

Exercise 15.4 (Internal and external drain). Explain why the Bank of England can stop an internal drain by lending freely at Bank rate but must raise Bank rate to stop an external drain. What does suspension change for a country that is not at the top of the hierarchy?

15.6 Toward a theory of exchange

After suspension. A student: if payments are suspended, does the model not break down, and should sterling not be strongly devalued? Two answers [► L15.6 @ 0:03]:

1. yes, the story of mint par ratios and outside spreads goes away; it is replaced by something else, next lecture [► L15.6 @ 0:24];
2. “depreciate against what?” Certainly against gold; but sterling may still be the best currency in the world, so not necessarily against other currencies: the whole currency system may depreciate against gold, but maybe not sterling against the rest [► L15.6 @ 1:05].

Two outside spreads gone. The gold standard had two central bank outside spreads: the gold points, giving FX dealers room to work, and Bank rate, giving discounting banks room to work [► L15.6 @ 2:08]. Moving away from gold removes both. Bank rate was a discount rate for 90-day bills; today the Fed sets an overnight rate, the Fed funds rate, and does not set 90-day bill prices except in a crisis (when it walked up the yield curve) [► L15.6 @ 2:49]. Next time the same two diagrams return, but the outside spreads will not be central banks [► L15.6 @ 3:51]. Why explain the gold standard with the dealer model? Because, unlike the usual textbook explanation, it extends to fiat money: going off gold does not mean “la la land” with no economics [► L15.6 @ 4:12].

Keynes, 1923. In *A Tract on Monetary Reform*, his first real money book, written after the war, while countries were trying to get back to gold, Keynes calls the gold standard “already a barbarous relic” (p. 172): central bankers now care about the stability of prices and employment and will not sacrifice them to the “outworn dogma” of a gold price of £3 17s 10½d per ounce [► L15.6 @ 4:53]. War finance had accustomed the world to paper money not backed by gold [► L15.6 @ 6:14]. His hope: each central bank stabilizes prices at home, exchange rates float, and through purchasing power parity they are stable [► L15.6 @ 6:55]. It did not work: the world went back to gold, which broke apart, and there was a world depression; this is the dream Mundell talked about [► L15.6 @ 7:17].

But Keynes also started to discuss forward exchange, and wrote an equation equivalent to covered interest parity (lecture 8), the key relationship for translating the two diagrams to a world without gold [► L15.6 @ 7:58]:

COURSE NOTE

In class the comparison is said to be between “lecture 10 and lecture 18”; it is between lecture 9 of these notes (“The World that Bagehot Knew”, which Mehrling calls lecture 10) and this lecture 15 [► L15.5 @ 10:32].

HISTORY: Suspensions of the Bank Charter Act

Under the Bank Charter Act of 1844 the Issue Department’s notes beyond a fixed fiduciary issue had to be backed by gold. In the crises of 1847, 1857 and 1866 the government authorised the Bank to exceed that limit (a “suspension” of the Act, though convertibility was kept). Convertibility itself was suspended from 1797 to 1821 (the Restriction period) and, in effect, at the outbreak of the First World War; Britain returned to gold in 1925 and left it in 1931.

References: J. Clapham, *The Bank of England: A History*, vol. 2, Cambridge UP, 1944; Bank of England, “A brief history of banknotes”.

KEY FORMULAS

Covered interest parity ▶ L15.6 @ 8:39]: with S the spot and F the forward rate (pounds per dollar), R and R^* the dollar and sterling term rates of interest,

$$\frac{F}{S} = \frac{1 + R^*}{1 + R}.$$

Key formulas and ideas

KEY FORMULAS

Trade finance	bill discounted in London for sterling notes; FX dealers in each country exchange local currency for sterling
Hierarchy	gold > Bank of England notes > other currencies (kept within the gold points)
Gold points	$x/y - \delta \leq E \leq x/y + \delta$: outside spread for FX dealers
Defense	deficit central bank: pay out reserves, sell assets, borrow reserves; balance sheet shrinks
Bank rate	outside spread for discounting banks; internal drain absorbed by expanding the balance sheet
External drain	raise Bank rate (world rate of interest) or suspend specie payments
Without gold	both outside spreads vanish; CIP $F/S = (1 + R^*)/(1 + R)$ is the bridge to the modern system

COURSE NOTE

The auto-captions give the gold price as “£3 17 shillings 102”; the British mint price of standard gold was £3 17s 10½d per troy ounce, as Keynes himself writes it, “£3:17:10½ per ounce”.
Checked against the scan (materials/readings/L15-Keynes_Tract_on_Monetary_Reform.pdf):

- “In truth, the gold standard is already a barbarous relic” is on p. 172.
- “Outworn dogma” follows on p. 173, with the remark that a regulated non-metallic standard “has slipped in unnoticed. *It exists.*”

The equation is written on the board in class and is garbled in the captions; it is reproduced here in the notation of lecture 8.
Reference: J. M. Keynes, *A Tract on Monetary Reform*, Macmillan, 1923, ch. 3 (forward exchange) and ch. 4 (“barbarous relic”).

Chapter 16

Foreign Exchange

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L16.1 FT: High frequency trading	4:41	EBS curbs “predatory” high-frequency traders.
L16.2 Uncovered interest parity	2:22	Three FX facts: CIP, UIP, EH; UIP and EH fail for the same reason.
L16.3 FX dealers under the gold standard, redux	5:03	The two gold-standard diagrams and their exogenous bounds.
L16.4 Private FX dealing system	10:41	The \$10 payment: matched-book and speculative dealers.
L16.5 Economics of the dealer function, speculative dealer	5:57	Forward rate below the expected spot rate: expected profit.
L16.6 Economics of the dealer function, matched-book dealer	6:15	F/S rises with the book; through CIP, foreign term rates are bid up.
L16.7 Digression: Why do UIP and EH fail?	9:45	Dealers must be paid: liquidity premia, time-varying with the pattern of payments.
L16.8 Central bank as FX dealer of last resort	16:09	Central bank swaps; the deficit central bank as speculative dealer; when intervention helps.
L16.9 Reading: McCauley on internationalization of renminbi	10:29	Three tracks of China’s financial system; FX as hybrid of state and market.

Files. Transcripts `materials/transcripts/L16-P*.txt`; Mehrling’s notes [Notes 16 p.1]. **Reading.** R. N. McCauley, “Renminbi internationalisation and China’s financial development”, *BIS Quarterly Review*, December 2011; Baba and Packer, BIS WP 267 (2008); Mehrling, “Essential hybridity” (2013) (both in `materials/readings/`).

Motivation

Under the gold standard the central bank “put its arms around the system”: the gold points and Bank rate were explicit outside spreads (lecture 15). The modern system has no such explicit bounds; the challenge is to understand how it works [► L16.3 @ 4:12]. Mehrling calls this “in a way the hardest lecture of the whole semester”: more wheels are turning than in any other [► L16.9 @ 8:46]. The payoff: the failures of uncovered interest parity and of the expectations hypothesis turn out to be the same fact [► L16.2 @ 1:25].

16.1 FT: high frequency trading

CASE STUDY: FT, 2012: “EBS hails success of HFT curbs”

High frequency trading (“cyborg trading”): algorithms, often written by computer scientists and engineers, run on servers co-located next to the exchange [► L16.1 @ 0:06]. HFT firms make money, but it is not clear they make markets: they take money from slower regular dealers, who may protect themselves by widening the bid–ask spread, making markets less liquid [► L16.1 @ 1:09]. EBS, one of the largest currency trading platforms, took measures against “predatory practices”: the fifth decimal of a quote must end in 0 or 5, and limits on orders cancelled before execution. Volumes fell, but the chief executive says the real traders, and liquidity, stayed [► L16.1 @ 2:11].

Why does it matter especially in FX? Dealers hedge a spot position by a forward position; if a human takes a second between the two trades, a computer can take both first and sell the hedge back at a profit. Banks that discover this back off [► L16.1 @ 1:30]. Platforms protect their other customers because otherwise those customers go elsewhere: an arms race, the “war of the cyborgs versus the humans”, in a market of \$4 trillion a day [► L16.1 @ 3:32].

Definition 16.1 (New concepts: high frequency trading). Automated trading by algorithms that place and cancel orders in fractions of a second, exploiting speed (*high frequency*: many trades per unit of time) [► L16.1 @ 0:06].

16.2 Three FX facts

KEY FORMULAS

With S the spot and F the forward exchange rate (FX per dollar), R and R^* the dollar and foreign term rates [► L16.2 @ 0:03]:

1. **covered interest parity** (CIP), a strict arbitrage condition that more or less defines the forward rate, since a synthetic forward can be built from deposits: $\frac{F}{S} = \frac{1 + R^*}{1 + R}$;
2. **uncovered interest parity** (UIP): the forward rate equals the expected spot rate, $F = \mathbb{E}[S_T]$; empirically false;
3. **expectations hypothesis** (EH): the long rate equals the expected rollover of short rates, $(1 + R_{0,T}) = (1 + R_{0,N}) \mathbb{E}[1 + R_{N,T}]$; empirically false.

The claim of the lecture: taking a money view of foreign exchange explains why UIP and EH fail, and shows that they fail for exactly the same reason: “these are not different facts, they’re actually the same fact” [► L16.2 @ 1:25].

16.3 The gold standard, redux

The two diagrams of last lecture [▶ L16.3 @ 0:24](#):

- exchange rate (pounds per dollar) against the dealers' liquidity risk: as the dollar depreciates the private dealers' quotes move down until the gold point $x/y - \delta$, where the US central bank must defend the dollar; the rate fluctuates between $x/y - \delta$ and $x/y + \delta$ even though there is a mint par [▶ L16.3 @ 1:47](#);
- the rate of interest at the centre (the UK), an upward-sloping curve: stress moves the market rate until Bank rate, where the Bank of England backstops the system, subject to convertibility; under an external drain it must raise Bank rate or suspend [▶ L16.3 @ 2:27](#).

These were exogenous bounds set by the central bank. The modern system does not have them, at least not explicitly [▶ L16.3 @ 4:12](#).

16.4 The private FX dealing system

Back to the balance sheets of lecture 13. A surplus country has a \$10 "due from" item on a deficit country; neither is the US, but trade is invoiced and settled in dollars, the world reserve currency, and the deficit country has no dollars [▶ L16.4 @ 0:24](#).

Step 1: the payment. An FX dealer buys 10S of foreign exchange (FX, the deficit country's domestic currency) from the deficit country and pays with a new liability, \$10 spot, which the deficit country transfers to the surplus country [▶ L16.4 @ 1:29](#). Both countries are done; the dealer is not happy: short dollars, long FX, exposed to the spot rate [▶ L16.4 @ 3:18](#).

Step 2: the hedge. The spot dealer hedges in the forward market, which is like a matched pair of term deposits, short and long, in the two currencies: it adds a 10S FX term deposit liability (at R^*) and a \$10 term deposit asset (at R). This is CIP at work [▶ L16.4 @ 4:19](#). If spot and forward rates move together it loses on one side what it gains on the other; some basis risk remains [▶ L16.4 @ 6:02](#). One payment required two trades: this is why FX trading is \$4 trillion a day [▶ L16.4 @ 6:22](#).

Step 3: the counterparty. Whoever is on the other side of the hedge, neither country, takes the opposite position: a \$10 term deposit liability at R and a 10S FX term deposit asset at R^* [▶ L16.4 @ 6:44](#).

Deficit country		Surplus country		Matched-book dealer	
–FX 10S	–Due to \$10	–Due from \$10		FX 10S spot	\$10 spot
		+\$10 spot		\$10 term (R)	FX 10S term (R^*)
Speculative dealer					
		FX 10S term (R^*)	\$10 term (R)		

Figure 16.1: The private FX dealing system: one \$10 payment, two dealers [► L16.4 @ 1:50].

Definition 16.2 (New concepts: *matched-book* and *speculative FX dealer* (recalled)).

The *matched-book dealer* hedges a long FX spot position with a short FX forward position: not exactly matched, but hedged. The *speculative dealer* is naked, speculating in the forward market, not in spot: it does not need to come up with spot dollars on demand. Its position is equivalent to borrowing at the dollar rate and lending at the foreign rate: a carry trade [► L16.4 @ 7:53].

This is what finance people are paid for: making mutually beneficial trade possible across currencies, and shaving off their share. One dealer may take all of these on its books; or no dealer will, and a central bank does it [► L16.4 @ 9:35].

Exercise 16.1 (The two dealers with numbers). The spot rate is $S = 1.25$ FX per dollar, the six-month rates are $R = 1\%$ and $R^* = 3\%$ (not annualized). Write the T-accounts of the four agents for the \$10 payment, compute F from CIP, and the speculative dealer's profit in dollars if the spot rate in six months is 1.25, 1.27 or 1.30.

16.5 The economics of the dealer function

The speculative dealer is exposed to price risk, the subject of Treynor's model [► L16.5 @ 0:03]. On the vertical axis put the *value* of FX (the inverse of the course's usual quote, so that going down means depreciation of the deficit country's currency) [► L16.5 @ 0:24]. What matters is the forward rate locked in today relative to the expected spot rate at maturity: being long FX, the dealer wants to buy it cheap. If the forward price of FX is below the expected spot, the difference is expected profit [► L16.5 @ 1:46]. Prices fluctuate and the dealer may be wrong, but right more than half the time it makes money. The speculative dealer is paid to take the exchange risk off the matched-book dealer's hands [► L16.5 @ 4:37].

The matched-book dealer has hedged price risk but holds liquidity risk: it borrows short and lends long in the world reserve currency, with dollar liabilities that can be withdrawn [► L16.6 @ 0:03]. It expands its book only if it buys spot more cheaply than it sells forward: in F/S space it has an upward-sloping bid–ask curve, higher the larger the book; in a crisis the spread also widens a lot [► L16.6 @ 1:08].

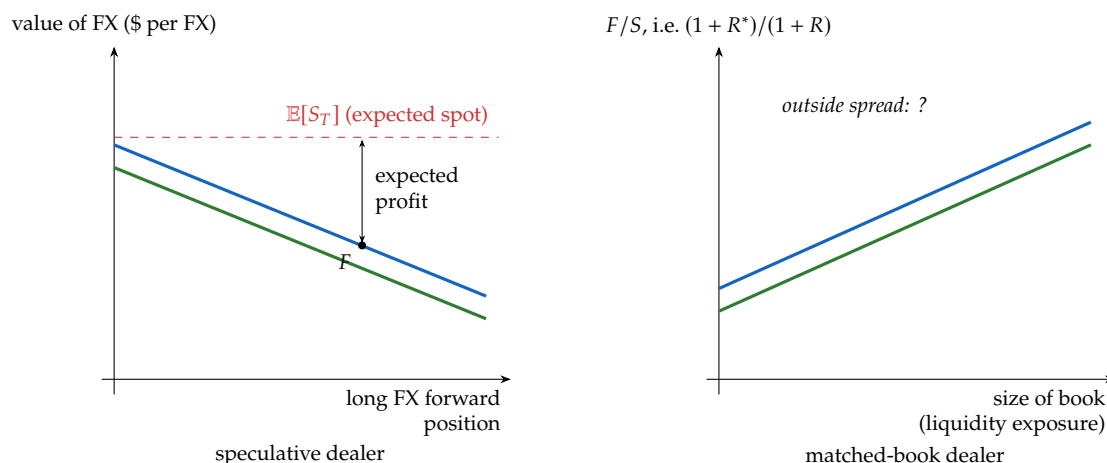


Figure 16.2: The two private FX dealers [► L16.5 @ 3:35][► L16.6 @ 1:30]. Unlike the gold standard, no outside spread is drawn: who sets it is the question [► L16.6 @ 2:58]. Schematic.

INTUITION: Expected spot as the new mint par

In the notes, the expected spot rate plays, under floating, the role the mint par ratio played under gold: deviations create dealer profit as long as there is some expectation of return [Notes 16 p.3]. But under floating there is no long-run anchor; and if people believe UIP, they read the forward depreciation that arises to induce dealers to absorb imbalances as a forecast of future spot depreciation: a source of volatility. “If that forecast leaks into expectations of actual future spot, then the whole system can get unhinged” [Notes 16 p.3][Notes 16 p.7].

Where is the outside spread? Under gold the central bank set the bounds; here nobody seems to. The key is CIP: $F/S = (1 + R^*)/(1 + R)$. If F/S is driven up by the matched-book dealer seeking profit, the term rates must move too; holding the US rate R constant, the foreign term rate R^* is bid up, an upward-sloping curve in interest-rate space [► L16.6 @ 3:39]. The deficit country’s central bank cares about those term rates. Central banks set overnight rates, not term rates; the conceptual link between the two is the expectations hypothesis: with no liquidity premia, term rates are expected rollovers of overnight rates, so a credible move in the overnight rate moves term rates [► L16.6 @ 5:10].

16.6 Why do UIP and EH fail?

INTUITION: One fact, two failures

- **EH fails:** to absorb the payment imbalance, term rates are bid up above what EH predicts. That is what we observe: on average the term structure is not flat but upward sloping. Dealers must make a profit to make markets, and this is where one kind of dealer’s profit comes from [► L16.7 @ 0:03];
- **UIP fails:** if the forward rate equalled the expected spot rate, no speculative dealer would take the risky trade with zero expected profit: “uncovered interest parity must be false”, or dealers will not make the market [► L16.7 @ 1:05].

Both failures come from the need to mobilize profit-making dealers of two kinds to handle international payment imbalances [► L16.7 @ 1:46].

An inverted yield curve? A student saw an inverted repo yield curve on a terminal. Inverted curves usually signal monetary tightening, the central bank pulling up the short rate; in repo just then, probably collateral shortage, unrelated to this argument [► L16.7 @ 2:27]. An inverted curve is a sign of severe money-market stress: people bid for spot money because they have payments to make. Central banks usually cap overnight rates unless they want stress; overnight rates pull the one-week and one-month rates through term-structure arbitrage, while at long horizons something other than liquidity sets prices [► L16.7 @ 3:49].

IN THE MARKETS: When CIP itself broke down

In the crisis the need to roll dollar funding was so strong that those who could not borrow Eurodollars borrowed yen and swapped them into dollars: the dollar funding crisis became an FX crisis, and covered interest parity broke down, because under stress people stopped doing the arbitrage and looked after themselves. The BIS documented it [► L16.7 @ 5:11]. The lecture's argument is about normal times [► L16.7 @ 5:52].

What risk premium? Abstracting from liquidity, standard economics has no story for either failure; it can invoke “time-varying risk premia”, but what is the risk? Here it is clear: exposure to liquidity, funding, turnover risk [► L16.7 @ 6:13]. The premia vary over time not because utility functions bounce around but because the pattern of payments does: with small imbalances the premia should nearly vanish, since dealers need no incentive [► L16.7 @ 7:14]. Inefficiencies appear even in the most liquid market in the world, \$4 trillion a day: it is not about volume but about the incentive for dealers [► L16.7 @ 7:54].

Pushing the survival constraint forward. The dealers move the deficit country's survival constraint on into the next day: “I don't have any dollars today, I'm going to have dollars tomorrow, I think”, and I pay dealers meanwhile. Since they do not know when the flow reverses, they must price as if they will hold to maturity; “that's where the prices come from” [► L16.7 @ 8:36].

Exercise 16.2 (UIP and expected profit). Show that if $F = \mathbb{E}[S_T]$ the speculative dealer's expected profit is zero. If its required expected return is 0.5% over six months, how far must F be from $\mathbb{E}[S_T]$? Translate this into a premium in R^* via CIP.

16.7 The central bank as FX dealer of last resort

The central bank comes in when private dealers are done: prices have moved, and they will take no more risk even if prices move dramatically [► L16.8 @ 0:03]. Add the two countries' central banks; for simplicity they deal with each other, not with a third country or the IMF [► L16.8 @ 0:50]. The private agent in the deficit country still needs to sell FX for dollars [► L16.8 @ 2:12]:

1. the deficit central bank borrows term dollars from the surplus central bank at a rate R_G^* (“G” for government, not necessarily a market rate), and the surplus central bank opens a \$10 spot deposit for it: a swap of IOUs, like all banking [► L16.8 @ 2:33];
2. the deficit central bank sells the \$10 to its resident for $10S_G$ of FX, at a rate that need not be the market rate (“we like you, you are an important industry, or a connected fellow”) [► L16.8 @ 4:46];
3. the resident pays the surplus country: the dollar deposit is transferred [► L16.8 @ 5:28].

Sterilization. The surplus central bank's balance sheet, and its money supply, expand; if it dislikes that it sterilizes by selling a Treasury bill. Net, it has sold an asset and lent the proceeds to another central bank: its government may be unhappy about lending to foreigners instead of holding Treasury bills, which “might ring a bell” from the crisis [► L16.8 @ 6:14]. The deficit central bank's sterilization is the mirror, an expansionary operation adding an FX term asset [► L16.8 @ 8:20].

COURSE NOTE

The BIS article is not named in class. Representative BIS studies of the 2007–08 dislocation of FX swap markets are N. Baba, F. Packer and T. Nagano, “The spillover of money market turbulence to FX swap and cross-currency swap markets”, *BIS Quarterly Review*, March 2008, and N. Baba and F. Packer, “Interpreting deviations from covered interest parity during the financial market turmoil of 2007–08”, BIS Working Paper 267, 2008. The second paper is on the course's reading list ([materials/readings/L16-BIS_work267_Baba_Packer.pdf](#)). Its abstract makes two points:

- The sharp and persistent deviations from CIP between dollar and euro were significantly associated with the difference in counterparty risk between European and US institutions.
- The ECB's dollar term auctions, funded by the Fed's swap lines, stabilised the FX swap market by lowering the volatility of those deviations.

Surplus central bank		Deficit central bank (net)	
+\$10 term loan to deficit CB (R_G^*)	(+\$10 spot deposit of deficit CB,	+FX $10S_G$ term	+\$10 term borrowing (R_G^*)
–T-bill (sterilization)	then transferred; net zero)		

Figure 16.3: The central bank as FX dealer of last resort, net positions after sterilization [► L16.8 @ 7:18].

The deficit central bank as speculative dealer. After the cancellations the deficit central bank borrows term dollars and lends term FX: it is a speculative dealer, with a naked forward position in its own currency, but without a profit motive; if there were expected profit, private dealers would be doing it [► L16.8 @ 9:04]. It ends up as FX dealer of last resort as a consequence of its commitment to peg overnight interest rates [► L16.8 @ 10:32].

Definition 16.3 (New concepts: FX dealer of last resort, sterilization). *FX dealer of last resort*: the central bank taking the naked forward position private speculative dealers will no longer take, as a backstop for the FX market [► L16.8 @ 10:32]. *Sterilization*: an offsetting operation (for example selling a Treasury bill) that cancels the effect of an FX operation on the domestic money supply (from Latin *sterilis*, “barren”: the operation is made to produce no monetary effect) [► L16.8 @ 6:34].

When is intervention efficient? Knowing it is the dealer of last resort, the central bank may intervene earlier: if private speculative dealers charge “an arm and a leg” for the naked exposure, pushing the forward rate away from its efficient level, the central bank can provide the forward hedge to the matched-book dealers itself [► L16.8 @ 10:53]. Typically these are central banks of poor countries, or countries without well-developed FX dealer markets, where premia can be very large: “I know I’m a minor currency, but my people are just as important as your people” [► L16.8 @ 11:54]. Prices are signals: of competitiveness, and, for term rates, of whether saving is scarce. But here it is liquidity, not saving, that is scarce, and the central bank can make it less scarce, accepting zero profit, or even losses for a national purpose such as industrialization [► L16.8 @ 12:36].

Exercise 16.3 (Central bank swap). Redo the central bank case with numbers: the deficit central bank borrows \$10 at $R_G^* = 2\%$ and sells it to a resident at $S_G = 1.20$ when the market rate is 1.25. Write all T-accounts, including both sterilizations, and identify who bears the exchange risk and who receives the subsidy.

COURSE NOTE

Not a blanket endorsement of currency intervention. A central bank offering FX hedges to the whole world would cause bubbles and “all kinds of problems”; with small distortions, why bother? Deep, liquid private markets need to make money, and a central bank that replaces them must decide politically who gets foreign exchange, or end up “offering Goldman Sachs their next bonuses”. The aim is to facilitate the underlying trade [► L16.8 @ 14:19].

16.8 McCauley on the internationalization of the renminbi

READING: R. N. McCauley, Renminbi internationalisation and China's financial development (BIS, 2011)

McCauley describes China's monetary, credit and bond markets as having three tracks

[► L16.9 @ 0:03]:

1. the **old track**, the controlled economy, still large: banks are told whom to lend to and at what rate [► L16.9 @ 0:24];
2. **peripheral markets**: loosened at the edges, a money market, a small interbank market, bond and loan markets with market rates; some market signals are better than none, in a controlled transition [► L16.9 @ 1:11];
3. a new **offshore track**, from the internationalization of the renminbi [► L16.9 @ 2:16].

The analogy in McCauley's mind is the Eurodollar market of the late 1950s and early 1960s: deposits and loans in dollars offshore, outside US regulation, often a way of evading capital controls and interest-rate ceilings (Regulation Q) [► L16.9 @ 2:16]. Offshore and onshore renminbi prices are not on top of each other, a sign of obstacles to arbitrage (as Eurodollar and Fed funds rates normally are, within "gold points" of a few basis points); in stress, the third and fourth quarters of 2011, they move farther apart, as Eurodollar and Fed funds did in the crisis [► L16.9 @ 3:38]. Offshore renminbi bonds yield less than onshore ones, even with lower credit quality, because holders of offshore renminbi deposits can switch only into offshore bonds [► L16.9 @ 5:01]. The Chinese authorities seem to allow it, as they allowed track two; but this is harder to turn off once developed [► L16.9 @ 5:41].

The notes place the offshore track in Hong Kong, money and capital markets both; the first step will be breaking down the barriers between the tracks, and in the process the central bank's role will shift "from a gold-standard like price setting function to a modern price backstopping function" [Notes 16 p.7].

A reserve currency? Behind the discussion is the renminbi as a world reserve currency. China is a surplus country: everyone has to make payments to it, and the survival constraint is asymmetric, so its liabilities are money, as US liabilities were after the two world wars. Today the Eurodollar market has little to do with the US trade balance [► L16.9 @ 6:03].

INTUITION: FX is inherently hybrid

Foreign exchange is a negotiated relationship between the state and the market, different in different countries. The Fed basically lets the private market work: there are deep spot and forward markets with the dollar on one side. Elsewhere there are no dealers and central banks make the markets; most countries are in between. These are not ideological choices: near the top of the hierarchy you look like the US, near the bottom like the central-bank case, in the middle a hybrid. "It's where states meet states and markets meet markets" [► L16.9 @ 7:04]. There is no general rule that intervention or capital controls are always bad; prices need not be at their efficient levels. Economists of all political stripes abstract from liquidity, "but liquidity doesn't abstract from them" [► L16.9 @ 9:08].

The notes widen the options [Notes 16 p.5][Notes 16 p.6]. Central banks that target overnight rates "are inevitably drawn into serving as FX dealers of last resort": first as speculative dealers selling hedges to matched-book dealers, then as matched-book dealers themselves through swaps with other central banks. The surplus central bank lending to a foreign central bank rather than its own government is a limiting case; other sources of reserves work too, regional liquidity pools or the IMF. And no other central bank is needed: the deficit central bank can serve as speculative dealer for private hedgers, or offer forward cover directly to its citizens; in the limit, forward cover for all comers at the current spot rate amounts to fixing the exchange rate. The rates involved (R_G^* , S_G) are policy rates, reflecting non-commercial relationships. The danger of trading with all comers at non-commercial prices is arbitrage by speculators; the notes quote DeRosa (2009) on currency crises, all preceded by large positions short dollars and long local currency that the whole market, "not counting the central bank", had to hedge at once. Hence: "the exchange rate is not a free variable left open for state choice, but neither is it a market price that always fully reflects fundamental valuation" [Notes 16 p.7].

READING: Mehrling, “Essential hybridity: a money view of FX” (*J. Comparative Economics* 41, 2013)

The written version of this lecture ([materials/readings/L16-Essential_hybridity.pdf](#)). Economics treats the exchange rate as the relative price of goods, finance as the relative price of assets. The paper treats it instead as the *relative price of moneys*, set in dealer markets by order flow.

- **A spectrum of regimes.** At one extreme the central bank quotes buy and sell prices and absorbs the order flow on its own balance sheet. At the other extreme private dealers set prices. In the general, hybrid case private dealers make the market and the central bank acts as “dealer of last resort”.
- **The central bank as speculative dealer.** The deficit country’s central bank borrows term dollars, for instance from the Fed through a swap line, and sells them spot to its citizens. It ends up with a naked forward position, the same position a private speculative dealer would take for profit; it shoulders FX risk that no private speculator will take. In the limit it offers forward cover to all comers at the current spot rate, which amounts to fixing the exchange rate (pp. 360–361).
- **The exchange rate is “essentially hybrid”.** It is neither a market price that always reflects fundamentals, nor a free policy instrument.
- **The hierarchy of currencies.** According to the BIS figures quoted, 84.9% of FX volume has the dollar as one leg. Of \$4 trillion a day, \$1.5 trillion is spot and \$1.8 trillion FX swaps. “The FX market is fundamentally a money market, not a capital market” (p. 361).
- **Majors, minors and crosses.** The majors all have the dollar as one leg; the minors trade against a major; most minor cross pairs do not trade at all. Every currency is ultimately a promise to pay dollars, through a regional key currency (Fig. 5, p. 362).

Exercise 16.4 (Offshore and onshore). Offshore renminbi bonds yield 3% and comparable onshore bonds 4%. Explain why arbitrage does not close the gap, what happens to the gap as controls are relaxed, and compare with the Eurodollar–Fed funds spread before and during 2008.

Key formulas and ideas

KEY FORMULAS

CIP	$F/S = (1 + R^*)/(1 + R)$: arbitrage, defines the forward rate (breaks down in crises)
UIP	$F = \mathbb{E}[S_T]$: fails, since speculative dealers need $F \neq \mathbb{E}[S_T]$ to earn expected profit
EH	long rate = expected rollover of short rates: fails, since term rates are bid up to pay matched-book dealers
Private system	matched-book dealer hedges spot with forward; speculative dealer holds the naked forward (carry trade)
Outside spread	no explicit bound; via CIP the deficit country's term rate, linked by EH to the overnight rate the central bank sets
Last resort	deficit central bank borrows term dollars, lends term FX: a speculative dealer without profit motive
Hybridity	FX regime reflects place in the hierarchy: markets at the top, central banks at the bottom

Chapter 17

Direct and Indirect Finance

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L17.1 FT: Shadow banking	4:15	Adair Turner: shadow banking is like cholesterol; traditional versus shadow bank.
L17.2 Bagehot's World	8:51	Money market and capital market as separate worlds.
L17.3 The New World	10:55	American banks: long-term assets, interbank repo, the origin of rating agencies.
L17.4 Funding liquidity versus market liquidity	2:33	Moulton's shiftability (1918).
L17.5 Digression: Schumpeter on banking	4:05	Banks as engines of development.
L17.6 Payment versus funding	5:31	A loan funded by deposits, then taken out by a bond.
L17.7 Reading: Gurley and Shaw	2:04	<i>Money in a Theory of Finance</i> (1960): direct and indirect finance.
L17.8 Financial evolution	13:12	Pension funds and insurers; from intermediation to mutual funds.
L17.9 Banking evolution	11:12	Deposits as cheap permanent funding; from Jimmy Stewart banks to shadow banks.
L17.10 Preview: Central banking and shadow banking	8:28	Liquidity puts, solvency versus liquidity, market liquidity provided by dealers.

Files. Transcripts `materials/transcripts/L17-P*.txt`; Mehrling's notes [[Notes 17 p.1](#)]. **Reading.** J. G. Gurley and E. S. Shaw, *Money in a Theory of Finance*, Brookings Institution, 1960 (passages).

Motivation

This lecture opens the fourth and last part of the course, which extends the money view from the money market to the capital market, to asset prices and the prices of risk [[► L17.1 @ 0:49](#)][[► L17.2 @ 0:03](#)]. The two subjects are usually taught separately: capital markets and CAPM in finance courses in the business school, money and banking (perhaps IS–LM) in economics departments [[► L17.2 @ 0:03](#)]. Shadow banking shows that this intellectual separation does not match the real world, where the two are more intertwined than at any time since World War II, perhaps World War I [[► L17.2 @ 1:04](#)]. The separation has a

historical origin, which the lecture reconstructs.

17.1 FT: regulators to tackle shadow banking

CASE STUDY: FT, November 2012: “Regulators to tackle shadow banking”

A report led by Lord Adair Turner, the UK regulator, calls for regulation of shadow banking; Turner: shadow banking “is like cholesterol”, there is good and bad [▶ L17.1 @ 0:07]. Mehrling, who was working on the subject, thinks shadow banking is in our future and hopes it will be the good kind [▶ L17.1 @ 0:28].

Traditional (“Jimmy Stewart”) bank		Shadow bank	
Loans	Deposits	RMBS (tranchéd)	Repo (RP)
		Interest rate swap	ABCP held by
		Credit default swap	MMMF

Figure 17.1: Traditional and shadow bank, as on Mehrling’s slide [▶ L17.1 @ 1:51]. The traditional bank has a solvency backstop (FDIC) and a liquidity backstop (Fed); the shadow bank has neither directly, only liquidity puts to traditional banks.

The **traditional bank**, deposits against loans, is the “Jimmy Stewart bank” of the holiday film with a run on his bank. It is backstopped by the FDIC, which insures the deposits (a solvency backstop), and by the Fed (a liquidity backstop); it is the balance sheet most people have in mind when they say “bank” [▶ L17.1 @ 1:51]. The world is increasingly like the **shadow bank**: mortgages packaged into residential mortgage-backed securities, tranchéd into risky and risk-free tranches, insured by interest rate and credit default swaps, and funded in the money market by repo or asset-backed commercial paper held by money market mutual funds [▶ L17.1 @ 2:53]. On both sides of its balance sheet the prices are market prices: the money market funding rate and the price of risk; hence the coming lectures on swaps [▶ L17.1 @ 3:34].

Definition 17.1 (New concepts: shadow banking). *Shadow banking*: money market funding of capital market lending, outside the traditional banking system and its backstops [▶ L17.1 @ 0:49](“shadow” because it does banking business outside the light of bank regulation).

17.2 Bagehot’s world: two separate markets

The money market: indirect finance. In Bagehot’s world banks discounted short-term bills that firms issued to finance goods on their way to final sale; a portfolio of bills maturing at various dates assured the bank’s liquidity [▶ L17.2 @ 2:06]. New in the course: the bank seen as an *intermediary*, between a primary lender (the household, for the first time in the course, holding deposits to make payments) and a primary borrower (a business financing working capital with bills): deposits fund loans. This is *indirect finance* [▶ L17.2 @ 2:47].

HISTORY: Jimmy Stewart’s bank

In Frank Capra’s *It’s a Wonderful Life* (1946) James Stewart plays George Bailey, who runs the Bailey Building and Loan in Bedford Falls; during a run he tells the depositors that their money is not in a vault but in each other’s houses. Reference: F. Capra (dir.), *It’s a Wonderful Life*, Liberty Films/RKO, 1946.

The capital market: direct finance. There was also a capital market: government bonds (including perpetual bonds paying a coupon forever), some corporate bonds for large projects, and equity, held directly by rich people as wealth: *direct finance*, no intermediary [► L17.2 @ 4:30]. The two markets touch when bonds are issued, since the household pays with deposits that the firm spends; afterwards the bond market seems separate. Bonds promise small coupons and principal only at maturity, perhaps 30 years later; the staggered-timing liquidity issue of bills does not arise [► L17.2 @ 5:54].

Household (primary lender)		Bagehot bank	
Deposits		Bills	Deposits
Bonds, equity			
		Business (primary borrower)	
		Working capital	Bills
		Fixed capital	Bonds, equity

Figure 17.2: Bagehot’s world: indirect finance in the money market, direct finance in the capital market [► L17.2 @ 2:47].

17.3 The new world

The US was a developing country, with an extreme need for long-term finance: railroads, buildings, factories, homes. American banks held corporate bonds and long-term loans, including “six-month” loans that were always rolled over, and few short-term bills [► L17.3 @ 0:03]. Before there was a central bank, how did they manage? By cash reserves, by bankers’ (correspondent) balances, and by interbank borrowing against bonds as collateral: essentially repo, a paired sale and repurchase [► L17.3 @ 1:26]. A deficit bank at the clearing pays in cash, in bankers’ balances, or borrows against a bond.

ASIDE: A cult of liquidity?

Some economic historians (among them some of Mehrling’s professors) argue that Britain’s hyper-developed short-term money market, the centre of world trade finance, came with underdeveloped long-term finance: banks were forbidden by a “cult of liquidity” from long-term lending, which may help explain Britain lagging after the industrial revolution. Mehrling does not vouch for it: “I’m not an expert on Britain” [► L17.2 @ 7:15].

Deficit bank		Surplus bank	
Bonds (collateral)	Deposits	Bonds, loans	Deposits
Loans	or +Repo	Cash reserves	Bankers’ balances
–Cash or	borrowing	+Cash or	
–Bankers’ balances		+Repo lending	

Figure 17.3: Interbank settlement in the US before the Fed [► L17.3 @ 1:50].

Deficit at the clearing. A student asks whether, in a developing country with many borrowers, the central bank acted as the surplus bank. There was no central bank; and

“deficit” refers to the clearing, not the whole system. Banks lend, as always, by creating deposits; when the borrower spends the deposit in New York, the bank must pay the New York bank, with reserves, with bankers’ balances at its correspondent, or by borrowing against a bond [► L17.3 @ 3:55].

The origin of rating agencies. In Mehrling’s account rating agencies arose in this period: banks paying each other did not want to analyse every bond offered as collateral, so they bought Moody’s book of ratings, which said a bond was unlikely to default in the next year, good enough for short-term collateral; the rating entered the terms of the repo [► L17.3 @ 5:38]. Unsecured lending through bankers’ balances needs a relationship; secured lending against a rated railroad bond can be done with a New York bank that has never heard of you [► L17.3 @ 7:20]. Capital market and money market are completely intertwined: “19th-century shadow banking” [► L17.3 @ 7:41].

The Fed and the shadow system. After the 1907 crisis the reformers tried to replace this system with discounting of short-term bills at a new Fed, but not enough bills were issued: the US needed long-term finance [► L17.3 @ 7:41]. Mehrling’s argument: the Fed’s mistake around 1929 was to let the shadow banking system of its day collapse, refusing to support mortgage credit and bonds as improper for banks, and so it brought down the US banking system and the world’s [► L17.3 @ 8:22]. In 2007 we were fortunate that the Fed did backstop the shadow banking system, much of it in Europe, through the liquidity puts linking shadow banks to Citibank and other traditional banks [► L17.3 @ 9:03]. The founders of the Fed “pasted a kind of central bank onto” a system it did not fit; today’s question is how to have a central bank that supports a system like this [► L17.3 @ 10:04]. The notes put it sharply: the banking collapse of the 1930s can be read as a run on the shadow banking system of the time, private funding liquidity through repo on non-Treasury securities; the Fed refused to support it, it collapsed, and the government had to take on long-term lending itself, most famously through the Reconstruction Finance Corporation: “this mistake was not, thankfully, repeated in 2007–8!” The transmission from funding liquidity to market liquidity and asset prices is, in the US, “not recent, but in fact ancient” [Notes 17 p.2][Notes 17 p.3].

17.4 Funding liquidity versus market liquidity

The British concept of liquidity was *funding liquidity*: guaranteed cash inflows. In the new world a bank’s liquidity was having bonds pledgeable or shiftable to someone else: *shiftability*, which the course calls market liquidity, the ability to buy or sell an asset quickly without moving the price much [► L17.4 @ 0:04].

Definition 17.2 (New concepts: shiftability). *Shiftability*: the liquidity of an asset that can be shifted to another holder (sold or pledged) without loss, as opposed to liquidity from self-liquidating short-term assets. Term of Harold Moulton, 1918 [► L17.4 @ 0:45].

Moulton argued that shiftability was the right concept for the US, and that the Fed should backstop it. The Fed did, not for all assets but for Treasury bills: that is where open market operations came from, supporting market liquidity in the capital market [► L17.4 @ 1:06]. This course uses both kinds of liquidity and their interlink, dealers who create market liquidity must fund their inventories, an understanding not yet there in 1918 [► L17.4 @ 1:46].

HISTORY: Moody’s, 1909

John Moody published *Moody’s Analyses of Railroad Investments* in 1909, assigning letter ratings to railroad bonds; Poor’s (1916), Standard Statistics (1922) and Fitch (1924) followed. The link to interbank collateral is Mehrling’s interpretation.

References: R. Sylla, “An historical primer on the business of credit rating”, in R. M. Levich et al. (eds), *Ratings, Rating Agencies and the Global Financial System*, Kluwer, 2002.

READING: Moulton, “Commercial Banking and Capital Formation” I–IV (*JPE* 26, 1918)

The four parts appeared in nos. 5, 6, 7 and 9, May to November 1918: pp. 484–508, 638–663, 705–731 and 849–881 (materials/readings/L17-Moulton_*.pdf).

Moulton’s argument against the conventional “commercial loan” theory of banking, inherited from early 19th-century England:

- American commercial banks in fact supply permanent working capital, and they hold bonds and stocks as investments and as collateral.
- A customer’s commercial paper is among the *least* reliable secondary reserves. It cannot be collected in a crisis, because customers must be “carried”. It cannot readily be shifted to other banks either.
- Listed securities are, “in normal times”, almost instantly convertible into cash, and so are more liquid than customers’ paper.
- The key sentence (part III, p. 723): liquidity comes from keeping assets “that can be shifted to other banks before maturity” and not from maturing loans. “Liquidity is tantamount to shiftability.”
- In a crisis almost no asset is liquid, and whatever liquidity exists comes from inter-bank accommodation. Under the Federal Reserve, liquidity becomes “a question of shiftability to the Federal Reserve banks” (p. 723, n. 1).

Exercise 17.1 (Permanent funding). A bank’s deposits have fluctuated between 800 and 1000 over ten years. How much of its balance sheet can it hold in long-term loans, and how much in reserves? What changes if a money market fund offers a close substitute for deposits?

Exercise 17.2 (Market liquidity needs funding). Using Treynor’s model, explain why a dealer in AAA mortgage securities stops quoting when its repo funding is cut, and what happens to the value of the same securities as repo collateral.

HISTORY: Harold G. Moulton

Harold Glenn Moulton (1883–1965) published “Commercial Banking and Capital Formation” in four parts in the *Journal of Political Economy* in 1918; he was the first president of the Brookings Institution (1927–1952).
References: H. G. Moulton, *JPE* vol. 26, 1918 (four parts); Brookings Institution, “History”.

17.5 Schumpeter on banking

Joseph Schumpeter, sympathetic to the American scene, emphasized all his life the importance of banks willing to do long-term development finance: money market funding of capital market lending [► L17.5 @ 0:04]. Banks create new money for entrepreneurs, who need not find someone who has saved: “created from nothing”, the alchemy of banking. To say so risks sounding like a monetary crank, and Schumpeter was not one [► L17.5 @ 1:08]. Expanding the balance sheet is not “bread from heaven”: something must discipline it (next section) [► L17.5 @ 1:50].

Schumpeter admired H. D. Macleod, who stressed this kind of banking in the development of Scotland. The Lewis model of development, with surplus labour in subsistence agriculture, also gives banking a role in mobilizing it towards cities and factories [► L17.5 @ 2:10]. Development economists are usually “real side” people; Mehrling would give half of a development course to finance, “but hey, I would say that” [► L17.5 @ 3:32].

INTUITION: There is such a thing as a free lunch

The notes warn that most self-styled monetary alchemists are cranks: “the ability to create money from nothing is not the same as the ability to create bread from nothing”, and economics has come close to making “there ain’t no such thing as a free lunch” a creed. “But there is.” Adam Smith (1776) already urged paper money to economize on gold, trading a sterile commodity for productive capital; the main source of the free lunch, though, is that banking can *mobilize unused resources*: if entrepreneurs spend new deposits on things that would otherwise go unsold, they raise activity, not prices [Notes 17 p.3].

17.6 Payment versus funding

A business wants to buy a machine from “society”. The bank lends by a swap of IOUs, +loan against +deposit, and the business pays with the deposit [► L17.6 @ 0:05]. The question: what *funds* the loan? If society is happy to add the deposit to its portfolio, the deposit funds the machine, even though it is a demand deposit and the machine a loom with a ten-year life [► L17.6 @ 1:53]. If society does not want to hold it, the business must fund the machine otherwise, by issuing a bond that society buys with the deposit, and repaying the loan [► L17.6 @ 2:55].

COURSE NOTE

In class Schumpeter’s “thesis” is dated 1912. The book meant is *Theorie der wirtschaftlichen Entwicklung* (published late 1911, dated 1912; English *The Theory of Economic Development*, 1934), not his doctoral thesis. Henry Dunning Macleod (1821–1902) was a Scottish economist, author of *The Theory and Practice of Banking* (1855–56). The Lewis model is W. A. Lewis, “Economic Development with Unlimited Supplies of Labour”, *Manchester School*, 1954, which discusses capital formation financed by credit creation [► L17.5 @ 0:04]. The course file (materials/readings/L17-3681ewistable.pdf) reprints the essay. In it Lewis argues that a community with idle resources can create capital by creating extra money to finance the extra employment. There is no loss of other output while the capital is made. Profit-financed and credit-financed capital differ “not in the ultimate effects on output, but in the immediate effects on prices and on the distribution of income”.

Bank		Business		Society	
+Loan	+Deposit	+Machine	+Loan	–Machine,	
(later	(later		(later	+Deposit	
–Loan)	–Deposit)		–Loan,	(later	
			+Bond)	–Deposit,	
				+Bond)	

Figure 17.4: Payment versus funding: bank credit as a bridge loan, the bond as take-out financing [► L17.6 @ 1:10].

The money market intermediation is a bridge loan: it gets the ball rolling, using the elasticity of the payment system; once the machine is running, the capital market provides permanent, take-out financing. First intermediation, as in Bagehot but long; then direct finance, bond for bond, the intermediary drops out [► L17.6 @ 4:17].

Exercise 17.3 (Bridge and take-out). A firm borrows 100 from a bank to buy a machine and, a year later, issues a 10-year bond of 100 bought by a household. Write the T-accounts of bank, firm and household at each step, and say which step is indirect and which direct finance.

17.7 Gurley and Shaw: intermediaries in the capital market

READING: J. G. Gurley and E. S. Shaw, *Money in a Theory of Finance* (1960)

Gurley and Shaw (Brookings, 1960) introduced the terms *direct finance* (the primary lender lends directly to the primary borrower) and *indirect finance* (an intermediary in between). They stressed intermediation not just in the money market but in the capital market, for long-term finance, following Moulton; and not only by banks but by the other intermediaries that had grown since his day [► L17.7 @ 0:04].

Definition 17.3 (New concepts: direct and indirect finance, financial intermediary).

Direct finance: the primary borrower's securities are held by the primary lender. *Indirect finance*: a *financial intermediary* (from Latin *inter*, between, and *medius*, middle) issues its own liabilities to lenders and holds the borrowers' securities [► L17.7 @ 0:31].

17.8 Financial evolution

The two main capital market intermediaries in the US are pension funds and insurance companies [► L17.8 @ 0:05]:

- a **pension fund**, typically a defined benefit plan in 1960, promises workers a fraction of final income for life, a long-term, annuity-like promise, and hedges it with long-term assets, especially equities [► L17.8 @ 0:46];
- an **insurance company** has contingent liabilities (fire, life policies) and holds bonds [► L17.8 @ 1:30].

Both stand between households and businesses, like Bagehot's bank, though their liabilities are not money; they became bigger than banks in funding American development [► L17.8 @ 2:11].

INTUITION: Gurley and Shaw's vision: intermediaries face both ways

Households want to hold liquid assets, money, insurance, pensions; businesses want to borrow long. How can the capital development of the nation be financed? An intermediary faces both ways: to households it says "I'll give you what you want", deposits, policies, pensions; to businesses, ten-year loans, or it buys their bonds and equity. Without it, the mismatch between what households want and what businesses issue could block development [► L17.8 @ 3:33].

Their liquidity: insurance has an actuarial quality (not every house burns at once), and in a hurricane the insurer sells bonds; pension funds turn equities into bonds (annuities) when a worker retires. Both depend on market liquidity, shiftability, as Moulton said for banks; Moulton became president of Brookings, which may be why Brookings hired Gurley and Shaw [► L17.8 @ 5:17].

The flaw: risk does not go away. The risk of the assets does not disappear because you promise something else: it falls on the liabilities, or on whoever guarantees them, typically the government. With risky assets, guaranteed benefits may not be paid; the risk is borne by the sponsor (often a corporation), by the government through the Pension Benefit Guaranty Corporation, or by the beneficiaries when promises are not kept. Finance

and arbitrage will make you appreciate it, through mispricing [► L17.8 @ 7:20].

From indirect to direct finance. Since 1960 indirect finance has been hollowed out in favour of mutual funds: stock funds hold equities and issue shares in the portfolio (Fidelity's Magellan), bond funds hold bonds (PIMCO): the risk passes right through to the holder, "this is not investment advice" [► L17.8 @ 9:07]. Pensions moved from defined benefit to defined contribution plans such as the 401(k), where you see your risk [► L17.8 @ 10:51]. In insurance, term policies replaced whole-life policies with their savings component: both trends reflect mutual funds competing with traditional indirect finance. A mutual fund looks like an intermediary, but it mainly pools risk through diversification, with little transformation: its shares have the risk of the underlying pool; the liquidity of buying back at NAV requires cash or credit lines, paid for by the shareholders. Bond funds replaced the savings in whole-life policies and bank time deposits, stock funds replaced defined-benefit plans [Notes 17 p.6][Notes 17 p.7]. Gurley and Shaw "overdid it", the notes say: modern finance stresses that intermediation eliminates no risk, only transfers it, sometimes opaquely, so capital markets equilibrate by price, and attempts to avoid this create arbitrage opportunities that have transformed the system [Notes 17 p.6].

KEY FORMULAS

Quantities versus prices [► L17.8 @ 11:32]. Indirect finance solves the mismatch between what lenders want and what borrowers want with *quantities*, on its own balance sheet. Modern finance solves it with *prices*: if you don't want to hold bonds, bonds fall in price until you do. Price does much more work than in 1960, which is why everyone watches the stock market.

The same evolution, from intermediation to direct finance, has happened in banking and the money market: shadow banking is its natural next step [► L17.8 @ 12:33].

17.9 Banking evolution

Banks as intermediaries are a new conception in the course, which has treated banking as a payment system and as market making, not as funding [► L17.9 @ 0:06]. Why can banks fund things? Because people want deposits not just to pay but as an asset, a liquidity reserve in their wealth [► L17.9 @ 0:26]:

- deposits are **permanent** funding for the system as a whole: money demand fluctuates but does not go to zero, so reserves need cover only the fluctuations; a banker who sees deposits never fall below 80% of the balance sheet can hold 80% in earning loans [► L17.9 @ 2:10];
- deposits are **cheap** funding, close substitutes for cash that pays no interest; even when banks compete by paying interest, it is less than on bonds [► L17.9 @ 1:30].

Liquidity at the margin. In the notes: because households and firms want to *hold* means of payment, not only make payments, the banking system can keep a permanent short position in cash and use it to fund long positions in non-cash assets. Liquid liabilities need liquid assets only at the margin: cash reserves plus the ability to replenish them (Fed funds, the discount window, "secondary reserve" assets that can be sold); the rest of the deposits can go into less liquid assets. Once the focus is intermediation, solvency risk, interest rate risk and credit risk, comes to the fore [Notes 17 p.7][Notes 17 p.8].

Sharing out the spoils. Everyone wants this cheap funding [► L17.9@3:52]: the government through the central bank (Treasury bills against cash), business through commercial banks (industrial loans against deposits), households through savings and loans, the Jimmy Stewart banks (mortgages against “deposits”, legally shares but used as money) [► L17.9@4:33].

Central bank		Commercial bank		Savings and loan	
Treasury bills	Cash	Business loans	Deposits	Household mortgages	“Deposits” (shares)

Figure 17.5: Cheap permanent funding shared out among government, business and households [► L17.9@4:33].

The risk passes through, to the taxpayer. The banks face both ways, but the risk does not vanish: in the Jimmy Stewart bank the FDIC guarantees solvency and the Fed liquidity, so the taxpayer bears it. Hence regulation (no risky loans, no interest on deposits), and hence a regulatory arbitrage to create non-banks doing the same thing: that is where shadow banking came from [► L17.9@6:17]. In the Jimmy Stewart bank the risk stays in the community: “your money is in your houses” [► L17.9@7:40].

Shadow banking strips the risks. In shadow banking the loans become traded securities whose price is the price of all their risks: a mortgage has duration risk (30 years) and default risk (household income). The interest rate swap and the credit default swap strip them off, leaving a synthetic Treasury bill funded in the money market: risk is not eliminated but hedged, sold off separately, rather than borne by a government that pretends it is not there, with the moral hazard that implies [► L17.9@8:22]. As pension funds evolved into stock funds and insurers into bond funds, banks evolved into money market mutual funds, many outside the US, holding dollars for sovereign wealth funds and central banks: “a global dollar system funding subprime mortgages in Cleveland” [► L17.9@9:44]. From indirect finance (Gurley and Shaw, 1960) to direct finance (2012), with all risks priced in markets and money and capital markets completely integrated [► L17.9@10:45].

Exercise 17.4 (Who bears the risk?). For a defined benefit pension fund, a defined contribution 401(k), a Jimmy Stewart bank and a shadow bank holding an RMBS hedged with swaps, say who bears the default risk and who the liquidity risk.

17.10 Preview: central banking and shadow banking

The central bank’s exposure. A student asks how the central bank becomes exposed to shadow banks [► L17.10@0:10]. To the traditional bank the Fed gives a liquidity backstop only; if it is insolvent the FDIC shuts it down and pays depositors from accumulated premia, and beyond that the government backstops solvency [► L17.10@0:52]. Nobody directly backstops the shadow bank, but there is the liquidity put: when the quality of the collateral came into question, money funds refused to roll their funding (“you told me these were just like Treasury bills with a yield advantage”), and the shadow banks drew on their parents’ promises, say Citibank’s, to roll their funding; so they came back onto

traditional banks' balance sheets [► L17.10 @ 1:33]. Regulators had endorsed the system to get risk out of banks: it removed most solvency risk but not liquidity risk [► L17.10 @ 2:35].

Definition 17.4 (New concepts: liquidity put). A commitment by a bank to provide funding to an off-balance-sheet vehicle if it cannot roll its money market funding: in effect an option to “put” the funding problem back to the bank [► L17.10 @ 1:33].

Where is solvency? Most banking courses are about solvency; this course stresses liquidity. In the shadow bank the solvency risk sits with the holders of the swaps and of the mezzanine and junior tranches. For AAA credit default swaps it was AIG, an insurance company with a lot of capital and bonds, the “FDIC equivalent” of the system, whose capital was sucked out very quickly by the counterparties, Société Générale and Goldman Sachs and their clients, which were acting as market makers [► L17.10 @ 2:55].

INTUITION: The fantasy of modern finance

The shadow banking system had no explicit lender of last resort, on the belief that removing solvency risk removes liquidity risk: a risk-free asset is like a Treasury bill, hence liquid by design. False: someone must still buy it. Market liquidity is created by dealers quoting bid–ask spreads, and dealers must fund their inventories; if they cannot, they stop making markets, prices go away, nobody knows what the collateral is worth, and the ability to fund the portfolio goes away. That is what happened in 2007–09 [► L17.10 @ 5:03].

A traditional bank has no market prices on either side: deposits were forbidden to pay interest, and loans are illiquid, specific to each borrower. A shadow bank lives on two prices, the market price of the RMBS and the world dollar funding rate [► L17.10 @ 6:26]. The system had moved off the Fed's balance sheet, “out of sight, out of mind”, Europe's problem; wrong: a global funding system came under threat almost immediately, and the expansion of the Fed's balance sheet shown in the first lecture bailed it out [► L17.10 @ 7:49].

Key formulas and ideas

KEY FORMULAS

Shadow banking	money market funding of capital market lending; prices on both sides
Bagehot's world	money market (bills, indirect) and capital market (bonds, equity, direct) separate
New world	long-term bank assets, interbank repo against rated bonds: capital and money markets intertwined
Liquidity	funding liquidity (British) versus shiftability = market liquidity (Moulton 1918)
Payment vs funding	bank credit as bridge loan; bonds as take-out finance
Gurley–Shaw	intermediaries face both ways; but risk passes through to someone
Evolution	intermediation → mutual funds; banks → MMMFs and shadow banks; quantities → prices
Backstops	traditional: FDIC (solvency), Fed (liquidity); shadow: liquidity puts, AIG as solvency

Chapter 18

Forwards and Futures

SOURCES FOR THIS CHAPTER		
Segment	Length	Content
L18.1 Preview: FT: Argentina in court to fight debt ruling	4:56	Holdouts, pari passu and collective action clauses.
L18.2 Banking as advance clearing	5:55	Changed expectations cause cash flows today: the survival constraint again.
L18.3 Forwards versus futures	13:41	Forward rate agreements as matched book; speculators and futures; $f > \text{futures} > \mathbb{E}[r]$.
L18.4 Futures contracts, fluctuations II	14:07	Convergence; the value of a forward contract over its life.
L18.5 Futures contracts, fluctuations II	6:07	Futures marked to market daily: cash flows all along.
L18.6 Cash and carry arbitrage, defined	7:28	Stigum's example; full carry pricing; implied repo rate.
L18.7 Cash and carry arbitrage, explained as liquidity risk	5:26	Fully hedged, but not in cash flow.
L18.8 Cash and carry arbitrage as counterparty risk	2:49	An alternative explanation.
L18.9 Cash and carry arbitrage, as natural banking business	3:22	Banks' comparative advantage in liquidity risk.
Files. Transcripts <code>materials/transcripts/L18-P*.txt</code> . (The two segments 4 and 5 carry the same title on the playlist, but their content differs.) Mehrling's notes [Notes 18 p.1]. Reading. M. Stigum, <i>The Money Market</i> , pp. 718–722 (cash and carry), as cited in class; J. R. Hicks, <i>Value and Capital</i> (1939), ch. XI; FT press cutting on Argentina (all in <code>materials/readings/</code>).		

Motivation

The last section of the course is called “banking as advance clearing” [► L18.2 @ 0:03]. The first section showed the payment system: cash inflows and outflows balanced each day, deficits pushed into the future, stress showing up as an interest rate, and defaults when it gets big enough [► L18.2 @ 0:26]. The economy is coherent not because it is an intertemporal general equilibrium but because the clearing constraint forces everyone to pay attention to their place in the system [► L18.2 @ 1:49]. Forward-looking financial markets do the same: changed expectations change asset prices (standard finance), and the money view adds

that *price changes cause cash flows today*, tightening or loosening the survival constraint [► L18.2 @ 2:30]. The dramatic case: the second round of funding is refused, or the bond meant to take out short-term funding cannot be sold, “then I’m dead”; the everyday case: pressure on money market prices [► L18.2 @ 4:39].

18.1 FT: Argentina in court

CASE STUDY: FT, November 2012: “Argentina in court to fight debt ruling”

In 2001 Argentina defaulted. Eventually 93% of bondholders by value accepted a restructuring (a reduction of the debt) and got new bonds; the other 7%, holdouts, mostly hedge funds, sued for full payment [► L18.1 @ 0:07]. A New York court ruled that unless the holdouts are paid in full, Argentina may not pay the restructured bonds, since all bondholders must be treated equally; this undermines restructuring itself [► L18.1 @ 1:29]. With a payment on the restructured bonds due the following month, Argentina faces a “monetary cliff”: pay the holdouts too, or default on everything [► L18.1 @ 2:51]. The FT’s editorial underlines the importance of collective action clauses: one holdout blocking agreement is a social loss, and taking a loss can be in the bondholders’ collective interest [► L18.1 @ 3:12].

The article in the course file (Jude Webber, *FT*, 26 November 2012, p. 6; materials/readings/L18-Financial_Times_-_Presscuttings.pdf) adds the details:

- The 2001 default was on nearly \$100bn. The restructurings were in 2005 and 2010.
- Judge Thomas Griesa ordered Argentina to pay \$1.3bn to the holdouts when it paid the restructured bonds on 15 December, and to deposit the money in escrow pending appeal. Third parties helping it evade the order would count as aiders and abettors.
- The dispute turns on the “*pari passu*” (“equal step”) clause.
- Argentina’s own law forbade reopening the restructuring. Failing to comply could mean a technical default “perhaps as early as mid-January”.

Definition 18.1 (New concepts: restructuring, holdout, collective action clause).

Restructuring: renegotiating a debt to less (longer terms, lower rates, lower principal).

Holdouts: creditors who refuse the deal. A *collective action clause* (CAC) lets a qualified majority of bondholders bind the minority, so there can be no holdouts; the 2001 Argentine bonds, issued under New York law, had none [► L18.1 @ 0:48].

18.2 Forwards versus futures

A forward is advance clearing. Recall the forward rate agreement as a swap of IOUs between a bank and firm A: a six-month loan against a three-month deposit, notional amounts, no money changing hands, locking in a three-month rate three months ahead, $f_{3,6}$, defined by forward interest parity [► L18.3 @ 0:24]:

HISTORY: The end of the Argentine hold-out saga

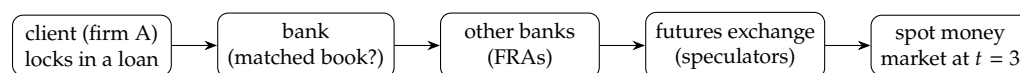
The ruling was by Judge Thomas Griesa (S.D.N.Y.), interpreting the bonds’ *pari passu* clause; it was upheld on appeal, and in June 2014 the US Supreme Court declined to hear Argentina’s appeal. Argentina defaulted again in July 2014 and settled with most holdouts in 2016. References: *NML Capital, Ltd. v. Republic of Argentina*, 699 F.3d 246 (2d Cir. 2012); IMF, “Strengthening the Contractual Framework to Address Collective Action Problems in Sovereign Debt Restructuring”, 2014.

KEY FORMULAS

Forward interest parity ▶ L18.3 @ 1:50: $(1 + r_{0,3})(1 + f_{3,6}) = 1 + r_{0,6}$, where $r_{0,3}$ and $r_{0,6}$ are today's three- and six-month rates: the forward rate makes the long rate equal to rolling over two short ones.

The bank, having promised to lend in three months, hedges with a firm B wanting the opposite, to lock in a deposit rate: the bank's positions offset. In practice these are forward contracts listed off balance sheet, not parallel loans ▶ L18.3 @ 2:54]. The bank has *cleared in advance*: in three months A will be a deficit agent and B a surplus agent, and the flows are already agreed; one way to evade the survival constraint is to take care of it ahead of time ▶ L18.3 @ 4:40]. The bank runs a matched book and earns the bid–ask spread ▶ L18.3 @ 5:42].

From matched book to speculators. There is no reason for equal numbers of fundamental customers on both sides; imbalance pushes prices until dealers take the other side onto their balance sheets. When the banking system as a whole has as much risk as it wants, banks find hedge funds and speculators in the *futures* market: a speculator goes long futures to help the bank square its book with a short futures position ▶ L18.3 @ 6:22]. The chain: client → bank → other banks (FRAs) → futures exchange (speculators) → spot money market, where a rate is realized in three months ▶ L18.3 @ 8:07].



$f_{3,6} > \text{futures rate} > \mathbb{E}[r_{3,6}]$: each hedge takes a piece of the premium

Figure 18.1: A sequence of hedging transactions ▶ L18.3 @ 8:31].

A stylized fact. The forward rate is not an unbiased forecast of the future spot rate (the expectations hypothesis fails), and one reason, in this course, is a mismatched book in the forward market: dealers must be paid to take the risk ▶ L18.3 @ 9:32]. The sequence of hedges makes sense only if the forward rate exceeds the expected spot rate, each taking a piece; so one also expects a gap between forward and futures rates ▶ L18.3 @ 10:33]:

KEY FORMULAS

Stylized fact (as stated in class) ▶ L18.3 @ 11:14]: $f_{3,6} > \text{futures rate} > \mathbb{E}[r_{3,6}]$. The forward–futures gap is small compared with the forward–expected spot gap; finance usually attributes it to something other than liquidity.

READING: Hicks, *Value and Capital* (1939), ch. XI “Interest”, pp. 141–152

The course reading (materials/readings/L18-Hicks_Value_and_Capital.pdf) is the classic source of this stylized fact.

- A loan is a transaction with only one side executed now. Any loan can be reduced to a money loan plus spot and forward transactions. With forward markets, “own” rates of interest are implicitly fixed: with money at 5% and coffee futures 3% above spot, the coffee rate is about 2%.
- Long rates are built up from *forward* short rates, determined like commodity futures prices by hedgers and speculators.

- The forward market for loans has a “constitutional weakness” on one side: most people prefer to lend short. Only speculators fill the gap, so the forward rate must exceed the rate they expect, by a risk premium which “corresponds exactly to the ‘normal backwardation’ of the commodity markets” (p. 147).
- Hence, if short rates are not expected to change, forward rates exceed current short rates by this premium.

This is the course’s mismatched forward book, with the dealer’s required compensation as Hicks’s premium.

The difference is cash flow. A forward causes no cash flow until maturity; futures are *marked to market* and cause cash flows every day, in or out, as prices move. So do most derivatives, through margin and collateral calls, “and those collateral calls cause people to go out of business”: the survival constraint [► L18.3 @ 12:16].

Definition 18.2 (New concepts: futures contract, marking to market). A *futures contract* is a standardized forward traded on an exchange, whose price is reset every day; *marking to market* means settling the change in value in cash each day (through margin), so that the contract’s value returns to zero [► L18.3 @ 12:38][► L18.5 @ 0:04]. (*Futures*: short for contracts for future delivery.)

Exercise 18.1 (Forward versus futures cash flows). A forward and a futures contract on the same bill have $K = F_0 = 98$. The futures price then goes 98.5, 97.0, 96.5, 98.0, 99.0 at maturity. Compute the daily cash flows of a short futures position and the single cash flow of a short forward. What is the largest cumulative cash outflow of the futures position?

18.3 The value of a forward over its life

Convergence. Rewrite forward interest parity as [► L18.4 @ 0:03]

$$\frac{1}{1 + f_{3,6}^0} = \frac{1 + r_{0,3}}{1 + r_{0,6}}, \quad \frac{1}{1 + f_{3,6}^1} = \frac{1 + r_{1,3}}{1 + r_{1,6}}, \quad \dots, \quad \frac{1}{1 + f_{3,6}^3} = \frac{1}{1 + r_{3,6}},$$

where $f_{3,6}^t$ is the forward rate quoted at time t ; in general they all differ. At maturity the forward rate equals the spot rate [► L18.4 @ 1:11]. So the contract between firm A and the bank, a zero-value swap of IOUs at inception, changes value: a zero-sum game in which, if rates behave as usual, the bank wins, having locked in a lending rate above the expected spot rate; the firm pays a premium for peace of mind about a serious survival constraint [► L18.4 @ 3:22].

INTUITION: Coordinating views of the future

In the notes’ framing [Notes 18 p.1]: in finance the future determines the present, but nobody knows the future, so there are many views of it. How is one path chosen and diverse views coordinated? By market pricing of the views, and by the effect of prices on behaviour through the survival constraint. At one extreme, if the market changes its mind about your view you cannot roll your funding; more subtly, changing ideas about the future change cash flows today, loosening the constraint for some and tightening it for others, as futures prices do.

The classic forward is the wheat farmer, naturally long wheat, and the baker, naturally short: by a forward contract both lock in the price, each hedging a natural exposure; at delivery one side “wins”, but in a forward the winnings are only notional until the end [Notes 18 p.3]. The forward price is an arbitrage condition: if $K > S_0 e^{rT}$, buy the bond spot and sell it forward, locking in a return above the borrowing rate r ; if $K < S_0 e^{rT}$, sell spot and buy forward [Notes 18 p.4]. (The notes follow J. Hull, *Options, Futures, and Other Derivatives*, 5th ed., equations 3.5 and 3.9.)

Continuous-time version. In price terms, with K the delivery (forward) price, S_t the spot price of the underlying (here the six-month bill) and r the interest rate ▶ L18.4 @ 5:46:

KEY FORMULAS

Forward price and value ▶ L18.4 @ 7:10:

$$K = S_0 e^{rT} \quad (1), \quad f_t = S_t - K e^{-r(T-t)} \quad (2), \quad F_t = S_t e^{r(T-t)} \quad (3).$$

K is fixed at inception; the value f_t of the forward to the long side moves with the spot price and with the shrinking discount as maturity approaches: $f_0 = 0$ and $f_T = S_T - K$. The futures price F_t (3) is K reset every day so that the contract’s value is again zero ▶ L18.5 @ 0:04.

Banking deals constantly with zero-value contracts that later acquire value, “a little mind-blowing” ▶ L18.4 @ 10:00. The same apparatus applies to commodities with storage costs and to coupon bonds with more complicated formulas; here we keep discount bonds of the money market ▶ L18.4 @ 11:28. With a forward there are no cash flows until the end, when the shorts pay the longs $S_T - K$ (in financial contracts by cash netting) or the reverse ▶ L18.4 @ 12:30.

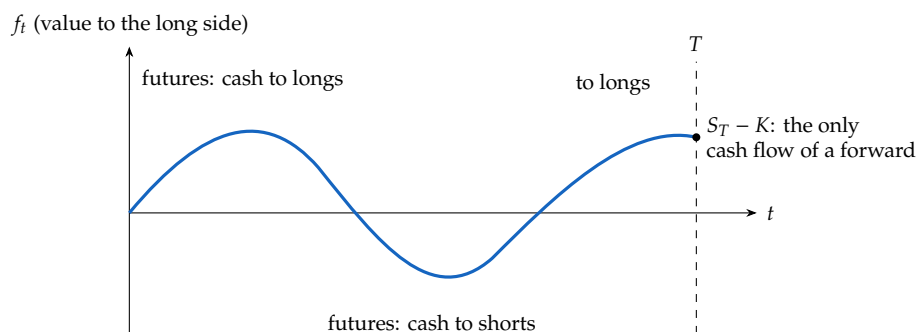


Figure 18.2: The same price history, two cash-flow profiles: a forward pays once at maturity, a futures contract pays every day as the value changes ▶ L18.4 @ 11:01]▶ L18.5 @ 1:47]. Schematic.

Margin. The payments go through margin accounts at the futures clearinghouse, and both sides post margin, since at inception both are equally likely to lose; the clearinghouse accepts liquid securities as margin, repriced every day, so collateral and contract are both marked to market. The cumulative futures payments equal the forward’s final payment, but they come every day: “a very concrete way in which views about the future are settled today” [Notes 18 p.6]. (The notes’ formula abstracts from interest on the fluctuating margin balances [Notes 18 p.5].)

Futures cash flows. Whenever the futures price changes, cash moves from losers to winners: while the forward’s value rises, from the shorts to the longs; when it falls, back [► L18.5 @ 1:06]. With a forward, who was winning in the middle does not matter; with futures the largest swings may come in the middle, and you must be ready to absorb them. Futures carry liquidity risk that forwards do not, which may be why their prices differ, though the formulas treat them as interchangeable [► L18.5 @ 2:50].

Exercise 18.2 (Forward interest parity). Three-month and six-month rates are 1.0% and 2.2% (not annualized). Compute $f_{3,6}$. One month later the two-month and five-month rates are 0.7% and 1.8%: compute the new forward rate and say whether the bank that lent forward has gained or lost.

18.4 Cash and carry arbitrage

The trade. Buy a six-month Treasury bill and finance it with a three-month repo: borrowing short, lending long; the price risk (the value of the then three-month bill in three months) is hedged by a short futures position [► L18.6 @ 0:03]. Long bill plus short repo is a synthetic forward (the six-month loan against the three-month deposit of the balance sheets): so the position is *long forward*, *short futures* for the same term. Why should it make any profit? [► L18.6 @ 2:08]

Cash and carry position	
6-month Treasury bill	3-month repo Short futures

ASIDE: Ten years of worrying
Stigum’s pages 718–722 show real cash and carry trades making money, which, by finance theory, should not be possible. “This lecture is a consequence of 10 years of me figuring this stuff out”: find something that does not seem right and keep worrying at it [► L18.5 @ 4:12].

Figure 18.3: Stigum’s cash and carry: long positions above, short positions below the line in class; long bill + short repo = synthetic long forward [► L18.6 @ 1:48].

KEY FORMULAS

Full carry pricing and the implied repo rate [► L18.6 @ 3:51]:
$$F_0 = S_0 e^{rT} \text{ (full carry), } F_0 > S_0 e^{rT} \text{ (cash and carry), } F_0 < S_0 e^{rT} \text{ (reverse).}$$

The *implied repo rate* ρ is defined by $F_0 = S_0 e^{\rho T}$: the rate that would make full carry pricing true. Cash and carry ($\rho > r$) is like borrowing at the actual repo rate r and lending at ρ : an interest rate spread.

READING: Stigum’s example, *The Money Market*, pp. 718–722

Stigum’s own numbers (materials/readings/L18-Pages_from_Stigum.pdf) are dated 28 October 1982, “when liquidity in T-bills was at its peak”.

1. Buy \$1 million of the 24 March 1983 bill, 147 days to maturity, at a discount rate of 8.26: price \$966,271.67.
2. Sell the December 1982 bill future, which expires in 56 days, at 91.72 (rate 8.28). Delivering the bill, then a 91-day bill, brings in \$979,070.00.
3. The holding-period yield over the 56 days, on a 360-day basis, is the *implied repo rate*: 8.51%.

4. With 56-day term repo at 8.25, the trade earns 26 bp. On \$10 million that is \$3,908.03.

Stigum lists the “slips twixt the cup and the lip”: commission, margin calls, convergence-price risk and back-office costs. He judges margin calls a small threat. Even a 100 bp rally on the first day would cost only $2\frac{1}{4}$ bp of the spread. In Mehrling’s reading this cash-flow timing is the whole point: what is left after the price risk is hedged is liquidity risk.

(Table 15.4 dates the futures sale “March 28, 1982”; the context shows it is 28 October 1982.)

Explained as liquidity risk. Look at the cash flows ▶ L18.7 @ 0:03:

- **long forward:** since the expectations hypothesis fails, the long side probably wins, but receives cash only at T ;
- **short futures:** probably loses, and pays all along, with margin calls until T ; if you cannot keep up, the clearinghouse liquidates your position ▶ L18.7 @ 1:45].

All price risk is gone; what is left is liquidity risk: the inflows will cover the outflows, “but not at the same time”. The position is risk-free only if one could somehow use the certain final inflow to pay the earlier outflows ▶ L18.7 @ 2:26]. The same banking view that explains the failure of EH (an imbalance in the forward market) explains Stigum’s confusing pages: forwards and futures do not trade at the same price ▶ L18.7 @ 3:28]. The notes stress that the negative flows might all come at once: the volatility of the bill’s spot price is what creates liquidity risk, and the two anomalies, the failure of EH and the profit of cash and carry, have the same origin, a mismatch in the natural forward interest rate positions of the real economy. A former teaching assistant of the course, Daniel Neilson, showed how forward–futures deviations can serve as an observable proxy for the unobservable deviation of forward rates from expected spot rates, a test of the liquidity premium theory of the term structure [Notes 18 p.8].

Or counterparty risk? A student suggests credit risk: a forward with a repo counterparty is one big payment at the end, never sure until paid, while futures are margined by an exchange, with warning from many small payments. More credit risk in the forward would call for compensation in the right direction ▶ L18.8 @ 0:03]. Typical of these debates: a spread exists, and one must list all candidate causes (cash-flow timing, counterparty risk, reinvestment) and do the empirical work to separate them; probably traders have done it and are making money, but “it’s not been published in the academic literature” ▶ L18.8 @ 1:46].

A natural banking business. Another student: futures need cash at a moment’s notice, so how does that affect banks? Banks use futures to hedge forward positions, so value fluctuations offset; the naked speculator on the other side must post margin. What the bank is not hedged against is the liquidity risk, but a bank has the discount window and interbank relationships, a comparative advantage in liquidity risk, so cash and carry is natural banking business; speculators bear liquidity risk and are paid the term premium, the difference between the forward rate and the expected spot rate ▶ L18.9 @ 0:01].

Exercise 18.3 (Implied repo rate). A six-month bill trades at $S_0 = 99$ and the three-month futures price on it is $F_0 = 99.6$; the three-month repo rate is 2% a year (continuous

COURSE NOTE

Mehrling notes that the standard explanation, reinvestment of daily futures cash flows, “goes in the wrong direction” and may even hide the size of the liquidity premium ▶ L18.7 @ 4:12]. In the literature this is the forward–futures price difference under stochastic interest rates (J. C. Cox, J. E. Ingersoll and S. A. Ross, “The relation between forward prices and futures prices”, *Journal of Financial Economics* 9, 1981); for interest-rate futures it implies a futures rate *above* the forward rate (the “convexity adjustment”), the opposite of the class’s stylized fact $f > \text{futures}$.

compounding). Compute the implied repo rate. Is a cash and carry trade profitable, and what liquidity must the trader hold?

Exercise 18.4 (Who should do it?). Explain why a bank with access to the discount window, rather than a hedge fund, is the natural holder of a cash and carry position, and what happens to the forward–futures spread if banks’ access to liquidity is restricted.

Key formulas and ideas

KEY FORMULAS

Advance clearing	forward contracts settle future payments today; asset price changes cause cash flows today
Forward parity	$(1 + r_{0,3})(1 + f_{3,6}) = 1 + r_{0,6}$; at maturity the forward rate equals the spot rate
Forward	$K = S_0 e^{rT}$, $f_t = S_t - K e^{-r(T-t)}$; one cash flow $S_T - K$ at T
Futures	$F_t = S_t e^{r(T-t)}$ reset daily; marked to market, cash flows all along
Stylized fact	$f_{3,6} > \text{futures rate} > \mathbb{E}[r_{3,6}]$: each hedger takes part of the liquidity premium
Cash and carry	long bill + short repo + short futures; profit if $\rho > r$; the residual risk is liquidity risk

Chapter 19

Interest Rate Swaps

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L19.1 FT: Sovereign debt crises	3:21	Argentina, Greece, Draghi; pension funds leaving long sovereign debt.
L19.2 Reading: FOMC Report (1952)	7:40	The Fed's secret 1952 plan: dealers, market liquidity, transmission along the yield curve.
L19.3 Treasury-swap spread, a puzzle	11:04	Negative swap spreads at the long end; what a swap is; swap lingo.
L19.4 What is a swap	2:59	A swap as a portfolio of forwards; a natural banking instrument.
L19.5 Why swap? An example from Stigum	10:37	AA and BBB share 35 basis points; exposure in a swap.
L19.6 Market making in swaps	8:42	The swap dealer's matched book; where capital market prices come from.
L19.7 Money market swaps, example	5:57	Stigum's one-year swap through J.P. Morgan.
L19.8 Life in Arbitrage Land	5:04	Stigum's "arbitrage land"; from corporate bonds to mortgages.
L19.9 Treasury-swap spread	7:21	Why nobody arbitrages the negative swap spread: liquidity risk.

Files. Transcripts `materials/transcripts/L19-P*.txt`; L19.2 has no YouTube captions and was transcribed locally from the audio (`faster-whisper, tools/whisper_segments.py`). Mehrling's notes [Notes 19 p.1]. The video published as L19.4 in the ColumbiaLearn playlist (`nCrxDIuCZ2Y`) is a copy of L19.5; the right segment (2:59) was found in a re-upload of the Coursera version (`eZk9A6Pw10g`, channel "Educational Documentaries") and transcribed locally; the links point to it. **Reading.** Federal Open Market Committee, report of the Ad Hoc Subcommittee on the Government Securities Market (1952); M. Stigum, *The Money Market*.

Motivation

Interest rate swaps are a private-sector yield curve, a kind of corporate bond term structure, usually seen as a spread over the Treasury curve [[► L19.3 @ 0:02](#)]. Since the crisis the spread has gone "a little crazy", negative at the long end: an apparent arbitrage nobody takes [[► L19.3 @ 0:46](#)]. To understand why, the lecture translates swaps into the course's balance sheets, shows the swap dealers who make the market, and finds that the prices of the capital market are made by dealers who fund themselves in the money market.

19.1 FT: sovereign debt crises

CASE STUDY: FT, November 2012: sovereign debt

(The file linked for this lecture, [materials/readings/L19-Financial_Times_-_Argentina_debt_Nov_27.pdf](#), is the European edition of the 26 November article summarised in [section 18.1](#), not the full page discussed here.) A full page on Argentina's "unforgiving debt" after the New York ruling on the holdouts ("vulture" bondholders), quoting people from a research project Mehrling was involved in; a full page on Greece, where, as Gavyn Davies writes, the eurozone has embarked on official debt forgiveness while avoiding outright cancellation of nominal debt: most Greek debt is now held by official European entities, so the renegotiation is with euro partners, not bondholders [[▶ L19.1 @ 0:07](#)]. Spanish and Italian yields have fallen after Draghi's announcement of Outright Monetary Transactions, though nothing has been bought yet, as the theory of the course would predict (a midterm question) [[▶ L19.1 @ 1:52](#)]. "This party is over": some pension funds are giving up long-dated sovereign debt, expecting yields to rise, possibly related to the Treasury and swap curves [[▶ L19.1 @ 2:32](#)].

19.2 Reading: the FOMC report of 1952

READING: FOMC, Report of the Ad Hoc Subcommittee on the Government Securities Market, 12 November 1952

"It doesn't sound interesting, does it?" A preamble says the document was secret, internal to the Fed, and came to light only through a congressional investigation. It is the Fed's own judgement of how it would run monetary policy from 1952 on, close to the money view and unlike the academic textbooks, which were written before it became public and picked up the public version [[▶ L19.2 @ 0:00](#)].

From dealer of first resort to dealers. The report asks whether operating practices "conceived in the atmosphere of war finance", made to keep a fixed pattern of prices and yields, inhibit the development of "depth, breadth and resiliency" in a market that operates without continuous support [[▶ L19.2 @ 1:24](#)]. During the war the Fed pegged Treasury rates, buying whenever a bill or bond yield tended to rise above its peg: not dealer of last resort but "dealer of first resort", in effect the underwriter, and it kept the pegs after the war so bondholders would not take capital losses [[▶ L19.2 @ 1:50](#)]. With the Treasury-Fed Accord it got authority over short-term rates back, the birth of post-war monetary policy; this document explains what it meant to do with that authority [[▶ L19.2 @ 2:43](#)].

Market liquidity through private dealers. By 1952 a dealer market in government securities had developed; "depth, breadth and resiliency" means market liquidity. Private, profit-driven dealers make markets at every maturity and arbitrage along the yield curve (with ideas such as forward interest parity), so the Fed no longer needs to support prices: it will intervene "around the edges" and keep the "tone" of the money market, knowing that dealers finance their long bonds by borrowing in the money market: leveraged dealers [[▶ L19.2 @ 3:08](#)]. Only government securities: bond and stock markets, closed down in the Depression and the war, were still in their infancy [[▶ L19.2 @ 4:57](#)].

INTUITION: Open market operations are not “changing the interest rate”

Textbooks treat changing the rate of interest and open market operations as equivalent; the report does not. Any purchase or sale by the Committee changes member banks' reserves and is “promptly reflected in the tone of the money market”: a small purchase of bills ordinarily finds a response in the long-term market, large purchases could scarcely fail to. The transmission is along the yield curve to asset prices, through leveraged dealers who are in effect shadow banks of the government securities market; nothing of the textbook story of reserves lent out by banks in a credit expansion

[► L19.2 @ 5:11].

COURSE NOTE

In class the wartime pegs are illustrative (“let’s say” 1% for bills and 2% for the ten-year). The actual pegs, from 1942, were $\frac{3}{8}\%$ on Treasury bills and $2\frac{1}{2}\%$ on long-term bonds; the bill peg was dropped in 1947 and the Treasury–Federal Reserve Accord of March 1951 freed the Fed from supporting bond prices [► L19.2 @ 1:57].

Reference: Federal Reserve History, “The Treasury–Fed Accord”; A. H. Meltzer, *A History of the Federal Reserve*, vol. 1, Chicago UP, 2003, ch. 7.

19.3 The Treasury–swap spread: a puzzle

On the day of the lecture: at one year the swap rate is a little above the Treasury rate; at ten years they are on top of each other; at 30 years the swap spread is negative [► L19.3 @ 1:07]. That suggests an arbitrage, short the “corporate” rate and long Treasuries. Why does nobody do it? The negative long-end spread has persisted throughout and since the crisis: something is seriously skewed in the structure of the capital market [► L19.3 @ 1:48].

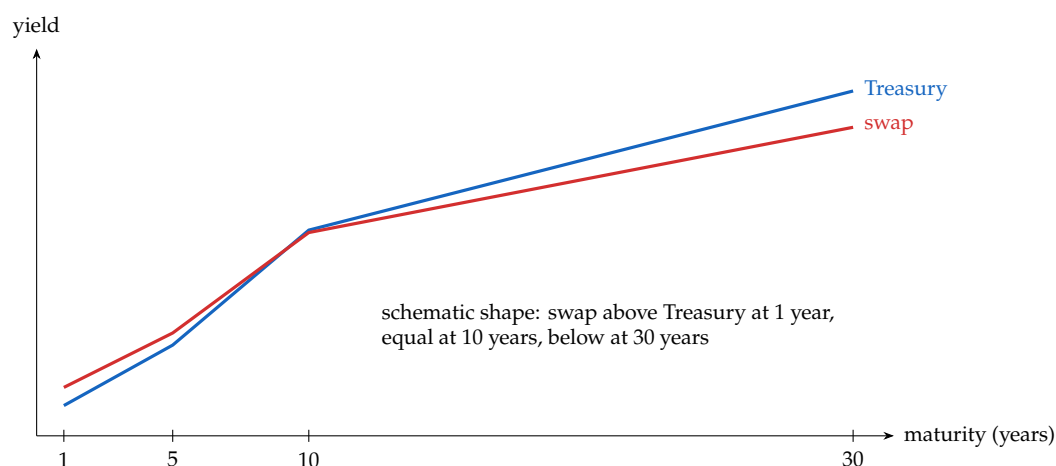


Figure 19.1: Treasury and swap yield curves, late 2012, as shown in class (four maturities joined by straight lines) [► L19.3 @ 1:07]. Shape only, not the actual numbers.

19.4 What is a swap

Stigum’s example: an AA firm can borrow relatively cheaply at a fixed long-term rate (a ten-year bond), a BBB firm most cheaply at a short-term floating rate (say three-month LIBOR); BBB worries about rolling over and would like to lock in long-term funding, AA is happy with short-term borrowing. They swap exposures [► L19.3 @ 3:11].

A swap is a parallel loan. As always, ask what IOUs are notionally swapped [► L19.3 @ 4:55]:

- AA, wanting floating: lends fixed and borrows at LIBOR, i.e. *receives fixed, pays floating*;
- BBB, wanting fixed: *pays fixed, receives floating*.

In reality there is no loan and no principal: the parties net the interest payments, so the swap is off balance sheet and does not show up as leverage [► L19.3 @ 7:42].

Definition 19.1 (New concepts: interest rate swap, swap rate). An *interest rate swap* is an agreement to exchange, on a notional principal, fixed-rate interest payments for floating-rate payments (e.g. six-month LIBOR) for the life of the swap; only the net payment changes hands. The *swap rate* is the fixed rate quoted against the floating leg [► L19.3 @ 5:59]. (*Swap*: exchange, from the old sense “to strike hands” on a bargain.)

IN THE MARKETS: Reading the FT quotes

The FT’s second section quotes swaps daily for euros, pounds, Swiss francs, dollars and yen, from one to 30 years, with a bid and an ask: for the ten-year dollar swap, 1.66–1.69, against six-month LIBOR as the footnotes say [► L19.3 @ 6:19].

Street lingo. AA, receiving fixed and paying floating, is the *seller* of the swap, “short” the swap; BBB, paying fixed, is the *buyer*, “long” the swap. Following the lingo, the course books a short swap on the liability side and a long swap on the asset side, though actual balance sheets have no such convention [► L19.3 @ 8:23].

INTUITION: A short swap is like a repo’d bond

A short swap position (receive fixed, pay floating) is like owning a fixed-rate bond and financing it short term in the repo market: a long-term repo in which a corporate bond is the collateral [► L19.3 @ 9:49].

A portfolio of forwards. A second analogy, with last lecture: a forward is like a six-month loan against a three-month deposit, one longer and one shorter term. Stigum “makes a big deal” of the point that a swap is a portfolio of forward contracts, one for each payment date, six months, one year, one and a half, two years and so on, each compared with six-month LIBOR [► L19.4 @ 0:00]. So a swap dealer can hedge in the forward market, and therefore in the futures market: the swap market is linked to the plain forward and futures markets, which is why they came first [► L19.4 @ 1:12]. A swap sounds exotic, but it is “just like things you’ve seen before”, stripped of principal payments: “a natural banking instrument”, a kind of long-term repo, locking in forward positions for ten years rather than three months, hence a market of trillions [► L19.4 @ 1:41].

INTUITION: A swap is a strip of forwards

Finance usually analyses a swap as a portfolio of forward interest rate contracts, one for each payment date (forwards, not futures, since payments are periodic, not marked to market): BBB locks in future borrowing rates, AA future lending rates. Comparing parallel loans, the first payment of a *short* swap is exactly a *long* forward, the later ones more distant forwards. Warning: being long a swap is like being short a portfolio of forwards, so a long swap is hedged with *long* futures, “rather counterintuitive (to say the least!)” [Notes 19 p.2]. The parallel loan structure itself would expand both balance sheets, and with them apparent leverage and counterparty exposure; netting the payments, and then the principal too, puts the whole thing off balance sheet [Notes 19 p.1].

Exercise 19.1 (Swap cash flows). A five-year swap on \$100 million pays 5.50% fixed against six-month LIBOR, settled semi-annually. LIBOR fixes at 5.0%, 5.6%, 6.2%. Compute the net payment each period and who pays it.

19.5 Why swap? An example from Stigum

AA and BBB ask their bankers for a floating-rate term loan and a five-year fixed-rate Eurobond [► L19.5 @ 0:01]:

	AA	BBB	difference
floating (term loan)	six-month LIBOR + $\frac{1}{8}$	six-month LIBOR + $\frac{1}{4}$	12.5 bp
fixed (five-year Eurobond)	5.375%	5.85%	47.5 bp
room for a deal	$47.5 - 12.5 = 35 \text{ bp}$		

Figure 19.2: Stigum’s quotes [► L19.5 @ 1:05].

Comparative advantage. AA has an absolute advantage at every maturity, but its advantage is much larger in the long market: BBB has a comparative advantage in short-term borrowing. Mehrling is “not sure that is really a very good way to understand this, but the math works that way” [► L19.5 @ 2:51].

The deal. AA issues fixed at 5.375% and does a swap with BBB at 5.50% against LIBOR flat; BBB borrows at six-month LIBOR + $\frac{1}{4}$ [► L19.5 @ 4:14]:

- BBB pays $(\text{LIBOR} + 0.25) + 5.50 - \text{LIBOR} = 5.75\%$ fixed: 10 bp less than its 5.85% [► L19.5 @ 5:17];
- AA pays $5.375 + \text{LIBOR} - 5.50 = \text{LIBOR} - \frac{1}{8}$: 25 bp less than its $\text{LIBOR} + \frac{1}{8}$ [► L19.5 @ 5:59].

The 35 bp are shared 25 to AA and 10 to BBB; the split is negotiated, and one would expect the better borrower to get most of it [► L19.5 @ 2:30].

AA		BBB	
(Fixed loan to BBB, 5.50) = short swap	Fixed bond 5.375%	(LIBOR loan to AA) = long swap	$\text{LIBOR} + \frac{1}{4}$ bank loan
	(LIBOR loan from BBB)		(Fixed loan from AA, 5.50)

Figure 19.3: Stigum’s swap as a notional parallel loan (in parentheses: not actual loans; only net payments are made) [► L19.5 @ 4:14].

Exposure. Why can AA get funding below LIBOR? It has an exposure to BBB. BBB has hedged LIBOR movements, not its own credit spread: if in six months its banker wants LIBOR +1, it still receives only LIBOR flat on the swap, may get into trouble rolling its loan, and then paying AA [► L19.5 @ 6:40]. But it is not credit risk on a principal: if BBB stops paying, AA stops paying and tears up the swap, losing only the change in value of net payments, far less than on an actual parallel loan [► L19.5 @ 8:27].

Market imperfections. Many swaps exist because one borrower has cheap local funding (the government likes it, capital controls, an immature bond market) and swaps it to others, breaking down inequalities across markets until the bond market grows up [► L19.5 @ 9:28]. Stigum tells of British capital controls evaded by parallel loans that were in effect currency swaps, and dates US interest rate swaps to 1981, in the Volcker tight-money period, when lesser credits were locked out of some markets completely [Notes 19 p.3].

Exercise 19.2 (Splitting the gains). With Stigum’s quotes, find the swap fixed rate at which AA and BBB split the 35 bp equally, and show each firm’s all-in cost. What limits the range of feasible swap rates?

19.6 Market making in swaps

The 35 bp are an incentive for a dealer to set the deal up and keep a couple of basis points. In the FT the bid–ask spread is 0.32–0.35 at one year, three basis points, and three basis points at two and 30 years too: the dealer’s spread [► L19.6 @ 0:01]. An investment bank stands between AA (short a swap) and BBB (long a swap), doing a swap with each, and runs a swap book, here shown as a matched book earning the spread [► L19.6 @ 1:24].

AA		Swap dealer (investment bank)		BBB	
	Short swap	Swap with BBB (bid)	Swap with AA (ask)	Long swap	

Figure 19.4: The swap dealer’s matched book [► L19.6 @ 1:24].

Where prices come from. The swap yield curve comes from dealers quoting bid and ask; customers decide whether to buy or sell and need not find a counterparty, which is the dealer’s problem; the dealer hedges imbalances with Treasuries, forwards or futures [► L19.6 @ 2:46]. With an investment bank in the middle, AA’s exposure is to Goldman Sachs, not to BBB; the dealer holds a diversified portfolio of low-rated counterparties on one side and high-rated on the other, in effect a diversified corporate bond portfolio in swap space that takes up no balance sheet room [► L19.6 @ 5:31]. Usually the fixed rate exceeds the floating, so BBB pays AA the difference over the swap’s life, while still paying its bank; if its six-month loan is not renewed, it is in crisis: that is why the market lent to it only for six months [► L19.6 @ 4:09].

KEY FORMULAS

The swap market is the government securities dealer market for corporate bonds [► L19.6 @ 6:55]. Interest rate swaps are where forward rates all along the yield curve come from; the Fed, by acting on the short end, also influences the swap curve, since the two curves

trade off each other and the same dealers often run both books, hedging one with the other. The dealers are highly leveraged and fund themselves in the money market: capital market prices are much closer to the money market than one thought.

19.7 Money market swaps

Stigum’s second example, a money market swap, mostly an interbank market [► L19.7 @ 0:02]. An AA bank receives a one-year deposit from a client but wants a floating-rate liability, to match its assets: it swaps with J.P. Morgan, one year fixed against three-month LIBOR, at 4.44% [► L19.7 @ 1:05]. Morgan, now exposed to interest rate risk, may hedge short term in futures or FRAs until it finds the opposite customer: a leveraged buyout firm (LBO) with a three-month floating bank loan that wants to swap into one year, at 4.47% [► L19.7 @ 2:56].

AA bank		J.P. Morgan (dealer)	
(3-month LIBOR loan to Morgan)	1-year deposit	(1-year, 4.44, to AA bank)	(3-month LIBOR from AA bank)
	(1-year loan from Morgan, 4.44)	(3-month LIBOR to LBO)	(1-year, 4.47, from LBO)
LBO			
	(1-year, 4.47, to Morgan)	3-month bank loan	(3-month LIBOR from Morgan)

Figure 19.5: A money market swap as notional parallel loans: Morgan receives 4.47 and pays 4.44, the LIBOR legs net out [► L19.7 @ 4:25].

Exercise 19.3 (A dealer’s mismatched book). Morgan has done the AA bank’s swap but finds no LBO for a week. Write its exposure and describe how it can hedge with three-month Eurodollar futures. What does it earn when the LBO arrives at 4.47%?

19.8 Life in arbitrage land

Swap market making happens at all maturities, one to 30 years. Liquidity lives in derivatives markets because they use little capital: no loans, “swaps of IOUs all the way down”, profit from selling one IOU for more than you buy the other, cash flows netted. “This is banking, but without the actual loans”: the core of the modern system is not Jimmy Stewart banking but swap dealers quoting two-sided markets and so making prices [► L19.8 @ 0:02].

COURSE NOTE

Students often object that 4.47 on a liability exceeds 4.44 on an asset, which does not look like profit. The lingo trips you up: think of the parallel loans behind the swaps. Morgan is paid 4.47 by the LBO and pays 4.44 to the AA bank, and the two LIBOR legs net: three basis points. Booking a “short swap” as a liability does not mean subtracting it from assets to get net worth [► L19.7 @ 4:25].

READING: M. Stigum, *The Money Market*, p. 900: arbitrage land

Stigum describes money market swaps as living in what could be called “arbitrage land”: traders arbitrage swaps against futures, against cash, against FRAs, FRAs against futures, and so on. An event that moves one rate sends a ripple through the related markets, creating basis points for every player, except perhaps the unhedged speculator in the futures pit, who usually is the one who loses [► L19.8 @ 1:24].

A hierarchy of arbitrages. The chain ends in the futures market, where price risk is hedged and liquidity risk is borne: hence last lecture on forwards and futures [► L19.8 @ 2:26]. The same idea is in the 1952 FOMC report: government securities dealers doing term structure arbitrage, buying the five-year when cheap, financing it in a primitive repo market, selling the ten-year. “1952 is the future”, extended from government securities to corporate securities and then to mortgages: an RMBS is a 30-year bond with option features (early repayment), and the swap apparatus built for corporate securities was moved into mortgage space, producing a global market in household bonds: “AA, LBO, subprime” [► L19.8 @ 2:47].

KEY FORMULAS

Swaps in the hierarchy (Mehrling’s notes) [Notes 19 p.4][Notes 19 p.5][Notes 19 p.6]. Stigum calls a swap a synthetic corporate bond (p. 906): a short swap is a long bond financed by repo, hedged with a short position in Treasuries or Eurodollar futures, leaving basis risk (the swap rate is a corporate rate) and counterparty risk on both sides. The swap dealer is a government securities dealer for corporate bonds: at every maturity a private rate and a government rate, kept in touch by speculators on the spread. The swap market, by notional principal larger than the bond market itself, may seem to rival the government market; in good times its liquidity matches, but in bad times “the hierarchical structure is restored”: swap dealers sit below government bond dealers, who sit below the banks that make markets in money, who sit below central banks. Swaps spread stress across the system and across time and unite separate markets, evading regulatory imperfections; but the sequence swap (over the counter) → FRA (interbank) → futures also concentrates stress: “everything comes back to futures”. In the money market swap, Morgan hedges temporarily with long futures, a position that carries liquidity risk, until the LBO swap closes it out (“a swap is a strip of FRAs, and a FRA is a single-set swap”, p. 897).

19.9 The Treasury–swap spread explained

To exploit a swap curve below the Treasury curve, one would borrow at the “corporate” rate and lend at the Treasury rate: be long a Treasury, financed in repo with the Treasury as collateral, and long a swap (pay fixed, receive floating) [► L19.9 @ 0:22]. Corporations that can borrow are in effect doing it on their own balance sheets, issuing 30-year debt and holding cash or Treasuries; it is not enough to close the gap [► L19.9 @ 2:27].

Arbitrageur	
30-year Treasury	Repo (short-term funding)
Long swap (pay fixed, receive floating)	

COURSE NOTE

On the 1952 report see [section 19.2](#). It is that of the FOMC’s Ad Hoc Subcommittee on the Government Securities Market (chaired by William McChesney Martin, November 1952), which recommended that open market operations be confined to the short end of the market, normally Treasury bills (“bills only”), leaving the rest of the yield curve to dealers’ arbitrage. Reference: Federal Reserve, *Federal Reserve Bulletin*, 1953; FOMC Minutes, March 1953.

Figure 19.6: The arbitrage that would close a negative swap spread: no credit risk, but borrowing short to lend long [► L19.9 @ 1:24].

INTUITION: Why the gap stays open

The position borrows short and lends long: no credit risk (it is the Treasury), but liquidity risk, since one does not know whether, or at what price, the funding can be rolled. Even with the market flooded by QE, people do not want that risk. Other stories (capital constraints coming with Basel III) are possible; any story must explain why nobody has done the arbitrage for four years [► L19.9 @ 3:09].

Where the mortgage rate comes from. The interest rate swap market is enormous, “trillions, trillions, trillions”. When a Jimmy Stewart banker quotes you a mortgage rate, it comes from here: the mortgage will be packaged and sold as a bond, so the banker looks at the swap curve for mortgage-backed securities. Prices are made in dealer markets in derivatives, with supply and demand for loans behind the scenes: “that’s the way modern banking works” [► L19.9 @ 5:54].

Exercise 19.4 (The negative swap spread). The 30-year Treasury yields 2.85% and the 30-year swap rate is 2.55%. Write the balance sheet of the arbitrage, the annual carry if three-month repo costs 0.20% and three-month LIBOR is 0.30%, and the risks that remain.

ASIDE: The fiscal cliff

A manufactured crisis: unless Congress agreed by a date, taxes would rise and spending fall, with large demand effects. Mehrling sees political brinkmanship; it may lead people to stay long cash, and firms to pay out cash as dividends ahead of tax rises, but it cannot explain a four-year anomaly [► L19.9 @ 4:11].

Key formulas and ideas

KEY FORMULAS

Swap	notional parallel loan, fixed against floating; only net payments; off balance sheet
Lingo	receive fixed = seller, short the swap; pay fixed = buyer, long the swap
Analogy	short swap $\approx$ fixed-rate bond financed by term repo
Stigum’s gains	$(5.85 - 5.375) - (\frac{1}{4} - \frac{1}{8}) = 0.35$: AA gets 25 bp ($\text{LIBOR} - \frac{1}{8}$), BBB 10 bp (5.75%)
Dealers	swap book earns the bid-ask (about 3 bp); swap curve made by leveraged dealers funded in the money market
Arbitrage land	arbitrages chain to the futures market, where liquidity risk ends up
Negative spread	long Treasury in repo + long swap: no credit risk, but funding liquidity risk

Chapter 20

Credit Default Swaps

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L20.1 FT: Internationalization of the Euro	5:07	Offshore euro trading in London; bonds replacing syndicated loans.
L20.2 Credit indices	3:40	The Markit CDX high-yield index.
L20.3 Fischer Black (1970), risk-free security	2:14	A risky bond = risk-free bond + interest rate risk + default risk.
L20.4 What is a Credit Default Swap (CDS)	5:12	Buyer and seller of protection on the balance sheet.
L20.5 Corporate bonds	3:43	Price as present value; spreads and ratings.
L20.6 CDS pricing	11:24	Bond + IRS = LIBOR + s ; CDS premium u ; arbitrage $s = u$; is it insurance?
L20.7 Market making in CDS	4:03	Dealers hedge single names with the index.
L20.8 Example: Negative basis trade and Liquidity Risk	10:20	UBS: AAA CDO tranches, insured, funded in the money market.
L20.9 Example: Private backstop of marketmaking in CDS	15:15	AIG, Goldman Sachs and collateral calls: “liquidity kills you quick”.
L20.10 Example: Synthetic CDO as Collateral Prepayment	6:43	Paulson, Abacus and IKB.

Files. Transcripts `materials/transcripts/L20-P*.txt`; L20.1 has no YouTube captions and was transcribed locally from the audio (faster-whisper). Mehrling’s notes [Notes 20 p.1].

Motivation

All semester Mehrling promised that the course’s balance sheets would handle credit default swaps; now he delivers [► L20.2 @ 0:06]. Think of a corporate bond with a lot of credit risk: a credit default swap peels that risk off and sells it separately, just as an interest rate swap peels off interest rate exposure [► L20.2 @ 3:11]. Three examples from the crisis (UBS, AIG, Abacus) then show that the trouble came not from defaults but from liquidity.

20.1 FT: the internationalization of the euro

Mehrling had just come back from Rome, where Italian central bankers and securities regulators asked his shadow banking team whether shadow banking could channel credit to small and medium-sized enterprises: they depend on banks, and banks, having bought government bonds, now need to build up capital and are not lending [► L20.1 @ 0:00]. He reads two articles of the day as symptoms of sea changes in the European financial system [► L20.1 @ 1:24].

CASE STUDY: FT, November 2012: two European evolutions

1. “France targets UK euro trade supremacy”: 40% of euro trading takes place in London, and France wants it inside the eurozone, under control [► L20.1 @ 1:35]. Mehrling recalls the rise of the Eurodollar market in London and the old American worry about losing control of the dollar; the US concluded it could not stop it, and the same holds for the euro: people who want to trade euro derivatives will find somewhere to do it, London or Hong Kong. A “little naive”, but a sign of the maturation of the euro as an international currency [► L20.1 @ 2:03];
2. “Syndicated bank loans in eurozone slump to ten-year low”: top-rated corporations no longer take syndicated loans (a bank making a loan much bigger than it could hold and parcelling out pieces to other banks) but issue bonds directly, in big amounts: the capital market replaces bank lending, as in the US evolution from indirect to direct finance (lecture 17) [► L20.1 @ 3:21].

One evolution is in the money market (London), the other in the capital market; link the two and you have shadow banking, money market funding of capital market lending, driven by market forces, while “none of the regulations are right for this” [► L20.1 @ 4:15].

20.2 Credit indices

IN THE MARKETS: The Markit CDX family

Markit’s CDX indices are, in its words, the standard tradable family of credit default swap indices for North America and emerging markets: they offer exposure to a diversified portfolio of credits, a hedge of credit risk, and liquidity that flows into the single-name CDS market [► L20.2 @ 0:27]. The CDX.NA.HY (North America high yield, i.e. junk) index: spreads of 450–580 basis points over a riskless rate between late September and the day of the lecture; the index is set at 100 and moves inversely to the spread (about 530 bp at that point). Its constituents are 100 companies, each with 1% weight, with ratings such as BB, B, CCC [► L20.2 @ 1:29].

COURSE NOTE

The figure of 40% is as said in class, quoting the FT. The “30 years ago” of the Eurodollar debate is approximate: the Eurodollar market grew in London from the late 1950s, and the US debate on controlling it was intense in the late 1960s and 1970s [► L20.1 @ 1:46].

20.3 Fischer Black’s vision (1970)

In an unpublished 1970 memo, which Mehrling read while writing Black’s biography, the young Fischer Black imagined a long-term corporate bond sold to three separate persons: one supplying the money, one bearing the interest rate risk and one the default risk, the last two putting up no capital but perhaps posting collateral. “That’s the world that came to be” [► L20.3 @ 0:05].

KEY FORMULAS

Black’s decomposition ▶ L20.3 @ 0:48:

risk-free security = risky (corporate) bond + interest rate hedge (IRS) + credit hedge (CDS).

Exercise 20.1 (Black’s decomposition). Write the balance sheet of an investor who holds a ten-year BBB bond, an interest rate swap and a CDS on it, and show that its net exposure is a Treasury bill. Who bears the interest rate risk and who the default risk?

HISTORY: Fischer Black

Fischer Black (1938–1995), co-author of the Black–Scholes option pricing formula (1973), worked at Arthur D. Little, the University of Chicago, MIT and Goldman Sachs. Mehrling’s biography draws on his unpublished papers, including the 1970 memo “Fundamentals of Liquidity”.
Reference: P. Mehrling, *Fischer Black and the Revolutionary Idea of Finance*, Wiley, 2005.

20.4 What is a credit default swap

There is a buyer and a seller of “insurance”. The buyer is *long the swap* and *short credit risk* ▶ L20.4 @ 0:04]. Balance sheet construction ▶ L20.4 @ 0:56]:

1. start from a risky corporate bond;
2. add (in brackets: notional) a long Treasury bond and a short position in the same corporate bond, of the same maturity: the net exposure is now a Treasury bond. The pair is the CDS: it carves off the credit risk, the 530 bp;
3. add a short Treasury bond and a long Treasury bill: the net exposure is a Treasury bill. This pair, long fixed and short floating, is an interest rate swap.

Buyer of protection	Seller of protection
Corporate bond	CDS (short swap, long credit risk)
[+Treasury bond, –corporate bond] = CDS (long swap)	IRS
[+T-bill, –Treasury bond] = IRS	

Figure 20.1: A CDS and an IRS as notional swaps of IOUs: the hedges are booked as contingent assets of the buyer and contingent liabilities of the seller ▶ L20.4 @ 2:50].

Definition 20.1 (New concepts: credit default swap, protection buyer and seller). A *credit default swap* (CDS) is a contract in which the *protection buyer* pays a periodic premium and, if the reference bond defaults, receives from the *protection seller* the difference between face value and the bond’s post-default value ▶ L20.4 @ 0:04][▶ L20.6 @ 4:28]. (Default: from Old French *defaute*, a failing or lack.)

The course books any hedge as a contingent asset, so the buyer lists both the long credit exposure (the bond) and the short one (the CDS) as assets; real-world practice is not consistent, often off balance sheet ▶ L20.4 @ 3:12]. The seller is long credit risk and short the

swap: the lingo is consistent between CDS and IRS. “Credit default swaps are not some weird exotic creature of Wall Street” [► L20.4 @ 4:33].

Exercise 20.2 (CDS cash flows). A protection buyer on \$10 million face pays $u = 400$ bp a year. The bond defaults at the end of year 3 and then trades at 35. Write the cash flows each year. What is the buyer’s position worth just before default if spreads have widened to 900 bp?

20.5 Corporate bonds

For the first time in the course, a corporate bond: a fixed coupon every six months or year for, say, ten years, and the principal at the end; unlike a mortgage it is not amortized [► L20.5 @ 0:04].

KEY FORMULAS

Bond price [► L20.5 @ 1:05]: with coupons C_t , face value F at maturity T , and a risk-weighted discount rate (the risk-free rate plus a credit spread δ , e.g. 530 bp),

$$P_0 = \sum_{t=1}^T \frac{C_t}{(1+r+\delta)^t} + \frac{F}{(1+r+\delta)^T}.$$

Bonds have ratings, and at any moment there are credit spreads between rating classes, which change over time. So a bond’s price changes when the spread for its rating changes and when its rating changes, a “double whammy” if a downgrade comes with wider spreads [► L20.5 @ 2:12].

20.6 CDS pricing

Step 1: strip the interest rate risk. Take a ten-year bond with constant coupon and swap fixed for LIBOR: what remains is a ten-year floating-rate bond paying LIBOR + s , with the credit spread s locked in [► L20.6 @ 0:02].

Step 2: the CDS as a swap. Each year the bond survives, one side pays LIBOR times face value and the other LIBOR + u : the protection buyer pays the net premium u times face value [► L20.6 @ 1:51]. If the bond defaults, say in period five, the buyer swaps it for a Treasury bond: it receives face value minus the bond’s price at that time [► L20.6 @ 4:05]. The contract follows the two parts of a bond: the annual (coupon) business and the terminal payment if it defaults [► L20.6 @ 5:35].

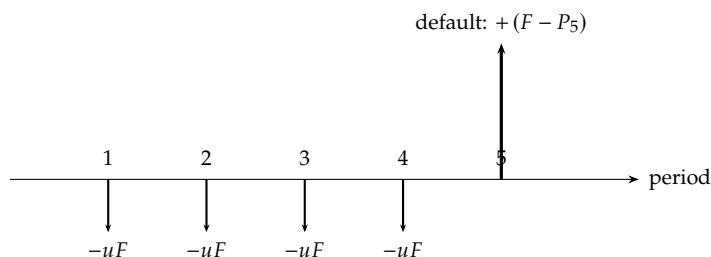


Figure 20.2: Cash flows to the protection buyer: the premium u times face value F while the bond survives, face value minus the bond's price P_5 at default [► L20.6 @ 3:45].

KEY FORMULAS

CDS pricing by arbitrage [► L20.6 @ 6:36]: the CDS premium u should equal the bond's credit spread s ; otherwise go long the bond and insure it (if $u < s$), or short the bond and sell insurance (if $u > s$). This is how CDS are priced, "assuming that there's no liquidity premium anywhere in this picture". The difference $u - s$ is the *CDS-bond basis*.

The notes call the protection seller's IOU a *mirror bond*: it promises LIBOR + U on face value as long as the issuer pays, and the liquidation value on default; the buyer's IOU promises LIBOR, and face value on default. On default the buyer recovers F in full: P_5 from the bond, $F - P_5$ from the swap. U is the premium that makes the two IOUs equal in value at inception; but since any change in the spread S moves its value, "CDS is not so much insurance against eventual default as it is insurance against change in the credit spread" [Notes 20 p.3][Notes 20 p.4]. Pooling CDS reduces idiosyncratic risk (the less correlated, the better), letting sellers charge less: a flourishing swap market should lower the price of credit risk overall, "and that is exactly what happened"; for subprime the index used for hedging was ABX, the index of top-tranche mortgage CDOs [Notes 20 p.1][Notes 20 p.4][Notes 20 p.5].

Not just insurance. A CDS is a promise to pay a fixed premium for its life, but its value fluctuates, inversely to the bond, whenever the warranted discount rate changes; so CDS are used not only as insurance against default but as a way to get exposure to credit risk, traded and reversed like any liquid derivative. Unlike fire insurance, you are not just waiting for the house to burn [► L20.6 @ 8:00].

INTUITION: Is a CDS insurance?

"Insurance" is used to remember which side of the balance sheet to book; but many people dislike the word, on both sides of the political spectrum. If it were insurance it would be regulated as insurance, with reserves and suitability standards; and insurance works by pooling independent risks (fires), while credit risk is systemic: corporate bonds and their spreads move together. The mathematicians and theorists say: it is not insurance, just a swap [► L20.6 @ 9:43].

20.7 Market making in CDS

Opposite the protection buyer is typically a dealer writing CDS on bonds $i, j, k, \dots$: a portfolio of CDS is like a diversified portfolio of corporate bonds [► L20.7 @ 0:03]. The dealer cannot find someone wanting exactly the opposite single-name positions, so it hedges with a CDS on the CDX index, which is correlated with its portfolio; on the other side may be an asset manager seeking exposure to the bond market as a whole [► L20.7 @ 1:08].

Hedger (single name)		CDS dealer (investment bank)		Asset manager	
CDS on bond i (long)		CDS on CDX index	CDS on bonds $i, j, k, \dots$		CDS on CDX index (long credit risk)

Figure 20.3: A CDS dealer hedging a portfolio of single names with the index [► L20.7 @ 1:31].

With a dealer in the middle, order flow pushes CDS prices around without necessarily moving bond prices; someone must arbitrage to bring them in line. So arbitrage opportunities appear whenever CDS prices are out of line with bond prices, “and sometimes it is” one [► L20.7 @ 2:55].

20.8 UBS: the negative basis trade and liquidity risk

UBS lost a lot of money on what it thought was a riskless trade, and its regulator made it write a public report explaining how; Mehrling wishes US regulators had required the same of Citibank and the TARP banks [► L20.8 @ 0:03]. UBS noticed that the spread s on a security exceeded the cost u of insuring it against default, at least from AIG [► L20.8 @ 0:45]:

- 1. it bought AAA tranches of CDOs on mortgage-backed securities, some on sub-prime mortgages, with the credit risk supposedly carved off into lower tranches [► L20.8 @ 1:26];
- 2. its European regulators required insurance to use them as bank collateral, so it bought CDS, from AIG among others: “double insurance” [► L20.8 @ 2:09];
- 3. it funded them in the money market, asset-backed commercial paper or repo, with the tranche as collateral: money market funding of capital market lending, shadow banking, but right on UBS’s balance sheet, unlike Citibank’s structured investment vehicles off balance sheet [► L20.8 @ 3:17].

UBS (negative basis trade)	
AAA CDO tranche (yield LIBOR + s)	Money market funding (ABCP, repo), collateral: the tranche
CDS bought from AIG (premium $u < s$)	

Figure 20.4: The negative basis trade [► L20.8 @ 1:48].

The numbers. The notes give them from UBS’s report: UBS paid AIG 11 basis points for 100% protection on a super-senior CDO tranche, financed the tranche in the wholesale money market, and netted an apparently riskless 20 basis points; “because it was apparently riskfree, they did massive amounts of it” [Notes 20 p.5]. In the notes’ notation, if the CDS premium U is below the spread S , the buyer of protection swaps out the credit risk and keeps $S - U$ over LIBOR.

Booking 20 years of profit today. The *negative basis trade* left money over each period for the life of the tranche, 10 or 20 years; UBS booked the whole stream as profit today and paid bonuses on it [► L20.8 @ 4:42].

The liquidity spiral. Then small doubts about the AAA tranche: its value fluctuated a little; the CDS moved the other way, fine; but the tranche, not the CDS, was the collateral, so the funding could not be rolled, the tranche had to be sold, pushing its price down further, and others doing the same trade made a downward liquidity spiral [► L20.8 @ 5:47]. Moreover, AIG’s insurance was expensive, so late in the game UBS insured only against a 2% fall (from 100 to 98) and kept the tail risk itself [► L20.8 @ 6:48]. They felt they took no risk at all and lost billions, because they assumed they could always roll their funding [► L20.8 @ 7:50].

Exercise 20.3 (Negative basis). A AAA tranche yields LIBOR + 60 bp, protection costs 20 bp, and repo funding costs LIBOR + 10 bp. Compute the annual carry on \$1 billion. The tranche’s price falls from 100 to 96 and the repo haircut rises from 2% to 10%: how much cash must be found, and from where?

20.9 AIG and Goldman Sachs: a private backstop

Goldman Sachs, a dealer in CDS selling protection to its clients (not completely matched book: short credit risk when it mattered), kept buying CDS from AIG, which held CDS liabilities and capital [► L20.9 @ 0:00]. When the CDS came into the money, AIG’s contingent liability, zero when written, became, say, \$30 billion owed to Goldman [► L20.9 @ 1:36]. Goldman had written the contracts so that if AIG’s rating fell below AAA it would post margin; Moody’s downgraded AIG, and it had to *pay* the \$30 billion, not just owe it [► L20.9 @ 2:19].

Goldman Sachs (dealer)		AIG	
CDS bought from AIG	CDS sold to clients	Capital (bonds)	CDS on AAA securities
(+\$30 bn when in the money: margin received)	(+\$30 bn owed to clients)	(–cash paid as margin)	(+\$30 bn contingent liability)

Figure 20.5: AIG as private backstop of CDS market making; “say \$30 billion” in class; the notes give about \$30 billion of collateral posted to Goldman’s segregated account by the time AIG failed [► L20.9 @ 1:16][Notes 20 p.6].

INTUITION: Liquidity kills you quick

AIG thought it was insuring against defaults that would never happen; but you can go bankrupt without any default if the CDS fluctuate in value and you must pay margin. The contractual promise to post collateral killed AIG; monoline insurers that wrote CDS without margin saw the marks go against them but paid nothing. “Insolvency you can survive for a long time. . . but liquidity kills you quick”: AIG, the anchor of the system, fell two or three days after Lehman in September 2008, and the system went into free fall [► L20.9 @ 2:40].

COURSE NOTE

A student asks why the lenders did not accept the CDS as collateral together with the tranche. Perhaps by analogy with Bagehot’s bills, where the acceptance was attached to the bill; here they are separate securities, and money market funds are restricted in the collateral they may accept. Mehrling recalls that when AIG was failing he proposed that the government write CDS itself, since a CDS is like an acceptance, which is what a liquidity problem needs [► L20.8 @ 8:11].

HISTORY: The UBS report

At the request of the Swiss Federal Banking Commission, UBS published in April 2008 a “Shareholder Report on UBS’s Write-Downs”, describing among other things its negative basis trades and the “Amplified Mortgage Portfolio” (AMPS), in which only a small first slice of losses on super-senior positions was hedged.
Reference: UBS AG, *Shareholder Report on UBS’s Write-Downs*, 18 April 2008.

COURSE NOTE: What the UBS report says

The report is in the course files (materials/readings/L20-ubs-shareholder-report.pdf). It confirms the 11 bp and the 20 bp, with some differences in detail.

- **Who sold the protection.** In the NegBasis trades “a counterparty, such as a monoline insurer” provided 100% loss protection (p. 14). The report does not name AIG as the seller; class and notes do.
- **The cost of hedging.** A NegBasis hedge cost about 11 bp, an AMPS hedge about 5–6 bp (p. 30).
- **The carry.** Several super-senior positions had a “thin positive carry of approximately 20 bp” over their funding cost (p. 30).
- **How much AMPS protection.** AMPS trades covered “typically between 2% and 4%” of the nominal position (“100 to 98” in class) (p. 14).
- **Sizes.** The super-senior book grew to

INTUITION: Was Goldman paid at par by the government?

The notes correct “a lot of loose talk” [Notes 20 p.6]: since the CDS were marked to market, Goldman already held the collateral, it had already been paid. When AIG could not meet further collateral calls the government took over, lending \$85 billion; that money was used to acquire the referenced securities at liquidation value to end the swaps. In the algebra of the pricing example, AIG had already paid $F - P_5$; the government lent AIG P_5 to buy the bonds, which went onto the Fed’s balance sheet as Maiden Lane II and III.

Should the government have clawed it back? Neil Barofsky (SIGTARP), whose book is on the final-exam reading, wrote a report on AIG: Goldman and, even bigger, Société Générale sucked the capital out of AIG, the government took over AIG, and the question was what to do about the money already paid to Goldman [► L20.9 @ 4:03]. But to the extent Goldman was a market maker with offsetting positions, taking back the \$30 billion would have left it unable to pay its clients and the whole thing unravels; “I don’t say that Geithner necessarily did the right thing”, but it was not simply unearned profit: the point of a CDS is that when the corporate bond goes south you replace it with a Treasury [► L20.9 @ 5:05].

No fraud, a misunderstanding of risk. Mehrling is not sure there was fraud, at AIG or at UBS: a systematic misunderstanding of risk, a supposedly risk-free arbitrage on which you bet the bank, and do more of it. “If it looks like free money, look again”, with the money view: is the arbitrage just a liquidity premium you think should be zero? Then you are becoming the lender of last resort to the whole system without being a central bank that can print money [► L20.9 @ 7:28].

Moral hazard? A student suggests they took the risk expecting a bailout. There is always some such calculation, but Mehrling thinks they were surprised. The one really relying on the government was Goldman: it chose AIG though it was more expensive (UBS found it too expensive), because AIG promised margin and had a huge capital base; and AIG, reserving nothing and paying bonuses on each sale, could not be allowed to fail, with fire insurance policies worldwide. But a counterparty that dies is not the right counterparty: Goldman surely did not expect to destroy AIG [► L20.9 @ 9:18].

What triggered payment? Without margin, the premium flows and a large payment comes only at actual default; for AAA tranches, with many tranches beneath, defaults rarely reached them. It was not default but the fluctuation in CDS values that caused the problem. That is why many underestimated the crisis: “there just aren’t a lot of subprime mortgages”, ignoring leverage, liquidity and the embedding in world money and capital markets. Those who took the low tranches and knew they bore risk mostly survived; those who thought they took no risk failed [► L20.9 @ 12:24].

Exercise 20.4 (Margin or no margin). Compare an AIG-style CDS seller that posts margin when downgraded with a monoline that posts none, if the marked-to-market value of the CDS rises by \$5 billion but no default occurs. Which one fails, and why?

about USD 50bn by September 2007, of which 27bn were AMPs or unhedged. Of the 20.8bn bought from third parties, 15bn were fully hedged NegBasis positions (p. 15).

- **Losses.** By the end of 2007 the NegBasis positions accounted for about 10% of super-senior losses, AMPs for 63% and unhedged positions for 27%. Total subprime-related losses for 2007 were USD 18.7bn.

So, by the report’s own breakdown, most of the loss came from the thinly hedged AMPs book, not from the fully insured trade.

COURSE NOTE

The AIG executive is not named in class. Joseph Cassano, head of AIG Financial Products, told investors in August 2007 that it was hard to see a scenario in which the unit would lose “one dollar” on these transactions (“a nickel” in class). The US government took a 79.9% equity stake in AIG in September 2008 (“80” in class).

References: Financial Crisis Inquiry Commission, *Final Report*, 2011, ch. 13; SIGTARP, *Factors Affecting Efforts to Limit Payments to AIG Counterparties*, November 2009.

COURSE NOTE: What the SIGTARP report says: the \$30 billion was not Goldman’s alone

The report is on the course reading list ([materials/readings/L20-SIGTARP_AIG_counterparties.pdf](#)). Its figures:

- **Collateral postings.** AIG’s collateral postings on its multi-sector CDO book grew from \$2.7bn at the end of 2007 to \$31.5bn at 30 September 2008, \$32.8bn in all (Table 1). So the “\$30 billion” of class and notes is the total posted to *all* counterparties.
- **Goldman’s share.** As of 7 November 2008 Goldman Sachs held \$8.4bn of collateral and received \$5.6bn from Maiden Lane III, \$14.0bn in all. Société Générale received the most: \$9.6bn of collateral plus \$6.9bn, \$16.5bn in all (Table 2, p. 20).
- **The rescue.** The Fed authorised an \$85bn loan on 16 September 2008. Maiden Lane III (November 2008) was authorised to borrow up to \$30bn from the New York Fed and actually borrowed \$24.3bn, plus \$5bn of AIG equity. It paid \$29.6bn for CDOs with a face value of \$62.1bn: \$27.1bn to the counterparties and \$2.5bn to AIGFP for overcollateralisation.
- **Paid at par.** Together with the \$35bn of collateral they kept, the counterparties received \$62.1bn, “effectively the par value” of the swaps.
- **No haircut.** The New York Fed asked eight of the largest counterparties to accept a haircut. Seven refused, and the eighth, UBS, would accept 2% only if the others made a similar concession (p. 15).

This supports the notes’ point that the counterparties already held most of the money as collateral.

20.10 Abacus: the synthetic CDO as collateral prepayment

John Paulson wanted to short subprime, expecting the housing bubble to end. He bought CDS on CDO tranches of mortgage-backed securities, insurance without owning the securities [► L20.10 @ 0:02]. Goldman Sachs set up an entity, Abacus, which took the other side of the CDS, credit exposure as if owning RMBS, “just a side bet” [► L20.10 @ 1:04]. Abacus sold the German bank IKB a CDO tranche paying a spread over the risk-free rate; IKB borrowed to buy it, and Abacus invested the proceeds in Treasury bills [► L20.10 @ 2:08].

Paulson		Abacus (synthetic CDO)		IKB	
CDS on CDO tranches (long protection)		Treasury bills	CDS sold to Paulson CDO tranche sold to IKB	CDO tranche	Borrowing

Figure 20.6: The synthetic CDO: T-bills plus a short CDS replicate a long position in mortgage-backed securities [► L20.10 @ 2:28].

Definition 20.2 (New concepts: synthetic CDO). A collateralized debt obligation (CDO) issues tranches backed by a pool of debt; a *synthetic* CDO owns no such debt but replicates the exposure with risk-free securities plus CDS sold on the reference assets [► L20.10 @ 2:49].

When the tranches fell, the CDS rose and the T-bills went to Paulson as margin. From a liquidity perspective that is the genius of it: IKB had paid 100% upfront, so there was no problem getting paid, no bill to send to IKB that it could dispute, no need to liquidate RMBS exposure [► L20.10 @ 3:30]. There were issues about Paulson’s role in choosing the tranches most likely to default and about Goldman’s representations, and a fine was paid; but no issue of payment. IKB was bailed out by the German government; AIG’s counterparties included Société Générale: the international aspects again [► L20.10 @ 4:55].

Key formulas and ideas

HISTORY: Abacus 2007-AC1

In April 2010 the SEC charged Goldman Sachs with misleading investors in the synthetic CDO Abacus 2007-AC1, whose reference portfolio Paulson & Co. had helped select while betting against it; Goldman settled in July 2010 for \$550 million. IKB, an early casualty of the crisis in 2007, was rescued with the support of the state-owned KfW.

References: SEC press releases 2010-59 (16 April 2010) and 2010-123 (15 July 2010); G. Zuckerman, *The Greatest Trade Ever*, 2009 (cited in Mehrling’s notes, which also recall, from G. Tett, *Fool’s Gold*, 2009, that the first CDS were built in this synthetic way for J. P. Morgan to hedge tail risk on corporate loans) [Notes 20 p.6].

KEY FORMULAS

Black (1970)	risky bond + IRS + CDS = risk-free bond
CDS	notional long Treasury + short corporate bond of the same maturity; buyer long the swap, short credit risk
Bond price	$P_0 = \sum C_t / (1 + r + \delta)^t + F / (1 + r + \delta)^T$; moves with spreads and ratings
Pricing	bond + IRS = LIBOR + s ; CDS premium u ; no-arbitrage $u = s$ if no liquidity premium
Negative basis	$u < s$: buy bond, buy protection, fund in money market; liquidity risk in the funding
AIG	margin calls on in-the-money CDS: "liquidity kills you quick"
Abacus	synthetic CDO = T-bills + short CDS: collateral prepaid

Chapter 21

Central Banking for Shadow Banking

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L21.1 FT: Regulation of shadow banking	3:34	Bail-in debt for banks; central counterparties for swaps.
L21.2 Shadow banking vs traditional banking	6:55	Pozsar’s diagram; a stylized shadow banking system without government backstops.
L21.3 Liquidity and solvency backstops	7:18	Immature private backstops: liquidity puts and CDS; collapse onto the Fed.
L21.4 Global dimension	5:16	Dollar funding of Korean banks and of shadow banks; a mature funding system, an immature risk transfer system.
L21.5 Evolution of modern finance	3:19	Securitization, globalization, derivatives: risk transfer.
L21.6 What is shadow banking?	12:22	Money market funding of capital market lending; a world of only shadow banking.
L21.7 Backstopping the market makers	7:46	The Fed’s balance sheet in December 2011 as a dealer in swaps.
L21.8 Regulation of systemic risk	6:27	Inherent instability of credit: boom and bust through dealers’ balance sheets.
L21.9 Regulation of Collateral and Payment	10:29	Normal fluctuations: collateral flows and payment flows.
L21.10 Private backstop, and public	8:20	Four principles; the central bank as outside spread.

Files. Transcripts `materials/transcripts/L21-P*.txt`; L21.4 has no YouTube captions and was transcribed locally from the audio (faster-whisper). Mehrling’s slides [[Notes 21 p.1](#)].

Motivation

This is the last lecture about banking: it draws together forwards and futures, FX swaps, interest rate swaps and credit default swaps, and asks what they mean for banking and central banking; the next lecture connects the course to economics [[► L21.2 @ 0:03](#)]. The question: why is shadow banking in our future, “given how desperately it has failed us”, and how should it be managed? [[► L21.2 @ 2:27](#)]

21.1 FT: old banking and new banking

CASE STUDY: FT, December 2012: two articles on reform

“Regulators in long-term debt plan for big banks”: more long-term debt, subordinated and “bail-inable” bonds that convert to equity when a bank runs into trouble, could lessen reliance on cheaper short-term funding. This is the traditional Basel approach (capital adequacy), now extended from the equity tranche to subordinated debt, for traditional banks [▶ L21.1 @ 0:07]. Three pages later, “Regulators wrestle with hidden risks in swaps shake-up”: standardized interest rate swaps and credit derivatives to be traded on exchanges and cleared through central counterparties, to increase price transparency, ensure adequate collateral and keep taxpayers from paying next time [▶ L21.1 @ 1:53]. Old banking and new banking, a few pages apart [▶ L21.1 @ 2:55].

Definition 21.1 (New concepts: central counterparty, bail-in). A *central counterparty* (CCP) interposes itself between the two sides of every trade, becoming buyer to every seller and seller to every buyer, and manages margin and netting. *Bail-in*: imposing losses on a bank’s creditors (by conversion to equity or write-down) instead of a taxpayer *bail-out* [▶ L21.1 @ 1:32].

21.2 Shadow banking versus traditional banking

Zoltan Pozsar’s famous diagram, made at the New York Fed, shows loans funded by deposits through seven stages of securitization, slicing and dicing, wrapped by the Fed’s U-shaped embrace of liquidity facilities during the crisis: a Rube Goldberg machine, built piece by piece; much of it fell apart, but the next stage can be simpler [▶ L21.2 @ 1:06].

Stylized comparison. The traditional (Jimmy Stewart) bank takes retail deposits, makes retail loans, holds cash reserves and a capital buffer; beyond them the Fed is the liquidity backstop and the FDIC the capital backstop [▶ L21.2 @ 2:48]. The stylized shadow banking system, three entities instead of seven: RMBS tranching into high, mid and low tranches; the high tranche held by a shadow bank as collateral for repo funding; the repo bought by a money market mutual fund issuing “deposits” [▶ L21.2 @ 3:49]. Differences: it is wholesale (pension funds, foreign central banks, corporations; a “small” deposit is \$10 million), its assets are securities in the capital market, and it may be offshore, on a computer in the Cayman Islands [▶ L21.2 @ 4:53]. Seen through traditional banking it looks similar, deposits funding loans, but no government backstop is anywhere to be seen; the Fed and the FDIC said “good riddance” to offshore risk [▶ L21.2 @ 6:14].

Traditional bank		Shadow bank		MMMF	
Loans	Deposits	High tranche RMBS	Repo (RP)	Repo (RP)	“Deposits” (shares)
Reserves	Capital				

Figure 21.1: Traditional bank, backstopped by Fed and FDIC, versus a stylized shadow banking chain with no government backstop [▶ L21.2 @ 3:08].

HISTORY: The shadow banking map

Z. Pozsar, T. Adrian, A. Ashcraft and H. Boesky, “Shadow Banking”, Federal Reserve Bank of New York Staff Report 458, July 2010, includes a large fold-out map of the shadow banking system and of the Fed’s crisis facilities.
Reference: FRBNY Staff Report 458, 2010 (revised 2012).

21.3 Immature backstops

A private liquidity backstop. Invented over the last 30 years or so, the system had an immature, private liquidity backstop: liquidity puts held by the shadow bank and by the money fund, liabilities of traditional banks. Citibank wrote in contracts that if its offshore entities could not roll their repo, for any reason, it would lend them what they needed; other backstops were only reputational [► L21.3 @ 0:02]. All money funds were run, but only the Reserve Primary Fund failed, because it had no larger parent to back it [► L21.3 @ 1:43]. A private backstop works only if the backstopper is healthy; in a large crisis it becomes the traditional bank’s crisis, and then the government backstops kick in: the government was indirectly on the hook, perhaps without realizing it [► L21.3 @ 2:25].

A private solvency backstop. Even more immature: the shadow bank bought CDS on its high tranche from AIG; mid tranches went to pension funds and insurers, low tranches to credit hedge funds, which might buy CDS from an investment bank (think Lehman or Bear Stearns) making markets in CDS on all tranches; AIG, writing naked CDS on the highest tranche, “picking up nickels”, was the ultimate support [► L21.3 @ 3:06]. Lehman went, AIG went, prices went into free fall; through the liquidity backstop everything dropped onto the Fed’s balance sheet, which tripled more or less overnight [► L21.3 @ 6:10].

Shadow bank		AIG		Traditional bank	
High tranche	Repo	Capital	CDS on high tranche		Liquidity put
CDS on high tranche					
Liquidity put					

Pension fund / insurer		Credit hedge fund		Investment bank (CDS dealer)	
Mid tranche	DB pensions, annuities	Low tranche	Capital	CDS on high	CDS on mid, low
CDS on mid		CDS on low			

Figure 21.2: Immature private liquidity and solvency backstops of the shadow banking system [► L21.3 @ 0:22].

Exercise 21.1 (Liquidity put). A sponsored vehicle holds \$50 billion of AAA RMBS funded by three-month ABCP, with a liquidity put from its parent bank. Write the balance sheets before and after the ABCP fails to roll, and then after the bank borrows from the Fed.

21.4 The global dimension

What Mehrling had not adequately appreciated: shadow banking tapped a *global* funding market, the dollar funding markets, which are mature and have been tested many times [► L21.4 @ 0:00]. His slides call this “shadow banking as global banking” [Notes 21 p.7].

Development lending in dollars. A Korean bank borrows short term in dollars from a global bank (say Deutsche Bank), which issues short-term money market liabilities held by a “dollar bank”, perhaps a money market mutual fund: since the dollar is the world reserve currency, a country that wants to invest more than its domestic savings can tap these markets [► L21.4 @ 0:27]. Lending in won and borrowing in dollars means FX risk, hedged with FX swaps or dollar reserves, perhaps by the central bank as counterparty; someone must bear it [► L21.4 @ 1:18]. When the money stops there is a banking crisis and probably a currency crisis: the Asian financial crisis, from which “we learned something”, though it did not turn out so well for the Tigers [► L21.4 @ 1:50].

Shadow banking used the same system for dollar loans. RMBS tranching, the high tranche held by a shadow bank funded by ABCP, the ABCP held by a money fund: borrowing in dollars to lend in dollars, so no FX risk, the risk behind the Asian crisis; credit and duration risk remained, but those existed in Korea too [► L21.4 @ 2:20].

HISTORY: The Reserve Primary Fund

On 16 September 2008, the day after Lehman’s bankruptcy, the Reserve Primary Fund, which held \$785 million of Lehman commercial paper, “broke the buck” (net asset value below \$1 per share, \$0.97); on 19 September the US Treasury announced a temporary guarantee of money market funds. *References:* Financial Crisis Inquiry Commission, *Final Report*, 2011, ch. 18; US Treasury press release HP-1147, 19 September 2008.

Korean bank		Global bank		Dollar bank	
Domestic currency loans	Dollar funding	Dollar lending	Wholesale money market	Wholesale money market	“Deposits”
FX swaps					
\$ reserves					

RMBS		Shadow bank		MMMF	
Mortgage loans	Hi tranche	Hi tranche	ABCP	ABCP	“Deposits”
	Mid tranche				
	Lo tranche				

Figure 21.3: Shadow banking as global banking: the same dollar funding chain for Korean development lending and for US mortgages [► L21.4 @ 0:35][Notes 21 p.7].

KEY FORMULAS

A mature funding system, an immature risk transfer system [► L21.4 @ 3:08]: shadow banking tapped a mature global funding system for a new purpose, while the risk transfer piece (the CDS) was new, “invented on the fly”. It was easy to borrow, hard to transfer risk.

This is why the crisis was global: it hit the global money market at once. The spread between LIBOR (Eurodollar rates) and term Fed funds, which should be about equal for

the same term, blew out to about 100 basis points before Bear Stearns and 300 or more after Lehman and AIG, with the Fed apparently stabilizing it in between but nowhere near zero [► L21.4 @ 3:44]. The shadow banks were in Europe, funded through the Eurodollar market, and the crisis touched every major money market, Japan's too [► L21.4 @ 4:36]. The next slide is titled "From domestic to international LOLR" [Notes 21 p.8]: the central bank swap lines of lecture 14. For financial globalization the slides cite H. S. Shin, "Global Banking Glut and Loan Risk Premium" [Notes 21 p.11].

21.5 The evolution of modern finance: risk transfer

Shadow banking used recent developments of modern finance: securitization, which came with globalization (individual small loans are not transferred across borders, but packaged bonds with a market price are bought by foreign asset managers); global capital markets; derivatives markets for moving risk, of which tranching was an awkward approximation [► L21.5 @ 0:04].

An idealized shadow banking system. A shadow bank holds RMBS funded in the money market and strips out all the risk with derivatives (interest rate swaps, FX swaps, CDS): what is left is a quasi-Treasury bill, which can be financed in the money market without price risk. On the other side, an asset manager gets its customers' risk exposure entirely through derivatives, without holding risky assets, equivalent to holding RMBS outright: Black's 1970 vision. That is what shadow banking was trying to do, and probably what its future looks like [► L21.5 @ 1:06].

21.6 What is shadow banking?

Definition 21.2 (New concepts: shadow banking as market-based credit). Shadow banking is *money market funding of capital market lending*, on whatever balance sheet it sits (a bank's, as at UBS, a non-bank's, or a special vehicle's), with three features [► L21.6 @ 0:03]:

1. global (dollar) wholesale funding of local lending;
2. market prices on both sides, for money and for capital, visible and changing over time;
3. market makers, dealers quoting bid–ask spreads according to inventories and risk tolerance: that is where the prices come from.

The existing literature starts from traditional banking and stresses the "shadow": banking by non-banks, evading regulation, without a direct government backstop, something a little fishy; so it would move the same regulations and backstops over, make it a bank [► L21.6 @ 2:06]. Mehrling disagrees: there was plenty of regulatory arbitrage and fraud, but the fundamental forces are financial globalization and the revolution in modern finance; removing the fraud will not remove shadow banking [► L21.6 @ 3:28].

A world of only shadow banking. His project since *The New Lombard Street*: imagine a world with only shadow banking and ask how to regulate it. It will never be literally true, but it frees us from Jimmy Stewart habits of thought; extending old regulations creates

new obstacles that modern finance will get around, “a disastrous way of thinking about the problem” [► L21.6 @ 4:10].

Capital funding bank		Global money dealer		Asset manager	
RMBS	Money market funding	Money market loans	Deposits	Deposits	Capital CDS, IRS, FXS
CDS, IRS, FXS					

Derivative dealer	
CDS, IRS, FXS (with asset manager)	CDS, IRS, FXS (with capital funding bank)

Figure 21.4: Mehrling’s idealized market-based credit system: the dealers stand between the asset manager and the capital funding bank in the funding market and in the risk market [► L21.6 @ 5:37].

The dealers are the key. The global money dealer, borrowing and lending three-month deposits, is where the price of money is set; the derivative dealer, on the other side of all the derivatives, is the risk market [► L21.6 @ 6:18]. Today these balance sheets are the big investment banks’ swap books; central counterparties would move those positions (Lehman’s, Bear Stearns’s) into one place where they can be seen and netted. The money funding system is relatively mature; the risk transfer system is very immature, and this is the part to clean up [► L21.6 @ 7:20]. If the dealers stop making markets the whole thing collapses; so regulation focused on the capital funding bank (“let’s have long-term funding”) looks in the wrong place [► L21.6 @ 8:44].

Where is the capital? In the idealization the only capital is the asset manager’s: the dealers are perfectly matched, the hedges perfect, no liquidity reserves anywhere. If risk is properly transferred, the capital backing the capital funding bank sits on the asset manager’s balance sheet, not the bank’s [► L21.6 @ 9:45]. The model abstracts from retail depositors, traditional banks, bad underwriting, reserves and speculative dealers, to focus on system interlinkages, normal liquidity risk and tail backstops [► L21.6 @ 11:07].

21.7 Backstopping the market makers

In a crisis, the backstop: a liquidity put to the global money dealer (lender of last resort, familiar from Bagehot, 1873) and one to the derivative dealer, a backstop for capital market prices (dealer of last resort, the new bit), both contingent liabilities of a central bank or a consortium of central banks [► L21.7 @ 0:03].

The Fed as a dealer in swaps. The Fed’s H.4.1 for 15 December 2011 [Notes 21 p.16]: about \$1.7 trillion of long-term Treasuries and almost \$1 trillion of MBS and GSE (Fannie and Freddie) securities, funded by about \$1 trillion of currency and \$1.6 trillion of excess reserves (against \$50 billion of reserves in normal times) [► L21.7 @ 1:47]. Now add the same items to both sides [► L21.7 @ 2:50]:

Federal Reserve (restated, \$ trillion)			
MBS, GSE	0.9	[T-bonds]	0.9
T-bonds		[T-bills]	2.6
(1.7 + 0.9)	2.6	Currency and	
[T-bills]	2.6	reserves	2.6

Figure 21.5: The Fed’s balance sheet, 15 December 2011, restated as exposures: risky securities against T-bonds (a CDS), T-bonds against T-bills (an IRS), T-bills against overnight money (a funding swap) [► L21.7 @ 3:10].

The bracketed pairs are swaps, parallel loans: risky securities against T-bonds is a CDS; long against short bonds an interest rate swap; T-bills against currency and reserves, three-month against overnight money, like the overnight index swap that blew out in the crisis [► L21.7 @ 3:53]. The out-of-the-money liquidity puts came into the money and onto the Fed’s balance sheet: in parallel-loan terms about \$6.4 trillion of notional swap exposure, the Fed acting both as global money dealer and as global derivative dealer [► L21.7 @ 4:36].

INTUITION: A Bagehot moment

What a central bank would have to do to backstop a collapsing shadow banking system is exactly what the Fed did: evidence that shadow banking is the centre of the system and traditional banking the edges. “We’re living in a Bagehot moment”: the role of the central bank is shifting, and monetary policy, bank regulation and inflation targeting, designed for a previous banking system, must be rethought from scratch. As Bagehot said, money does not manage itself [► L21.7 @ 5:37].

Exercise 21.2 (Restating the Fed). Using the December 2011 figures, redo the restatement of the Fed’s balance sheet and check the \$6.4 trillion of notional swap exposure. Which swap corresponds to QE in mortgage securities and which to QE in Treasuries?

21.8 Regulating systemic risk

Start from the inherent instability of credit, not from a self-equilibrating system [► L21.8 @ 0:02][Notes 21 p.18]. Earlier authors located its endogenous sources: Hawtrey coined “the inherent instability of credit” for the British banking system (1919); Minsky, *Stabilizing an Unstable Economy*, focused on business credit; Adrian and Shin, “Liquidity and Leverage”, on dealer balance sheets [► L21.8 @ 0:43]. Mehrling uses Treynor’s model: dealers create market liquidity by moving order-flow imbalances onto their own balance sheets, positions that must be funded [► L21.8 @ 1:24].

Boom. Asset managers pile up cash and seek credit risk exposure faster than capital funding banks absorb it; dealers absorb the imbalance, which pushes money market rates

down and risk premia down: exactly what makes setting up capital funding banks very profitable. Prices are distorted and more shadow banking is built than matched-book equilibrium models would suggest [► L21.8 @ 2:05].

Bust. Once overbuilt, the asset managers' flows fall short of the shadow banks' funding needs; the dealers' imbalance reverses, rates and risk premia rise, shadow banks look unprofitable; losses shrink the dealers' capital and financing limits, leaving excess exposure to liquidate, and in Treynor's model liquidation happens at the outside spread: the central bank [► L21.8 @ 4:08]. A Hawtrey–Minsky story updated for modern times, “a starting place” [► L21.8 @ 5:53].

READING: Adrian and Shin, “Liquidity and Leverage” (FRBNY Staff Report 328, 2008, rev. 2010)

The course file (materials/readings/L21-sr328.pdf) is the dealer-balance-sheet version of the Hawtrey–Minsky story.

- With balance sheets marked to market, asset price changes show up at once as changes in net worth, and intermediaries respond by adjusting the size of their balance sheets.
- For US security dealers and brokers, *leverage is procyclical*: it is high in booms and low in busts. Leverage and total assets grow together, and the margin of adjustment is repo.
- Changes in dealer repos forecast changes in market risk, measured by innovations in the VIX.
- “Aggregate liquidity can be seen as the rate of change of the aggregate balance sheet of the financial intermediaries.”

21.9 Normal times: collateral and payments

In normal times the issue is the payment system: the survival constraint, “liquidity kills you quick”. But besides payment flows there are collateral flows, the subject of the paper by Singh and Aitken [► L21.9 @ 0:03][Notes 21 p.25].

The model with numbers. The capital funding bank holds RMBS of 100, funded by 100 of secured money market funding, the collateral perhaps passing through the money dealer to secure the asset manager's deposit; CDS and IRS are worth 0 at inception everywhere; the asset manager has plenty of deposits as cash collateral [► L21.9 @ 1:06]. No counterparty risk by construction: the bank is exactly hedged, and the asset manager's capital (the pensioners') can absorb even a total loss [► L21.9 @ 2:52].

A temporary fall: collateral. RMBS fall to 90, CDS rise to 10. Collateral must flow from the asset manager through the derivative dealer; and the money market funding of 100 is now backed by only 90 of RMBS, so the CDS, or the collateral behind it, must be usable for the funding. Otherwise the asset manager may pull its deposits from a money dealer lending against inadequate collateral, and a liquidity spiral can start from collateral problems alone [► L21.9 @ 4:14]. The central bank can help as lender of last resort to the dealers, or as dealer of last resort in RMBS or CDS, putting a floor under the collateral market [► L21.9 @ 6:20].

HISTORY: Hawtrey and the instability of credit

Ralph Hawtrey (1879–1975), a Treasury economist, argued in *Currency and Credit* (1919) that credit is inherently unstable, a cumulative process that the banking system must restrain. Hyman Minsky's *Stabilizing an Unstable Economy* appeared in 1986.

References: R. G. Hawtrey, *Currency and Credit*, Longmans, 1919; H. P. Minsky, *Stabilizing an Unstable Economy*, Yale UP, 1986; T. Adrian and H. S. Shin, “Liquidity and Leverage”, *Journal of Financial Intermediation* 19(3), 2010.

Definition 21.3 (New concepts: *rehypothecation*). *Rehypothecation*: reusing collateral received from one party as collateral for one's own borrowing (from *hypothecation*, Greek *hypothēkē*, a pledge) [► L21.9 @ 7:02]. Making it illegal, as some propose, would block the collateral flows through the dealers and push the work onto the central bank [► L21.9 @ 7:23].

READING: Singh and Aitken (2010), IMF WP/10/172

What the paper says (materials/readings/L21-Singh_Aitken_wp10172.pdf):

- **The rule.** A hedge fund's collateral posted to its prime broker can be reused by the broker for its own funding.
- **US cap.** In the US, Regulation T caps rehypothecation at 140% of the customer's debit balance. Owing \$200, a customer lets the broker reuse up to \$280.
- **UK.** In the UK there is no cap and there are no customer protection rules, so the UK "provides a platform for higher leveraging".
- **Collapse after Lehman.** Collateral that the large US broker-dealers were permitted to repledge fell from about \$4.5 trillion at end-2007 to \$2.1 trillion at end-2009.
- **Churning factor.** About \$1 trillion of hedge fund securities was rehypothecated at end-2007. About 40% of the \$10 trillion of pledgeable collateral received by the ten largest global banks came from hedge funds. The "churning factor", or re-use, of collateral is therefore about 4.
- **Conclusion.** Counting rehypothecation, the shadow banking system was "at least 50 percent bigger than documented so far".

A permanent fall: payments. If the loss is permanent, the system deleverages: RMBS 90, funding 90, CDS back to 0. The asset manager pays 10 to the derivative dealer, who pays the capital funding bank, which repays 10 of money market funding: a circle of payments, and if someone says "I'm not paying until I get paid", everything stops [► L21.9 @ 7:43].

COURSE NOTE

The paper is given in the captions as by "Aiken Singh". It is M. Singh and J. Aitken, "The (sizable) role of rehypothecation in the shadow banking system", IMF Working Paper 10/172, July 2010 [► L21.9 @ 0:44].

Capital funding bank		Derivative dealer		Asset manager	
RMBS: 100, 90	Funding: 100, 90	CDS: 0, 10, 0	CDS: 0, 10, 0	Deposits: 100, 90	Capital CDS: 0, 10, 0

Figure 21.6: A fall in RMBS from 100 to 90: first collateral flows, then, if permanent, a circle of payments of 10 [► L21.9 @ 4:35].

Outside spread, not inside. Collateral flows and payment flows are both key to keeping normal fluctuations from becoming crises. Having become dealer of last resort, the central bank's job, as in Bagehot's day, is to set up the system so that it is not called upon every day, because of moral hazard: "I'm willing to be outside spread, but not inside spread" [► L21.9 @ 9:27].

Exercise 21.3 (Collateral and payments). In the 100/90 example, write all balance sheets after the temporary fall, assuming collateral flows work, and after the permanent fall. Which payment, if refused, stops the circle?

21.10 Private backstop, and public

To keep the central bank from acting daily, make the dealers robust, especially the new one, the derivative dealer: matched-book dealers need liquidity reserves, speculative (proprietary) dealers need capital to absorb losses [► L21.10 @ 0:04].

KEY FORMULAS

Four ideas for regulating the shadow banking system of the future (“they might change tomorrow”) [► L21.10 @ 1:27][Notes 21 p.30]:

1. focus on the dealers, money dealers and derivative dealers, not on the shadow banks;
2. distinguish matched-book from proprietary (speculative) dealers;
3. liquidity for the matched-book dealers, capital for the proprietary ones (“liquidity kills you quick”);
4. the survival constraint concerns collateral flows as well as payment flows: a system of secured funding; a collateral call by Goldman and Société Générale killed AIG, perhaps not insolvent on fundamental valuations [► L21.10 @ 2:50].

“I think we’re at a Bagehot moment, 1873. We’re building it from scratch” [► L21.10 @ 3:53].

How to keep dealers from leaning on the Fed? A student asks. First, adequate capital for the speculative dealers, so their financing limits do not shift (and, since the line between matched-book and speculative dealing is not sharp, probably for matched-book dealers too) [► L21.10 @ 4:34]. Second, the central bank should create the outside spread, not the inside spread: “insure freely but at a high premium”, put a floor under capital markets “at 80 cents on the dollar”, uneconomic except in need. Some programs, the self-liquidating ones, did this; mostly the Fed played catch-up, forced to act on weekends [► L21.10 @ 5:35]. The modern analogue of Bagehot’s “lend freely at a high rate against good security” is the outside spread: be away from the market, but act boldly when you are; make it a known principle, expensive for those who come, who must come “begging” [► L21.10 @ 6:36]. There will be another crisis: the inherent instability of credit [► L21.10 @ 7:58].

Exercise 21.4 (Outside spread). Compare Bagehot’s rule with “insure freely at a high premium”. Design a central bank backstop bid for AAA RMBS at a price that is an outside spread, and explain why it should rarely be used.

Key formulas and ideas

KEY FORMULAS

Shadow banking	money market funding of capital market lending: global funding, market prices on both sides, dealers
Immature backstops	private liquidity puts (traditional banks), private solvency backstop (AIG CDS)
Ideal system	capital funding bank + asset manager linked by a money dealer and a derivative dealer
Fed, Dec. 2011	restated as CDS + IRS + funding swap: dealer of last resort in money and risk
Instability	order-flow imbalances on dealers' balance sheets: boom (low rates, low premia), bust (outside spread = central bank)
Normal times	collateral flows and payment flows must not stop
Principles	regulate dealers; liquidity for matched book, capital for proprietary; central bank as outside spread

Chapter 22

Touching the Elephant: Three Views

SOURCES FOR THIS CHAPTER

Segment	Length	Content
L22.1 FT: Future of banking	4:46	Banking union, cross-border fragmentation, resolution of global banks.
L22.2 Three world views	15:14	Money, economics and finance views; Black's 1976 manifesto.
L22.3 Economics View: Commodity Exchange	8:54	Relative prices, the quantity theory pasted on; Fisher and IS-LM.
L22.4 Finance View: Risk	15:41	CAPM in balance sheets; Black's banking in a world without money.
L22.5 The education of Fischer Black	4:52	"Noise" (1985); Treynor's dealer model.
L22.6 Steps from finance view to money view	7:45	Market liquidity, funding liquidity, the central bank.
L22.7 A money view of economics and finance	5:12	Selective control over liquidity allocation.

Files. Transcripts `materials/transcripts/L22-P*.txt`; Mehrling's notes [Notes 22 p.1].

Motivation

Lecture 1 promised a course that would not be applied microeconomics (utility functions for banks) nor applied macroeconomics (what lies behind the LM curve), but would start from the world, the actual operations of money markets, and learn inductively [► L22.2 @ 0:03]. The last lecture connects that money view to the economics and finance that students know: three blind men touching the same elephant.

22.1 FT: the future of banking

CASE STUDY: FT, December 2012: three data points in one day

1. "Debate rages on how to set up bank supervisor": the ECB, with no supervisory experience, no dedicated staff and hostility from some of its future charges, is to supervise the eurozone's 6000 banks in a banking union [► L22.1 @ 0:07]. Mehrling's reading of the unspoken resistance: common collateral rules would end the sweetheart deals by which national central banks lean on local banks to

buy their government's bonds [► L22.1 @ 1:09];

2. "Cross-border lending falls sharply": US banks stop lending to Europe and separate their European and US books; core-periphery lending within Europe shrinks: a market-driven "return to autarky" [► L22.1 @ 2:10];
3. an op-ed by Martin Gruenberg (FDIC chairman) and Paul Tucker (Bank of England deputy governor), "Global banks need global solutions when they fail": the resolution authority would take control of the parent, impose losses on shareholders and unsecured creditors, and remove management. They say they will no longer support institutions, not yet what they will do instead [► L22.1 @ 2:51].

Details from the two articles in the course files (*FT*, 10 December 2012; materials/readings/L22-*.pdf):

- **Cross-border lending** (Claire Jones, BIS quarterly data). Cross-border claims fell by \$575bn (1.9%) between the end of March and the end of June 2012, and interbank claims by \$581bn (3.1%). This was the second sharpest retrenchment since early 2009, "mainly the result of reduced inter-office positions". Dollar interbank lending fell by the most since the fourth quarter of 2008.
- **Global resolution** (Gruenberg and Tucker). Of the world's 28 G-SIFIs, 12 are headquartered in the US or the UK. In a failure, the resolution authority would take control of the parent and wipe out shareholders. It would convert part of the unsecured debt into equity to recapitalise the group, and shrink, break up or close the businesses responsible.

22.2 Three world views

The money view: the present determines the present. The course has focused on now: the day's clearing, with past investments and future promises fixed as data. Cash outflows (discipline) meet cash inflows (elasticity); if outflows exceed inflows people borrow from the future and drive up the money rate of interest [► L22.2 @ 0:45]. It is a flow view; stocks sit unchanged in the background [► L22.2 @ 3:12].

The economics view: the past determines the present. Output today is a function of the capital stock accumulated by past investment, sunk and unchangeable [► L22.2 @ 4:19]. Very much Adam Smith's view: worry about parsimony and profligacy, princes plundering the capital stock, slow "turtle" growth by accumulation [► L22.2 @ 5:21]. Through most of the history of economic thought the money view, natural to bankers and central bankers, was a minority view against this one, which seems more natural and more ethical (the virtue of saving) [► L22.2 @ 6:24].

The finance view: the future determines the present. After World War II a new view: the key variable is the *price* of capital, the expected discounted value of future cash flows, not the labour that built it. "Radically subversive": if capital is not productive it is worth zero; and an idea about the future, once capitalized, gives purchasing power today, the tech bubble mentality [► L22.2 @ 7:49]. Promises based on those dreams are promises to pay cash: back to the money view [► L22.2 @ 10:24].

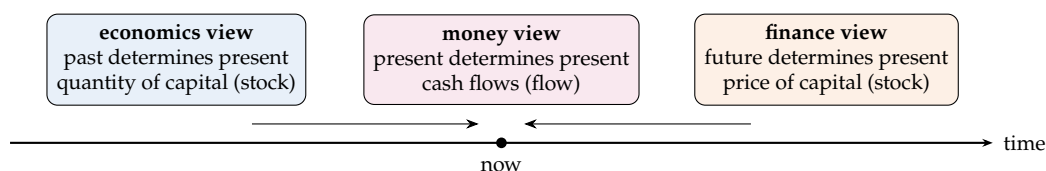


Figure 22.1: Three views of the same elephant [► L22.2 @ 3:33].

Past investments generate cash inflows, the way they were financed generates cash outflows; dreams about the future require outlays greater than inflows; either someone takes the other side or the dreams do not get their chance [► L22.2 @ 10:45]. Economics and finance are stock views and mirror images of each other; the money view is a flow view [► L22.2 @ 11:25]. Mehrling's assertion: the big intellectual battle of the last 30 years has been economics versus finance, settled by a *modus vivendi* (finance in the business school, economics in the economics department, each a minor subfield of the other); neither has much room for money. His life project: make the money view talk to the finance view, as in the 19th century it talked to economics [► L22.2 @ 12:06].

READING: F. Black, "What a Non-Monetarist Thinks"

The challenge that started Mehrling on this road: in a country with a smoothly working financial system, Black argued, the government cannot control the money stock; money responds passively to the private sector's demand, and the Fed's monetary policy pronouncements are "a charade" with almost no effect on output, employment or inflation. If true, "we have just been wasting our time" this semester [► L22.2 @ 13:27].

22.3 The economics view: commodity exchange

The standard theory of value: apples and oranges, a production possibilities frontier, a budget line tangent to an indifference curve, its slope the relative price p_O/p_A [► L22.3 @ 0:01]. Two observations: there is no money (a barter relation between goods), and only a relative price is determined, not the price level [► L22.3 @ 1:08].

KEY FORMULAS

The quantity theory, pasted on [► L22.3 @ 1:48]:

$$MV = p_O q_O + p_A q_A = PQ.$$

Read left to right, money determines the price level; read right to left (the banking school, money as endogenous credit), transactions determine how much money is needed.

The velocity equation always had an uneasy relationship with the theory of value: an accounting identity turned into a theory, the closest the field came to science [► L22.3 @ 3:11]. Irving Fisher's breakthrough: treat the two goods as consumption at dates 1 and 2, so the relative price involves $1/(1+r)$, and extend the quantity equation to financial assets (the transactions version): more money means higher prices of goods and of assets, i.e. lower interest rates. From this tradition came IS-LM: in the short run money affects output and interest rates, in the long run only the price level [► L22.3 @ 3:52].

COURSE NOTE

In class the piece is dated 1976 and said to be published. It was reprinted in F. Black, *Business Cycles and Equilibrium*, Basil Blackwell, 1987; the original 1976 date is as given in class and in Mehrling's biography of Black.

♦ **BEYOND THE CORE: Attempts to put money into general equilibrium**

The notes list three approaches, following Frank Hahn's complaint (*Money and Inflation*, 1982) [Notes 22 p.3][Notes 22 p.4]:

1. overlapping generations, money as a store of value (N. Wallace, "The Overlapping Generations Model of Fiat Money", 1980);
2. search and matching (N. Kiyotaki and R. Wright, "On Money as a Medium of Exchange", 1989);
3. Walrasian general equilibrium with transactions costs (R. Starr, "Monetary General Equilibrium with Transactions Costs", 2005).

All study commodity exchange with no financial asset but money; their results do not generalize to a world with financial assets, where money, paying no interest, is a dominated asset. The most successful response to finance so far, Michael Woodford's *Interest and Prices*, synthesizes the Taylor-rule consensus but looks only at goods prices, not asset prices: what remains is to do for the modern model what Fisher did for the 19th-century one, link it to finance [Notes 22 p.8].

INTUITION: A patch

The same structure extends to n periods, intertemporal Arrow–Debreu general equilibrium. General equilibrium theorists, Frank Hahn above all, complained that it has no place for money, and were unhappy with Fisher's patch. Post-war monetary economics can be told as attempts to squeeze money in (search theory and others); Mehrling bet that they would not get far, and 25 years later is glad he chose finance instead [► L22.3 @ 6:58].

22.4 The finance view: risk

CAPM. (Say "cap-M": "CAPM model" says model twice.) Every capital asset is characterized by its expected return and its volatility; finance domesticates risk, which general equilibrium, bringing in time without risk, did not; it is probabilistic risk, not true uncertainty, an assumption, but a step ahead [► L22.4 @ 0:03]. Portfolios of securities form a frontier; investors like return and dislike risk; the tangency of the line from the risk-free rate is the market portfolio, held by everyone, combined with more or less of the risk-free asset according to risk aversion; beyond the tangency investors borrow [► L22.4 @ 2:10].

KEY FORMULAS

CAPM [► L22.4 @ 5:19]:

security market line: $\mathbb{E}[R_i] = R_f + \beta_i (\mathbb{E}[R_m] - R_f),$

capital market line: $\mathbb{E}[R_p] = R_f + (\mathbb{E}[R_m] - R_f) \frac{\sigma_p}{\sigma_m}.$

Definition 22.1 (New concepts: capital asset pricing model). The CAPM (Treynor, Sharpe, Lintner, Mossin, 1960s) prices each asset by its covariance with the market portfolio (β_i) times the market risk premium; "capital asset" because it values the claims to capital [► L22.4 @ 0:03].

CAPM in balance sheets. Two people with wealth 100 each: the risk-tolerant one holds 150 of the market portfolio and borrows 50; the risk-averse one holds 50 of the market portfolio and 50 risk-free [► L22.4 @ 7:03]. Black's twist, not standard in teaching CAPM: the 50 lent and the 50 borrowed are the same 50, inside borrowing intermediated by a bank,

loans on one side, deposits on the other [► L22.4 @ 8:48].

Risk tolerant		Bank			Risk averse	
Market portfolio 150	Loan 50	Loan 50	Deposit 50	Market portfolio 50	Net worth 100	
	Net worth 100			Deposit 50		

Figure 22.2: Black’s “banking in a world without money”: the risk-free asset is a bank deposit funding a loan to the risk-tolerant investor [► L22.4 @ 7:45].

READING: Black, “Banking and Interest Rates in a World Without Money” (1970)

The course file (materials/readings/L22-World_Without_Money.pdf) is the reprint as ch. 1 of Black’s *Business Cycles and Equilibrium* (1987), from the *Journal of Bank Research*, Autumn 1970.

- **The thought experiment.** Banks may offer checking accounts on any terms and pay interest on demand deposits; there are no reserve requirements. Then there is no reason for open market operations, and no reasonable definition of the quantity of money. Neither the quantity theory nor liquidity preference applies (p. 1).
- **Predecessors.** Black sides with Vickrey, Tobin, Gurley–Shaw and Patinkin: the price level is indeterminate. He rejects Johnson and Friedman–Schwartz, who predicted an unlimited rise in prices (pp. 1–4).
- **Accounts.** Every individual, business or government has one account that may be positive (a deposit) or negative (a loan) up to a maximum set by the bank. Deposits earn a “standard wholesale money rate”, and an active inter-bank market evens out imbalances (pp. 4–6).
- **No government bonds.** The government simply borrows from the banks like everyone else. Its borrowing is limited by Congress; “massive government spending that is not balanced by taxation would cause the financial system to break down” (p. 6).
- **No quantity of money.** Any candidate (positive balances, net balances, unused credit lines) is either arbitrary or equal to bank capital or to total wealth. Hence “the quantity of money cannot affect national income, employment, or the rate of inflation, because it does not exist” (pp. 7–8).
- **Currency.** Currency is issued on demand and so is passive (p. 9).
- **Evolution of payment.** Black then builds the system up step by step: commodity money, then shares of a standard portfolio, then personal notes, then banks that guarantee notes (pp. 9–11).

Two conclusions. The bank intermediates between risk-averse and risk-tolerant; with diversification and limits to lending its loans might be nearly risk-free [► L22.4 @ 9:34]:

1. the risk-free rate is the rate that clears this loan market, and the risk premium is determined in the capital market too: no degree of freedom is left for the Fed to peg the short rate, hence “a charade” [► L22.4 @ 10:35];
2. if the price of capital rises, both gain, the risk-tolerant three times as much: it wants to borrow more and buy more of the market portfolio, while the risk-averse wants to sell risky assets and lend more. The bank’s balance sheet grows on both sides: the money supply must be endogenous, rising with asset prices, for asset markets to clear. Money is passive; read the quantity theory from right to left [► L22.4 @ 11:57].

This made Black the *bête noire* of monetarists and economists; he did not mind being a minority of one [► L22.4 @ 15:06].

The notes complete Black’s picture [Notes 22 p.8][Notes 22 p.9]. Once people use banks for riskless borrowing and lending, they inevitably use them for payments too: a payment from a negative to a positive account expands credit, from positive to negative contracts it, the elasticity of lecture 5. But in the finance view that elasticity comes from the elasticity of credit in capital markets, not the other way round: Black starts at the bottom of the hierarchy, the money view at the top, and both arrive at the intertwining of payments and credit. The big difference: for Black the only constraint is wealth, allocated by risk tolerance in efficient markets; nothing about the survival constraint or dealers. (Black liked to say there is nothing we can call the quantity of money, hence nothing to control [Notes 22 p.8].)

Exercise 22.1 (Black’s banking world). In the two-person example the price of the market portfolio rises by 20%. Compute both net worths, and the new loans and deposits if each restores its original ratio of risky assets to net worth. By how much does the money supply grow?

22.5 The education of Fischer Black

Black taught at Chicago and MIT, then joined Goldman Sachs, remarking that the world looks different from the banks of the Hudson than from the banks of the Charles [► L22.5 @ 0:04]. As president of the American Finance Association he gave the address “Noise” at its New York meeting on 30 December 1985: an efficient market might be defined as one in which price is “within a factor of 2 of value”, and by that definition almost all markets are efficient almost all of the time (at least 90%) [► L22.5 @ 0:25]. A bombshell: prices need not be closely related to fundamentals [► L22.5 @ 1:07].

INTUITION: Treynor added to CAPM

Black, who never took an economics or finance course, had been a strong believer in efficient markets, and Chicago hired him. Mehrling argues, in his biography, that the “factor of two” came from his friend Jack Treynor’s dealer model: prices are set by dealers quoting bid–ask spreads, and when inventories are pushed far to one side prices can deviate as far as the outside spread, without outsiders knowing unless they see dealers’ inventories. Ironically Black learned CAPM itself from Treynor, whose correct unpublished version preceded Sharpe’s. The dealer model is the beginning of a bridge between finance and the money view [► L22.5 @ 1:28].

In the notes, Black when he died still thought of the constraint on market liquidity as dealer *capital*, not dealer *funding*; the collapse of LTCM in 1998, whose relative-value strategy he had warned about, “loading up on liquidity risk” and unable to refinance

HISTORY: Black’s first paper

Fischer Black’s first publication was “Banking and Interest Rates in a World without Money: The Effects of Uncontrolled Banking”, *Journal of Bank Research* 1, 1970, pp. 8–20, describing in words a world where banks intermediate between borrowers and lenders and money has no special role. Reference: P. Mehrling, *Fischer Black and the Revolutionary Idea of Finance*, Wiley, 2005.

when liquidity dried up, taught the second lesson. “The education of Fischer Black had proceeded through Lesson 2, but not yet Lesson 3”; and Lesson 3 implies that “monetary policy is not a charade. The answer to the challenge of finance is the money view” [Notes 22 p.9][Notes 22 p.10].

By his death in 1995 Black had moved far from efficient markets, though not all the way to the money view; Mehrling imagines him in the audience: “you came part of the way; now the next logical step is here” [► L22.5 @ 4:13].

Exercise 22.2 (A factor of two). Using Treynor’s model, explain how a dealer’s price can deviate from value as far as the outside spread, and what determines the width of that spread in an equity market and in a money market.

22.6 Steps from the finance view to the money view

Perfect liquidity, buying and selling at the efficient price, is an assumption that liquidity is a free good; in reality a dealer takes imbalances onto his balance sheet, and trading itself moves prices, not just fundamentals [► L22.6 @ 0:03].

KEY FORMULAS

Three steps [► L22.6 @ 1:25]:

1. market liquidity (buying or selling fast, in volume, without moving the price) depends on the dealer system: it is not free, and prices deviate from fundamentals;
2. dealers’ ability to make markets depends on their funding liquidity (and capital);
3. the ultimate source of funding liquidity is the central bank.

Funding, not just capital. The finance literature studies how dealers’ capital limits market making; but losing capital would not matter if dealers could borrow. The money view stresses funding liquidity, which can vanish even if capital does not: “I’m scared, I’m not going to fund you”, and the financing limits close in. Black surely got to step one and probably to step two [► L22.6 @ 3:13].

The third step, the one Black would have resisted most, is a long way from “money can only be passive”. But it is not all the way to IS–LM and fine-tuning: in a financially developed economy markets and dealers play a large role in money market rates, and the central bank changes them essentially by being a dealer, taking pressure onto its own balance sheet, which works because its liabilities are money, better than dealers’ [► L22.6 @ 5:21]. The hierarchy of money is antithetical to the finance view, where assets differ only by price and risk, “all flat”; the money view is hierarchical, the best money, the next best, then securities: where lectures 1 and 2 began [► L22.6 @ 7:06].

22.7 A money view of economics and finance

Banks, or money dealers more generally, can relax the survival constraint of anyone who comes to them, but need not: they sell access to their balance sheet [► L22.7 @ 0:04]. To the debtor bound by “the dead hand of the past” the banker may say “pay me tomorrow, I will carry you”, taking a liquidity risk the debtor cannot bear, or “no way, you’re out of

HISTORY: Black and Treynor

Black (1938–1995), PhD in applied mathematics at Harvard, met Jack Treynor at Arthur D. Little in the 1960s; Treynor’s manuscript “Toward a Theory of Market Value of Risky Assets” (1962) anticipated the CAPM of W. F. Sharpe (1964). Black taught at Chicago (1971–75) and MIT (1975–84) and joined Goldman Sachs in 1984. “Noise” was published in the *Journal of Finance* 41(3), July 1986. References: F. Black, “Noise”, *Journal of Finance* 41, 1986, 529–543; P. Mehrling, *Fischer Black and the Revolutionary Idea of Finance*, 2005.

business”; to the entrepreneur with a new app, needing cash outflows before any inflow, the banker may lend or refuse [► L22.7 @ 0:45].

KEY FORMULAS

Selective control over liquidity allocation [► L22.7 @ 1:46]: the banking system decides who gets spending power, who rolls over debt, who issues debt in the first place; by endorsing others’ liabilities it turns them into means of payment. It chooses discipline or elasticity, for individuals and the whole economy; the central bank influences it. This is where the money view connects with what economics cares about (investment) and what finance cares about (asset prices).

Economics and finance abstract from money as the oil that greases transactions, and end up seriously incomplete; the money view does not contradict them, but “the present determines the present” is a genuine aspect of the system, with real effects in crises and in normal times [► L22.7 @ 3:31]. It has always been a minority view and may remain one, “but if you want to understand money markets, it’s the point of view you want to be grappling with” [► L22.7 @ 4:34].

INTUITION: The future of banking

The notes close with an image [Notes 22 p.10][Notes 22 p.11]: the dead hand of the past and the ephemeral dream of the future meet in the survival constraint of the present. Banks let agents with obligations they cannot now meet put off the problem, and agents with brilliant plans they cannot now finance issue obligations; in doing so they make payments the agents cannot (yet) make. “The essential core of the banking function is the selective bearing of liquidity risk”: as long as there is liquidity risk, banks have a crucial allocative role. Regulation is about the externalities, the ways the private profit motive in bearing liquidity risk deviates from the social good: the more risk banks bear, the more they charge for more, until they will bear no more and the system breaks down. Central banks can help as lender and dealer of last resort, and so also by intervening earlier, when private and social begin to diverge.

Key formulas and ideas

KEY FORMULAS

Three views	economics: past → present (quantity of capital); finance: future → present (price of capital); money: present → present (cash flows)
Economics	relative prices from GE; $MV = PQ$ pasted on; Fisher → IS–LM
CAPM	$\mathbb{E}[R_i] = R_f + \beta_i(\mathbb{E}[R_m] - R_f)$; Black: risk-free asset = bank deposit, money endogenous and passive
Noise	price within a factor of 2 of value: the dealer’s outside spread
Three steps	market liquidity ← dealers ← funding liquidity ← central bank
Money view	hierarchy of money; banks’ selective control over liquidity allocation

Exercises

Standalone exercises follow the section they practise; connected or integrative ones are at the end.

Exercise 22.3 (Reading the quantity equation). Explain the difference between reading $MV = PQ$ from left to right and from right to left, and which reading fits Black's world, the banking school and the money view.

Exercise 22.4 (Three views of one event). Describe the collapse of AIG in September 2008 from the economics view, the finance view and the money view. Which features does each view see, and which does it miss?